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Effect of Structural Changes in Marine Derived Sulphated Polysaccharides on Growth Factor Binding and Cellular Responses

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Submitted in total fulfilment of the requirements of the degree of

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Department of Biomedical Engineering

Faculty of Engineering and Information Technology

The University of Melbourne

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COVID-19 Impact Statement

My PhD project was markedly disrupted by the COVID-19 pandemic.

From March 2020 to June 2020, there was no lab access due to lockdown restrictions in Melbourne. This caused a significant delay in my progress since my project is 100 % laboratory based. At this time, I was in the stage before the confirmation of my PhD candidature, and having no lab access created a lot of difficulties for me regarding learning experimental techniques and developing experimental protocols. Moreover, at this stage of my candidature I had not collected a complete set of data for a thesis chapter/publication to carry out analysis on and write up and hence further data collection was required.

From June 2020 to January 2021, I was able to go into the Biomedical Engineering (BME) PC2 laboratory which was the main lab I worked in but with limited capacity due to physical distancing requirements. During this period, only 5 to 7 people were allowed at a time in the lab that 20 to 25 people utilise, and it is also their main experimental lab. In addition, the time spent in the lab over this period was limited to no more than 4 hours per day. This allowed me to conduct some experiments in the BME PC2 lab, however I still had no access to required instruments located within external facilities, such as the Nuclear Magnetic Resonance (NMR) Facility at Bio21 Institute, and the Elemental Analysis instrument at the Materials Characterisation and Fabrication Platform, for several reasons: they were open to existing users only; there was a shortage of staff running testing services; and they prioritised COVID research at that time. Training opportunities therefore could not be provided, which caused a delay in the experimental progress of my PhD thesis.

In 2021, there were several circuit breaker lockdowns (13 – 17 February, 28 May – 10 June, 16 – 27 July) and a 3-month lockdown (5 August – 21 October) in Melbourne, resulting in 108 cumulative days of lockdown. This intermittently disrupted access to the BME lab and other external facilities including the Surface Plasmon Resonance (SPR) instrument at the Walter and Eliza Hall Institute (WEHI) and the NMR facility at Bio21 Institute. This resulted in the cancellation of instrument bookings and trainings as no visitors were allowed onsite in those facilities and there was a large backlog of bookings from existing users.

Although I was able to carry out experiments in the labs that I had access in whenever COVID restrictions were eased, some experiments could not be carried out to completion due to limited or no access to labs outside of the BME PC2 lab, causing significant delays and making it difficult and challenging to complete data collection in a timely manner. Consequently, the aims and directions of my PhD thesis were significantly modified multiple times in the first two years of my candidature due to circumstances outside of my control such as lockdown restrictions issued by the government and the access policy of external facilities mentioned.

Dedication

This thesis is dedicated to my mother for her sacrifice and endless love I received in the first 13 years of my life. You taught me the true value of determination, perseverance, and resilience that I still carry with me to this day.

Abstract

Glycosaminoglycans (GAGs) are components of the extracellular matrix that control physiological processes in cells and tissues. Heparan sulphate (HS) is a GAG known for its role in modulating growth factor interaction and signalling that impacts multiple cellular functions including cell proliferation, cell migration, and angiogenesis. The interaction between HS and growth factors is highly dependent on the structure of HS. However, the use of HS as a therapeutic material is limited by its heterogeneous chemical structure and the low yield obtained from animal tissue extraction. To overcome this, different approaches have been attempted to create HS mimetics through chemical and chemoenzymatic synthesis, polysaccharide modification, and recombination expression of HS proteoglycans. However, these methods have disadvantages such as complex synthesis steps, difficult control of specific sulphation, and heterogeneous structure of recombinant HS. This thesis explored an alternative source of materials from seaweed-derived sulphated polysaccharides (SPs) including kappa-, iota-, lambda-carrageenan, and ulvan, and investigated their ability to mimic HS functions. The specific aims of this thesis were to (1) characterise the degree of sulphation and position of sulphate groups of the SPs and examine how these sulphation characteristics impact growth factor binding, (2) investigate the effect of different sulphation characteristics of SPs on growth factor signalling and cellular responses, and (3) evaluate the influence of varying sulphur level within each type of SPs on growth factor signalling.

Results from the first aim showed that iota-carrageenan (ICAR) and lambda-carrageenan (LCAR) had higher degree of sulphation than kappa-carrageenan and ulvan, and sulphate groups were

detected at different positions in the sugar ring of carrageenan and ulvan. The interaction between SPs and fibroblast growth factor-2 (FGF2) was investigated using surface plasmon resonance. It was demonstrated that LCAR had a higher amount of bound FGF2 compared to other types of SPs, indicating that higher degree of sulphation resulted in higher level of FGF2 binding, and 6-O-sulphation was attributed to the increased amount of bound FGF2.

For the second aim, the ability of SPs to signal FGF2 was evaluated using FGF receptor (FGFR)-transfected cells, and only LCAR was found to have the capacity to signal FGF2 due to having similar position of sulphate groups to HS. The ability of SPs to facilitate FGFR activation was investigated by detecting the presence of phospho-ERK1/2 in endothelial cells and fibroblasts as an indirect measure of growth factor signalling. It was shown that SPs did not influence the FGF2-induced phosphorylation of ERK1/2; however, the level of phospho-ERK1/2 detected was different between the two cell types. Regarding cellular responses, all SPs except LCAR were found to decrease the migration of endothelial cells in the presence of FGF2 as compared to FGF2 only, while having no impact on the migration of fibroblasts. Furthermore, the SPs exhibited inhibitory effect on the sprouting activity of endothelial spheroids. These results indicate that different sulphation characteristics of SPs affected the ability to signal growth factor and modulate cellular activities in different cell types.

The third aim investigated the variation in sulphur content within each type of SPs and its impact on growth factor signalling. To achieve this, the SPs were separated into different fractions based on the level of negative charge, and the sulphur content of each fraction was determined. Results revealed that carrageenan contained two fractions while ulvan had three fractions with different levels of sulphur, and these fractions either enhanced, inhibited, or had no effect on FGF2

signalling. These findings indicate that the variation in sulphur content of the fractionated SPs altered the level of growth factor signalling, and the structural heterogeneity existing within each type of SPs may account for the variability in the biological function of SPs both in unfractionated and fractionated form.

This thesis contributes a fundamental understanding of how different sulphation characteristics of seaweed derived SPs influence their biological functions. Specifically, the findings in this work reveal the ability to modulate growth factor binding and signalling and cellular activities by varying sulphation properties of the SPs, as well as demonstrate the potential of SPs as materials that mimic the functions of HS. This provides a basis for the development of HS mimetics using seaweed derived SPs for various biomedical applications.

Declaration

This is to certify that:

- the thesis comprises only my original work towards the PhD;
- due acknowledgement has been made in the text to all other material used;
- the thesis is fewer than 100,000 words in length, exclusive of tables, maps, bibliographies and appendices.

Vuong Hoang Long Nguyen

February 2024

Preface

In this thesis, all experimental results in Chapter 3, 4, and 5 and the writing in Chapter 2 (Literature Review) are unpublished materials that are in preparation for publication. The following information describes the work which the candidate claims as original, and any work conducted by others that contributed to this thesis.

- **Chapter 2**: Marine derived sulphated polysaccharides for the development of biomimetic glycosaminoglycans: A review
  - All the writing is my own work.

- **Chapter 3**: Characterisation of degree of sulphation and position of sulphate groups of seaweed derived sulphated polysaccharides and how sulphation impacts growth factor binding.
  - All the results in this chapter are my own work.
  - Dr Anthony De Girolamo (Faculty of Engineering Technical Services, Monash University) performed CHNS elemental analysis.
  - Dr Karyn Jarvis (Swinburne University) performed X-ray photoelectron spectroscopy.
  - Dr Richard Birkinshaw and Dr Katherine Davies (Walter and Eliza Hall Institute) assisted with surface plasmon resonance training.

- **Chapter 4**: Effect of structure of seaweed derived SPs on growth factor signalling and cellular responses.
    - All the results in this chapter are my own work.

- **Chapter 5**: Effect of varying sulphur level of fractionated SPs on growth factor signalling.
    - All the results in this chapter are my own work.
    - Dr Karyn Jarvis (Swinburne University) performed X-ray photoelectron spectroscopy.

The following list shows the poster and oral presentations delivered by the candidate during the course of this PhD study.

- **Long V-H. Nguyen**, Daniel E. Heath, Sally L. McArthur, Brooke L. Farrugia. "Investigating the effect of varying sulphation degree of marine derived sulphated polysaccharides on growth factor interactions". Poster presentation at the 27th Annual Australasian Society for Biomaterials and Tissue Engineering (ASBTE) Conference, Melbourne, Australia, 20 - 24 April 2022.

- **Long V-H. Nguyen**, Daniel E. Heath, Sally L. McArthur, Brooke L. Farrugia. "Marine derived sulphated polysaccharides for improved chronic wound healing". Oral presentation at the Australasian Wound and Tissue Repair Society (AWTRS) Scientific Meeting, Sydney, Australia, 12 – 14 September 2022.

No work towards the thesis was carried out prior to enrolment. No work has been submitted for other qualifications. No third-party editorial assistance was employed in the preparation of the thesis.

This doctoral degree was funded by the Melbourne Research Scholarship awarded by the University of Melbourne. Conference travel and attendance was funded by Graeme Clark Institute and Faculty of Engineering and Information Technology at the University of Melbourne.

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Doing a PhD is like a rollercoaster ride with many ups and downs. Throughout this journey, I have grown both as a researcher and an individual. As I approach the end of my PhD thesis and look back, this thesis would not have been achievable without the kind support of many people to whom I owe my thanks.

First and foremost, I would like to express my deepest gratitude to my principal supervisor, Associate Professor Brooke Farrugia, for your tremendous support, encouragement, and patience since day one. Thank you for guiding me through the hardest times in my PhD and motivating me to achieve my goals. While doing a PhD in a foreign country during a global pandemic was challenging for me, you still had faith in my abilities and understood my hardships, and I am truly grateful for that.

I would like to extend my sincere thanks to my co-supervisors, Associate Professor Daniel Heath and Professor Sally McArthur. Thank you for your guidance and for providing me with constructive and valuable feedback that helped to enhance the quality of my thesis. I also wish to thank my advisory committee chair, Professor David Grayden, for your helpful advice and encouragement over the course of this PhD.

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I acknowledge the technical support that I received from my collaborators: Dr Karyn Jarvis at Swinburne University of Technology for helping me with XPS measurements; Professor Peter Czabotar, Dr Richard Birkinshaw, and Dr Katherine Davies at Walter and Eliza Hall Institute of Medical Research for providing me with the opportunity to conduct SPR experiments in your lab and supporting me during my time there; and Dr Sunnia Rajput at Bio21 Institute for aiding me with NMR analysis. Many thanks to the staff members at the Materials Characterisation and Fabrication Platform and the Faculty of Engineering Technical Services at Monash University for their help with sample measurements.

I would like to extend my warmest thanks to my partner, Nhat Huy Vo, for your unwavering support and encouragement. You were there witnessing me struggling with my thesis, patiently listening to my research, and helping me navigate through difficult emotions. I cannot thank you enough for your love, empathy, and compassion. I am so blessed to have you in my life, and I look forward to spending more time with you and celebrating our achievements.

To my friends in Melbourne and from afar, thank you for spreading positive energy over the years, and for sharing your inspiring stories that have motivated me to become better every day. Special thanks to Xuan for helping me with everything outside of work.

Last but not least, I would like to thank my family for their enormous support and profound belief in my work. Thank you for being by my side during many challenges life throws at me. I could not have gone this far without your love and encouragement. I hope I have made you proud.

Table of Contents

COVID-19 Impact Statement i
Dedication iii
Abstract iv
Declaration vii
Preface viii
Acknowledgements xi
Table of Contents xiii
List of Figures xvi
List of Tables xx
List of Abbreviations xxi

Chapter 1 1
Introduction 1
1.1. Research motive 2
1.2. Summary of thesis aims and chapter outline 4

Chapter 2 7
Literature review 7
2.1. Wound healing 8
2.2. Growth factors 10
2.2.1. Growth factor interaction with cofactors and receptors 10
2.2.2. Growth factor signalling 13
2.3. Glycosaminoglycans 14
2.3.1. Biosynthesis of heparin/HS 17
2.3.2. Biological functions of heparin/HS 19
2.4. Heparan sulphate mimetics 21
2.5. Marine derived sulphated polysaccharides 24
2.5.1. Carrageenan 25
2.5.2. Fucoidan 27

2.5.3. Ulvan 28
2.6. Growth factor binding and signalling of marine derived sulphated polysaccharides. 30
2.7. Application of seaweed derived sulphated polysaccharides in developing growth factor delivery systems. 39
2.8. Research questions and thesis aims 44

Chapter 3 47

Characterisation of degree of sulphation and position of sulphate groups of seaweed derived sulphated polysaccharides and how these sulphation characteristics impact growth factor binding 47

3.1. Introduction 48
3.2. Materials and Methods 55
3.2.1. Carbon-Hydrogen-Nitrogen-Sulphur (CHNS) elemental analysis 55
3.2.2. X-ray Photoelectron Spectroscopy (XPS) 56
3.2.3. Attenuated Total Reflection-Fourier Transform Infrared Spectroscopy (ATR-FTIR) 57
3.2.4. Surface Plasmon Resonance (SPR) 57
3.2.5. Statistical analysis 60
3.3. Results 61
3.3.1. CHNS elemental analysis 61
3.3.2. X-ray photoelectron spectroscopy (XPS) 66
3.3.3. Attenuated Total Reflection-Fourier Transform Infrared Spectroscopy (ATR-FTIR) 72
3.3.4. Surface plasmon resonance (SPR) 75
3.4. Discussion 82
3.5. Conclusion 87

Chapter 4 88

Effect of different sulphation characteristics of seaweed-derived SPs on growth factor signalling and cellular responses 88

4.1. Introduction 89
4.2. Materials and Methods 93
4.2.1. Growth factor signalling activity of SPs 93
4.2.2. Growth factor receptor activation 95
4.2.3. Cell migration assay 98

4.2.4. Cell spheroid sprouting assay 99
4.2.5. Statistical analysis 100
4.3. Results 101
4.3.1. BaF32 proliferation assay 101
4.3.2. Growth factor receptor activation 103
4.3.3. Cell migration assay 112
4.3.4. Cell spheroid sprouting assay 120
4.4. Discussion 125
4.5. Conclusion 132

Chapter 5 133
Effect of varying sulphur level within each type of SPs on growth factor signalling 133
5.1. Introduction 134
5.2. Materials and methods 137
5.2.1. Anion exchange chromatography 137
5.2.2. X-ray photoelectron spectroscopy (XPS) 139
5.2.3. Growth factor signalling activity of SP fractions. 139
5.2.4. Statistical analysis 141
5.3. Results 142
5.3.1. Anion exchange chromatography of sulphated polysaccharides 142
5.3.2. Sulphur content and growth factor signalling of fractionated SPs 150
5.4. Discussion 171
5.5. Conclusion 175

Chapter 6 176
Thesis summary and future directions 176
6.1. Summary of main findings, significance, and contributions to research field 177
6.2. Limitations and future directions 180
6.3. Conclusion 185

References 186
Appendix 211

List of Figures

Figure 2.1. Mechanism of growth factor activity 10

Figure 2.2. Activation of tyrosine kinase domain of FGF receptor. 12

Figure 2.3. Formation of FGF-FGFR-HS ternary complex leading to the activation of FGF receptor. 13

Figure 2.4. Illustration of A) Heparin and B) HS disaccharide unit showing possible modifications of sulphation and/or acetylation at various positions 16

Figure 2.5. Biosynthesis process of heparin/HS showing the formation and modification of polysaccharide backbone, and the enzymes involved in each biosynthetic step 18

Figure 2.6. Chemical backbone of HS consisting of N-acetylated (NA) domains and N-sulphated (NS) domains. 19

Figure 2.7. Chemical structures of repeating disaccharide units of A) Kappa-carrageenan, B) Iota-carrageenan, and C) Lambda-carrageenan 26

Figure 2.8. Chemical structures of repeating disaccharide units of fucoidan. 27

Figure 2.9. Chemical structures of main repeating disaccharide units of ulvan. 29

Figure 3.1. Chemical structures of disaccharide units of A) Heparin and B) Heparan sulphate .. 49

Figure 3.2. Chemical structures of disaccharide units of A) Kappa-carrageenan, B) Iota-carrageenan, and C) Lambda-carrageenan 52

Figure 3.3. Chemical structures of two main types of disaccharide units of ulvan. A) Ulvanobiuronic acid, B) Ulvanobiose 53

Figure 3.4. Thiolated SPs and control solutions after mixing with Ellman's reagent solution. 59

Figure 3.5. CHNS elemental analysis of heparin, kappa- (KCAR), iota- (ICAR), lambda-carrageenan (LCAR), and ulvan.. 65

Figure 3.6. XPS analysis of heparin, kappa- (KCAR), iota- (ICAR), lambda-carrageenan (LCAR), and ulvan 68

Figure 3.7. High resolution curve fitted C1s XPS spectra of A) heparin, B) kappa-carrageenan (KCAR), C) iota-carrageenan (ICAR), D) lambda-carrageenan (LCAR), and E) ulvan. 71

Figure 3.8. ATR-FTIR analysis of A) heparin and B) ulvan. 73

Figure 3.9. ATR-FTIR analysis of A) kappa-carrageenan (KCAR), B) iota-carrageenan (ICAR), and C) lambda-carrageenan (LCAR). 74
Figure 3.10. Illustration of SPR results..... 76
Figure 3.11. SPR results of FGF2 flowing over a gold chip surface containing BSA only..... 77
Figure 3.12. SPR sensorgrams of FGF2 bound to A) heparin, B) kappa-carrageenan (KCAR), C) iota-carrageenan (ICAR), D) lambda-carrageenan (LCAR), or E) ulvan, and F) Representative SPR curves of FGF2 bound to each SP. 80
Figure 3.13. Amount of FGF2 bound to each SP presented in response units (RU). 81
Figure 4.1. ERK signalling pathway of FGF2..... 91
Figure 4.2. FGF2 signalling of SPs determined by the BaF32 proliferation assay.. 102
Figure 4.3. Western blot membranes showing the presence of A) GAPDH, B) ERK1/2 and C) phospho-ERK1/2 in endothelial cells incubated in carrageenan with or without FGF2..... 105
Figure 4.4. Western blot membranes showing the presence of A) GAPDH, B) ERK1/2 and C) phospho-ERK1/2 in endothelial cells incubated in ulvan with or without FGF2 106
Figure 4.5. FGF2 signalling of various SPs in endothelial cells determined by Western blotting. 107
Figure 4.6. Western blot membranes showing the presence of A) GAPDH, B) ERK1/2 and C) phospho-ERK1/2 in fibroblasts incubated in carrageenan with or without FGF2..... 109
Figure 4.7. Western blot membranes showing the presence of A) GAPDH, B) ERK1/2 and C) phospho-ERK1/2 in fibroblasts incubated in ulvan with or without FGF2 110
Figure 4.8. FGF2 signalling of various SPs in fibroblasts determined by western blotting. 111
Figure 4.9. Merged images and grey scale images of phalloidin channel of migrated endothelial cells in Transwell migration assay. Cells were incubated in EBM supplemented with 2 % FBS or heparin in the presence or absence of FGF2 for 24 hours 113
Figure 4.10. Merged images and grey scale images of phalloidin channel of migrated endothelial cells in Transwell migration assay. Cells were incubated in various SPs in the presence or absence of FGF2 for 24 hours 114
Figure 4.11. Quantification of number of migrated endothelial cells in Transwell migration assay. 115

Figure 4.12. Merged images and grey scale images of phalloidin channel of migrated fibroblasts in Transwell migration assay. Cells were incubated in DMEM or heparin in the presence or absence of FGF2 for 8 hours 117
Figure 4.13. Merged images and grey scale images of phalloidin channel of migrated fibroblasts in Transwell migration assay. Cells were incubated in various SPs in the presence or absence of FGF2 for 8 hours..... 118
Figure 4.14. Quantification of number of migrated fibroblasts in Transwell migration assay. . 119
Figure 4.15. Sprouting analysis of a spheroid..... 121
Figure 4.16. Sprouting of endothelial cell spheroids embedded in fibrin gels. The spheroids were incubated in EBM supplemented with 2 % FBS or heparin in the presence or absence of FGF2 for 24 hours for sprout formation. 122
Figure 4.17. Sprouting of endothelial cell spheroids embedded in fibrin gels. The spheroids were incubated in various SPs in the presence or absence of FGF2 for 24 hours for sprout formation. 123
Figure 4.18. Measurement of A) sprout number and B) sprout length of endothelial cell spheroids. 124
Figure 5.1. Anion exchange chromatography (AEC) profiles of heparin.. 145
Figure 5.2. Anion exchange chromatography (AEC) profiles of ulvan..... 146
Figure 5.3. Anion exchange chromatography (AEC) profiles of kappa-carrageenan (KCAR).. ... 147
Figure 5.4. Anion exchange chromatography (AEC) profiles of iota-carrageenan (ICAR)..... 148
Figure 5.5. Anion exchange chromatography (AEC) profiles of lambda-carrageenan (LCAR).. . 149
Figure 5.6. Characterisation of sulphur content of fractionated heparin (HEP). 152
Figure 5.7. FGF2 signalling activity of fractionated heparin determined via the proliferation of BaF32 cells transfected with FGFR1c in the presence of fractionated heparin and FGF2. 154
Figure 5.8. Characterisation of sulphur content of fractionated kappa-carrageenan (KCAR). .. 156
Figure 5.9. FGF2 signalling activity of fractionated KCAR determined via the proliferation of BaF32 cells transfected with FGFR1c in the presence of fractionated KCAR and FGF2. 158
Figure 5.10. Characterisation of sulphur content of fractionated iota-carrageenan (ICAR)..... 160

Figure 5.11. FGF2 signalling activity of fractionated ICAR determined via the proliferation of BaF32 cells transfected with FGFR1c in the presence of fractionated ICAR and FGF2.. 162
Figure 5.12. Characterisation of sulphur content of fractionated lambda-carrageenan (LCAR). 164
Figure 5.13. FGF2 signalling activity of fractionated LCAR determined via the proliferation of BaF32 cells transfected with FGFR1c in the presence of fractionated LCAR and FGF2.. 166
Figure 5.14. Characterisation of sulphur content of fractionated ulvan..... 168
Figure 5.15. FGF2 signalling activity of fractionated ulvan determined via the proliferation of BaF32 cells transfected with FGFR1c in the presence of fractionated ulvan and FGF2.. 170
Figure A.1. Full AEC profiles of A) heparin and B) ulvan showing the fractions eluted at salt concentrations of 0.25 M, 0.5 M, 0.75 M, 1 M, and 2 M NaCl. 211
Figure A.2. Full AEC profiles of A) KCAR, B) ICAR, and C) LCAR showing the fractions eluted at salt concentrations of 0.25 M, 0.5 M, 0.75 M, 1 M, and 2 M NaCl. 212

List of Tables

Table 2.1. Summary of structural characteristics of SPs from various seaweed species and their modulating effects on growth factor binding..... 32

Table 2.2. Summary of structural characteristics of SPs from various seaweed species and their modulating effects on growth factor downstream functions 33

Table 2.3. Growth factor delivery systems fabricated from various types of seaweed derived polysaccharides and their tissue engineering applications. 40

Table 3.1. Mass concentration (%) of carbon, hydrogen, nitrogen, and sulphur of heparin, KCAR, ICAR, LCAR, and ulvan measured by elemental analysis. 63

Table 3.2. Atomic concentration (%) of chemical elements detected in heparin, KCAR, ICAR, LCAR, and ulvan using XPS 67

Table 3.3. Percent (%) of peak area in C1s spectra and O/C ratio of heparin, KCAR, ICAR, LCAR, and ulvan..... 70

Table 5.1. Description of names of fractions obtained from anion exchange chromatography. 138

Table 5.2. Percent (%) of peak area in C1s spectra and O/C ratio of HEP0.25, HEP0.5, and HEP0.75 fractions. 153

Table 5.3. Percent (%) of peak area in C1s spectra and O/C ratio of KCAR0.25 and KCAR0.5 fractions 157

Table 5.4. Percent (%) of peak area in C1s spectra and O/C ratio of ICAR0.25 and ICAR0.5 fractions. 161

Table 5.5. Percent (%) of peak area in C1s spectra and O/C ratio of LCAR0.25 and LCAR0.5 fractions. 165

Table 5.6. Percent (%) of peak area in C1s spectra and O/C ratio of Ulvan0.25, Ulvan0.5, and Ulvan0.75 fractions. 169

List of Abbreviations

AEC	Anion exchange chromatography
ATIII	Anti-thrombin III
ATR-FTIR	Attenuated total reflection-Fourier transform infrared spectroscopy
CS	Chondroitin sulphate
DMEM	Dulbecco's modified eagle's medium
DS	Dermatan sulphate
EBM	Endothelial basal medium
ECM	Extracellular matrix
EGF	Epidermal growth factor
EGM	Endothelial growth medium
ERK	Extracellular signal-regulated kinase
EXT	Exostosin
FBS	Foetal bovine serum
FGF	Fibroblast growth factor
FGFR	Fibroblast growth factor receptor
GAGs	Glycosaminoglycans
GAPDH	Glyceraldehyde 3-phosphate dehydrogenase
GlcA	D-glucuronic acid
GlcN	D-glucosamine
GlcNAc	*N*-acetyl glucosamine
GlcNS	*N*-sulphate glucosamine
HA	Hyaluronic acid
HEP	Heparin
HS	Heparan sulphate
HUVECs	Human umbilical vein endothelial cells
ICAR	Iota-carrageenan
IdoA	L-iduronic acid

KCAR	Kappa-carrageenan
KS	Keratan sulphate
LCAR	Lambda-carrageenan
MAPK	Ras-mitogen-activated protein kinase
NA	*N*-acetylated
NDST	*N*-deacetylase/*N*-sulfotransferase
NMR	Nuclear magnetic resonance
NS	*N*-sulphated
PAPS	3’-phosphoadenosine 5’-phosphosulphate
PGs	Proteoglycans
Phospho-ERK	Phosphorylated extracellular signal-regulated kinase
PI3K/AKT	Phosphoinositide 3-kinase–Akt
PLCγ1	Phospholipase Cγ1
PS	Penicillin-streptomycin
SPR	Surface plasmon resonance
SPs	Sulphated polysaccharides
UDP	Uridine diphosphate
VEGF	Vascular endothelial growth factor
VEGFR	Vascular endothelial growth factor receptor
XPS	X-ray photoelectron spectroscopy

Chapter 1

Introduction

1.1. Research motive

Heparan sulphate (HS) is a component of the extracellular matrix (ECM) that is known to have an essential role in modulating cell growth and development by regulating the signalling activities of growth factors, thus contributing to tissue development and repair [1]. However, the use of HS in tissue engineering and regenerative medicine applications has been hindered by the insufficient yield of HS obtained from mammalian tissues, or the complex and expensive production of HS mimetics. Therefore, this thesis focuses on evaluating the ability to mimic the biological function of HS using marine derived sulphated polysaccharides (SPs) with the aim to understand how the structure of SPs affects their function.

HS is a glycosaminoglycan (GAG) which is a linear polysaccharide composed of repeating disaccharide units [2]. The chemical backbone of HS is decorated with sulphate groups at various positions [2]. HS occurs in the form of proteoglycan which comprises HS chains that are covalently attached to a protein core. HS proteoglycans are located on the surface of cells and in the ECM and interact with growth factors mainly via the HS chains, which facilitates growth factor signalling and inducing a variety of cellular responses including proliferation, migration, and differentiation [3]. The interaction between HS and growth factors requires a defined structure of HS to bind and signal the growth factor [4]. Being able to produce HS of a defined chemical structure will therefore be an advantageous factor for the application of HS in developing growth factor delivery systems. However, this is difficult to achieve since the natural synthesis of HS is not a template-driven process but rather dependent on cell type and cell environment, making the structure of HS heterogeneous in terms of sulphate modifications. Other strategies have been explored to produce HS biomimetics, including chemical and chemoenzymatic synthesis,

chemical modification of polysaccharides, and recombinant expression of HS proteoglycans for therapeutic applications such as wound healing, inflammation, and regenerative medicine [5-7]. However, these methods still have challenges to overcome, including numerous complex reactions, low yields, and heterogeneous structure of HS products [8]. Hence, it is essential to seek alternative sources of materials and approaches to create new HS biomimetic materials.

Marine derived sulphated polysaccharides (SPs) are considered to be a potential candidate for the development of HS biomimetic materials. These polysaccharides are extracted from seaweeds in abundant quantities and contain sulphate groups along their chemical backbone, thus mimicking the structural characteristic of HS. Hence, utilising the SPs can help to overcome the issues of low yields and complex chemical reactions of current HS mimetics. There are three main types of marine derived SPs, including carrageenan, fucoidan, and ulvan. These SPs contain varying degree of sulphation and position of sulphate groups on the sugar ring [9-11]. The chemical structure of marine derived SPs is heterogeneous not only due to seaweed species but also extraction methods, and this structural heterogeneity affects the biological function of SPs [12]. The SPs have been shown to possess various biological properties such as anticoagulant, antibacterial, antiviral, antitumour, and immunomodulatory [13], hence they have high therapeutic potential in the field of tissue engineering and regenerative medicine. It is known that the interaction between HS and growth factor requires specific sulphation of HS to bind and signal the growth factor [4]. To mimic the function of HS, the seaweed derived SPs are expected to interact with growth factors and modulate their signalling activity. It is hypothesised that the variation in the degree of sulphation and the position of sulphate groups of SPs can have an impact on the interaction with growth factors and downstream biological functions. In the

context of developing HS biomimetic materials, the relationship between chemical structure and biological function of seaweed derived SPs including carrageenan and ulvan requires further investigation. Taking these gaps into account, this thesis is going to answer the following research questions.

Q1: How does the variation in degree of sulphation and position of sulphate groups of SPs affect growth factor binding?

Q2: How does the variation in degree of sulphation and position of sulphate groups of SPs impact growth factor signalling and cellular responses?

Q3: How does the degree of sulphation vary within each type of SPs and how does this variation influence growth factor signalling?

To answer these research questions the section below details the aims that will be undertaken in this thesis together with an outline of each thesis chapter.

1.2. Summary of thesis aims and chapter outline

This thesis investigates the impact of structural variation of seaweed derived SPs on their ability to interact with growth factors and induce downstream cellular responses. The following aims were therefore formulated:

Aim 1: To characterise the degree of sulphation and position of sulphate groups of seaweed derived SPs and examine how these sulphation characteristics impact growth factor binding.

Aim 2: To investigate the effect of different sulphation characteristics of seaweed derived SPs on growth factor signalling and cellular responses.

Aim 3: To evaluate the influence of varying sulphur level within each type of SPs on growth factor signalling.

An outline of each thesis chapter is detailed below:

Chapter 2 provides background information regarding wound healing process and biological functions of growth factors and heparan sulphate, and structural characteristics and biological activities of seaweed derived SPs. A review of the literature was conducted with an emphasis on approaches to create HS mimetics, investigations in the structure-function relationship of SPs, and application of SPs in developing growth factor delivery systems.

Chapter 3 presents the characterisation of the degree of sulphation and position of sulphate groups of seaweed derived SPs including kappa-, iota-, lambda-carrageenan, and ulvan, using elemental analysis, X-ray photoelectron spectroscopy, and attenuated total reflection-Fourier transform infrared spectroscopy. The binding of fibroblast growth factor-2 (FGF2) to each type of SPs was evaluated using surface plasmon resonance.

Chapter 4 reports on the effect of various sulphation characteristics of SPs on growth factor signalling and cellular responses. The ability of SPs to form a ternary complex with FGF2 and its associated receptor and induce intracellular signalling pathway was evaluated using BaF32 cell proliferation assay and Western blot. The effect of SPs on cell migration and cell sprouting was also evaluated using a Transwell migration assay and spheroid sprouting assay, respectively.

Chapter 5 details research that examined the effect of varying sulphur level of fractionated SPs on growth factor signalling. Each type of SPs was separated into different fractions based on the overall level of negative charge using anion exchange chromatography. The sulphur percent of

fractionated SPs was measured using X-ray photoelectron spectroscopy, and their ability to form a ternary complex with FGF2 and its associated receptor for growth factor signalling was examined using BaF32 cell proliferation assay.

Chapter 6 concludes the thesis by summarising the findings and highlighting the contribution to the research field, as well as presenting limitations and recommendations for future work.

Chapter 2

Literature review

2.1. Wound healing

Wound healing is the response of the body to injury, which involves a complex series of biological events initiated by the onset of haemostasis (or blood clotting), followed by inflammation, proliferation, and finally matrix remodelling [14]. Upon injury to the skin, blood components such as platelets are recruited to the wound site to activate the clotting cascade [15]. The resulting clot provides a matrix for the invasion of inflammatory cells including neutrophils and monocytes [15]. Platelets secrete various inflammatory cytokines such as interleukin (IL)-8 which attracts neutrophils and promotes the differentiation of monocytes into macrophages for defending against contaminating microorganisms [16]. These macrophages also supply essential growth factors required for the proliferative phase of wound healing. At this stage, macrophages produce epidermal growth factor (EGF) and transforming growth factor-alpha (TGF-α) which stimulate the proliferation of epithelial cells for epithelialisation [17]. They also release fibroblast growth factor (FGF) and vascular endothelial growth factor (VEGF) to promote the proliferation of endothelial cells, which leads to the formation of new blood vessels in the healing wound [18]. Fibroblasts from neighbouring areas migrate to the wound site and deposit extracellular matrix (ECM) molecules such as fibronectin, glycosaminoglycans, and collagen, forming new tissue known as granulation tissue [19]. Finally, the remodelling phase occurs where granulation tissue is replaced by the new network of collagen fibres [19].

Angiogenesis is a critical step in the proliferative stage of wound healing process in which new blood vessels are established to supply oxygen and nutrients to the wounded tissue [20]. Endothelial cells are the main cell type involving in angiogenesis [21]. The cells undergo a series of events which are controlled by a number of angiogenic factors including FGF, VEGF, TGF-α and

TGF-β, in which FGF and VEGF play crucial regulatory roles in all cellular events [21]. When exposed to a proangiogenic stimulus, endothelial cells start to degrade the basement membrane of the existing blood vessels and penetrate the extracellular matrix using secreted proteases [22]. They then proliferate and migrate to the surrounding tissues via chemotaxis in which the cells follow a gradient of proangiogenic mediator such as VEGF [23]. The proliferation and migration of endothelial cells result in the formation of new capillary sprouts, followed by cell differentiation to create the lumen of the sprouts [21]. Newly formed sprouts will connect with the neighbouring sprouts to allow blood to flow [23].

Fibroblasts are also the cell type that has important roles in different stages of wound healing. Following injury, they migrate to the wounded site from surrounding tissues and proliferate, which is regulated by epidermal growth factor (EGF) and platelet-derived growth factor (PDGF) [15]. The cells then start synthesising matrix molecules such as collagen type 1 and 3, fibronectin, and proteoglycans [18]. When extracellular matrix accumulates, fibroblasts transform their phenotype into myofibroblasts in response to the effect of TGF-β [19]. Myofibroblasts contain α-smooth muscle actin protein which allows them to exert contractile force and mediate wound contraction [24].

Many cellular activities in wound healing process are regulated by growth factors. The next section will describe the biological functions of growth factors regarding how they interact with their receptors on the cell surface and initiate signalling pathways to induce cellular responses.

2.2. Growth factors

2.2.1. Growth factor interaction with cofactors and receptors

Growth factors are proteins that transmit signals between cells and guide specific cellular responses in various biological environments [25]. These responses can induce cell actions including cell migration, differentiation and proliferation [26]. Growth factors are produced by cells and subsequently secreted to the extracellular matrix. Figure 2.1 demonstrates the mechanism of growth factor activity. Growth factors bind to their receptors on the surface of target cells in the presence of cofactors such as glycosaminoglycans and deliver biological signals to the intracellular environment, which initiates signalling pathways and eventually induces cellular responses.

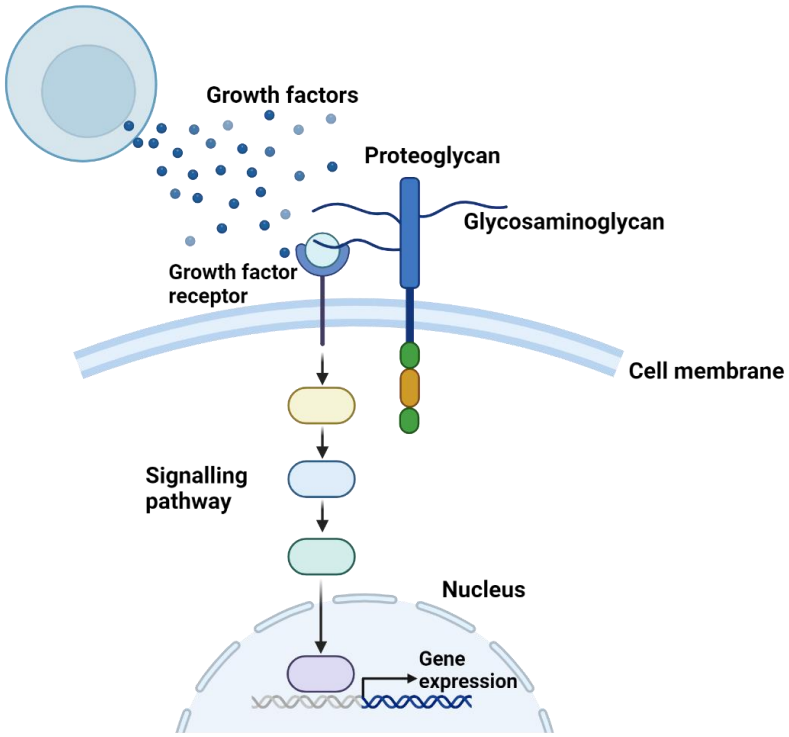


Figure 2.1. Mechanism of growth factor activity. Created with BioRender.com

Fibroblast growth factor (FGF), vascular endothelial growth factor (VEGF), epidermal growth factor (EGF), and transforming growth factor (TGF) are the primary growth factors that coordinate vital cellular events during the wound healing process [27]. Among them, FGF is the largest family of growth factors consisting of 23 members which are produced by fibroblasts, endothelial cells, keratinocytes, mast cells, smooth muscle cells, and chondrocytes [28]. FGF members regulate various cellular activities including migration, proliferation, and differentiation, which contribute to tissue development, metabolism, and homeostasis [29]. Specifically, it has been shown that FGF2, a member of FGF family, promotes granulation tissue formation and keratinocyte migration during re-epithelialisation [27, 30], and increases tissue remodelling by promoting the production of collagenase [31], thereby playing important roles in wound healing and tissue repair.

FGF exhibits biological functions through binding to its specific FGF receptor (FGFR) which belongs to a family of tyrosine kinase receptors [32]. These receptors are essentially transmembrane proteins comprising an extracellular domain, a transmembrane domain, and an intracellular tyrosine kinase domain [33]. The binding of FGF to its FGFR will induce the conformation changes of the FGFR, resulting in dimerisation and activation of the kinase domain by phosphorylating the tyrosine residues of the FGFR. Activated FGFR kinase then triggers the phosphorylation of its intracellular substrates and initiates various downstream signalling pathways which lead to distinct cellular responses [34]. Figure 2.2 illustrates the activation of tyrosine kinase domain of FGF receptor.

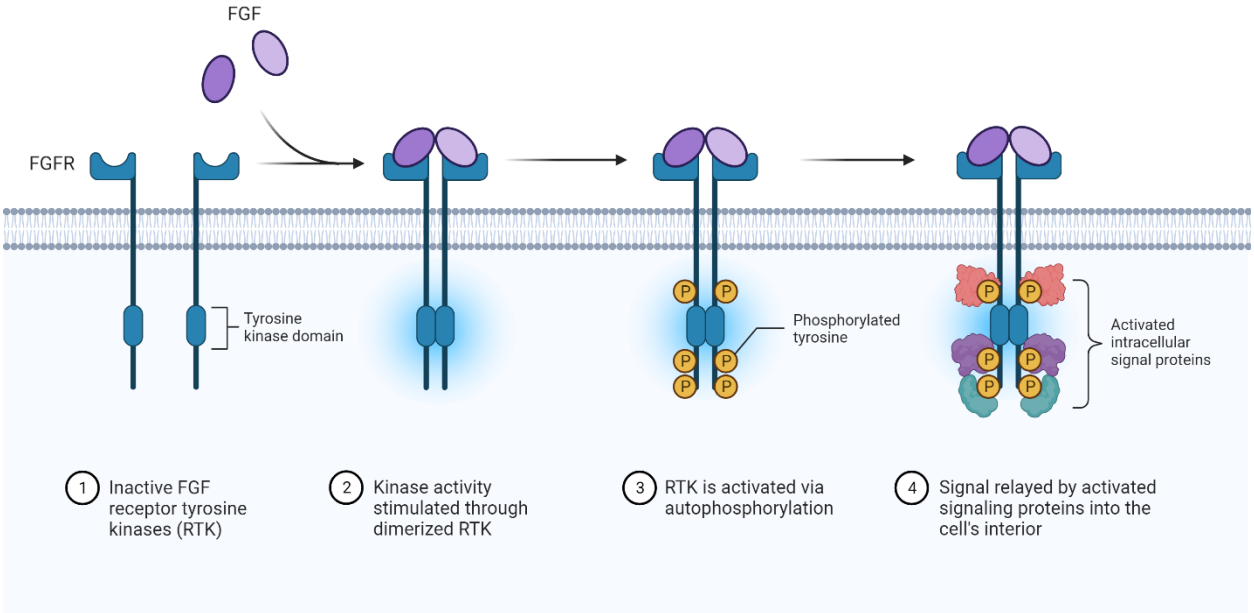


Figure 2.2. Activation of tyrosine kinase domain of FGF receptor. Created with BioRender.com.

The ability of growth factors to induce various cellular responses depends on their biochemical identity, type of cells and receptors that they bind, and intracellular signal transduction pathways following growth factor binding [25]. However, the short lifetime of growth factors can affect the progress of transmitting signals and reduce their activities. In such cases, interacting with ECM molecules such as glycosaminoglycans (GAGs) as a cofactor can accommodate the binding of growth factors to cellular receptors, thereby enhancing the ability to induce downstream biological functions [35, 36]. Heparan sulphate (HS), a type of GAG, is the cofactor of FGF2 that facilitates the ability of FGF2 to bind and activate its receptor, FGFR1. It has been shown that HS promotes the formation of a symmetric dimeric 2:2:2 FGF2-FGFR1-HS ternary complex, which

induces dimerisation of receptor and activation of downstream signalling pathways [37], as illustrated in Figure 2.3.

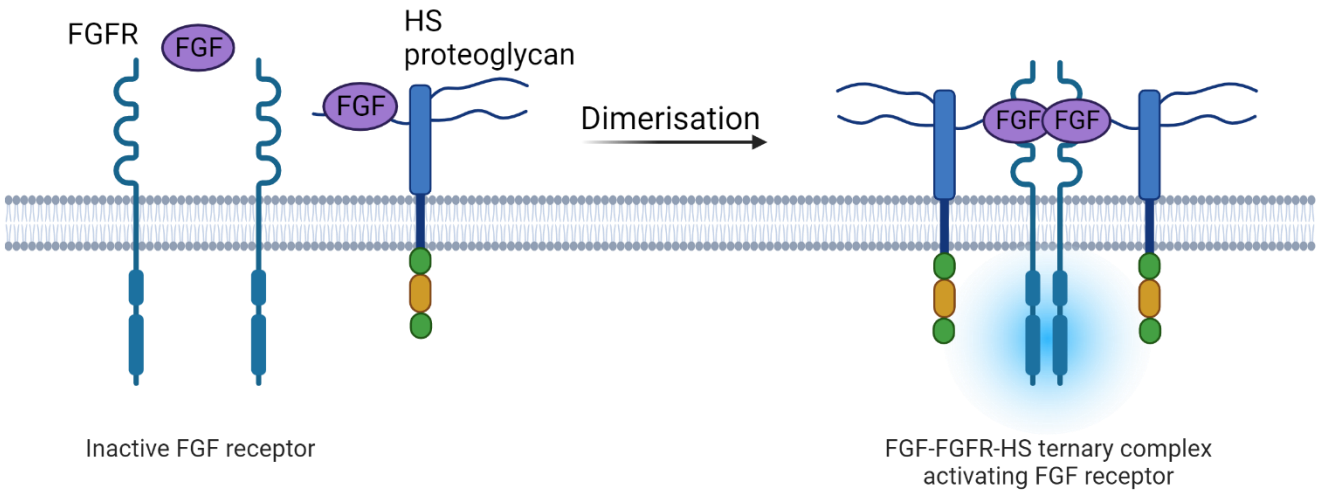


Figure 2.3. Formation of FGF-FGFR-HS ternary complex leading to the activation of FGF receptor. Created with BioRender.com.

2.2.2. Growth factor signalling

The signalling activity of FGF occurs after its receptor has been activated. The tyrosine kinase domain of FGF receptor undergoes phosphorylation subsequent to the formation of FGF-FGFR-HS ternary complex, which initiates downstream signalling pathways. Typical signalling pathways of FGF include Ras mitogen-activated protein kinase (RAS/MAPK), phosphoinositide 3-kinase-Akt (PI3K/AKT), and phospholipase Cγ1 (PLCγ1) [38].

RAS/MAPK signalling pathway consists of serine/threonine-specific protein kinases that regulate cell proliferation and differentiation [39]. Typical MAP kinase effectors include extracellular signal-regulated kinase (ERK), c-Jun N-terminal kinase (JNK), and p38 mitogen-activated kinase [40]. The binding of FGF to its receptor leads to the phosphorylation of fibroblast growth factor receptor substrate 2α (FRS2α) which is the docking protein in the tyrosine residues of FGFR. The phosphorylated FRS2α then forms a complex consisting of GRB2-associated binding protein 1 (GAB1), guanine nucleotide exchange factor 2 (GRB2), tyrosine phosphatase (SHP2), and the son of sevenless (SOS), which results in the activation of RAS/MAP kinases [41].

PI3/AKT signalling pathway is initiated following the formation of the FRS2α complex [38]. The GAB1 protein in the complex forms a connection between the activated FGF receptor and PI3 kinase, which then stimulates downstream AKT kinase [38]. This pathway mainly controls the survival of cells [42].

PLCγ1 signalling pathway is initiated by the interaction between the PLCγ molecule and the phosphorylated Tyr766 of the FGF receptor. Diacylglycerol (DAG) and inositol triphosphate (IP3) are produced following the hydrolysis of activated PLCγ [38]. Together with the calcium ions in the cytosol, they activate protein kinase C (PKC), which leads to a variety of cellular responses such as changes in cell morphology, cell migration, and adhesion [32, 43].

2.3. Glycosaminoglycans

Proteoglycans (PGs) are a component of the extracellular matrix. They are composed of a protein core to which glycosaminoglycan (GAG) chains are linked via covalent bonds. There are various

types of PGs including hyaluronan-binding PGs (aggrecan, versican), basement membrane PGs (perlecan, agrin and collagen type XVIII) [44], cell surface PGs (syndecans, glypicans), and intracellular PG (serglycin) [45]. The biological function of PGs is partly attributed to the attached GAG chains that regulate the biological activity of cells and tissues [45, 46].

GAGs are long, polydisperse, linear polysaccharides consisting of repeating disaccharide units that comprise an amino sugar (glucosamine or galactosamine that is N-acetylated or N-sulphated) and a uronic acid (glucuronic acid or iduronic acid) or galactose. GAGs can be categorised into four main groups including hyaluronic acid or hyaluronan (HA), keratan sulphate (KS), chondroitin/dermatan sulphate (CS/DS), and heparin/heparan sulphate (HS). They differ from each other in terms of monosaccharide components, glycosidic bonds, position and degree of saccharide functionalisation. Of the four types of GAGs, HA differs from the other three as it is not decorated with sulphate groups along the polysaccharide backbone. The repeating disaccharide units of HA consist of glucuronic acid and N-acetylglucosamine. KS contains repeating disaccharide units composed of galactose and 6-O-sulphated N-acetylglucosamine. CS and DS are composed of repeating disaccharide units containing 4-O-sulphated N-acetylgalactosamine residues and either glucuronic acid or iduronic acid residues, respectively.

Heparin and HS are composed of disaccharide units including D-glucosamine (GlcN) and uronic acid residues that are linked through (1→4) bonds. The uronic acid is either α(1→4) linked L-iduronic acid (IdoA) or β(1→4) linked D-glucuronic acid (GlcA). Chemical modifications within the disaccharide units usually occur at the positions of oxygen and/or nitrogen atoms, including 2-O-sulphation of uronic acid, and N-sulphation, N-acetylation, 3-O- and 6-O-sulphation of

glucosamine unit. Figure 2.4A and 2.4B demonstrate the positions within the disaccharide unit of heparin and HS, respectively, where modifications can occur.

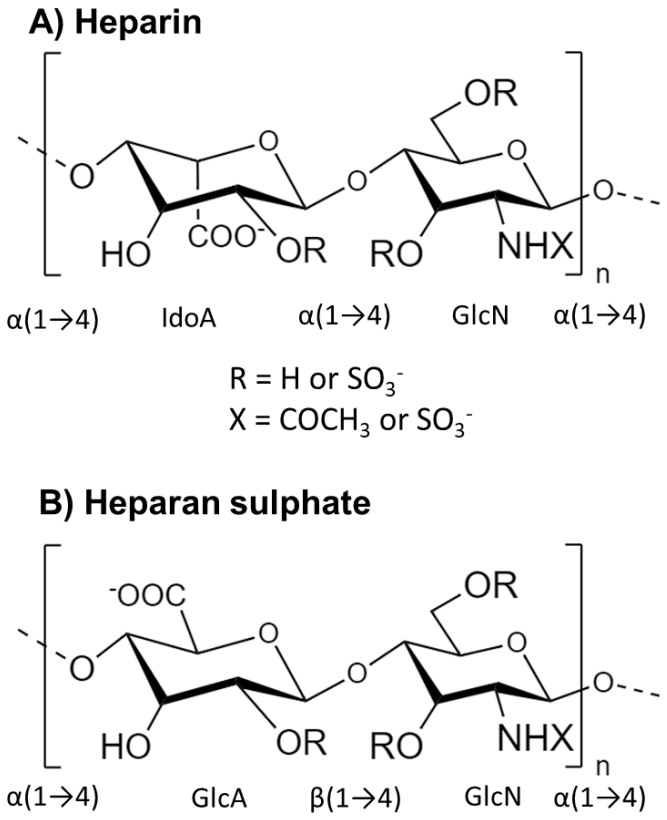


Figure 2.4. Illustration of A) Heparin and B) HS disaccharide unit showing possible modifications of sulphation and/or acetylation at various positions.

HS is known as the cofactor of growth factors when they bind to their corresponding receptors [47]. This plays an essential role in regulating the signalling activity of growth factors to induce cellular responses. The following sections will focus on the biosynthesis of heparin/HS and the interaction between HS and growth factors.

2.3.1. Biosynthesis of heparin/HS

The biosynthesis of heparin/HS occurs in Golgi apparatus and endoplasmic reticulum where various enzymes are involved in several biosynthetic steps including chain initiation, polymerisation, and modification [48, 49]. Chain initiation includes the formation of a linkage tetrasaccharide consisting of a xylose, two galactose and a glucuronic acid molecule, all of which are covalently linked to the serine residues of the protein core through an O-glycosidic linkage [50]. The chain is then built up by the supplementation of disaccharide units comprising glucuronic acid (GlcA) and N-acetyl glucosamine (GlcNAc) to the non-reducing end of the linkage tetrasaccharide. This process is sequentially performed by two exostosin (EXT) glycosyl transferases (EXT1 and EXT2) and the polysaccharide backbone $[GlcA\text{-}(1,4)\text{-}GlcNAc]_n$ is formed. Subsequently, the formed backbone undergoes modifications through a series of enzymes, including N-deacetylase-N-sulfotransferase (NDST), C5-epimerase, 2-O, 3-O and 6-O sulfotransferase [51]. NDST removes the acetyl groups on some GlcNAc residues and replace them with sulphate groups, forming N-sulphate glucosamine (GlcNS) residues. C5-epimerase has the role of epimerising some of the GlcA residues to iduronic acid (IdoA). 2-O-sulfotransferase has the ability to add sulphate groups to the C2 position of IdoA residues. 3-O-sulfotransferase and 6-O-sulfotransferase add sulphate groups to the C3 and C6 position of GlcNS residues, respectively. The biosynthesis process of heparin/HS is illustrated in Figure 2.5.

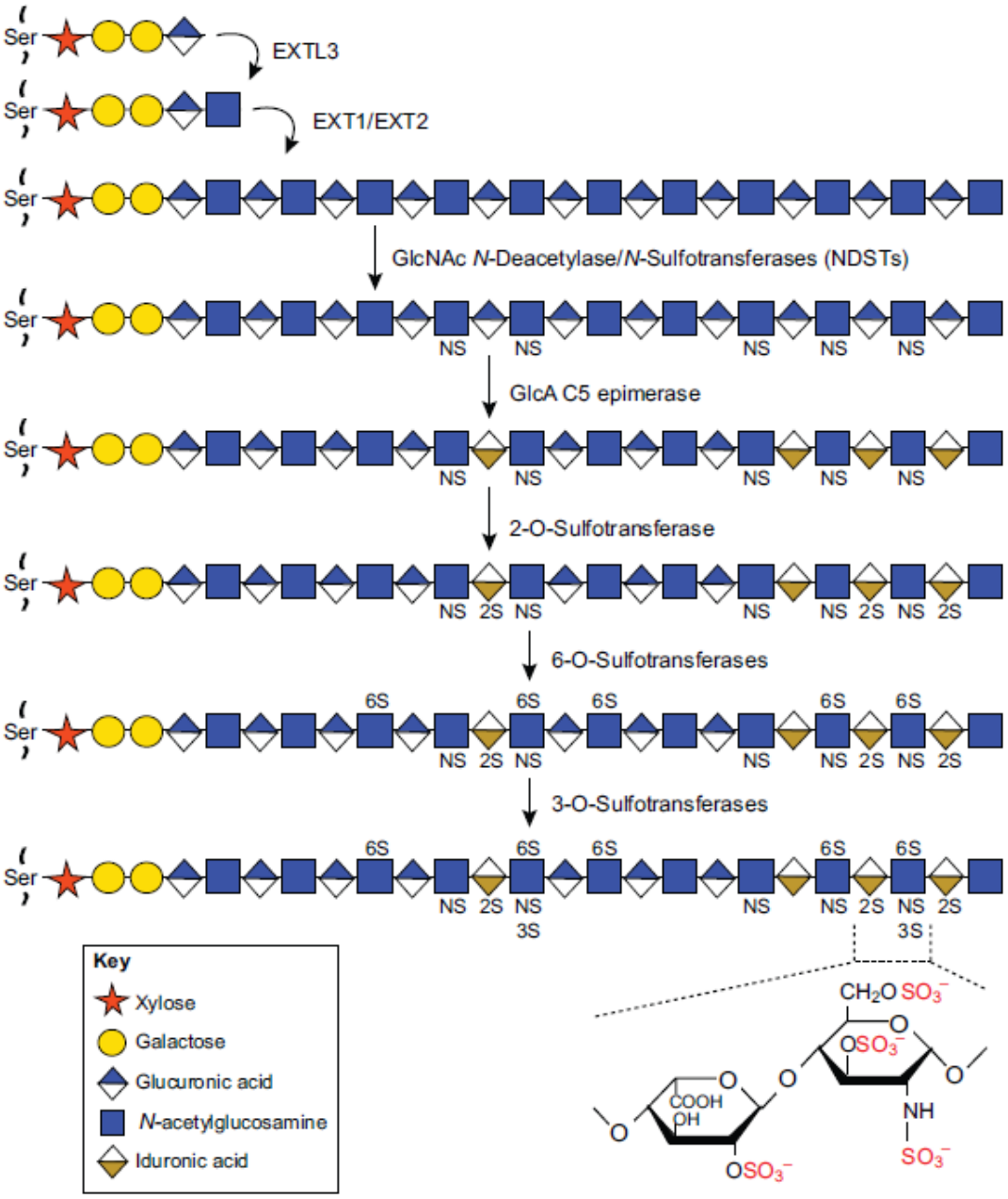


Figure 2.5. Biosynthesis process of heparin/HS showing the formation and modification of polysaccharide backbone, and the enzymes involved in each biosynthetic step. Image adapted from [49]. Copyright 1987 from The Company of Biologists Ltd. Adapted with permission.

The synthesis of heparin/HS is not template-driven as it involves various enzyme-catalysed reactions. These reactions are dependent on the expression level and specificity of participating enzymes which may vary in different cell types, thereby influencing the ability of the monosaccharide units to be modified and eventually contributing to the heterogeneity of heparin/HS structure regarding chain length, degree of sulphation, epimerisation, size and spacing of the sulphated domains [35, 52]. The chemical backbone of HS and heparin differs in a way that HS chains usually contain more GlcNAc residues which are located in unmodified N-acetylated (NA) domains, while heparin chains contain larger N-sulphated (NS) domains in which GlcNS residues are located [50]. Moreover, HS possesses fewer O-sulphate groups as compared to heparin, with the range of sulphate/disaccharide being 0.6-1.8 and 1.8-2.6, respectively [52]. Figure 2.6 illustrates the polysaccharide backbone of HS consisting of different NA and NS domains.

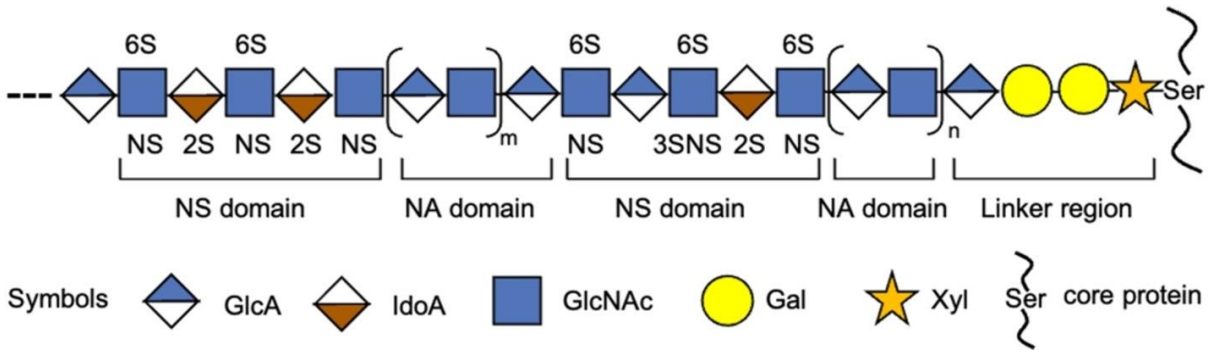


Figure 2.6. Chemical backbone of HS consisting of N-acetylated (NA) domains and N-sulphated (NS) domains. Image adapted from [53]. Copyright 2022 by John Wiley and Sons. Adapted with permission.

2.3.2. Biological functions of heparin/HS

Heparin is well-known for its therapeutic use as an anticoagulant [54]. This is due to the presence of a pentasaccharide sequence containing the specific 3-O-sulphate group, GlcNS6S-GlcA-

GlcNS**3S**,6S-IdoA2S-GlcNS6S that drives the interaction between heparin and anti-thrombin III (ATIII) and further inhibits thrombin and other coagulation cascade proteases [55].

The main biological function of HS is to interact with growth factors, protect them from proteolytic degradation, and modulate their activities in various physiological processes, including cell attachment, migration, differentiation, morphogenesis, inflammation, and responses to injury [52]. The interaction between HS and growth factors predominantly occurs between the negatively charged sulphate and carboxyl groups of the HS chains, and growth factors' positively charged residues [56]. The vast heterogeneity of HS structure plays a key role in its ability to interact with a wide range of signalling molecules, and consequently modulate the specific interactions and downstream biological functions. Variation in HS structures, such as chain length, degree and pattern of sulphation, and polysaccharide sequence, can result in distinct binding abilities between HS and growth factors, which leads to the alteration of signalling pathway and in turn produces distinct biological responses [57].

HS is a cofactor in FGF/FGFR signalling. It has been shown that FGF2 requires the assembly of the ternary complex consisting of FGF2, FGFR1c and HS to initiate the signal transduction. To interact with FGF2 and FGFR1c, it is essential that 2-O-sulphate, 6-O-sulphate, and N-sulphate groups are present within HS oligosaccharide sequences [58], which illustrates the structural specificity in the interaction between HS and growth factors. Similarly, 2-O-, 6-O, and N-sulphation of HS are reported to contribute to the interaction with VEGF165, in which 6-O-sulphation appears to be particularly important. It has been shown that the number and position of 6-O-sulphate groups within NS domains of HS chain determined HS affinity for VEGF165 [59, 60]. The binding of HS to VEGF165 prolonged the growth factor bioactivity and protected it from physiological degradation

[61]. These findings indicate that the interaction between HS and growth factors requires specific structure of HS to facilitate the signalling activity of growth factors.

2.4. Heparan sulphate mimetics

The extraction of HS from animal tissues yields a few milligrams of materials with high levels of structural heterogeneity, which is insufficient for further applications [62, 63]. To overcome this issue, heparin has been used as an alternative model to study the specificity of HS-binding growth factors owing to the abundant amount of commercial heparin. However, since the interactions between HS and growth factors require a specific structure of HS, the binding of a growth factor to heparin does not necessarily imply that it will bind to HS with the same affinity, resulting in different biological functions of heparin and HS. Therefore, alternative approaches have been used to develop biomimetic HS with well-defined saccharide sequences to study the relationship between structures and functions, and hence develop materials for tissue engineering and therapeutic applications.

One approach to produce HS is via chemical synthesis [64] which involves using GlcNAc-GlcA disaccharide precursors as building blocks to obtain HS oligosaccharides [65]. Sulphation of disaccharides can occur at N-position of glucosamine and C2 position of uronic acid [66]. However, the chemical synthesis of HS still requires numerous reaction steps and purification of product after each step, making this mimetic approach relatively challenging. An alternative to chemical synthesis is chemical modification of polysaccharide backbone. Here, polysaccharide backbones are modified to incorporate sulphate groups via chemical reactions where sulphur-containing complexes are employed as sulphating agent [67]. Various synthetically sulphated

polysaccharides have been synthesised, including sulphated cellulose and sulphated chitosan where they were shown to potentiate the activities of several growth factors including BMP-2 and VEGF, and induce biological functions such as bone and vessel formation [68, 69]. Sulphated hyaluronan and sulphated alginate were also produced for the delivery of stromal cell-derived factor 1 and the control of stem cell behaviour, respectively [70, 71]. Regarding *in vivo* wound healing effect, sulphated chitosan has been used to develop hydrogels that promoted wound healing in mice models by enhancing wound closure rate and vascularisation [72, 73]. In addition, OTR4120, a sulphated carboxymethyl dextran, has been shown to increase collagen content and angiogenesis in diabetic rat models, thus improving diabetes impaired wound healing [74]. This type of HS mimetics has also been used in clinical setting for the treatment of skin injury, demonstrating the therapeutic advantage of this material [75]. Despite being bioactive, sulphated polysaccharides obtained by chemical modification are mainly heterogeneous populations due to the regioselectivity of sulphation and different reactivity of hydroxyl groups. Moreover, the harsh reaction conditions required for the sulphating reagents also add more complexity to the synthesis process.

An alternative to chemical synthesis is the use of enzymes to create sulphated polysaccharides to be used as HS biomimetic materials. The enzymes used are similar to those present in the biosynthesis of HS, including glycosyltransferases, N- and O-sulfotransferases, and epimerase. Glycosyltransferases are used to add monosaccharides from uridine diphosphate (UDP)-sugar donors to construct the backbone structure of HS which comprises a disaccharide repeating unit of GlcA and GlcNAc [76]. Epimerase is responsible for the conversion of GlcA to IdoA. Sulfotransferases are utilised to transfer sulfo groups from 3'-phosphoadenosine 5'-

phosphosulphate (PAPS) to the HS oligosaccharides [76]. Due to the selectivity of these enzymes in modifying HS monosaccharide units, this approach requires a carefully optimised design to obtain desired HS structures [77]. The chemoenzymatic synthesis is shorter as compared to chemical synthesis regarding the number of steps required; however, there is still a need to synthesise several UDP-sugars to build up HS oligosaccharides as they are not available in natural forms [64]. HS mimetics produced by chemoenzymatic approach have been used as antagonists of receptor of advanced glycation end-products (RAGE) which are linked to many diseases including inflammatory-related diseases, neurodegenerative disorders, and cancer [78].

Another method to synthesise HS biomimetics is to apply recombinant expression of HS proteoglycans, using mammalian cell lines such as Chinese hamster ovary (CHO) and human embryonic kidney (HEK-293 and HEK-293T). Since these cells express the enzymes that are involved in the synthesis of GAGs, the recombinant core protein of HS proteoglycans can be decorated with HS chains post translation [79]. The expression of recombinant domain V of human perlecan, a type of HS proteoglycan, was achieved in HEK-293 cells and shown to promote endothelialisation for vascular bioengineering applications [80]. Syndecan-1, another type of HS proteoglycan, was expressed in CHO cells and demonstrated to exhibit antitumour activity [81]. Though, recombinant production of HS proteoglycans has several disadvantages. Since the biosynthesis of HS is not a DNA/RNA template driven process and the expression level of involving enzymes is different in each cell type, the HS produced varies greatly not only in chain length and sulphation pattern but also in the amount obtained [82], making it difficult to control the structure of HS products.

In short, various methods have been attempted to mimic the structure of HS with the desire of replicating its biological functions. However, these different methods still have their limitations from producing heterogenous structures to low yield and complex synthesis process.

2.5. Marine derived sulphated polysaccharides

Naturally derived materials are promising candidates for the research and development of new types of biomaterials owing to their broad chemical and biological diversity [83]. Nature has rapidly evolved to provide a wide range of materials with different characteristics and functions [84]. Chitosan and alginate are two well-known marine polymers utilised to fabricate various types of biomaterials for different applications in tissue engineering. Chitosan-based membranes/porous scaffolds or alginate-based microspheres are examples of biomaterials from marine-origin polymers for wound healing and cartilage tissue engineering, respectively [85, 86]. Despite the growing development in biomaterials fabrication from those substances, there are still other marine materials that remain to be investigated. One example of this is sulphated polysaccharides (SPs) from macroalgae (also known as seaweeds). Seaweed is a type of marine plant that is available in large amounts and capable of producing a rich source of SPs that can be used in food, cosmetic, pharmaceutical, and biomedical applications [87-89]. There are three major groups of seaweeds including red, brown, and green seaweed, and the main types of SPs they produce consist of carrageenan, fucoidan, and ulvan, respectively. The backbone of these SPs contains sulphate-decorated sugars linked by glycosidic bonds, similar to the structural characteristic of HS. This provides opportunities to understand the structures and functions of SPs to develop novel types of HS biomimetic materials that can overcome the drawbacks of traditional mammalian-derived HS and its existing biomimetics.

The following sections will discuss the three common types of seaweed derived SPs regarding their chemical structures and biological activities. The use of these SPs in developing growth factor delivery systems will also be reviewed, including the relationship between chemical structures and biological functions of SPs.

2.5.1. Carrageenan

Carrageenan is a sulphated polysaccharide occurring as a matrix material of red seaweeds. The backbone units consist of β-(1→4) and α-(1→3) linked D-galactose and 3,6-anhydrogalactose residues with the proportion of sulphate groups ranging from 15 % to 40 % [90]. According to the number of sulphate groups per disaccharide unit, carrageenan can be classified into three main categories: kappa, iota, and lambda [91]. Figure 2.7 illustrates the chemical structures of kappa, iota, and lambda carrageenan. Kappa-carrageenan has sulphate groups at C4 position of the galactose units. Iota-carrageenan contains sulphate groups at C4 position of the galactose residues and C2 position of the 3,6-anhydrogalactose residues. Lambda-carrageenan has the highest number of sulphate groups among three types of carrageenan, with sulphate groups at C2 position of one galactose unit and C2 and C6 position of the other galactose unit.

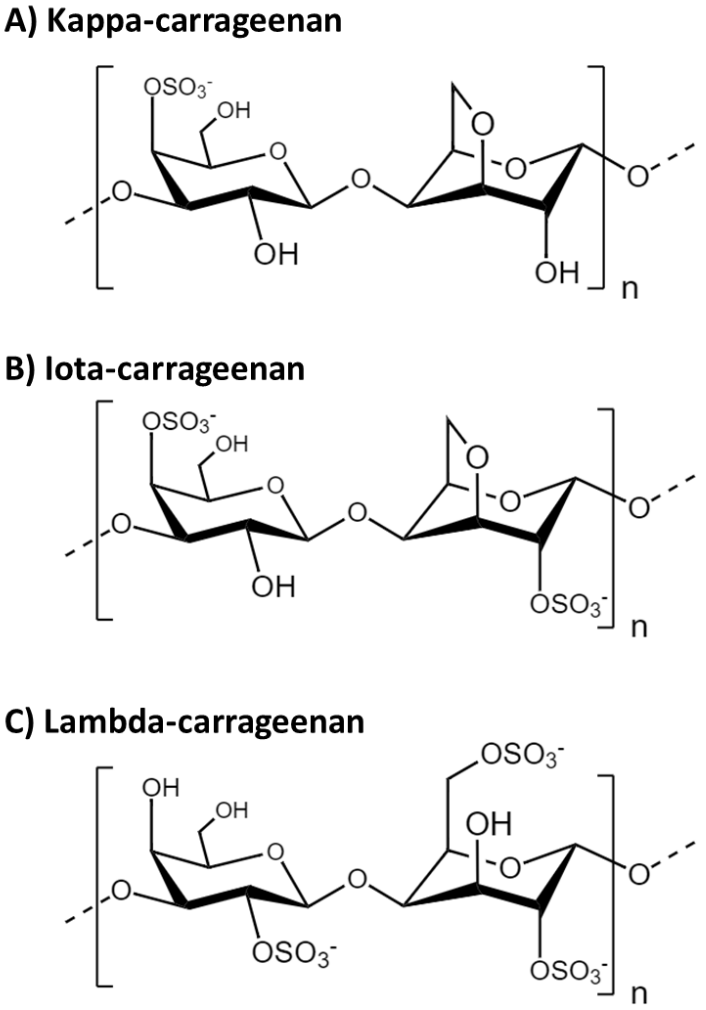


Figure 2.7. Chemical structures of repeating disaccharide units of A) Kappa-carrageenan, B) Iota-carrageenan, and C) Lambda-carrageenan, each composed of alternating β-(1→4) and α-(1→3)-linked galactose units.

Carrageenan has been shown to have anticoagulant, anti-tumour and antiviral activities which vary according to molecular weight and level of sulphate content [92-94]. In addition, it has been employed in various biomedical applications. Yegappan et al. [95] fabricated carrageenan-based nanocomposite hydrogel that supported angiogenesis and osteogenesis *in vitro*, which has shown potential in bone tissue engineering. In addition, the group of Popa et al. [96] developed a hydrogel system from kappa-carrageenan for the delivery of adipose stem cells to injured

cartilage sites. Results show that the hydrogel demonstrated appropriate physicochemical properties which promoted cell proliferation and differentiation. Moreover, research from Rocha et al. [97] indicated that carrageenan could be used to synthesise a thermo-responsive hydrogel formulation for cell and growth factor encapsulation. In terms of *in vivo* effects, carrageenan has been utilised to develop injectable hydrogels or hydrogel films that provided support for the healing of full thickness wounds in rat models [98, 99], demonstrating its potential in the development of HS mimetics for wound healing application.

2.5.2. Fucoidan

Fucoidan, found in the cell wall of brown seaweeds, is a sulphated polysaccharide composed of (1→3)-linked L-fucose units (type I) or alternating (1→3)- and (1→4)-linked L-fucose units (type II) showed in Figure 2.8. Additional sugar residues including xylose, mannose, galactose, and uronic acid units are also detected in fucoidan due to the extraction of fucoidan from various seaweed species [100, 101]. The amount of sulphate groups in fucoidan ranges from 7 to 38 % [102].

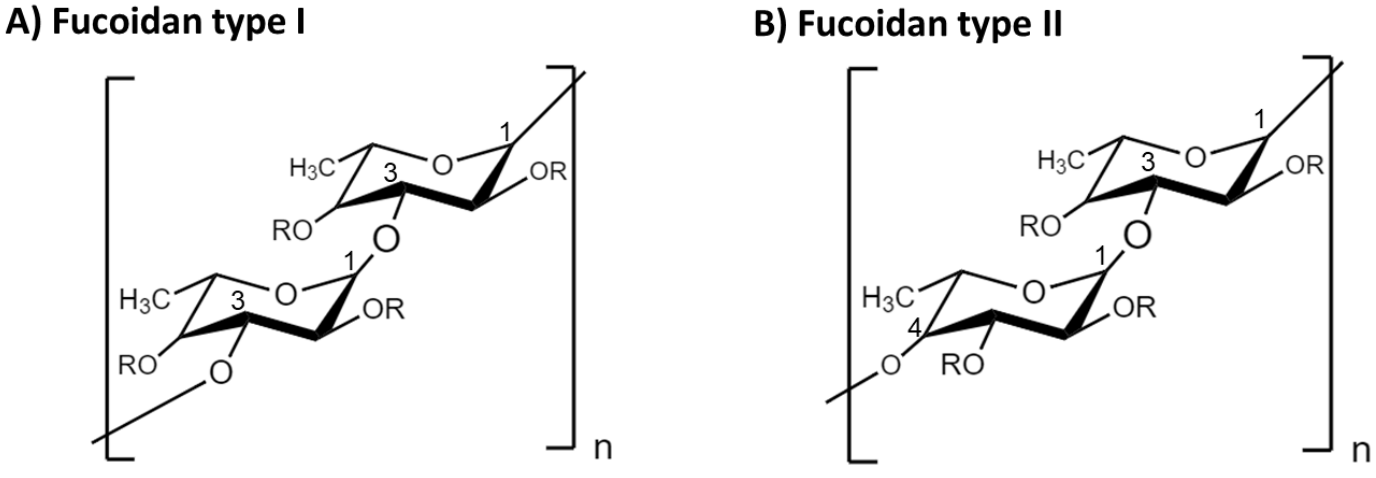


Figure 2.8. Chemical structures of repeating disaccharide units of fucoidan. A) Type I consists of (1→3)-linked L-fucose units. B) Type II consists of alternating (1→3)- and (1→4)-linked L-fucose units.

Key biological properties of fucoidan include anticoagulant and antithrombotic activities [103]. Similar to carrageenan, the properties have been shown to be dependent on molecular weight and sulphate content [104]. Homogeneous low molecular weight fucoidans with more than 20 % of both fucose units and sulphate groups can increase the inhibition of thrombin [105]. Moreover, low molecular weight fucoidans were reported to have pro-angiogenic activity which enhances proliferation and migration of endothelial cells, while fucoidans with high molecular weight possess anti-angiogenic property [106]. Fucoidans have also been shown to exhibit antiviral activity where they inhibited the replication and stimulation of virus through oral administration in a mice model [107]. Regarding tissue engineering applications, fucoidan has been employed in drug delivery and wound healing areas. It has been combined with different polymers like alginate, chitosan, or polycaprolactone (PCL) to produce scaffolds, hydrogels, and nanofibers. Fucoidan-chitosan film-forming hydrogels were fabricated as wound dressing materials and they were shown to promote wound healing and contraction [108]. Fucoidan-PCL porous scaffolds were also created in which the release of fucoidan from the scaffold composite was demonstrated to enhance bone tissue regeneration [109]. In addition, fucoidan was used to synthesise nanoparticles for the delivery and controlled release of stromal cell-derived factor 1 (SDF-1). The nanoparticles protected SDF-1 from *in vitro* environmental effects while maintaining its mitogenic activity and migratory effect on mesenchymal stem cells at the target sites [110].

2.5.3. Ulvan

Ulvan is a sulphated polysaccharide extracted from the cell wall of green seaweed, which accounts for about 8-29 % of the seaweed dry weight. Two main types of repeating disaccharide units of ulvan include ulvanobiuronic acids (type A and B) and ulvanobioses (type U) showed in

Figure 2.9. Type A_{3s} is composed of D-glucuronic acid (1,4)-linked to L-rhamnose 3-sulphate, while type B_{3s} is composed of L-iduronic acid (1,4)-linked to L-rhamnose 3-sulphate. Type U_{3s} consists of D-xylose (1,4)-linked to L-rhamnose 3-sulphate, whereas type $U_{2's3s}$ consists of D-xylose 2-sulphate (1,4)-linked to L-rhamnose 3-sulphate. The degree of sulphation of ulvan varies from 2.3 to 40 % [111].

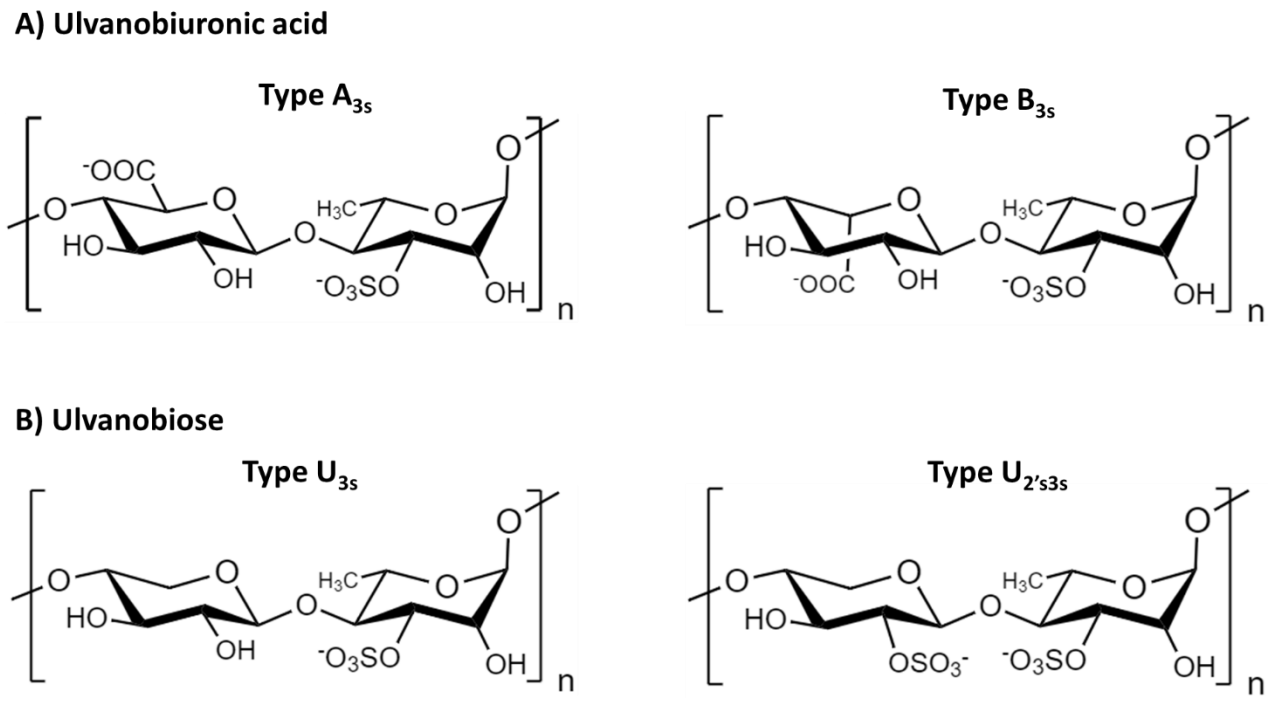


Figure 2.9. Chemical structures of main repeating disaccharide units of ulvan. A) Ulvanobiuronic acid type A_{3s} consists of glucuronic acid linked to rhamnose 3-sulphate, while type B_{3s} consists of iduronic acid linked to rhamnose 3-sulphate. B) Ulvanobiose type U_{3s} is composed of xylose linked to rhamnose 3-sulphate, while type $U_{2's3s}$ is composed of xylose 2-sulphate linked to rhamnose 3-sulphate.

Ulvan and its oligosaccharides have been shown to possess antiviral activity [112], in which the avian influenza viruses were inhibited by ulvan in a dose-dependent and strain-specific manner. Moreover, ulvan also exhibited anticoagulant effect where this property was compared with

heparin in terms of activated partial thromboplastin time using normal human plasma [113]. The antilipidemic activity of ulvan has also been reported since it reduced total serum cholesterol and low-density lipoprotein cholesterol [114]. In terms of biomedical applications, ulvan has been used as an agent for therapeutics or drugs delivery [115]. Furthermore, it has been employed to produce ulvan-based hydrogels for enhanced wound healing owing to appropriate physiochemical properties and biocompatibility [116]. Epidermal growth factor (EGF) was incorporated into the hydrogels, and the sustained release of EGF was shown to promote cell proliferation and wound healing. No information regarding the interaction between ulvan and EGF was reported in this study [116]. Another study also created hydrogels for wound dressing application using ulvan as one of the hydrogel components [117]. The hydrogels were shown to enhance wound healing in diabetic mice models, highlighting the *in vivo* therapeutic effects of ulvan [117].

2.6. Growth factor binding and signalling of marine derived sulphated polysaccharides.

The structures of SPs extracted from seaweeds vary remarkably due to seaweed species, seasonality, and extraction and/or purification conditions [118]. Furthermore, their abilities to exhibit biological activities require specific structural characteristics such as molecular weight, sulphate pattern, degree of sulphation [119], which is similar to HS. Therefore, understanding the structural variation of SPs could give insights into the structure-function relationships to design HS biomimetic materials [120].

SPs are extracted from various seaweed species using chemical or enzymatic methods, followed by purification step by dialysis and chromatographic techniques [121]. SPs are characterised in terms of their constituent monosaccharides and sulphate content using spectroscopic methods such as nuclear magnetic resonance (NMR) and Fourier-transform infrared spectroscopy (FTIR), elemental analysis, or chemometric methods including colorimetric and turbidimetric assay [111, 122, 123].

Various studies have been conducted to understand the correlation between the structural features of SPs and their biological activities. While investigations mainly focus on the impact of molecular weight on the biological activities of SPs such as anticoagulant [124], antiviral [101], and anti-tumour [92], the impacts of sulphate content and pattern remain to be explored. In terms of developing HS biomimetic materials from marine derived SPs, the study of SPs - growth factors interactions and downstream biological functions is critical to evaluate the ability of SPs to mimic HS in modulating growth factors activities. Information in the following pages summarises results from recent studies that investigated the structural characteristics of SPs from various seaweed species and their modulating effects on growth factor binding (Table 2.1) and downstream functions (Table 2.2).

Table 2.1. Summary of structural characteristics of SPs from various seaweed species and their modulating effects on growth factor binding

SPs and seaweed species	Structural characteristics	Modulating effects on growth factor binding	References
Fucoidan from Fucus vesiculosus	Molecular weight (MW): 100-130 kDa Sulphate content: 32.6 %	Bound VEGF165 with affinity of 368 nM	[125]
Fucoidan from Fucus vesiculosus	MW: 100-130 kDa Sulphate content: 56.7 %	Bound VEGF165 with affinity of 52 nM	[125]
Fucoidan from Sargassum thunbergii and Sargassum japonica	MW: 10 -100 kDa	Exhibited strong affinity towards FGF-1 and FGF-2 with affinity of 69 nM and 3.9 nM, respectively.	[126]
Lambda-carrageenan (Sigma-Aldrich product)	No structural characteristics were reported	Bound FGF-1 and FGF-2 and increased their thermal stability. No information regarding affinity was reported	[127]
Ulvan from Enteromorpha prolifera	MW: 7.7 kDa Sulphate content: 16 %	Interacted with FGF-1 and FGF-2. No information regarding affinity was reported.	[128]

Table 2.2. Summary of structural characteristics of SPs from various seaweed species and their modulating effects on growth factor downstream functions

SPs and seaweed species	Structural characteristics	Modulating effects on growth factor downstream functions	References
Fucoidan from Fucus vesiculosus	Molecular weight (MW): 100-130 kDa Sulphate content: 32.6 %	Moderately inhibited FGF-2-induced migration and tube formation of human umbilical vein endothelial cells (HUVEC) (100 µg/mL) Decreased the binding of VEGF165 to its cell receptor, thereby inhibiting the proliferation and migration of HUVECs induced by VEGF165	[125, 129]
Fucoidan from Fucus vesiculosus	MW: 100-130 kDa Sulphate content: 56.7 %	Inhibited HUVEC migration and tube formation Inhibited the migration and tube formation of HUVECs induced by FGF2 (100 µg/mL) Promoted FGF-2-receptor interaction for autophosphorylation in the presence of HS; Inhibited FGF-2-receptor interaction when HS was removed. Inhibited the proliferation and migration of HUVEC induced by VEGF165 Prevented the binding of VEGF165 to its receptor, lowering receptor phosphorylation	[125, 129, 130]

SPs and seaweed species	Structural characteristics	Modulating effects on growth factor downstream functions	References
Fucoidan from Undaria pinnatifida	MW: ~ 95.2 kDa Sulphate content: 21 %	Inhibited the proliferation, migration, tube formation and vascular network formation of HUVEC at the concentration of 400 µg/mL, suppressed ex vivo micro vessel formation at 100 µg/mL Down-regulated VEGF165 and VEGF121 expression	[131]
Fucoidan from Sargassum fusiforme	MW: 47.5 kDa Sulphate content: 20.8 %	Inhibited tube formation of human microvascular endothelial cells (HMEC-1) (25 µg/mL) Inhibited cell migration (0.4 mg/mL)	[132]
Fucoidan from Laminaria japonica	Medium MW fucoidan 1 (MMWF-1): 15-20 kDa, 8.2 % sulphate. MMWF-2: 30 kDa, 32.2 % sulphate. MMWF-3: 30 kDa, 33.2 % sulphate.	MMWF-2 and -3 had similar MW, hence the results of MMWF-1 and -3 were mainly discussed. MMWF-3 inhibited HUVEC tube formation and micro vessel growth in an ex vivo model, while MMWF-1 had no impact on tube formation Both MMWF-1 and -3 stimulated HUVEC migration in the presence of VEGF The authors suggested MW of 20–30 kDa seemed to be a critical point for the effect of fucoidan on endothelial cells functions and angiogenesis	[133]

SPs and seaweed species	Structural characteristics	Modulating effects on growth factor downstream functions	References
Fucoidan from Ascophyllumm nodosum	MW: ~16 kDa Sulphate content: 29 %	Had no impact on HUVEC proliferation in the presence of FGF-2 (5 ng/mL) Enhanced α6, β1 and PECAM-1, and inhibited $\alpha_V\beta_3$ integrin expression in the presence of FGF-2 Enhanced FGF-2-induced vascular tube formation by HUVEC	[134]
Fucoidan from Ascophyllumm nodosum	MW: ~5.1 kDa	Increased the migration of endothelial cells by promoting VEGF165 binding to VEGFR-2	[135]
Fucoidan from Ascophyllumm nodosum	MW: 4 kDa Sulphate content: 4 %	Enhanced endothelial progenitor cells (EPC) proliferation (1 μg/mL) Enhanced FGF-2-induced vascular tube formation of EPC and HUVEC Enhanced HUVEC attachment to laminin by enhancing α6 integrin expression Promoted EPC migration in the presence of VEGF	[136]
Fucoidan from Laminaria japonica	Sulphate content: 23.5 %	Promoted the proliferation, migration, and tube formation of HUVEC in the presence of FGF-2	[137]
Fucoidan from Saccharina japonica	MW: 10.9 kDa Sulphate content: 41.34 %	Activated FGF-1, -2, -7, -8, -9, or -10/FGFR1c signalling activity, demonstrated by BaF3 cell proliferation	[138]

SPs and seaweed species	Structural characteristics	Modulating effects on growth factor downstream functions	References
Fucoidan from Laminaria japonica	Fraction 1 MW: 7.4 kDa Sulphate: 34.02 % Fraction 2 MW: 34 kDa Sulphate: 29.57 %	Activated FGF-1, -2, -7, -8, -9, or -10/FGFR1c signalling activity, demonstrated by BaF3 cell proliferation. The fraction with lower MW promoted endothelial protection in an in vivo model	[139]
Lambda-carrageenan (Sigma-Aldrich product)	Different oligosaccharides with different degree of polymerisation (DP)	Oligosaccharide with DP7 had high affinity to FGF-2, and inhibited cell proliferation induced by FGF-2 and heparanase activity	[127]
Lambda-carrageenan (Sigma-Aldrich product)	No structural characteristics were reported	Protected FGF-1 biological activity in prolonged incubation	[140]
Ulvan from Ulvan ohnoi	Various fractions with MW ranging from 7 kDa to 209 kDa	Modulated the expression level of inflammatory-mediating molecules. High MW ulvan fractions exhibited better immunomodulatory effect than low MW fractions	[141]
Ulvan from Enteromorpha prolifera	MW: 7.7 kDa Sulphate content: 16 %	Inhibited A549 cancer cell proliferation.	[128]

From Table 2.1 and Table 2.2, fucoidan has been investigated with the key focus being the effects of molecular weight and sulphate content on the pro- and anti-angiogenic activities of fucoidan. Briefly, fucoidan with MW higher than 20 kDa demonstrated anti-angiogenic effect, which inhibited the migration, proliferation and tube formation of endothelial cells induced by either FGF-2 or VEGF165. Some fractions were shown to inhibit the binding of VEGF165 to its receptor [125, 129]. In contrast, fucoidan with MW lower than 20 kDa was likely to show pro-angiogenic effect by enhancing vascular tube formation and the expression of integrins involved in angiogenesis [134, 136]. In terms of sulphate content, results from studies [126, 138, 139] imply that fucoidan with sulphate content higher than 30 % or with intermediate and high level of negative charge had high affinity for FGF-1 and FGF-2, and promoted the growth factors binding to FGFR1c for signalling activity. Similar trend was also observed when fucoidan with sulphate content of 43 % was incorporated into chitosan/alginate scaffolds [142]. The loaded fucoidan had the capacity to protect FGF-2 from degradation, control its release, and enhance fibroblast migration [142]. However, fucoidan with the sulphate content of 56.7 % was shown to prevent the binding of VEGF165 to its receptor, thereby lowering receptor phosphorylation and inhibiting HUVEC proliferation and migration [125, 129, 130]. This contrasts with fucoidan having lower sulphate content of 23.5 % which promoted the proliferation of HUVEC in the presence of FGF-2 [137]. Together these findings indicate a correlation between the sulphate content of fucoidan and its ability to induce cellular responses.

Regarding carrageenan and ulvan, investigations into the impact of structural variation on growth factor interactions and subsequent biological functions have been conducted, as can be seen from Table 2.1 and Table 2.2. The main findings show that these two types of SPs could interact

with FGF-1 and FGF-2 and inhibit cell proliferation. However, adequate information about their structural features was not provided, and thus the correlation between molecular weight or sulphate content and bioactivities was not well established.

Recent studies regarding the relationship between the structures and the biological activities of SPs still has several limitations. Firstly, the SPs used in some studies were degraded by hydrolysis to obtain low molecular weight fractions, or modified by oversulphation to achieve high sulphate content [125, 127]. The hydrolytic treatment or chemical modification might affect the spatial conformation of the polymer chains, resulting in the alteration of biological functions. Alternatively, a thorough analysis on the structural features of intact polysaccharides should be performed. Secondly, the biological activities of SPs were usually compared with that of heparin in lieu of heparan sulphate [134, 138, 139]. This could be because heparin is available in abundant amount and the structure of heparin is presumably similar to HS. In fact, HS has a heterogeneous domain structure which consists of partially sulphated NS domains that are separated by NA domains, whereas heparin chains contain more NS domains with higher degree of sulphation. This structural difference between heparin and HS can lead to distinct binding affinities with growth factors and downstream biological functions. Moreover, since HS is mainly involved in the interactions with a variety of growth factors, using only heparin as a control sample in the growth factor interaction study might not provide accurate information regarding how a GAG binds and signals growth factors. Therefore, it is necessary to utilise both HS and heparin to compare with the functions of SPs for the development of HS biomimetic materials. Lastly, almost every mentioned study focused on examining only one type of SP but did not compare between

different types of SPs or SPs within the same seaweed group. Hence, comprehensive explorations and comparisons between the biological effects of various types of SPs remain to be investigated.

Understanding the relationship between structures and functions of seaweed derived SPs is a crucial step in the process of creating new HS mimetics. As they hold the ability to interact and modulate growth factor biological activities, utilising SPs to develop growth factor delivery systems has become a promising approach. The next section will provide a summary of some recent studies that particularly focus on the fabrication of growth factor delivery systems that are composed of seaweed derived SPs.

2.7. Application of seaweed derived sulphated polysaccharides in developing growth factor delivery systems.

Growth factors are components of the ECM that control various cellular processes during development and wound healing [143]. However, the biological activities of growth factors are often hindered by short half-life, slow diffusion to the target site and proteolytic degradation in chronic wound environment [25]. Hence, it is of great importance to develop strategies for protecting and delivering growth factors to the target site to overcome the mentioned limitations and enhance the therapeutic efficiency of growth factors. Growth factor delivery systems can be fabricated in various forms, such as scaffolds, nanoparticles, and hydrogels and can be utilised in various tissue engineering applications, namely bone tissue engineering, cartilage tissue engineering and wound repair. Materials employed for the fabrication of these systems mainly include synthetic polymers, inorganic composites, or naturally derived materials [144]. Among them, natural materials are particularly suitable for growth factor delivery due to their general

biodegradability and bioactive ability which facilitates growth factor storage and diffusion [144, 145].

When selecting materials for the development of growth factor delivery systems, one of the most important properties to consider is the ability of the material to interact with growth factors to form complexes. Growth factors involved in wound healing process typically include FGF2, VEGF and TGF, all of which are cationic growth factors [145]. In this case, anionic polymers appear to be appropriate candidates for the fabrication of biomaterials that act as carriers in the delivery of such growth factors. Carrageenan, fucoidan, and ulvan are three types of anionic polymers that are naturally derived from seaweeds. They are negatively charged polymers due to the presence of sulfate groups, which makes them potential for the delivery of positively charged growth factors. Table 2.3 details various growth factor delivery systems manufactured from the seaweed derived polysaccharides as well as the growth factors encapsulated with corresponding applications.

Table 2.3. Growth factor delivery systems fabricated from various types of seaweed derived polysaccharides and their tissue engineering applications.

SP-based composites	**Degree of sulphation and position of sulphate groups**	**Growth factor encapsulated**	**Biological effects**	**Application**	**References**
Kappa-carrageenan based hydrogel beads	Not mentioned	Platelet derived growth factor (PDGF-BB)	Promoted high encapsulation efficiency and controlled release of	Bone tissue engineering	[146]

SP-based composites	Degree of sulphation and position of sulphate groups	Growth factor encapsulated	Biological effects	Application	References
			PDGF-BB in the first 14 days of the test period. Demonstrated biocompatibility against L929 fibroblasts after 7 days of culture.		
Kappa-carrageenan hydrogels	Not mentioned	Transforming growth factor-β1	Promoted the biological activity of TGF-β1 for enhancing the chondrogenic differentiation of adipose-derived stem cells and the expression of cartilage extracellular matrix molecules.	Cartilage tissue engineering	[97]

SP-based composites	Degree of sulphation and position of sulphate groups	Growth factor encapsulated	Biological effects	Application	References
Fucoidan based 3D scaffolds	Not mentioned	Vascular endothelial growth factor (VEGF)-165	Controlled the release of VEGF165 and directed vascularisation in an angiogenesis model to promote the formation of blood vessels and their functionality.	Vascularisation	[147]
Fucoidan based nanoparticles	Not mentioned	Basic fibroblast growth factor (bFGF)	Protected bFGF from deactivation by physiological factors such as enzymes. Controlled the release of bFGF and reduced the amount of bFGF required for neurite extension.	Nerve tissue engineering	[148]

SP-based composites	Degree of sulphation and position of sulphate groups	Growth factor encapsulated	Biological effects	Application	References
Fucoidan based scaffolds	43 % sulphation degree	Basic fibroblast growth factor (bFGF)	Maintained the activity and release of bFGF Elevated the viability of L929 fibroblasts and promoted their migration.	Wound repair	[142]
Ulvan based hydrogels	Not mentioned	Epidermal growth factor (EGF)	Supported the delivery and sustained release of EGF at wound site for 15 days, Accelerated granulations tissue formation and collagen deposition.	Wound healing	[149]

From Table 2.3, only kappa-carrageenan was reported while other types of carrageenan such as iota- and lambda-carrageenan have not been utilised for the fabrication of GF delivery systems. According to study [147], fucoidan was shown to bind VEGF165 with high affinity and induce a synergistic effect that promoted neovessel formation in mice models. Similarly, the signalling of

FGF2 was potentiated by the presence of fucoidan in the delivery systems to increase neurite extension and fibroblast migration, as described in study [148] and [142], respectively. Regarding ulvan, it has only been documented once where ulvan based hydrogels were created for the delivery of EGF to be used in wound healing applications [149]. Specifically, this study indicates that EGF released from the hydrogel system was responsible for enhancing fibroblast migration and wound healing. In most of the studies listed in Table 2.3, little is known about the structural characteristics of SPs used in the GF delivery systems. Particularly, the degree of sulphation and the position of sulphate groups were not clearly mentioned. Since there is a strong relationship between the chemical structures of SPs and their biological functions, it is essential to understand the structural features of SPs regarding the degree of sulphation and position of sulphate groups prior to employing them in the creation of GF delivery systems. This will help to better control the interaction between SPs and GFs and tailor their downstream biological effects to specific biomedical applications.

2.8. Research questions and thesis aims

Up to now, the biological functions of fucoidan have been investigated in the literature regarding the effects of varying structural features (molecular weight and sulphate content) on growth factor activities and downstream cellular responses. However, little is known about the structural variation of carrageenan and ulvan and its influence on the biological consequences. Investigating into the structure-function relationship of these marine based SPs will further provide a fundamental understanding of how different structures of SPs exhibit distinct functions.

When it comes to the interaction between HS and growth factors, varying sulphation level and position of sulphate groups in HS has higher impact on the growth factor activities than varying molecular weight. This is because of the negatively charged sulphate groups that are predominantly responsible for the interactions with positively charged lysine and arginine residues in the growth factors, not to mention that some HS-growth factor interactions also require highly specific HS structures for activating growth factor receptors. Therefore, it is worthwhile investigating how the change in the sulphation level and position of sulphate groups of seaweed derived SPs modulates growth factor activities. It is also important to examine the downstream biological functions of these SPs to fully understand their effect on cellular responses. Investigations of structure-function relationships of marine derived SPs would increase our knowledge about the structural characteristics and biological activities of these materials, which subsequently feeds into the development of new HS biomimetic material technologies.

Taking the current research gaps, this thesis is going to explore the following research questions.

Q1: How does the variation in degree of sulphation and position of sulphate groups of SPs affect growth factor binding?

Q2: How does the variation in degree of sulphation and position of sulphate groups of SPs impact growth factor signalling and cellular responses?

Q3: How does the degree of sulphation vary within each type of SPs and how does this variation influence growth factor signalling?

To answer these questions, the following aims were formulated.

Aim 1: To characterise the degree of sulphation and position of sulphate groups of seaweed derived SPs and examine how these sulphation characteristics impact growth factor binding.

Aim 2: To investigate the effect of different sulphation characteristics of seaweed derived SPs on growth factor signalling and cellular responses.

Aim 3: To evaluate the influence of varying sulphur level within each type of SPs on growth factor signalling.

Chapter 3

Characterisation of degree of sulphation and position of sulphate groups of seaweed derived sulphated polysaccharides and how these sulphation characteristics impact growth factor binding

3.1. Introduction

Heparin and heparan sulphate (HS) are glycosaminoglycans (GAGs) that have important functions in regulating the biological activities of cells and tissues [150]. Heparin is found within the intracellular granules of mast cells and is available commercially as an anticoagulant. Meanwhile, HS is present on the surface of mammalian cells or in the extracellular matrix (ECM) and one of its main functions is modulating the biological activity of different molecules such as chemokines and growth factors [151]. Heparin and HS are both composed of repeating disaccharide units of glucosamine (GlcN) units, and either glucuronic acid (GlcA) or iduronic acid (IdoA) units, which form the chemical backbone of heparin/HS (Figure 3.1A and 3.1B). These units are modified with sulphate groups at various positions of oxygen and/or nitrogen atoms on the saccharide ring, including 2-O-sulphation of uronic acid, N-sulphation, 3-O- and 6-O-sulphation of glucosamine unit [152].

The degree of sulphation, position of sulphate groups, and saccharide unit of heparin/HS are dependent on the modification reactions occurring during their biosynthesis process. These reactions are catalysed by different enzymes including N-deacetylase-N-sulfotransferase (NDST), C5-epimerase, 2-O, 3-O and 6-O sulfotransferase. NDST removes the acetyl groups on N-acetyl glucosamine (GlcNAc) residues and replaces them with sulphate groups, creating N-sulphate glucosamine units (GlcNS). C5-epimerase epimerises some of the glucuronic acid (GlcA) residues to iduronic acid (IdoA). 2-O sulfotransferase helps to add sulphate groups to the C2 position of IdoA units, while 3-O and 6-O sulfotransferase add sulphate groups to the C3 and C6 position of GlcNS units, respectively. The expression level of these enzymes varies in different cell types and tissue environment, which contributes to the heterogeneous structure of heparin/HS in terms

of length of polysaccharide chains, degree of sulphation, epimerisation, and size and spacing of sulphated domains [153, 154]. The chemical backbone of heparin and HS differs in a way that HS contains more GlcNAc residues located in unmodified N-acetylated (NA) domains while heparin contains larger N-sulphated (NS) domains where GlcNS residues are located [50]. Furthermore, heparin has more sulphate groups as compared to HS [52]. These differences contribute to not only differing biological functions between heparin and HS but also differing biological functions between different HS structures.

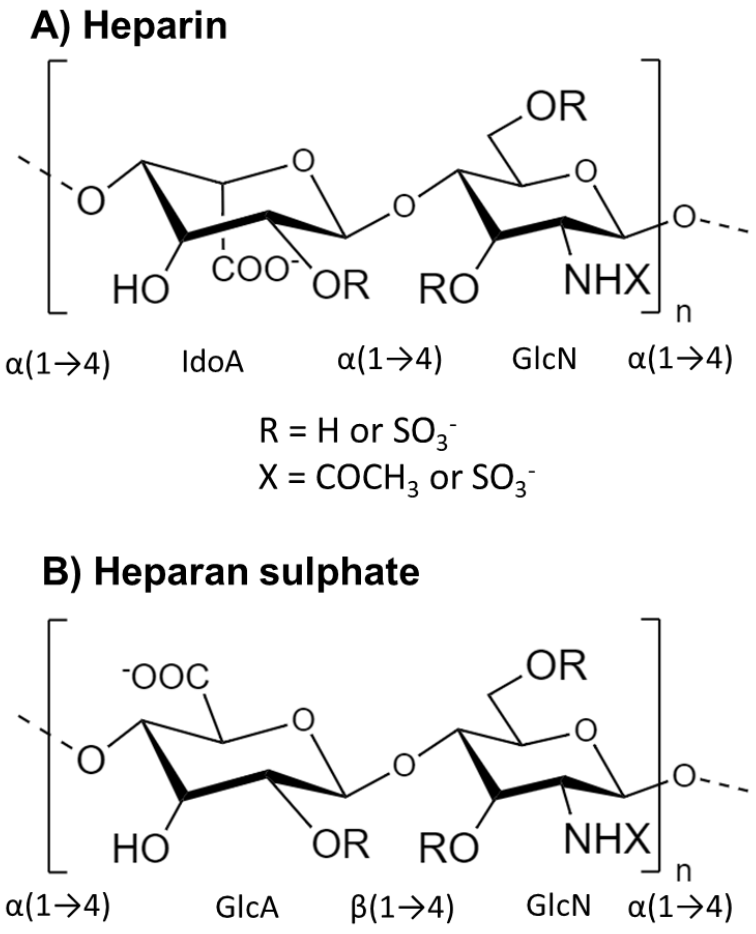


Figure 3.1. Chemical structures of disaccharide units of A) Heparin and B) Heparan sulphate

HS plays a critical role in modulating the biological activity of biomolecules in the body, which makes it important in different biological processes such as wound healing and tissue repair. Therefore, the production of HS as potential therapeutics is essential for biomedical applications. Both HS and heparin can be extracted from animal tissues. However, while heparin is extracted from porcine intestinal mucosa with the yield ranging from 180 to 260 mg per animal [155], the extraction of HS from animal tissues faces challenges with low yield and structural heterogeneity because the structure of HS is dependent on cell type and cell environment [150, 156]. Since the extraction of HS from mammalian tissues is limited, other methods have been explored to produce HS mimetic materials, including incorporation of sulphate groups in the chemical backbones of different types of polysaccharides [157-159], chemical and chemoenzymatic synthesis of HS oligosaccharides, and *in vitro* recombinant expression of HS proteoglycans [7, 160-162]. However, these methods involve numerous and complex synthesis steps, and the product yield was not sufficient for biomedical research, as previously described in Chapter 2.

To overcome the drawbacks in the production of HS mimetics, sulphated polysaccharides have been used as an alternative source of materials. Seaweeds are known to contain an abundant amount of bioactive compounds including sulphated polysaccharides (SPs). Seaweed derived SPs are already commercially available and used in many industries such as food and cosmetics [163, 164]. These SPs also have been shown to exhibit biological activities including anticoagulant, anticancer, and antiviral activities [165] which vary greatly depending on the structure of SPs. Thus, we need to further understand how SP structure is related to function for future use of these materials in biomedical applications. In the context of developing HS mimetics, carrageenan and ulvan are two potential candidates as they have similar structural characteristics

to HS in which they contain sulphate groups in their chemical backbone. These SPs are extracted from the cell wall of red and green seaweed, respectively [166]. There are three main types of carrageenan, including kappa-carrageenan (Figure 3.2A), iota-carrageenan (Figure 3.2B), and lambda-carrageenan (Figure 3.2C). Kappa- and iota-carrageenan consist of galactose and anhydro-galactose units. Kappa-carrageenan has sulphate groups at C4 position in the galactose units, while iota-carrageenan has sulphate groups at C4 position in the galactose units and C2 position in the anhydro-galactose units. Lambda-carrageenan is composed of galactose units only. In a repeating disaccharide unit of lambda-carrageenan, sulphate groups are decorated at C2 position in one galactose unit, and C4 and C6 position in the other galactose residue [167]. Regarding ulvan, there are two common ulvan types which are distinguished based on their repeating disaccharide units. Ulvanobiuronic acid contains rhamnose and glucuronic acid or iduronic acid with sulphate groups at C3 position in rhamnose units (Figure 3.3A). Ulvanobiose contains rhamnose and xylose with sulphate groups at C3 position in rhamnose residues or C2 position in xylose residues (Figure 3.3B) [168]. The sulphation degree and position of sulphate groups of carrageenan and ulvan can vary greatly based on the seaweed species and extraction conditions used, which will impact on their biological activities including the ability to interact with biological molecules.

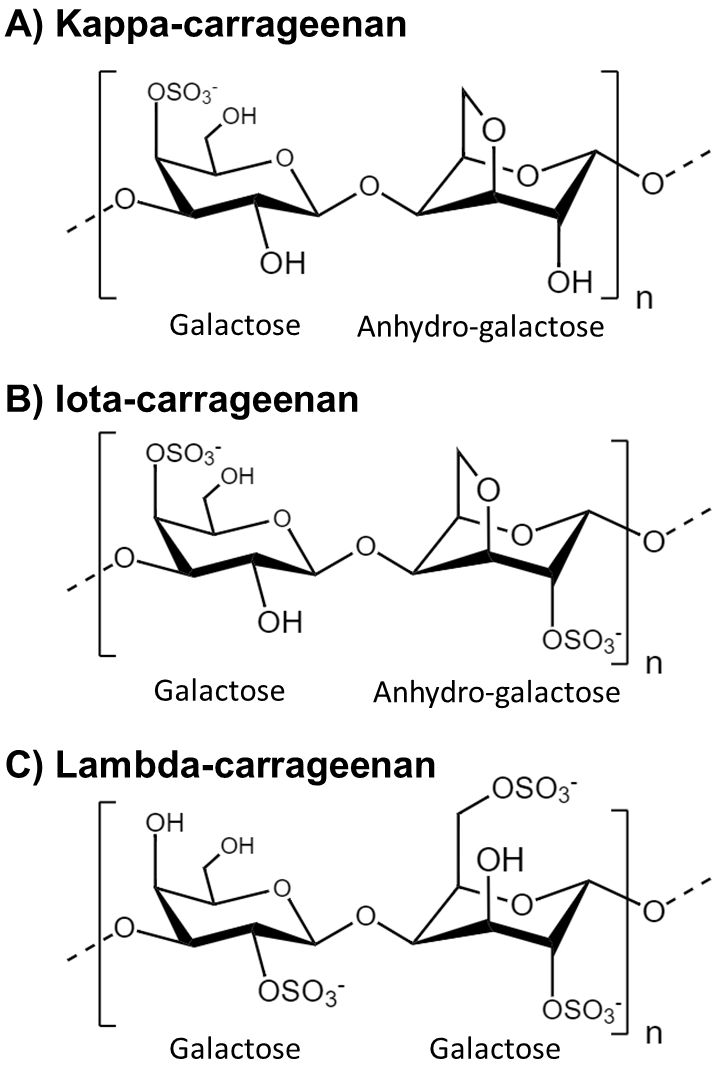


Figure 3.2. Chemical structures of disaccharide units of A) Kappa-carrageenan, B) Iota-carrageenan, and C) Lambda-carrageenan

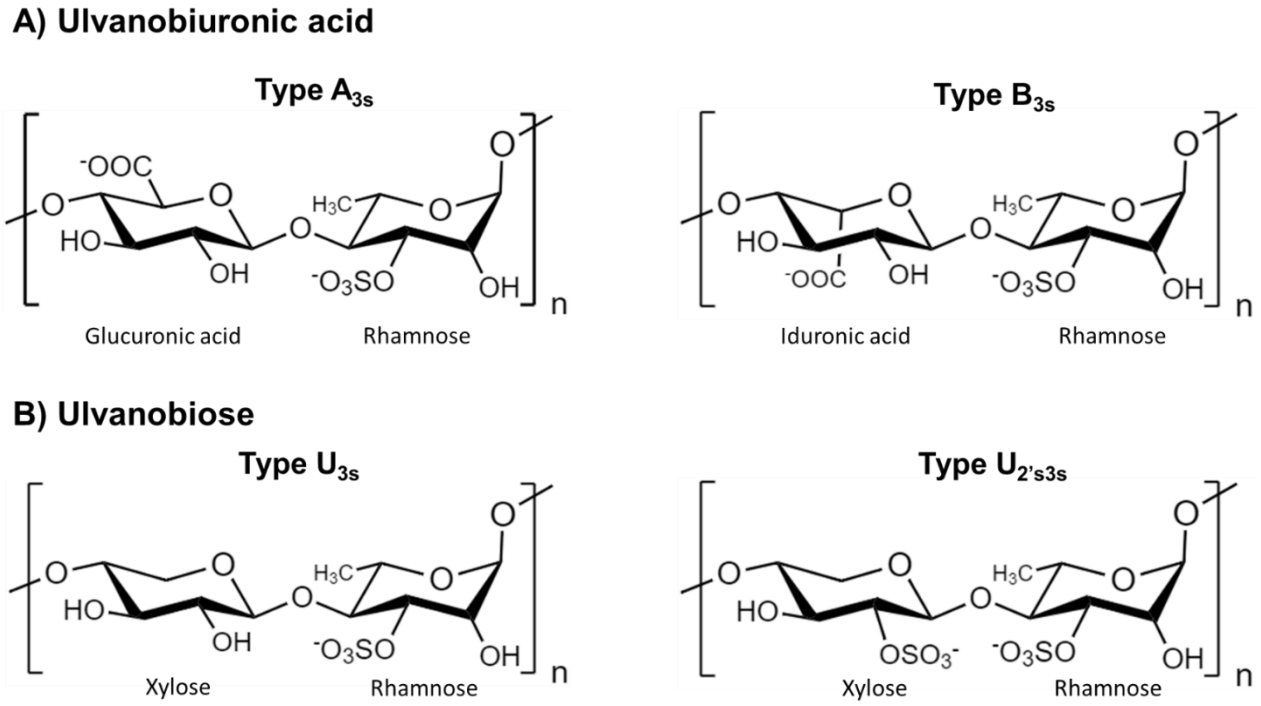


Figure 3.3. Chemical structures of two main types of disaccharide units of ulvan. A) Ulvanobiuronic acid, B) Ulvanobiose

As mentioned previously, heparin is known for its anticoagulant activity, whereas HS is most well-known for its interaction with a wide range of biomolecules, such as chemokines, cytokines, and growth factors [52], which is important for sequestration, protection, and signalling of these molecules in biological processes. In this thesis, we are specifically interested in wound healing, and growth factors are critical regulators of this process. Growth factors are proteins produced by cells with the purpose of guiding specific cellular responses in a physiological environment. There are specific receptors on the cell surface to which growth factors bind prior to activating downstream signalling pathways that control cellular behaviours [35]. Fibroblast growth factor 2 (FGF2) is a growth factor that plays important roles in controlling various cellular activities associated with wound healing and tissue repair such as cell proliferation, cell migration, and

angiogenesis [169]. HS has been known to be the co-factor of FGF2 which facilitates the binding of FGF2 to its receptor on the cell surface for receptor activation and downstream signalling [170]. The interaction between FGF2 and HS requires the presence of 2-O-sulphate and N-sulphate groups in the chemical backbone of HS [52]. Regarding carrageenan and ulvan, the presence of negatively charged sulphate groups in their molecular chains could aid in the interaction with positively charged amine residues in FGF2 through ionic interactions. However, much uncertainty still exists about the impact of varying degree of sulphation and position of sulphate groups of these seaweed derived SPs on the interaction with FGF2.

This study hypothesised that the variation in the chemical structures of seaweed derived SPs will lead to differences in interactions with FGF2. By understanding how changes in sulphation of seaweed derived SPs affect FGF2 binding, we could identify materials that mimic key biological behaviours of HS. These materials could then be explored for use in growth factor signalling, and potentially for use as therapeutics to promote wound healing. This chapter will investigate the difference in the structure of carrageenan and ulvan by characterising their degree of sulphation and position of sulphate groups, using carbon hydrogen nitrogen sulphur (CHNS) elemental analysis, X-ray photoelectron spectroscopy (XPS), and attenuated total reflection-Fourier transform infrared spectroscopy (ATR-FTIR). The interaction between seaweed derived SPs and FGF2 was then evaluated using surface plasmon resonance (SPR) to understand the influence of structural variation of SPs on growth factor binding.

3.2. Materials and Methods

In this thesis, heparin was used in all experiments as a control sample in place of HS. This is due to the commercial availability of heparin at a low cost, whereas HS can only be obtained in a few milligrams at a much higher cost. Moreover, this PhD conducted an initial screening of seaweed derived SPs to see how these materials exhibited their biological functions of modulating growth factor activities. Therefore, using heparin as a control sample at this stage was an appropriate option. HS could be used in future studies to further compare with the functional properties of the SPs, thereby selecting the material that can mimic the functions of HS for specific applications.

Heparin sodium salt from porcine intestinal mucosa (H3393), kappa-carrageenan (22048), iota-carrageenan from red seaweed Eucheuma spinosum (C1138), and lambda-carrageenan (22049) were purchased from Sigma-Aldrich (Castle Hill, Australia). Ulvan from green seaweed Ulva armoricana (YU11689) was obtained from Biosynth Ltd (United Kingdom). Human fibroblast growth factor-2 (FGF2, PHG0024) was supplied by Thermo Fisher in lyophilised form. The growth factor was reconstituted with MilliQ water at 0.1 mg/mL prior to being used in experiments. All other chemicals were purchased from Sigma-Aldrich (Castle Hill, Australia) unless stated otherwise.

3.2.1. Carbon-Hydrogen-Nitrogen-Sulphur (CHNS) elemental analysis

The mass percent of sulphur in the SPs was determined by CHNS analyser (Thermo Fisher) using cystine as the standards and K-factor as the calibration method. The samples were measured in triplicate and repeated three times. The degree of sulphation (DS) was calculated from the sulphur and carbon percent using Equation (1) as follows:

$$\mathrm{DS} = \frac{\%\mathrm{S}/32}{\%\mathrm{C}/12} \times 12 \quad (1)$$

where DS is defined as the number of sulphate groups per disaccharide unit, 32 and 12 are the atomic weight of sulphur and carbon, respectively, and 12 is the number of carbon atoms in a disaccharide unit of SPs.

3.2.2. X-ray Photoelectron Spectroscopy (XPS)

In addition to elemental analysis, XPS was used to determine the atomic percent of elements available in the SPs with a particular focus on sulphur element. An X-ray Photoelectron Spectrometer (Kratos, Manchester, UK) was employed to obtain the XPS spectra of SPs. This instrument was equipped with a monochromated Al Kα source (source energy 1486.69 eV) that was operated at a power of 150 W with a charger neutraliser. A piece of double-sided tape, approximately 8 mm x 8 mm, was placed on the XPS platen before the samples were tipped on the tape. A spatula was used to spread the samples out to achieve a single layer of adhered particles. Any excess particles not adhered to the tape were removed from the platen by a brush. Survey spectra were obtained using a pass energy of 160 eV and a step size of 1 eV, while high-resolution spectra of C1s were collected at a pass energy of 20 eV and a step size of 0.1 eV. Each sample was measured in triplicate, and obtained data was transformed into VAMAS format and subsequently analysed using CasaXPS software, version 2.3.16. A Shirley background and Gaussian-broadened Lorentzian functions (30 GL) were utilised to curve fitted the high-resolution C1s spectra. The constraint placed on the full-width-half-maximum of all components was based on that of the C–C, C–H peak. Charge correction was conducted for all spectra according to the relative position of the C–C, C–H peak to 285 eV.

3.2.3. Attenuated Total Reflection-Fourier Transform Infrared Spectroscopy (ATR-FTIR)

The presence and position of sulphate groups in the SPs was detected using ATR-FTIR. The FTIR spectra of SPs were recorded using the Bruker LUMOS II FTIR microscope equipped with an ATR sampling device. Samples in powder form were placed on the ATR crystal stage before the scanning process took place. The FTIR spectra were then acquired at room temperature using transmission mode with the wavenumber ranging from 4000 to 600 cm^{-1} and a sample scan time of 32 scans at a resolution of 4 cm^{-1}. A background spectrum of air was scanned using the same set of conditions applied before the samples were measured in triplicate. OPUS 7.8 software was used to acquire and process the obtained spectra.

3.2.4. Surface Plasmon Resonance (SPR)

SPR was employed to evaluate the binding between SPs and FGF2.

3.2.4.1. Thiolation of SPs

Prior to performing SPR, to enable the binding of SPs to the SPR gold sensor chip, the reductive amination method was utilised to incorporate single thiol groups at the reducing terminus of the SPs as previously reported [67, 171]. Briefly, 5 mg of each SP was dissolved in 1 mL MilliQ water and 500x equivalent of cysteamine hydrochloride dissolved in 0.6 mL methanol was added to the SP solution. Acetic acid (0.3 mL) was then supplemented, and the mixture was stirred for 40 minutes at 40 ^{o}C. Next, 5x equivalent of 2-picoline borane (Tokyo Chemical Industries) was dissolved in 0.1 mL of methanol before adding to the reacting solution, followed by stirring for a

further 40 minutes at 40 ^{o}C. The volume ratio of methanol, MilliQ water, and acetic acid in this experiment was 35:50:15. The resulting solution was dialysed against MilliQ water for two days using dialysis tube (2 kDa cut-off) and lyophilised to obtain the dry products.

3.2.4.2. Ellman's assay

Ellman's assay was conducted to confirm the thiolation of SPs. In this assay, Ellman's reagent (5,5-dithio-bis-(2-nitrobenzoic acid)) reacts with thiol groups to yield a yellow-coloured product, indicating the presence of thiol group in the SP samples after thiolation. To begin with, reaction buffer containing 0.1 M sodium phosphate, 1 mM EDTA, pH 8.0 was prepared, and 4 mg of Ellman's reagent was dissolved in 1 mL of reaction buffer to obtain Ellman's solution. Thiolated SPs were then dissolved in reaction buffer at the concentration of 1 mg/mL before adding Ellman's solution to allow the reaction between Ellman's reagent and thiol groups. Cysteamine hydrochloride was also dissolved in the reaction buffer and used as positive control solution, while reaction buffer only was used as the negative control. The change in the colour of thiolated SP solutions after the addition of Ellman's reagent was recorded and compared with positive and negative control solutions. Figure 3.4 confirms the thiolation of SPs using Ellman's solution. The yellow colour of the positive control and thiolated SP solutions indicated the presence of thiol groups, while negative control solution appeared to be in clear colour since no thiol groups were present.

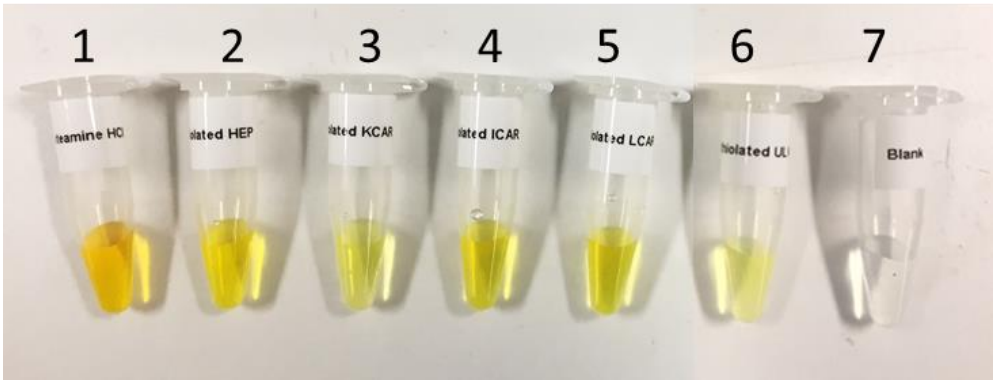


Figure 3.4. Thiolated SPs and control solutions after mixing with Ellman's reagent solution. 1) Cysteamine hydrochloride (positive control), 2) Thiolated heparin, 3) Thiolated kappa-carrageenan, 4) Thiolated iota-carrageenan, 5) Thiolated lambda-carrageenan, 6) Thiolated ulvan, 7) Reaction buffer (negative control).

3.2.4.3. Surface Plasmon Resonance

The interaction between SPs and FGF2 was assessed using SPR (Biacore S200, Cytiva). All the solutions were degassed prior to use in experiments. To begin with, PBS pH 7.4 was injected to the gold sensor chip at the flow rate of 5 µL/min until a stable baseline was achieved. Thiolated SPs (50 µg/mL in PBS) was immobilized on gold chip surface at 5 µL/min for 20 minutes. The chip was blocked with 0.1 % (w/v) bovine serum albumin at 5 µL/min for 5 minutes to prevent the non-specific binding of the growth factor. Any loosely bound materials to the gold sensor chip were removed by flowing 2M NaCl at 20 µL/min for 30 seconds until the baseline was stable. Then FGF2 (200 nM, Thermo Fisher) was flowed over the SPs bound to the gold chip at 20 µL/min for one minute and the response unit was monitored for a further 15 minutes until the amount of FGF2 bound reached equilibrium. After completion of the experiment, the chip was regenerated with 10 µL of 2 M NaCl to remove any remaining growth factor. The amount of growth factor bound to SPs was defined as the response unit (RU) value detected at 15-minute

point after FGF2 dissociated from the SPs. To compare the amount of FGF2 bound to each SP, the amount of FGF2 bound was a ratio of the amount of SP bound to the surface of the gold sensor chip and was normalised to 100 RU.

3.2.5. Statistical analysis

Significant differences were determined by one-way analysis of variance (ANOVA) followed by Tukey's multiple comparison test using GraphPad Prism 9.5.1 (GraphPad Software, San Diego, CA, USA). Differences between groups were considered significant at $p < 0.05$ and shown in the figures as *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, and ****$p < 0.0001$.

3.3. Results

3.3.1. CHNS elemental analysis

To understand the variation in the degree of sulphation of different seaweed based sulphated polysaccharides (SPs), elemental analysis was used to measure the sulphur mass percent and determine the degree of sulphation of the SPs with heparin used as the control sample.

Table 3.1 shows the mass concentration (%) of carbon (C), hydrogen (H), nitrogen (N), and sulphur (S) of heparin, KCAR, ICAR, LCAR, and ulvan as measured by elemental analysis. Unexpectedly, nitrogen was detected in ICAR, LCAR, and ulvan while these SPs do not contain nitrogen in their chemical structure. For ICAR and LCAR, detected nitrogen was considered as an artefact since the percent of nitrogen was less than the limit of detection of the equipment (0.35 %). For ulvan, the presence of nitrogen could be attributed to contaminants from the extraction process during the commercial production of ulvan, which has also been reported in a previous study [172]. Since this study investigated the variation in the sulphation of SPs, the change of sulphur percent in different SPs was of interest. Figure 3.5A illustrates the sulphur percent of heparin, KCAR, ICAR, LCAR, and ulvan as measured elemental analysis. According to the manufacturer of the equipment used in this experiment, the limit of detection of sulphur element is 0.3 %. Data presented here showed that the percent of sulphur detected in all samples was higher than 0.3 %, indicating that the SPs contained sulphur at a detectable level. Overall, heparin had significantly higher ($p<0.0001$) sulphur content (9.9 ± 0.2 %) compared to other SPs except ICAR. Regarding the three types of carrageenan, ICAR contained highest sulphur percent (9.8 ± 0.2 %) followed by

LCAR (9.3 ± 0.1 %) and KCAR (5.4 ± 0.1 %). Ulvan had significantly lower ($p<0.0001$) sulphur percent than heparin and carrageenan samples with the detected amount of 4.5 ± 0.0 %.

Degree of sulphation of the SPs was then calculated from the sulphur and carbon mass percent and defined as the number of sulphate groups per disaccharide unit of SPs. Figure 3.5B demonstrates the degree of sulphation of heparin, KCAR, ICAR, LCAR, and ulvan. Heparin had significantly higher ($p<0.0001$) degree of sulphation than all SPs tested with 2.21 ± 0.04 sulphate groups per disaccharide. This is in agreement with the reported degree of sulphation of heparin in the literature which is in the range of 1.8 - 2.6 sulphate groups per disaccharide [52, 173]. The degree of sulphation of carrageenan samples was consistent with its sulphur content, which appeared to be in decreasing order from ICAR (1.87 ± 0.02) to LCAR (1.75 ± 0.02) and finally KCAR (0.81 ± 0.02). Ulvan was shown to have a significantly lower ($p<0.05$) degree of sulphation than other samples with the calculated value of 0.78 ± 0.00. The trend in the calculated degree of sulphation of ICAR and LCAR is different from the trend in the theoretical number of sulphate groups per disaccharide unit of these carrageenan samples. Technically, ICAR has a lower degree of sulphation than LCAR since it contains two sulphate groups while LCAR has three sulphate groups per disaccharide unit [123]. The results here demonstrate differences between the four types of seaweed derived SPs evaluated in terms of sulphur mass percent and the degree of sulphation, as well as distinctions between the trend of theoretical and actual degree of sulphation of ICAR and LCAR.

Table 3.1. Mass concentration (%) of carbon, hydrogen, nitrogen, and sulphur of heparin, KCAR, ICAR, LCAR, and ulvan measured by elemental analysis.

			Mass concentration (%)			
Sample	Repeat	Replicate	C	H	N	S
Heparin	1	1	20.0	3.6	1.8	9.8
		2	20.2	3.3	1.8	9.7
		3	20.2	3.9	1.9	10.1
	2	1	20.1	3.3	1.8	10.0
		2	20.2	3.7	1.8	9.5
		3	20.2	3.8	1.8	10.0
	3	1	20.2	3.9	1.8	10.0
		2	20.1	3.8	1.9	9.9
		3	20.1	3.5	1.8	10.0
KCAR	1	1	30.1	4.8	0.0	5.3
		2	30.0	4.8	0.0	5.5
		3	30.1	4.7	0.0	5.3
	2	1	29.9	4.7	0.0	5.6
		2	30.0	4.7	0.0	5.4
		3	29.9	4.7	0.0	5.3
	3	1	29.8	4.5	0.0	5.4
		2	30.2	4.6	0.0	5.3
		3	29.5	4.6	0.0	5.5
ICAR	1	1	23.2	3.9	0.1	9.7
		2	23.1	3.9	0.1	9.4
		3	23.2	4.0	0.1	9.7
	2	1	24.0	4.0	0.1	10.0
		2	23.9	4.0	0.1	10.0
		3	23.8	4.1	0.1	10.0
	3	1	23.6	4.1	0.1	9.9
		2	23.4	4.2	0.1	9.7
		3	23.4	4.0	0.1	9.8
LCAR	1	1	23.9	4.5	0.2	9.1
		2	23.8	3.9	0.2	9.2
		3	23.8	4.1	0.2	9.3
	2	1	23.9	4.4	0.2	9.3
		2	23.9	4.1	0.2	9.3
		3	23.8	4.2	0.2	9.3
	3	1	23.9	4.0	0.2	9.3
		2	23.9	4.8	0.2	9.3
		3	23.8	4.2	0.2	9.3

			Mass concentration (%)			
Sample	Repeat	Replicate	C	H	N	S
Ulvan	1	1	26.5	5.0	1.2	4.6
		2	26.3	5.1	1.2	4.5
		3	26.0	4.6	1.1	4.5
	2	1	26.3	5.0	1.2	4.5
		2	26.2	5.2	1.2	4.5
		3	26.1	5.3	1.2	4.5
	3	1	26.0	4.9	1.1	4.5
		2	26.1	5.1	1.2	4.5
		3	26.1	5.2	1.2	4.5

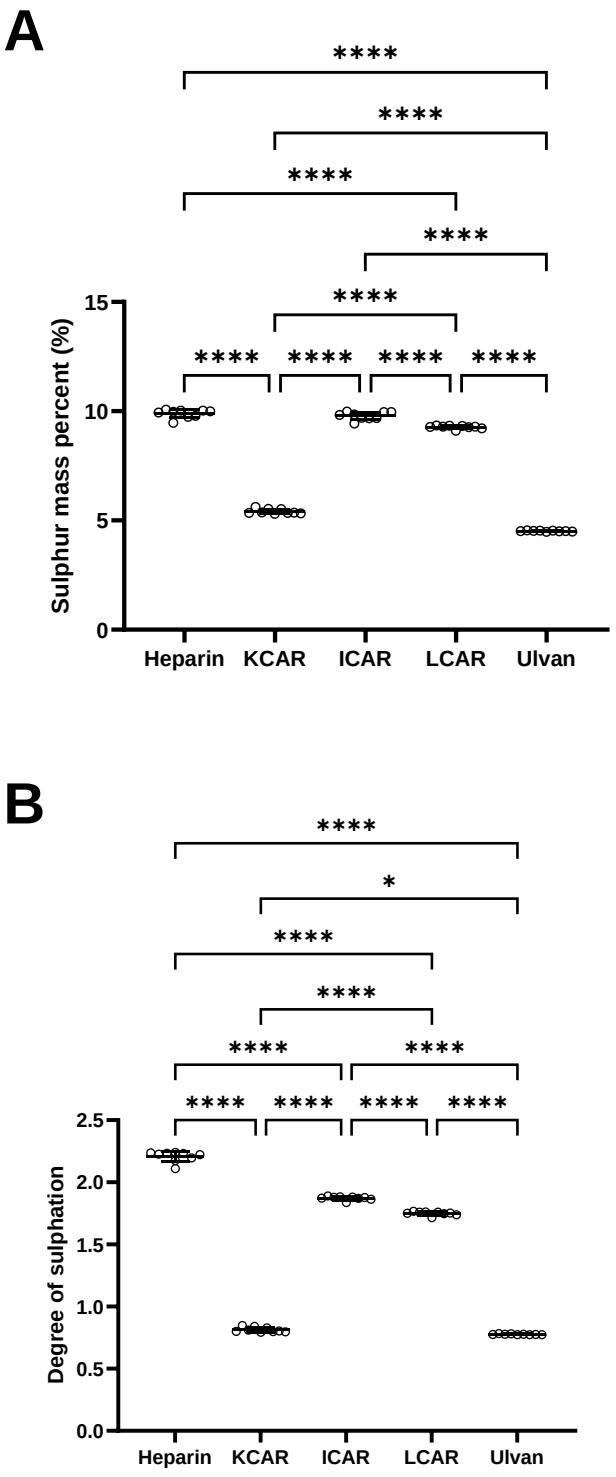


Figure 3.5. CHNS elemental analysis of heparin, kappa- (KCAR), iota- (ICAR), lambda-carrageenan (LCAR), and ulvan. A) Sulphur mass percent (%), and B) Degree of sulphation calculated from surphur and carbon percent. * represents statistically significant difference (*$p<0.05$, ****$p<0.0001$). n=9, lines represent mean ± standard deviation.

3.3.2. X-ray photoelectron spectroscopy (XPS)

In addition to the CHNS elemental analysis, XPS was also conducted to confirm the degree of sulphation of SPs. XPS is an analytical method that utilises X-ray radiation for the characterisation of chemical elements available in the samples evaluated. By measuring the electron energy released from the X-ray excited photoelectrons, qualitative and quantitative information regarding atomic percent of elements and chemical bonding within the sample can be obtained. Table 3.2 shows the atomic concentration (%) of elements detected in heparin, KCAR, ICAR, LCAR and ulvan as determined by XPS. Results show there was a variation in the amount of carbon, oxygen, and sulphur between the samples tested. Similar to the CHNS elemental results, nitrogen was unexpectedly detected in ulvan which could be due to impurities [172]. The presence of sodium (Na), potassium (K), and chloride (Cl) in the seaweed derived SPs could be attributed to contaminants. Silicone (Si) was due to the double-sided tape underneath the samples. As we focused on the variation in the degree of sulphation of SPs, the sulphur atomic percent (%) of heparin, KCAR, ICAR, LCAR, and ulvan was compared and showed in Figure 3.6A. Compared to heparin (5.9 ± 0.3 %), ICAR and LCAR possessed significantly higher ($p<0.05$) sulphur atomic percent which were 6.5 ± 0.2 % and 6.5 ± 0.1 %, respectively. KCAR and ulvan had the lowest sulphur percent among the SPs evaluated which was 2.5 ± 0.2 % and 2.6 ± 0.3 %, respectively.

The ratio between sulphur and carbon atomic percent (S/C ratio) was then calculated to understand the relative level of sulphation in each material. Figure 3.6B demonstrates the S/C ratios of heparin, KCAR, ICAR, LCAR, and ulvan. ICAR was shown to have significantly higher ($p<0.05$) S/C ratio (0.15 $\pm$ 0.03) than other samples except LCAR. There was no significant difference in the S/C ratio between LCAR (0.13 $\pm$ 0.00) and heparin (0.11 $\pm$ 0.01). Additionally, KCAR was shown to have similar S/C ratio to ulvan which was approximately 0.04. The S/C ratio

of SPs had some differences as compared to the degree of sulphation. While heparin had significantly higher degree of sulphation than ICAR and LCAR, it had significantly lower S/C ratio than ICAR and similar S/C ratio to LCAR. Moreover, although ulvan had lowest degree of sulphation among all samples tested, it showed a similar S/C ratio to KCAR. These results demonstrate the difference in the amount of sulphate groups detected by XPS and CHNS elemental analysis. In short, XPS analysis results suggest that a small amount of sulphur was detected in heparin and the seaweed derived SPs, and differences in the trend of level of sulphation were likely due to two different analytical techniques.

Table 3.2. Atomic concentration (%) of chemical elements detected in heparin, KCAR, ICAR, LCAR, and ulvan using XPS. Each sample was analysed at three different positions.

		Atomic Concentration (%)							
Sample	Position	C	N	O	Na	Si	S	K	Cl
Heparin	1	60.0	1.6	29.1	3.0	0.8	5.5	0.0	0.0
	2	54.1	2.0	33.2	3.7	0.9	6.1	0.0	0.0
	3	54.5	2.1	32.7	3.7	0.9	6.0	0.0	0.0
KCAR	1	55.2	0.0	36.2	0.3	3.8	2.5	1.4	0.5
	2	57.0	0.0	35.0	0.3	3.8	2.2	1.3	0.5
	3	56.5	0.0	34.9	0.2	3.7	2.7	1.4	0.7
ICAR	1	34.8	0.0	34.8	0.4	4.1	6.3	0.8	0.0
	2	49.1	0.0	36.9	0.5	3.9	6.6	1.2	0.0
	3	49.7	0.0	39.5	0.6	3.8	6.6	1.1	0.0
LCAR	1	52.0	0.0	35.4	2.0	4.2	6.4	0.0	0.0
	2	51.7	0.0	35.6	2.0	4.1	6.5	0.0	0.0
	3	51.0	0.0	36.1	2.0	4.3	6.6	0.0	0.0
Ulvan	1	60.1	4.5	31.5	0.4	0.7	2.9	0.0	0.0
	2	62.8	4.4	28.6	0.3	1.3	2.6	0.0	0.0
	3	64.0	4.0	27.6	0.3	1.8	2.3	0.0	0.0

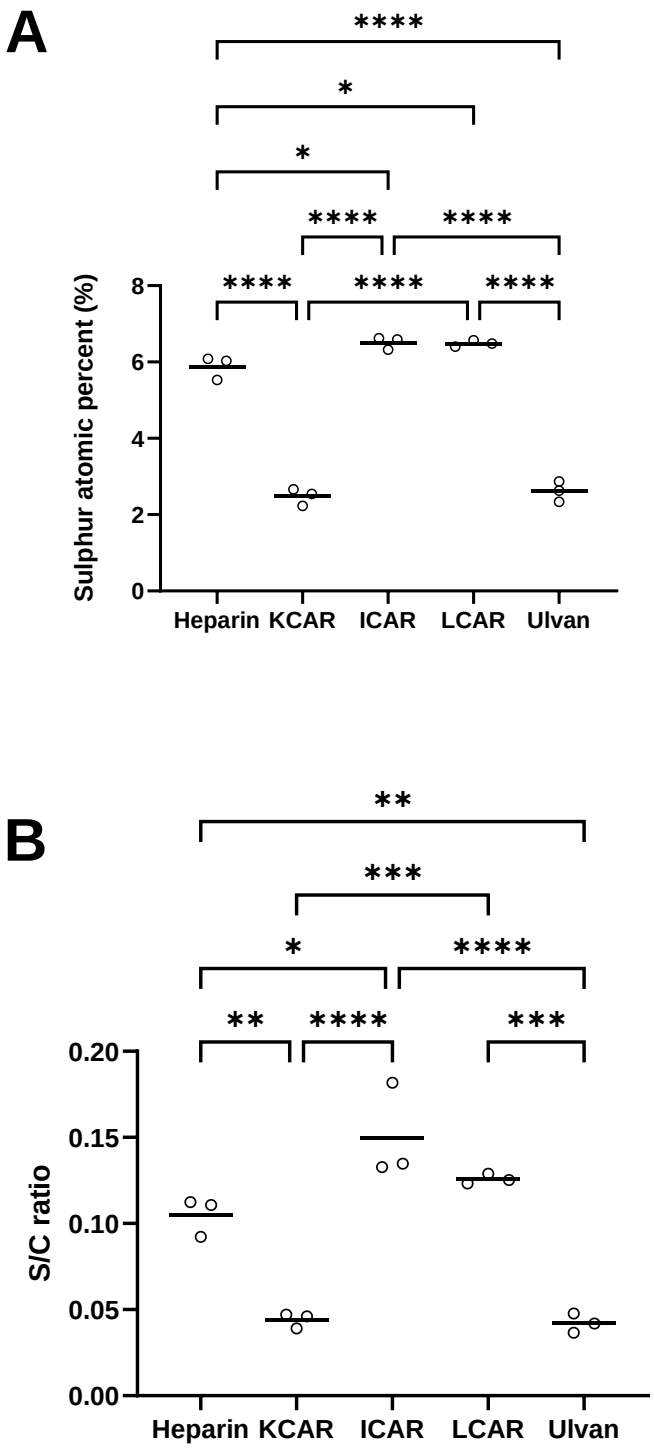


Figure 3.6. XPS analysis of heparin, kappa- (KCAR), iota- (ICAR), lambda-carrageenan (LCAR), and ulvan. A) Sulphur atomic percent (%), and B) S/C ratio calculated from sulphur and carbon atomic percent. * represents statistically significant difference (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001). n=3, line represents mean value.

Another structural feature of SPs that can be detected from XPS analysis is the chemical bonding within the polysaccharides. Figure 3.7 shows the high resolution C1s spectra of heparin, KCAR, ICAR, LCAR, and ulvan. Three peaks were fitted in the C1s spectra of all SPs at different binding energies, which are attributed to three main chemical linkages involving carbon atoms: C-C/C-H at ~ 285 eV, C-O/C-O-C at ~286.6 eV, and O-C=O at ~288.5 eV. The area under the peak is related the level of chemical bonds present in the samples. A comparison of the percentage of peak area for each SP is shown in Table 3.3. Regarding the C-C/C-H peak, heparin had a significantly higher ($p<0.01$) peak area than the other SPs, indicating it contained more C-C/C-H linkages. There were no significant differences in the area of C-C/C-H peak between KCAR, ICAR, LCAR, and ulvan. In terms of the C-O/C-O-C peak, heparin had significantly lower peak area than KCAR ($p<0.001$), ICAR ($p<0.05$), and LCAR ($p<0.05$). The chemical structure of KCAR and ICAR contained anhydro-galactose residues which had more C-O-C linkages as compared to heparin. This may explain why KCAR and ICAR had higher area of C-O-C peak than heparin. Moreover, ulvan had a significant lower ($p<0.01$) peak area than KCAR. There were no significant differences in the area of C-O/C-O-C peak between ICAR, LCAR, and ulvan. Regarding O-C=O peak, ulvan showed a significant higher peak area than heparin ($p<0.05$), ICAR ($p<0.05$), and LCAR ($p<0.01$). The chemical backbone of ulvan contained uronic acid residues which had carboxyl groups (COOH), whereas these groups were not present in ICAR and LCAR. This could explain why ulvan had higher area of O-C=O peak than ICAR and LCAR. No significant differences were observed between heparin, KCAR, ICAR, and LCAR.

The ratio of the sum of C-O/C-O-C and O-C=O peak area to C-C/C-H peak area (ratio of peak area) was calculated to understand the ratio of oxygen-containing linkages to carbon-carbon/carbon-hydrogen linkages (no oxygen). The O/C ratio of heparin, KCAR, ICAR, LCAR, and ulvan was also

calculated to determine the ratio of oxygen atoms to carbon atoms and the correlation with the ratio of peak area. Regarding the ratio of peak area, heparin had a significantly lower ($p<0.01$) ratio than KCAR, while no significant differences were observed between the seaweed derived SPs. In terms of the O/C ratio, ICAR had a significantly higher O/C ratio than heparin ($p<0.01$), KCAR ($p<0.05$), and ulvan ($p<0.001$). Moreover, LCAR had a significantly higher ($p<0.05$) O/C ratio than ulvan. For all samples, a high O/C ratio generally correlated with a high ratio of peak area and vice versa. Unexpectedly, KCAR had a lower O/C ratio than ICAR but possessed a higher ratio of peak area, possibly due to KCAR having lower mean percent of oxygen atoms than ICAR but higher mean percent of oxygen-containing linkages (C-O/C-O-C).

Table 3.3. Percent (%) of peak area in C1s spectra and O/C ratio of heparin, KCAR, ICAR, LCAR, and ulvan. Each sample was analysed at three different positions. The O/C ratio was calculated from the atomic concentration (%) of oxygen and carbon of each sample.

Sample	Position	% of peak area in C1s spectra			[(2) + (3)]/(1) ratio	O/C ratio
		C-C, C-H (1)	C-O, C-O-C (2)	O-C=O (3)		
Heparin	1	59.3	29.9	10.8	0.7	0.5
	2	49.1	37.2	13.8	1.0	0.6
	3	50.8	36.1	13.2	1.0	0.6
KCAR	1	31.3	56.3	12.5	2.2	0.7
	2	36.4	50.5	13.1	1.8	0.6
	3	35.8	51.1	13.1	1.8	0.6
ICAR	1	46.5	42.6	11.0	1.2	1.0
	2	41.1	47.2	11.7	1.4	0.8
	3	40.7	47.0	12.3	1.5	0.8
LCAR	1	45.2	44.5	10.3	1.2	0.7
	2	43.2	45.6	11.2	1.3	0.7
	3	39.9	48.3	11.8	1.5	0.7
Ulvan	1	36.3	45.1	18.6	1.8	0.5
	2	47.2	38.2	14.6	1.1	0.5
	3	48.6	36.3	15.0	1.1	0.4

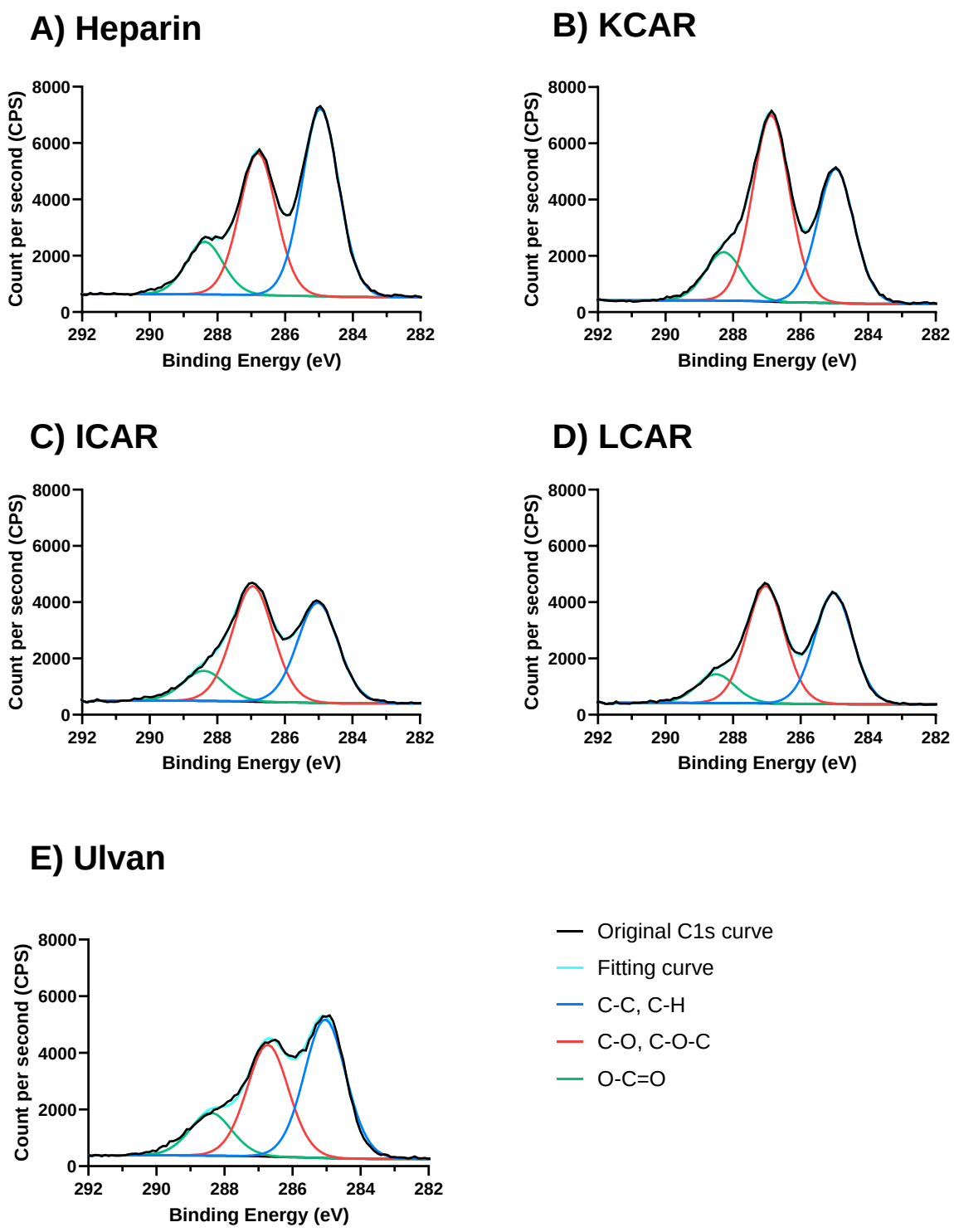


Figure 3.7. High resolution curve fitted C1s XPS spectra of A) heparin, B) kappa-carrageenan (KCAR), C) iota-carrageenan (ICAR), D) lambda-carrageenan (LCAR), and E) ulvan.

3.3.3. Attenuated Total Reflection-Fourier Transform Infrared Spectroscopy (ATR-FTIR)

In addition to investigating the variation in the degree of sulphation of SPs, the position of sulphate groups in the polymer chains of these materials was also evaluated using ATR-FTIR technique. Figure 3.8 shows the FTIR spectra of heparin and ulvan with full spectra and zoomed spectra from 2000 to 500 cm^{-1} presented for visualising the characteristic peaks. For heparin, the peak at 1223 cm^{-1} was attributed to the stretching vibration of S=O bonding in the N-sulphate glucosamine units [174]. Meanwhile, the peaks detected at 818 and 797 cm^{-1} were assigned to the C-O-S stretching on C6 of the glucosamine units and C2 of the iduronic acid units, respectively, indicating the presence of sulphate groups at these positions [175]. In terms of ulvan, the peaks appearing at 1214 and 846 cm^{-1} were characteristic peaks of sulphate groups on C3 of rhamnose residues, which were caused by the S=O stretching and C-O-S stretching vibrations, respectively [176, 177].

Figure 3.9 demonstrates the FTIR spectra of kappa- (KCAR), iota- (ICAR), and lambda-carrageenan (LCAR) with full and zoomed spectra from 2000 to 500 cm^{-1} presented. Regarding KCAR, the peak found at 845 cm^{-1} was caused by the C-O-S stretching vibration of sulphate groups on C4 of galactose units [178]. While this peak was also detected in ICAR, there was another peak at 802 cm^{-1} which was assigned to the C-O-S stretching of sulphate groups on C2 of 3,6-anhydrogalactose units [179]. This indicates ICAR contained sulphate groups at two different positions in its repeating disaccharide units. In terms of LCAR, the peak at 838 cm^{-1} stemmed from the C-O-S stretching of sulphate groups on the C2 of both galactose residues in a disaccharide unit. In addition, a shoulder peak was observed at 812 cm^{-1} which might indicate

the presence of sulphate groups at C6 position of galactose units [180]. Together this confirms the presence of sulphate groups at three different positions per disaccharide unit of LCAR.

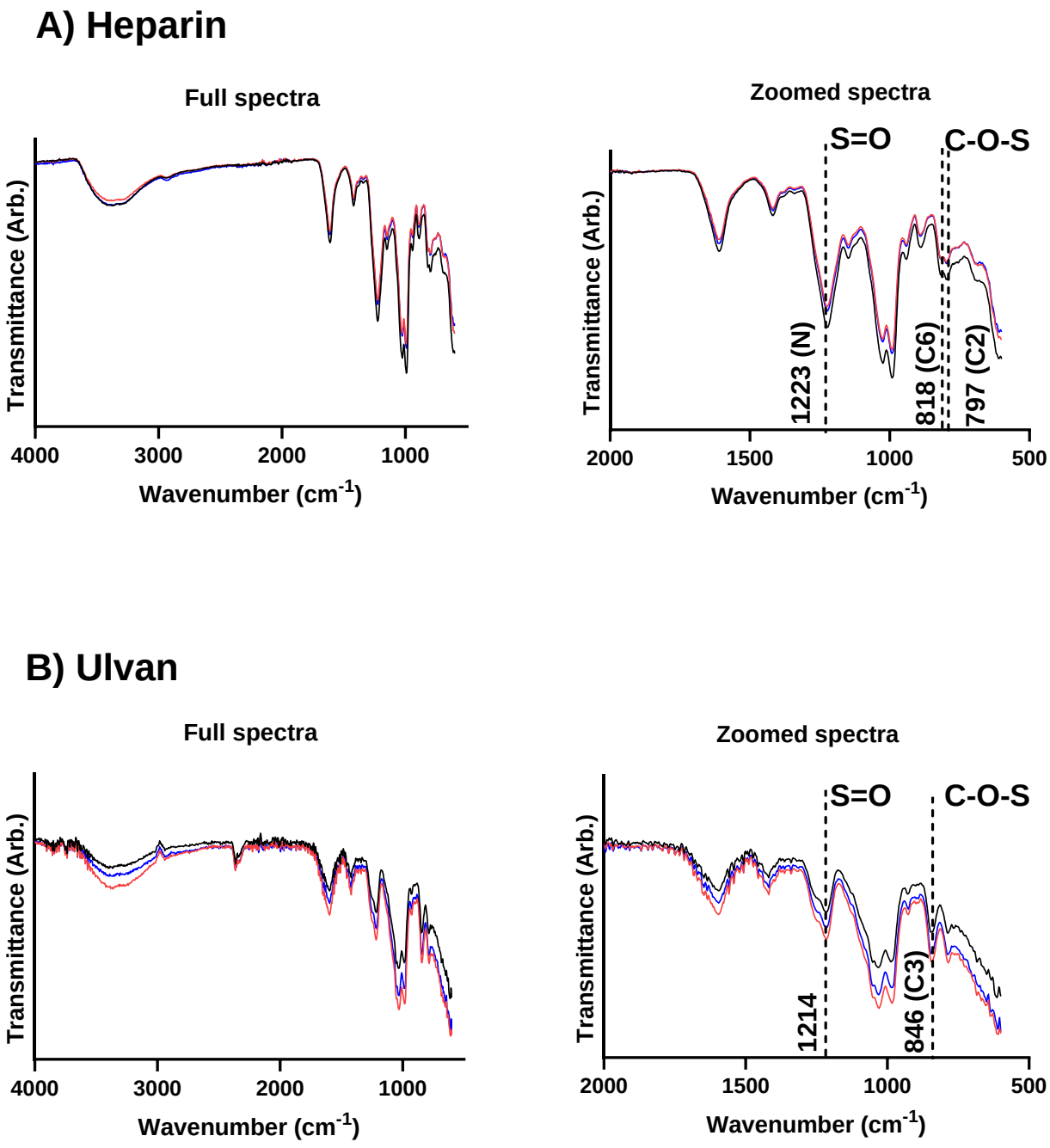


Figure 3.8. ATR-FTIR analysis of A) heparin and B) ulvan. For each material, full and zoomed spectra are shown for detailing peak appearance, and three repeating spectra are displayed to illustrate triplicate measurements.

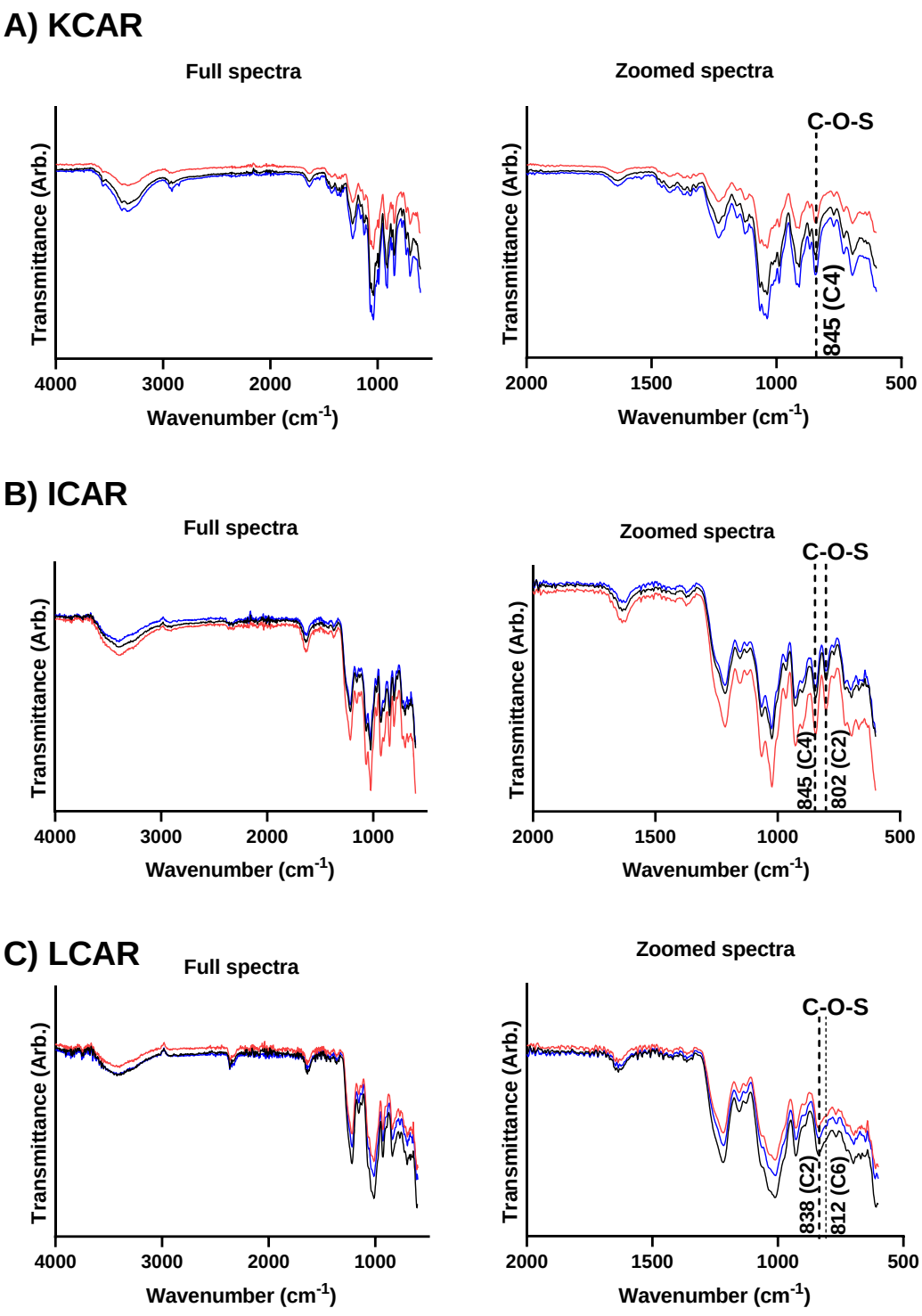


Figure 3.9. ATR-FTIR analysis of A) kappa-carrageenan (KCAR), B) iota-carrageenan (ICAR), and C) lambda-carrageenan (LCAR). For each material, full and zoomed spectra are shown for detailing peak appearance, and three repeating spectra are displayed to illustrate triplicate measurements.

3.3.4. Surface plasmon resonance (SPR)

SPR was conducted to investigate the interactions between SPs and fibroblast growth factor-2 (FGF2). It was hypothesised that different sulphation characteristics of SPs will impact the level of FGF2 binding. A typical SPR sensorgram is shown in Figure 3.10A. To begin with, SP was injected into the flow channel of the SPR sensor chip, and the amount of SP bound to the surface of the chip was illustrated by the increase of response unit (RU). For the SPs evaluated, the amount of SP bound to the gold sensor chip was approximately 500 RU. BSA was then injected to cover any empty space on the chip surface and block non-specific binding of FGF2, and the RU level was increased by around 300 to 600 RU. A salt wash was conducted to remove any unbound materials on the chip surface before FGF2 was flowed to evaluate interaction and binding to the immobilised SP. Figure 3.10B shows the binding curve between FGF2 and SP. FGF2 was injected for one minute to bind to the SP which caused the RU level to increase. As the injection stopped, the growth factor dissociated from the SP, resulting in the decrease of RU level.

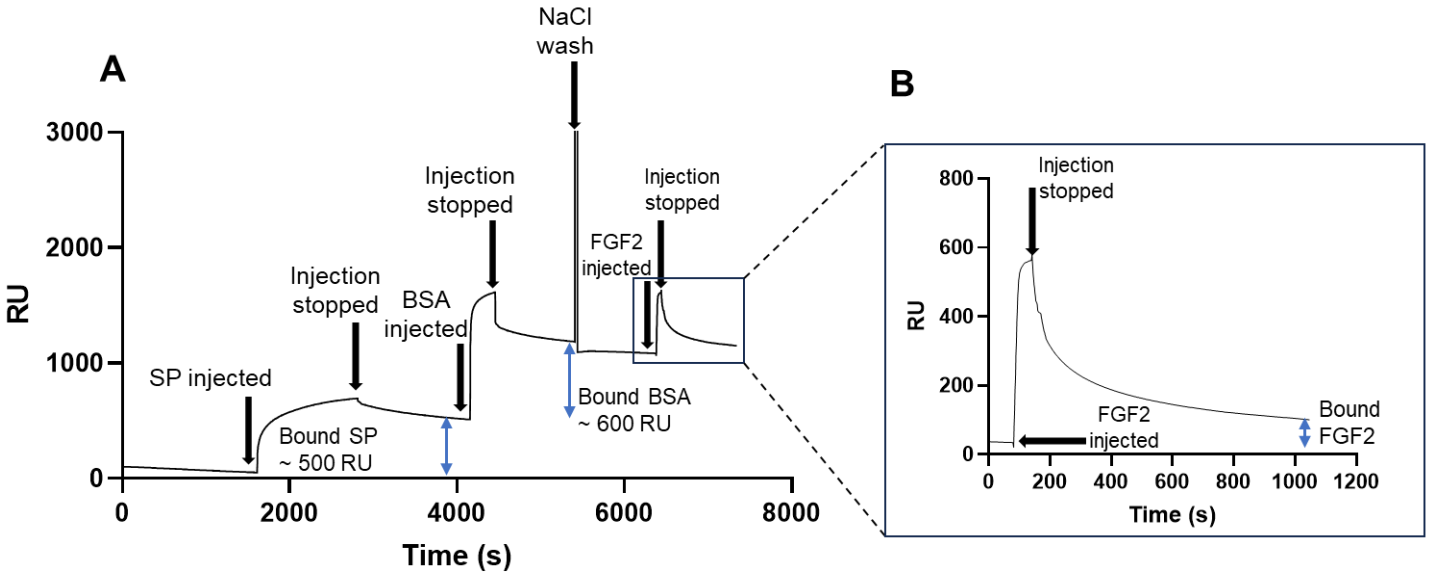


Figure 3.10. Illustration of SPR results. A) SP was injected into the flow channel of the SPR sensor chip which was demonstrated by the increase of response unit (RU) by approximately 500 RU. BSA was then injected to cover any empty space on the chip surface, resulting in the increase of RU level by around 600 RU. A salt wash was conducted to remove any unbound materials on the chip surface before FGF2 was flowed to assess binding to the immobilised SP. B) Zoomed image showing the binding curve between FGF2 and SP. FGF2 was injected for one minute to bind to the SP, causing the RU level to increase. As the injection stopped, the growth factor was dissociated from the SP, resulting in the decrease of RU level.

The amount of SPs bound to the surface of SPR sensor chip was around 500 RU for all SPs evaluated but did vary slightly between SPs. Therefore, to be able to compare the amount of FGF2 bound to different SPs, the amount of FGF2 bound was calculated as a ratio of the amount of SP bound to the surface and was normalised to 100 RU. Moreover, since BSA covered approximately half of the surface of the sensor chip, an SPR experiment was conducted to check if FGF2 bound to BSA on the chip. Figure 3.11 shows the SPR results of FGF2 flowing over a BSA only surface. When FGF2 was injected into the flow channel of the chip containing BSA, the RU value increased as FGF2 flowed over the sensor as expected, but when the injection stopped the

RU dropped down under the baseline then went back to equilibrium at zero RU, confirming FGF2 did not bind to BSA.

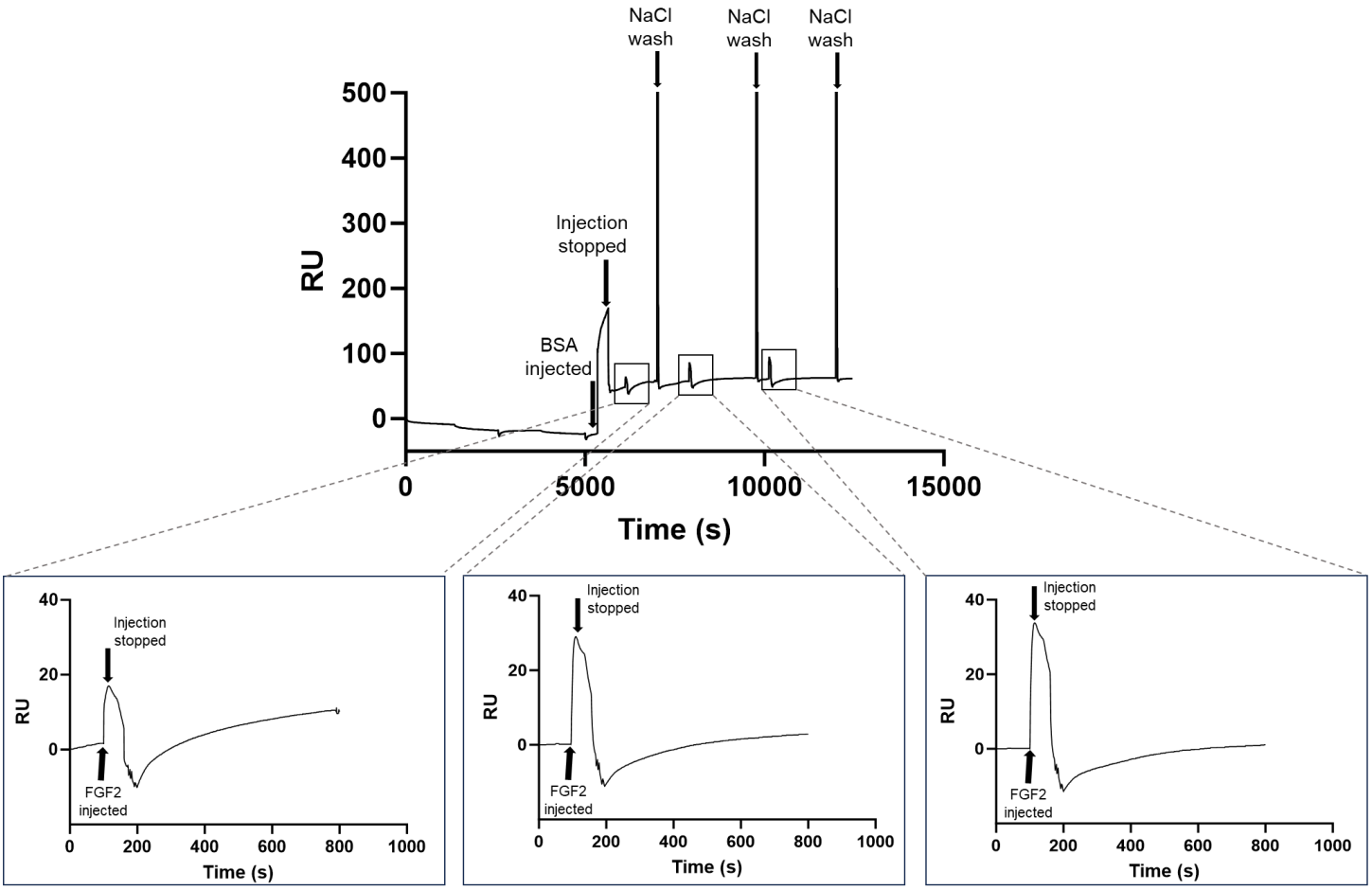


Figure 3.11. SPR results of FGF2 flowing over a gold chip surface containing BSA only.

Figure 3.12 shows the SPR sensorgrams of FGF2 binding to heparin (Figure 3.12A), kappa-(KCAR, Figure 3.12B), iota-(ICAR, Figure 3.12C), lambda-carrageenan (LCAR, Figure 3.12D), and ulvan (Figure 3.12E). Each sensorgram demonstrates the change in the amount of FGF2 bound to each material as measured in response unit (RU) over time. The amount of bound FGF2 was a ratio of

the amount of SP bound to the gold surface of the SPR sensor chip and was normalised to 100 RU. There are two phases in each sensorgram, the association and dissociation phases. The association phase is defined as the period when FGF2 flowed over the sensor chip and bound to the immobilised SPs. The dissociation phase follows the association phase and is defined as the period when FGF2 solution was replaced by running buffer and thus FGF2 dissociated from the SPs. At t = 0 second, the association phase began as FGF2 was injected onto the SP-immobilised surface of the sensor chip. There was a rapid increase in response unit on the sensorgram of the SPs, indicating FGF2 bound to the SPs. Furthermore, a variability in the change of RU during association phase was observed in the repeated sensorgrams of each SPs, which can be shown by the difference in RU value detected when the FGF2 injection stopped. This value ranged from 110 to 185 for heparin, 35 to 106 for KCAR, 41 to 97 for ICAR, 43 to 147 for LCAR, and 44 to 102 for ulvan. In several SPR sensorgrams, it was observed that the change in RU reached a plateau in the association phase, which indicates the maximum amount of FGF2 interaction had been reached. Moreover, time to reach a plateau was also varied within each SP. For heparin, it took 47 to 54 seconds from the point FGF2 was injected to reach a plateau. The time it took for KCAR, ICAR, LCAR, and ulvan was 44 to 54, 40 to 56, 57 to 59, and 40 to 54 seconds, respectively. After the FGF2 injection stopped, the flow was switched to a buffer solution to begin the dissociation phase. In this phase, the dissociation of FGF2 from the SPs was monitored and demonstrated by the decrease in RU. For all samples, the RU decreased rapidly in the first 50 seconds from the point FGF2 injection stopped, then it started to drop at a slower rate until reaching equilibrium. Figure 3.12F shows the representative SPR sensorgram for each SP. While having a similar trend in the dissociation phase, these SPs were different in the association phase regarding the RU

value detected when FGF2 injection stopped and the time to reach a plateau. Specifically, after 1-minute injection of FGF2, the amount of FGF2 bound to heparin, KCAR, ICAR, LCAR, and ulvan was 152 ± 28, 78 ± 22, 72 ± 18, 100 ± 35, 73 ± 17 RU, respectively. In terms of the time required to reach a plateau in the association phase, heparin took 50 ± 3 seconds while KCAR and ulvan needed 48 ± 4 and 48 ± 5 seconds, respectively. ICAR required slightly shorter time which was 47 ± 5 seconds. LCAR took the longest time among the samples tested for the RU to reach a plateau, which was 58 ± 1 seconds.

Figure 3.13 shows the amount of FGF2 bound to each of the SPs, which was determined after FGF2 dissociated from the SPs for 15 minutes. Heparin was utilised as the control sample. Overall, the amount of FGF2 bound to heparin was significantly higher than all other materials ($p<0.0001$). Moreover, LCAR was shown to have a significantly higher amount of FGF2 bound than KCAR, ICAR, and ulvan ($p<0.0001$). Meanwhile, FGF2 bound to KCAR, ICAR, and ulvan at low levels with no significant differences between these SPs. Together, SPR results show that the materials used in this study had lower capacity to bind FGF2 than heparin with LCAR binding more FGF2 than any other seaweed derived SPs. The results also demonstrate that the degree of sulphation of SPs had an impact on the amount of bound FGF2. Heparin had higher degree of sulphation than the seaweed derived SPs, which was in correlation with higher amount of bound FGF2 as compared to the SPs. Similarly, the degree of sulphation of LCAR was higher than KCAR and ulvan which resulted in higher level of bound FGF2. Interestingly, ICAR possessed higher degree of sulphation than LCAR but had lower amount of FGF2 bound. This finding suggests that in addition to degree of sulphation, position of sulphate groups may have influenced the level of FGF2 bound to the SPs evaluated.

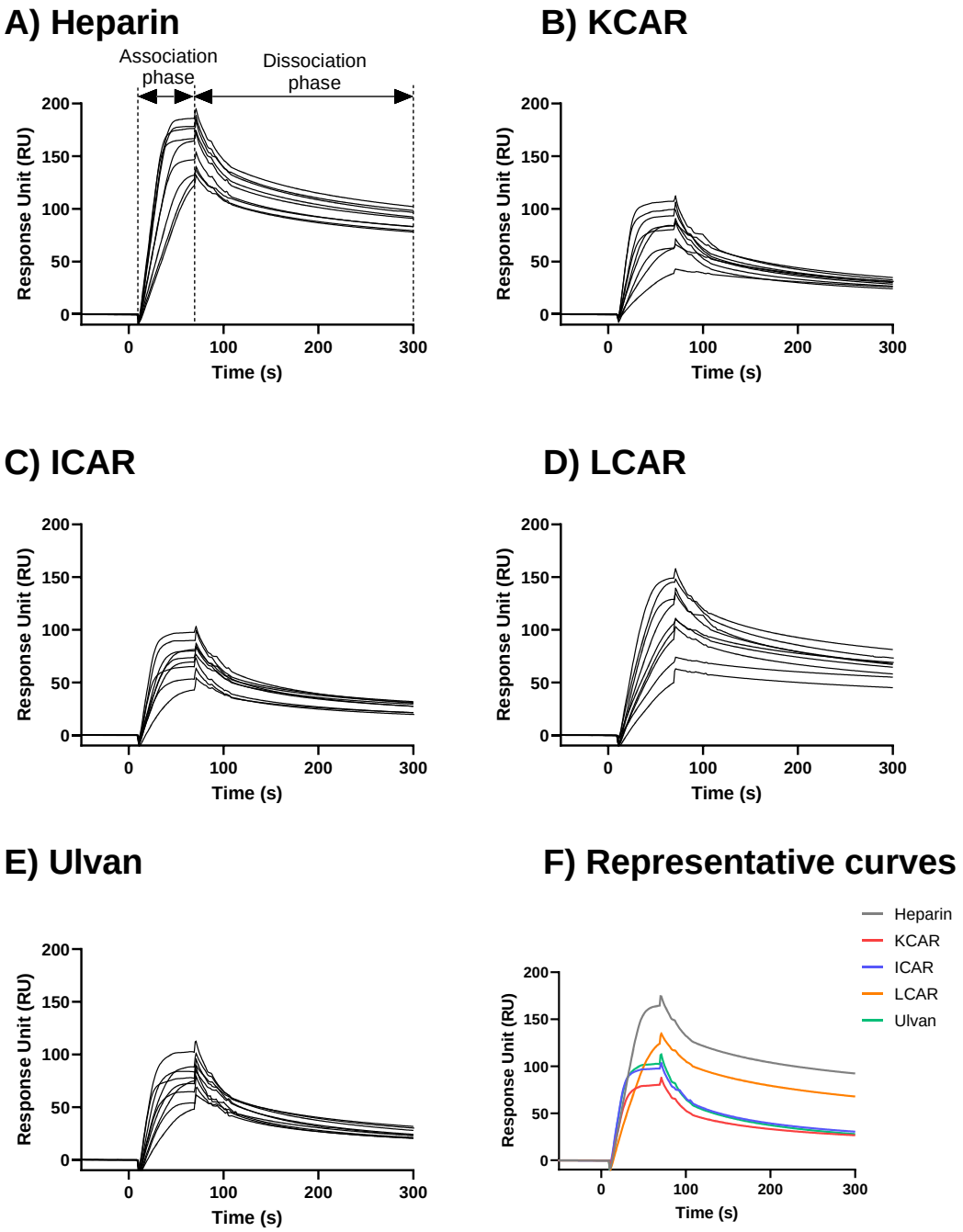


Figure 3.12. SPR sensorgrams of FGF2 bound to A) heparin, B) kappa-carrageenan (KCAR), C) iota-carrageenan (ICAR), D) lambda-carrageenan (LCAR), or E) ulvan, and F) Representative SPR curves of FGF2 bound to each SP. FGF2 was injected at t = 0 second and flowed to bind to SP for one minute, followed by dissociation phase for 15 minutes. The association and dissociation phases were illustrated in the SPR sensorgrams of heparin. In this figure, all sensorgrams were shown from 0 to 300 seconds to demonstrate the variability in the interaction between FGF2 and each material tested. The experiment was performed in triplicate and repeated three times, resulting in 9 sensorgrams for each material. The amount of FGF2 bound was a ratio of SP bound to the surface of gold sensor chip and was normalised to 100 RU.

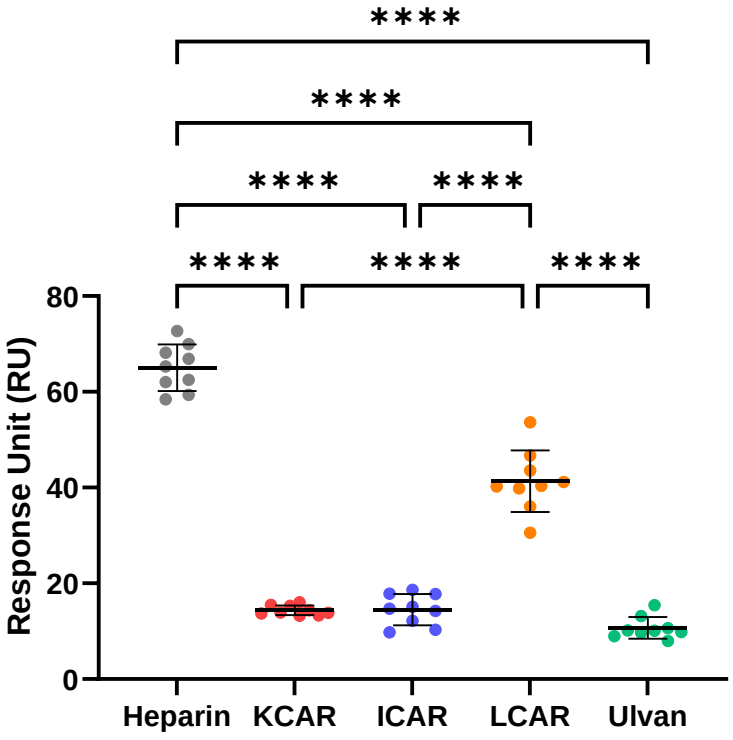


Figure 3.13. Amount of FGF2 bound to each SP presented in response units (RU). The amount of FGF2 bound to each material is defined as the RU value detected at 15-minute point after FGF2 dissociated from the SPs. * represents statistically significant difference (****p<0.0001). n=9, lines represent mean ± standard deviation.

3.4. Discussion

The aim of this chapter was to characterise the degree of sulphation and position of sulphate groups of seaweed derived SPs and investigate how different sulphation characteristics affected FGF2 binding. It was hypothesised that the variation in the chemical structures of SPs would lead to differences in the interaction with FGF2.

This study demonstrated that the degree of sulphation and position of sulphate groups were different between the different types of seaweed derived SPs evaluated. This was confirmed by CHNS elemental analysis, XPS, and ATR-FTIR results. The sulphur content of the SPs was measured using two different techniques, elemental analysis and XPS, and the obtained results were different in numerical values but similar in the trends. This could be explained as follows. In CHNS elemental analysis, the samples are combusted at a high temperature to allow the reaction of carbon, hydrogen, nitrogen, and sulphur to their combustion products. The amount of these products is recorded, and corresponding element contents is calculated as mass percentages. Meanwhile, XPS uses X-ray radiation to detect any elements existing at the top 10 nm depth of the surface of materials and quantify the elemental content in the form of atomic percentages. The sensitivity of XPS has been previously reported to be approximately 0.1 % [181], whereas CHNS elemental analysis has the sensitivity of 0.3 to 0.5 % according to the equipment manufacturer. The difference in the sensitivity of the two techniques as well as how the elemental compositions were measured may contribute to the differences in the elemental percentages obtained. Moreover, it was assumed that the chemical composition in the top 10 nm of the SPs was similar to that in the bulk of materials since samples analysed were in powder form, which contributes to the relative similarities in the trend of results. Previous studies have

reported the use of CHNS elemental analysis and XPS in the characterisation of elemental compositions of different materials and shown that values obtained by the two techniques were different but had similar trends, which supports the findings in the current study [182, 183].

Since sulphur is a component of the sulphate residues in the SPs, quantifying the amount of sulphur was used to determine the degree of sulphation. In this study, carbon and sulphur mass percent obtained from CHNS elemental analysis was employed to calculate the degree of sulphation of heparin, carrageenan, and ulvan. Interestingly, ICAR had higher degree of sulphation than LCAR, which contradicts with the theoretical amount of sulphate groups per disaccharide unit of these two types of carrageenan. Technically, ICAR contains two sulphate groups in each disaccharide unit while LCAR has three sulphate groups in each disaccharide unit [123]. The difference between actual and theoretical value suggests the natural variation in the chemical structure of carrageenan. A previous study by Rodiahwati et al. [184] reported the sulphur percent of ICAR and LCAR to be 10.07 ± 0.15 % and 10.03 ± 0.06 %, respectively, of which the trend is similar to results obtained in the current study. Rochas et al. [185] also showed the degree of sulphation of ICAR (1.62) was higher than LCAR (1.56). These findings support the idea that ICAR and LCAR exhibit natural variability in their structures. The sulphur percent of ulvan is also consistent with previously reported data which was approximately 4.33 % [172]. It has been shown that HS extracted from porcine intestine had the sulphur mass percent of 4.44 – 5.51 % [186-188], which is lower than the sulphur mass percent of heparin (9.9 %), ICAR (9.8 %), and LCAR (9.3 %) but similar to KCAR (5.4 %) and ulvan (4.5 %) evaluated in this study. This suggests that KCAR and ulvan may resemble the sulphur content of HS while ICAR and LCAR are likely to mimic the sulphur content of heparin.

Regarding XPS results, high resolution spectra of C1s demonstrated different amounts of chemical bonding involving carbon atom between different materials. Since sulphate groups are linked to carbon atoms in the chemical backbone of SPs, the difference in the amount of chemical bonding may suggest the variation of degree of sulphation in these materials. Furthermore, there was a correlation between sulphur content (atomic %) and O/C ratio of the SPs. High sulphur content indicates high degree of sulphation, which means that more S=O linkages are present in the SPs. As oxygen is a component of the S=O linkage, this implies more S=O linkages lead to a higher O/C ratio [189].

The position of sulphate groups of carrageenan and ulvan were investigated by ATR-FTIR. This method is useful to detect the functional groups of chemical compounds. From the FTIR results, the position of sulphate groups of the SPs is different between samples and is consistent with the findings in literature as described in the result section. Regarding LCAR, the small peak at 812 cm^{-1} associated with the sulphate groups at C6 position of galactose residues may indicate that there was only small amount of 6-O-sulphate present in this type of carrageenan. The position of sulphate groups of SPs can also be detected using 1H-13C heteronuclear single quantum coherence nuclear magnetic resonance [190, 191] in future studies to obtain information about the correlation between carbon and hydrogen linkages within the SPs, thus providing complementary results regarding the position of sulphate groups in the chemical backbone of SPs.

The present study examined the interaction between FGF2 and different types of seaweed derived SPs including carrageenan and ulvan using surface plasmon resonance. This technique investigates the interaction between two molecules, the ligand immobilised on the surface of a

sensor chip and the analyte that is flowed across the ligand, and the interactions between the two molecules are monitored. This study evaluated the interaction between SPs, heparin or seaweed derived, immobilised on the SPR sensor surface, and the growth factor FGF2. Heparin was used as a control sample in place of HS due to the commercial availability of heparin at a low cost. Results in this study show that there were differences in the degree of sulphation and position of sulphate groups between the seaweed derived SPs, which in turn affected the level of FGF2 binding. ICAR and LCAR showed a higher level of sulphur as compared to ulvan and KCAR. However, this trend did not correlate regarding FGF2 binding to the materials. LCAR was shown to have a higher degree of sulphation than KCAR and ulvan, which resulted in a high amount of bound FGF2 to LCAR as compared to KCAR and ulvan. Meanwhile, the amount of FGF2 bound to ICAR was significantly lower than LCAR and was comparable to the amount bound to KCAR and ulvan, indicating that both degree of sulphation as well as position of sulphate groups play an important role in growth factor binding. The effect of degree of sulphation and position of sulphate groups of HS mimetics on FGF2 binding has been reported in previous studies. It has been shown that synthetic HS oligomers with higher degree of sulphation exhibited higher affinity to FGF2 as compared to the ones with lower degree of sulphation [192]. The affinity to FGF2 of the high sulphated HS oligomers was still slightly lower than that of a native HS hexasaccharide that was used as a control. Other types of HS mimetics also showed similar trend in the relationship between degree of sulphation and level of FGF2 binding. An increase in the degree of sulphation of sulphated alginate [193, 194], sulphated hyaluronic acid [195], or sulphated chitosan-arginine [67] has been shown to cause an increase in the level of FGF2 binding. It is also noted that ICAR had lower amount of FGF2 bound as compared to LCAR despite having

higher degree of sulphation. This could be attributed to the difference in the position of sulphate groups in the chemical backbone of ICAR and LCAR. While ICAR had 4-O and 2-O-sulphate groups, LCAR contained 2-O and 2,6-O-sulphate groups. Previous studies regarding the effect of position of sulphate groups on FGF2 binding has revealed that sulphated cellulose [196] or sulphated chitosan [197] with high level of 6-O-sulphation along the polymer chain could bind FGF2 at a comparable or even higher level than natural heparin. These findings indicate the importance of 6-O-sulphate groups in the structure of SPs that promoted FGF2 binding, and also help to explain why LCAR had higher amount of FGF2 bound than ICAR. The amount of FGF2 bound to heparin was the highest among all tested samples, which correlates with its highest sulphur content and implies that electrostatic interaction played an important role in the binding of FGF2 to heparin. Moreover, ATR-FTIR spectra of heparin showed that 2-O-sulphate and N-sulphate groups were present in the chemical structure of heparin. These sulphate groups have been shown to be necessary for the interaction between HS and FGF2, which therefore explains why heparin had high level of FGF2 binding [52]. In the current study, the level of FGF2 bound to the seaweed derived SPs was lower than heparin. It has been shown that HS bound to FGF2 at a lower level as compared to heparin [198, 199]. This indicates that the SPs may have similar FGF2 binding properties to HS given the lower amount of FGF2 binding compared to heparin. Together the present study has demonstrated that different degree of sulphation and position of sulphate groups of SPs impacted FGF2 binding, and these materials were shown to mimic the growth factor binding activity of HS.

3.5. Conclusion

This study revealed that three types of carrageenan and ulvan had distinct degree of sulphation and position of sulphate groups. Growth factor binding study showed that higher degree of sulphation resulted in higher level of FGF2 binding, and 6-O-sulphation was considered to be a contributing factor to increased amount of bound FGF2. These findings demonstrated that both degree of sulphation and position of sulphate groups of seaweed derived SPs used in this study influenced their ability to interact with FGF2. This provides a fundamental understanding of the structural requirements of SPs in the interaction with growth factors, which paves the way for further studies on the downstream biological functions of these materials.

Chapter 4

Effect of different sulphation characteristics of seaweed-derived SPs on growth factor signalling and cellular responses

4.1. Introduction

Wound healing is a process that involves the regeneration of tissues after injury. The wound healing process comprises four main stages, including haemostasis, inflammation, proliferation, and remodelling [200]. A key event in the proliferation stage is the formation of new blood vessels, a process known as angiogenesis. This event involves the proliferation and migration of endothelial cells that form capillary sprouts which later extend, branch, and develop into vascular networks at wound sites [201, 202]. These cellular events are governed by a variety of growth factors including members of the fibroblast growth factor (FGF) family and vascular endothelial growth factor family [202]. In the early stage of new blood vessel formation, FGF is crucial in the recruitment of endothelial cells and stimulation of their angiogenic activities [203]. In addition to endothelial cells, fibroblasts also have important roles in the wound healing process, especially during the proliferation stage. They migrate into the provisional fibrin clot created by platelets and secrete matrix molecules which support angiogenesis, reepithelialisation, and the formation of granulation tissue [204, 205]. The migration of fibroblasts has been shown to be regulated by FGF in the wound healing process [206].

The FGF family comprises 23 different members in which their signalling plays an important role in controlling various cellular processes including cell proliferation, migration, differentiation, and angiogenesis during wound healing and tissue repair [30]. The signalling activity of many FGFs is modulated by heparan sulphate (HS) chains that decorate HS proteoglycans, where FGF interacts with HS in a structure-specific manner followed by the formation of ternary complex with FGF receptor (FGFR), resulting in the activation of the FGFR [207]. Upon activation, the FGFR dimerises and autophosphorylates to initiate various intracellular signalling pathways including

Ras-mitogen-activated protein kinase (MAPK), phosphoinositide 3-kinase–Akt (PI3K/AKT), and phospholipase Cγ1 (PLCγ1) pathway [208]. Among them, MAPK pathway has been shown to be involved in angiogenesis which is a crucial step in wound healing [209, 210]. MAPK is a family of serine/threonine kinases that consists of three main members including extracellular signal-regulated kinases 1/2 (ERK1/2), p38 and c-Jun N-terminal kinase (JNK) [211]. Specifically, in ERK signalling pathway, the FGF-induced activation of FGFR can lead to the phosphorylation of ERK1/2 and the presence of phosphorylated ERK1/2 is used as a sign to demonstrate whether FGFR is activated to induce cellular responses [212-214]. Figure 4.1 illustrates the involvement of ERK1/2 in the ERK signalling pathway of FGF2.

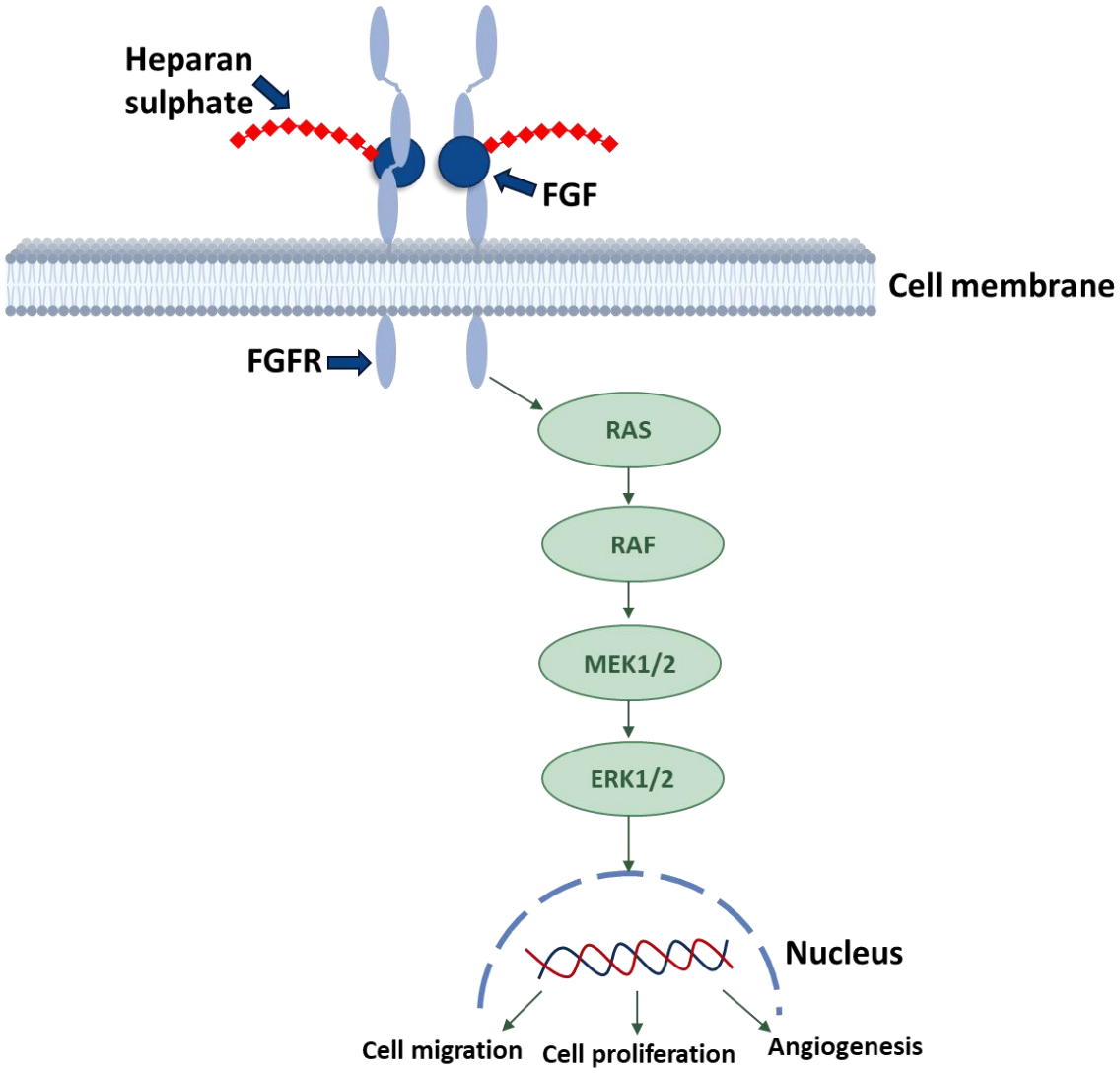


Figure 4.1. ERK signalling pathway of FGF2. ERK1/2 is a protein kinase that participates in the ERK signalling cascade. The phosphorylation of ERK1/2 is indicative of the activation of FGF receptor.

The initiation of ERK signalling pathway leads to various cellular activities that are important in the formation of new blood vessels during the wound healing process, including cell migration and cell sprouting [215]. It has been shown that these cellular activities are dependent on the level of FGF2 signalling which is modulated by HS proteoglycans on the cell surface [216]. In the context of developing HS mimetic materials, it has been reported that chemically synthetic HS

oligosaccharides or sulphated polysaccharides with varying degree of sulphation and position of sulphate groups had the ability to modulate FGF2 signalling and FGF2-mediated cell migration and vascular tube formation, which was essential for controlling angiogenesis process [66, 217-219]. However, the effect of different sulphation characteristics of seaweed derived SPs including carrageenan and ulvan on the signalling activity of FGF2 and FGF2-induced cell migration and cell sprouting remains to be investigated. In Chapter 3, the degree of sulphation and position of sulphate groups of SPs were shown to have an impact on the level of FGF2 binding. In this chapter, it was hypothesised that different sulphation characteristics of SPs could alter FGF2 signalling and modulate cell migration and cell sprouting. By gaining insight into the modulating impact of SPs on growth factor signalling and cellular behaviours, we can understand how these materials could mimic the biological functions of HS in regulating physiological processes in the body such as angiogenesis, thereby developing therapeutic strategies for wound healing and other tissue engineering applications. Therefore, the aim of this chapter was to investigate the ability of carrageenan and ulvan to modulate FGF2 signalling via the activation of FGF2 receptor, and downstream cellular activities including the migration of endothelial cells and fibroblasts and the formation of sprouts from endothelial spheroids.

4.2. Materials and Methods

Similar to Chapter 3, this chapter used heparin in all experiments as a control sample in place of HS. Results obtained will be put into context of HS mimetic materials to understand the biological functions of SPs as compared to HS.

Heparin sodium salt from porcine intestinal mucosa (H3393), kappa-carrageenan (22048), iota-carrageenan from red seaweed Eucheuma spinosum (C1138), and lambda-carrageenan (22049) were purchased from Sigma-Aldrich (Castle Hill, Australia). Ulvan from green seaweed Ulva armoricana (YU11689) was obtained from Biosynth Ltd (United Kingdom). Human fibroblast growth factor-2 (FGF2, PHG0024) was supplied by Thermo Fisher in lyophilised form. The growth factor was reconstituted with MilliQ water at 0.1 mg/mL prior to being used in experiments. All other chemicals were purchased from Sigma-Aldrich (Castle Hill, Australia) unless stated otherwise.

4.2.1. Growth factor signalling activity of SPs

To investigate the ability of SPs to signal FGF2, one approach was testing the formation of a ternary complex between SP, FGF2, and FGF2 receptor (FGFR1c) as a measure of FGF2 signalling. BaF32 cells, a cell line that does not produce endogenous HS and is transfected with FGFR1c, was used for this experiment. Information about this cell line and how the assay was performed is detailed as follows.

4.2.1.1. Culture and maintenance of murine interleukin-3 dependent pro-B cell line (BaF32 cells)

BaF32 cells are a myeloid B cell line that is dependent on Interleukin-3 (IL-3)-dependent and deficient in heparan sulphate proteoglycan. These cells have been transfected with fibroblast growth factor receptor (FGFR) 1c. In the presence of FGF and heparin/heparan sulphate (HS), FGF signalling is induced due to the formation of ternary complexes between heparin/HS, FGF, and FGFR on the cell surface which leads to the proliferation of BaF32 cells. Meanwhile, in the absence of the ternary complex being formed, the BaF32 cells do not proliferate. Thus, BaF32 cells were used to evaluate the ability of SPs to induce FGF2 signalling.

BaF32 cells transfected with FGFR1c were kindly provided by Professor David Ornitz (Washington University, USA). The cells were cultured in complete RPMI1640 medium (Media Preparation Unit, University of Melbourne) containing 10 % (v/v) Fetal Bovine Serum (FBS), 1 % (v/v) Penicillin-Streptomycin (PS) and 10 % (v/v) conditioned medium of WEHI-3BD cells at 37 °C with 5 % CO_2. BaF32 cells were passaged at 90 % confluency and transferred into a new flask at a concentration of 10^6 cells per mL. To eliminate untransfected cells prior to the assay, 200 µg/mL of G418 antibiotic solution was added into the complete medium during the culture of the BaF32 cells for two weeks. Medium was changed every four days, and the viability of the cells was checked using Trypan blue.

WEHI-3BD cells were kindly provided by the Ludwig Institute (Melbourne, Australia) and were cultured in RPMI1640 medium supplemented with 10 % (v/v) FBS and 1 % (v/v) PS at 37 °C with 5 % CO_2. The collection of conditioned medium of WEHI-3BD cells was performed twice per week

and the collected medium was preserved at a temperature of -20 °C. The medium was defrosted and sterile filtered prior to adding to BaF32 cell culture medium.

4.2.1.2. BaF32 proliferation assay

BaF32 cells expressing the FGF receptor FGFR1c were cultured in RPMI 1640 medium supplemented with 10 % (v/v) FBS, 10 % (v/v) WEHI-3BD conditioned medium and 1 % (v/v) PS. Prior to the assay, cells were starved in RPMI1640 medium without WEHI-3BD conditioned medium for 24 hours before being seeded into a 96-well plate at a density of 2 x 10^4 cells per well. Cells were incubated with phenol red-free RPMI medium (Thermo Fisher) only, and test samples including 30 nM heparin, kappa-, iota-, lambda-carrageenan, or ulvan either in the presence or absence of 0.3 nM FGF2. Cells treated with heparin and FGF2 were utilised as a positive control while cells exposed to heparin, FGF2, or medium only were used as negative controls. Test substances in medium without cells were also used for background absorbance readings. Cells were cultured for 72 hours at 37 °C and 5 % CO_2. Thereafter, the Cell Titer 96 Aqueous One Solution Reagent (MTS reagent, Promega) was added to each well and incubated for six hours at 37 °C, followed by the measurement of absorbance at 490 nm to assess BaF32 cell proliferation as a measure of growth factor signalling activity of SPs.

4.2.2. Growth factor receptor activation

In addition to BaF32 assay, the ability of SPs to signal FGF2 can be evaluated via the activation of FGF2 receptor on the cell surface of endothelial cells and fibroblasts. In this study, Western blot was utilised to detect the presence of phospho-ERK1/2 in the cells treated with SPs and FGF2 as an indirect measurement of receptor activation.

4.2.2.1. Culture and maintenance of endothelial cells and fibroblasts

Pooled donor Human Umbilical Vein Endothelial Cells (HUVECs, C2519A) were purchased from Lonza, Australia. Cells were maintained in Endothelial Cell Growth Medium (EGM, CC-3162, Lonza) containing Endothelial Cell Basal Medium (EBM, CC-3156, Lonza) supplemented with FBS, hydrocortisone, human FGF2, VEGF, R3-insulin-like growth factor-1 (R3-IGF-1), ascorbic acid, human epidermal growth factor (hEGF), Gentamicin sulphate -Amphotericin-1000 (GA-1000), and heparin (SingleQuots Supplements, CC-4176, Lonza) at 37 ^{o}C and 5 % CO_2. Culture medium was changed every two days. Cells were passaged at 80 % confluency and used between passage 3 and 5 for all experiments.

NIH 3T3 mouse fibroblasts (ATCC CCL-92, VA, USA) were cultured in Dulbecco's Modified Eagle Medium (DMEM) containing 10 % (v/v) FBS and 1 % (v/v) PS at 37 °C with 5 % CO_2. Cells were passaged at 80 % confluency.

4.2.2.2. Growth factor receptor activation assay

The activation of FGFR1 in endothelial cells and fibroblasts was analysed using Western blotting. Endothelial cells were seeded in 6-well plates at 1x10^6 cells per well in EGM at 37 ^{o}C and 5 % CO_2 and were starved in EBM for 18 hours prior to incubation with test substances. Fibroblasts were seeded in 6-well plates at 1x10^6 cells per well in DMEM supplemented with 10% (v/v) FBS and 1% PS at 37 ^{o}C and 5 % CO_2 and were starved in DMEM only for 18 hours prior to incubation with test substances. Cells were incubated for 10 minutes with control conditions including EBM (endothelial cells) or DMEM (fibroblasts) only, FGF2 (2 ng/mL), heparin (0.1 µg/mL), or FGF2 (2

ng/mL) and heparin (0.1 μg/mL) complex at 37 °C and 5 % CO_2. Test conditions consist of SPs only (0.1 μg/mL) or SPs (0.1 μg/mL) and FGF2 (2 ng/mL). After incubation, cells were washed with ice-cold phosphate buffered saline (PBS, Thermo Fisher) and lysed on ice with 0.1 mL lysis buffer (Abcam) per well. The lysates were agitated for 30 minutes at 4 °C, followed by centrifuging at 12,000 rpm for 20 minutes at 4 °C. The supernatant was then collected and stored at -80 °C.

Protein concentration of the cell lysates was determined using the Bradford protein assay (Thermo Fisher). A volume of cell lysate containing 10 μg of protein was combined with lithium dodecyl sulphate (LDS) sample buffer and reducing agent containing dithiothreitol (DTT), and the final volume was made up to 25 μL with MilliQ water. The mixture was then heated at 70 °C for 10 minutes. Cell lysate was electrophoresed on 4-12 % Bis-Tris Plus gels (Thermo Fisher) in 3-(N-Morpholino)propane sulfonic acid (MOPS) sodium dodecyl sulphate (SDS) running buffer (50 mM MOPS, 50 mM Tris Base, 0.1 % SDS, 1 mM ethylenediaminetetraacetic acid (EDTA), pH 7.7) at 200 V for 35 minutes using XCell SureLock Mini-Cell Electrophoresis System (Thermo Fisher). Molecular mass standards with molecular weights from 20 to 220 kDa (MagicMark XP Western Protein Standard, Thermo Fisher) were also electrophoresed on each gel. Samples were then transferred onto polyvinylidene fluoride (PVDF) membrane using Bis-Tris transfer buffer (25 mM Bicine, 25 mM Bis-Tris, 1 mM EDTA, 10 % (v/v) methanol, pH 7.2) using a semi-dry blotter (Novex, Thermo Fisher) at 300 mA and 20 V for 1 hour. The membrane was then blocked with 1 % (w/v) bovine serum albumin (BSA) in Tris buffered saline (TBS) (20 nM Tris base, 150 nM NaCl, pH 7.6) containing 0.1 % (v/v) Tween 20 (TBST) for 2 hours at room temperature, followed by incubation with primary antibodies including rabbit monoclonal anti-phospho-ERK1/2 antibody (1:2000, clone D13.14.4E, Cell Signalling Technology), rabbit monoclonal anti-ERK1/2 antibody (1:1000,

clone 137F5, Cell Signalling Technology), or rabbit polyclonal anti-GAPDH antibody (1:1000, Invitrogen) in 1 % w/v BSA in TBST at 4 ^{o}C overnight. Following this, membranes were rinsed with TBST, incubated with anti-rabbit HRP-conjugated secondary antibody (1:10000, Invitrogen) in 1 % w/v BSA in TBST at room temperature for 45 minutes, and rinsed with TBST and TBS three times before being treated with chemiluminescent substrate (SuperSignal West Pico PLUS, Thermo Fisher) for 5 minutes. Finally, membranes were imaged using a chemiluminescence imaging system (FluorChem E, Protein Simple). The density of the bands in each membrane was quantified using ImageJ software (NIH, USA) to compare between samples.

4.2.3. Cell migration assay

The effect of SPs on the migration of fibroblasts or endothelial cells was examined using a Transwell migration assay that comprised polycarbonate inserts with pore size of 8 µm in 24-well plates (Corning Costar, USA). Prior to commencing the assay, fibroblasts were starved in serum free DMEM, while endothelial cells were starved in EBM for 18 hours before being seeded in the upper chambers of the Transwell inserts at a density of $3x10^4$ cells per insert. The lower chambers were filled with 600 µL of DMEM or EBM with 2 % (v/v) FBS for fibroblasts or endothelial cells, respectively, containing 0.1 µg/mL heparin, kappa-, iota-, or lambda-carrageenan either in the presence or absence of 2 ng/mL FGF2. The cells were incubated at 37 ^{o}C and 5 % CO_2, and cell migration was evaluated following 8 hours or 24 hours for fibroblasts or endothelial cells, respectively. Cells that had migrated onto the lower side of the insert membrane were fixed with 4 % paraformaldehyde (ProSciTech, Australia) for 15 minutes, permeabilised with 0.1 % Triton-X (Chem Supply, Australia) in PBS for 4 minutes and stained with Phalloidin (1X in PBS, Abcam) and DAPI (1 µg/mL in PBS, Thermo Fisher) at room temperature. After staining, a cotton swab was

used to remove cells that had not migrated and remained on the upper side of the insert membrane. Migrated cells were imaged at four fields (top, bottom, left, right) per insert using a fluorescent microscope (Axio Vert. A1, Zeiss). The number of cells in four fields was manually counted using ImageJ software (NIH, USA) and compared between samples.

4.2.4. Cell spheroid sprouting assay

4.2.4.1. Spheroid formation

The influence of SPs on the sprouting activity of endothelial cells was investigated using a cell spheroid sprouting assay. To begin with, endothelial cells were suspended in EBM supplemented with 2 % (v/v) FBS and 0.4 % (w/v) methyl cellulose (Methocel A4M). The cell suspension was transferred into a low attachment round bottom 96-well plate (Corning, USA) at the concentration of 750 cells per well and cultured at 37 ^{o}C, 5 % CO_2 overnight for spheroid formation. Spheroids were harvested by centrifugation and suspended in EBM supplemented with 2 % FBS at the concentration of 100 spheroids per mL.

4.2.4.2. Fibrin hydrogel formation and spheroid embedding

For the sprouting assay, the endothelial spheroids were embedded in fibrin gels. Fibrin gels were formed by combining thrombin (5 U/mL), fibrinogen (20 mg/mL) and cell spheroids at a ratio of 1:1:1 by volume, and 250 µL of the mixture was placed in each well of a 48-well plate, and the fibrin gel was allowed to polymerise for three minutes at room temperature. Next, 500 µL of EBM containing 2 % (v/v) FBS and 0.1 µg/mL heparin, kappa-, iota-, or lambda-carrageenan either in the presence or absence of 2 ng/mL FGF2 was added on top of the polymerised gels. Spheroid

embedded gels were incubated for 24 hours at 37°C and 5 % CO_2, followed by imaging to observe sprout formation. Sprouting from the spheroid bodies was quantified by measuring the number and length of sprouts of each spheroid using ImageJ software (NIH, USA).

4.2.5. Statistical analysis

Significant differences were determined by two-way analysis of variance (ANOVA) followed by Dunnett's or Tukey's multiple comparison test using GraphPad Prism 9.5.1 (GraphPad Software, San Diego, CA, USA). Differences between groups were considered significant at $p < 0.05$ and indicated in the figures as *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, and ****$p < 0.0001$.

4.3. Results

4.3.1. BaF32 proliferation assay

In this study, BaF32 cells transfected with FGFR1c were employed to investigate the ability of SPs to form ternary complexes with FGF2 and FGFR1c as a measure of FGF2 signalling activity of SPs. The readout of this assay is the proliferation of BaF32 cells which was monitored using MTS assay with higher absorbance indicating higher number of cells (Figure 4.2). All the test substances were incubated in medium without cells and their absorbance values were similar to that of medium without any additives, indicating that the MTS reagent was not impacted by the presence of the materials. Cells treated with the positive control, heparin and FGF2, showed a significantly ($p<0.0001$) higher absorbance value than cells exposed to heparin only, demonstrating the formation of ternary complex between heparin, FGF2 and FGFR1c and the ability for heparin to signal FGF2. In the presence of FGF2, neither KCAR, ICAR, nor ulvan appeared to significantly enhance cell proliferation compared to cells exposed to these materials only, suggesting that they were incapable of signalling FGF2. Meanwhile, the exposure of cells to LCAR and FGF2 resulted in a significant ($p<0.01$) increase in absorbance value compared to cells that were exposed to LCAR only, indicating that LCAR formed an active ternary complex with FGF2 and FGFR1c. These data suggested that among the SPs investigated, only LCAR could mimic the biological activity of HS in terms of signalling FGF2. This trend was also in correlation with the FGF2 binding results in Chapter 3 where LCAR had significantly higher amount of bound FGF2 compared to KCAR, ICAR, and ulvan.

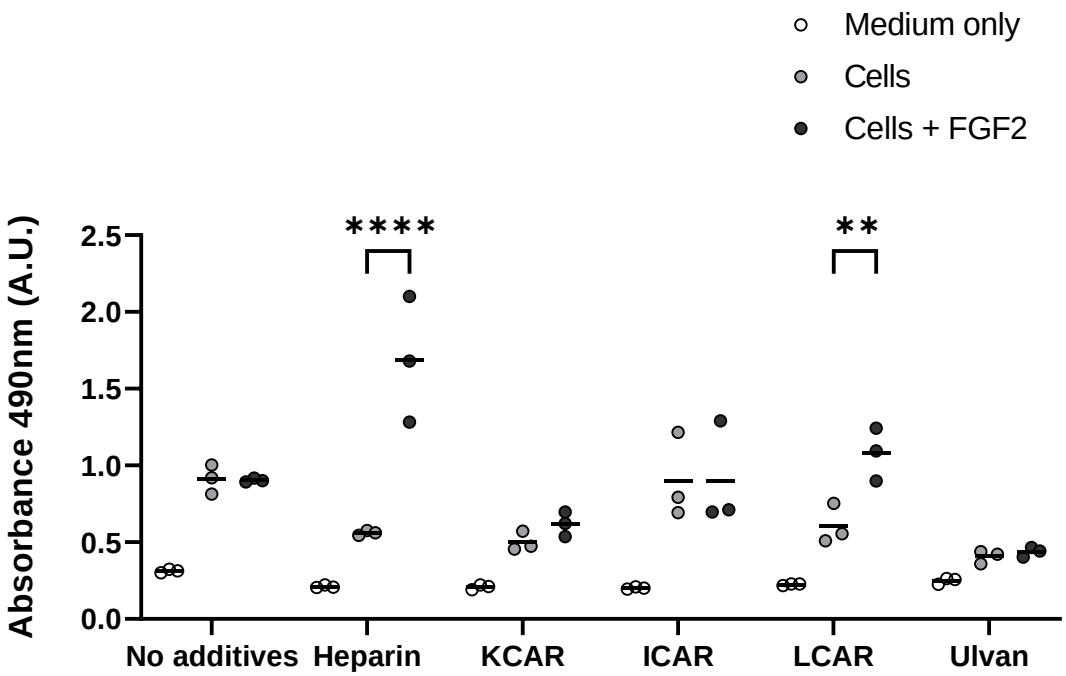


Figure 4.2. FGF2 signalling of SPs determined by the BaF32 proliferation assay. BaF32 cells transfected with FGFR1c were exposed to various SPs including kappa-, iota-, lambda-carrageenan, or ulvan either in the presence or absence of FGF2. Cells exposed to heparin and FGF2 were used as a positive control while cells exposed to heparin, FGF2, or medium only were regarded as negative controls. The ability of SPs to signal FGF2 was determined via the proliferation of BaF32 cells when exposed to both SPs and FGF2. MTS reagent was used to evaluate cell proliferation by measuring and comparing the absorbance of samples at 490 nm. * represents statistically significant difference (**$p < 0.01$ and ****$p < 0.0001$). KCAR = kappa-carrageenan, ICAR = iota-carrageenan, LCAR = lambda-carrageenan.

4.3.2. Growth factor receptor activation

The ability of SPs to signal FGF2 was also evaluated using Western blot. ERK1/2 is involved in the mitogen-activated protein kinase (MAPK) signalling pathway of FGF2, and the phosphorylation of ERK1/2 indirectly indicates the activation of FGFR1c by FGF2. In this study, both endothelial cells and fibroblasts, known to express FGFR1 [220, 221], were used to investigate FGF2 signalling of SPs to understand the effect of growth factor signalling activity of SPs on different cell types that are important in the wound healing process.

Figure 4.3 and Figure 4.4 show the Western blot membranes that were probed with anti-ERK1/2 and anti-phospho-ERK1/2 antibodies to detect the presence of ERK1/2 and phospho-ERK1/2 in endothelial cells exposed to SPs with or without FGF2, respectively. Each membrane contained control samples and test samples and was probed with either anti-ERK1/2 or anti-phospho-ERK1/2 antibody. An additional membrane was probed with GAPDH antibody which was used as the loading control for all conditions at the molecular weight of 37 kDa (Figure 4.3A and Figure 4.4A). Figure 4.3B and Figure 4.4B demonstrate that ERK1/2 was detected in endothelial cells and detected at the molecular weight of 42/44 kDa. Meanwhile, there was a difference in the level of phospho-ERK1/2 between cells incubated in the presence of FGF2 (lane 2, 4, 6, 8, and 10 in Figure 4.3C and lane 2, 4, and 6 in Figure 4.4C) and in the absence of FGF2 (lane 1, 3, 5, 7, and 9 in Figure 4.3C and lane 1, 3, and 5 in Figure 4.4C) as illustrated by the difference in the band intensity.

The intensity of the Western blot bands was quantified using ImageJ software to compare between samples. All band intensity values for ERK1/2 and phospho-ERK1/2 were normalised to

the band intensity of GAPDH. The ratio between GAPDH-normalised phospho-ERK1/2 and ERK1/2 was calculated to normalise the intensity value of phospho-ERK1/2 to that of ERK1/2 as shown in Figure 4.5B. Endothelial cells cultured in heparin and FGF2 were utilised as the positive control. Cells exposed to ICAR, LCAR, and ulvan in the presence of FGF2 showed a significant decrease ($p<0.05$) in the ratio of phospho-ERK1/2 and ERK1/2 compared to the heparin control. However, cells incubated with KCAR and FGF2, or with FGF2 only did not show a significant difference in the level of phospho-ERK1/2 as compared to the positive control. Cells cultured in each SP and FGF2 did not show any significant differences in the ratio of phospho-ERK1/2 and ERK1/2 compared to cells cultured in FGF2 only. It is also noticeable that the level of phospho-ERK1/2 detected in each SP was variable between repeats, indicating a variation in the growth factor signalling activity within each SP which could be related to the structural heterogeneity of these materials.

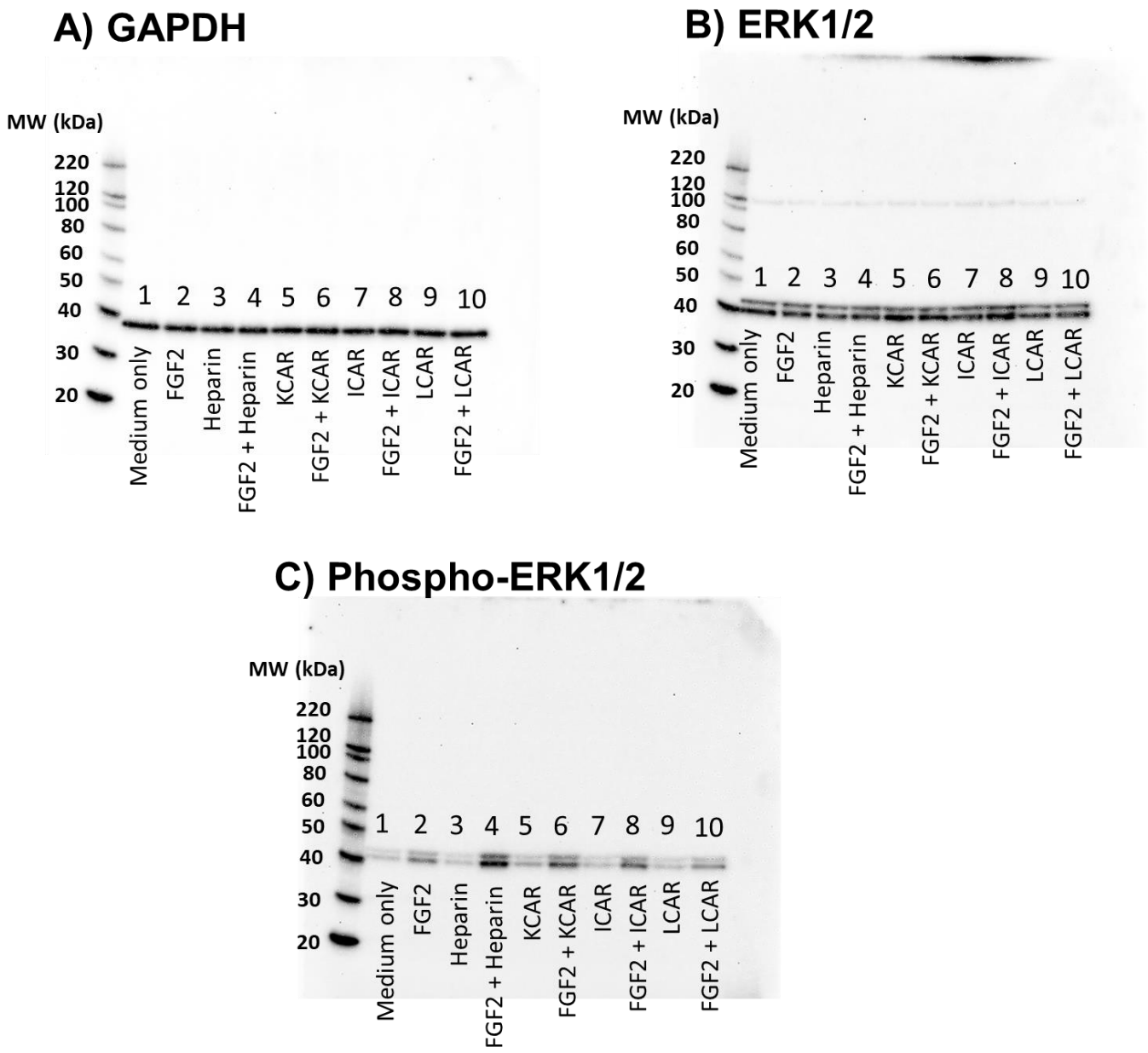


Figure 4.3. Western blot membranes showing the presence of A) GAPDH, B) ERK1/2 and C) phospho-ERK1/2 in endothelial cells incubated in various conditions. The proteins were detected using anti-GAPDH, anti-ERK1/2, and anti-phospho-ERK1/2 antibodies, respectively. In each membrane, the test conditions consist of 1: Medium only; 2: FGF2 (2 ng/mL); 3: Heparin (0.1 µg/mL); 4: FGF2 (2 ng/mL) and heparin (0.1 µg/mL); 5: kappa-carrageenan (KCAR, 0.1 µg/mL); 6: FGF2 (2 ng/mL) and kappa-carrageenan (0.1 µg/mL); 7: iota-carrageenan (ICAR, 0.1 µg/mL); 8: FGF2 (2 ng/mL) and iota-carrageenan (0.1 µg/mL); 9: lambda-carrageenan (LCAR, 0.1 µg/mL); 10: FGF2 (2 ng/mL) and lambda-carrageenan (0.1 µg/mL).

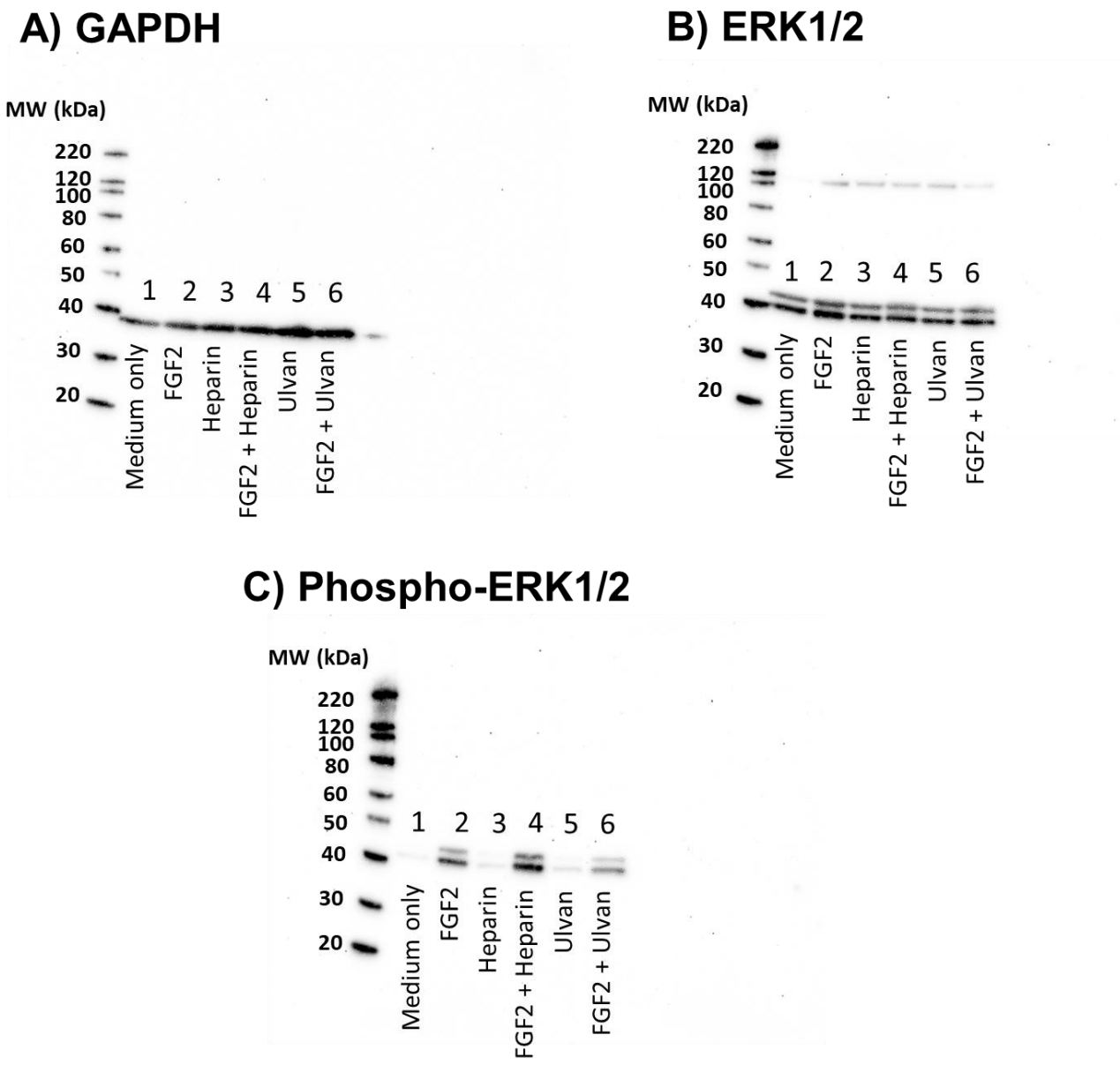


Figure 4.4. Western blot membranes showing the presence of A) GAPDH, B) ERK1/2 and C) phospho-ERK1/2 in endothelial cells incubated in various conditions. The proteins were detected using anti-GAPDH, anti-ERK1/2, and anti-phospho-ERK1/2 antibodies, respectively. In each membrane, the test conditions consist of 1: Medium only; 2: FGF2 (2 ng/mL); 3: Heparin (0.1 µg/mL); 4: FGF2 (2 ng/mL) and heparin (0.1 µg/mL); 5: ulvan (0.1 µg/mL); 6: FGF2 (2 ng/mL) and ulvan (0.1 µg/mL).

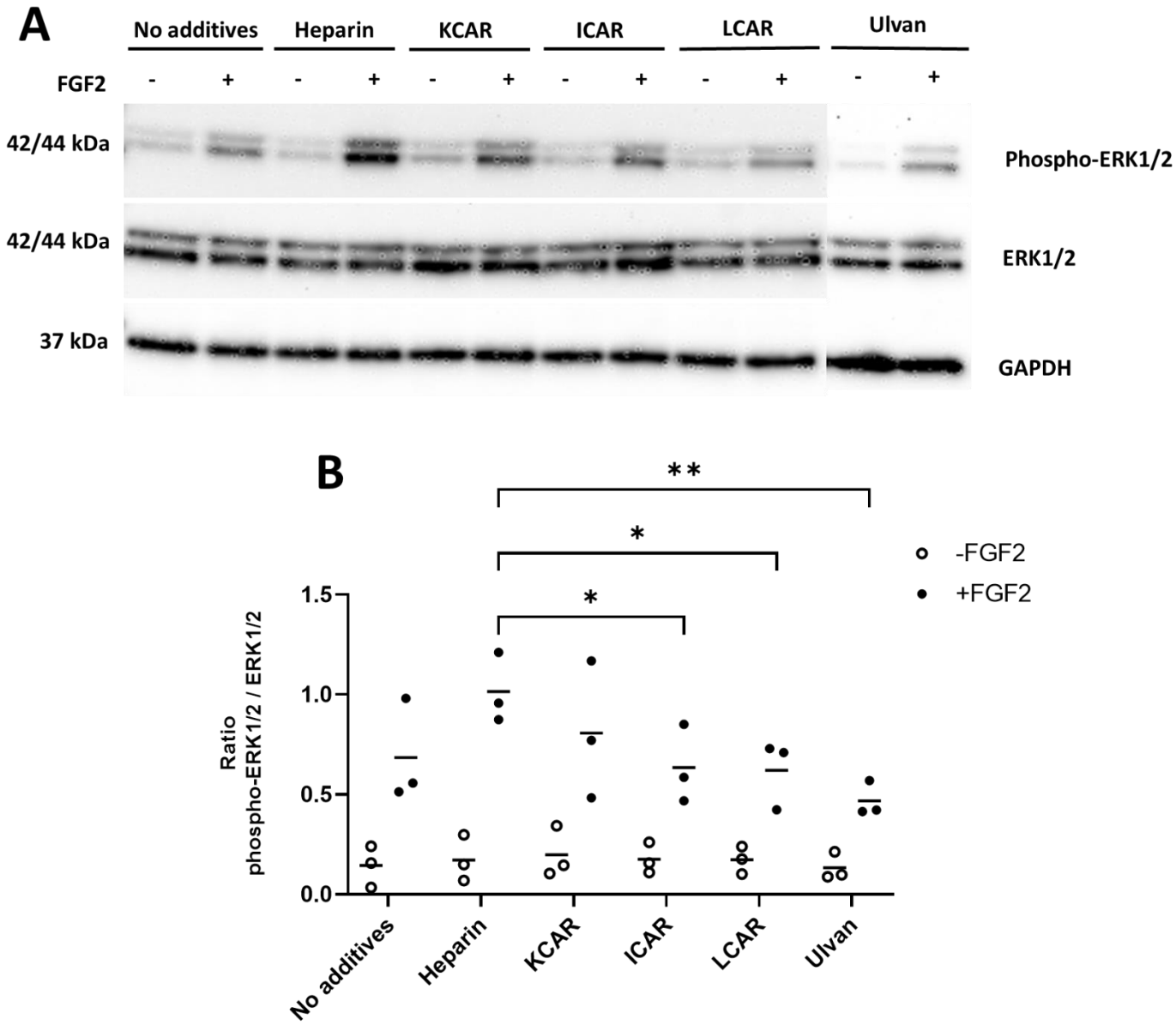


Figure 4.5. FGF2 signalling of various SPs in endothelial cells determined by Western blotting. A) Western blot bands showing the presence of GAPDH, ERK1/2 and phospho-ERK1/2 in endothelial cells incubated in various SPs with or without FGF2 using anti-GAPDH, anti-ERK1/2, and anti-phospho-ERK1/2 antibodies, respectively. B) Analysis of intensity of western blot bands in figure A showing the ratio of phospho-ERK1/2 and ERK1/2. All data points are normalised to GAPDH band intensity. * represents statistically significant difference (*$p<0.05$, **$p<0.01$).

In addition to endothelial cells, fibroblasts were also utilised to monitor the FGF2 signalling of SPs using western blotting. Figure 4.6 and Figure 4.7 show the western blot membranes with detected bands of GAPDH, ERK1/2, and phospho-ERK1/2 in fibroblasts. Each membrane was incubated with either ERK1/2 or phospho-ERK1/2 antibody, and GAPDH antibody was used in an additional membrane as the loading control. The band intensity of phospho-ERK1/2 appeared to be different in samples with FGF2 (lane 2, 4, 6, 8, and 10 in Figure 4.6C and lane 2, 4, and 6 in Figure 4.7C) compared to samples without FGF2 (lane 1, 3, 5, 7, and 9 in Figure 4.6C and lane 1, 3, and 5 in Figure 4.7C). All band intensity was quantified using Image J software and normalised to GAPDH band intensity. The band intensity of phospho-ERK1/2 was normalised to that of ERK1/2 and presented as the ratio of phospho-ERK1/2 and ERK1/2 in Figure 4.8B. The ratio of phospho-ERK1/2 and ERK1/2 of cells incubated in heparin and FGF2 appeared to be lower than cells incubated in FGF2 only. There were no significant differences in the level of phospho-ERK1/2 detected in cells incubated with each SP and FGF2 compared to cells incubated with heparin and FGF2 or with FGF2 only. This result show SPs had no or minimal effect on FGF2 signalling in fibroblasts at concentration and incubation time evaluated; however, there was variability in the signalling activity of SPs as demonstrated by the variable level of phospho-ERK1/2 detected in cells incubated with each SP and FGF2. It is also interesting that trends in how SPs affected FGF2 signalling as compared to heparin were different in the two cell types evaluated. Notably, the ratio of phospho-ERK1/2 and ERK1/2 of endothelial cells incubated in the presence of FGF2 was lower than that of fibroblasts incubated in the same condition.

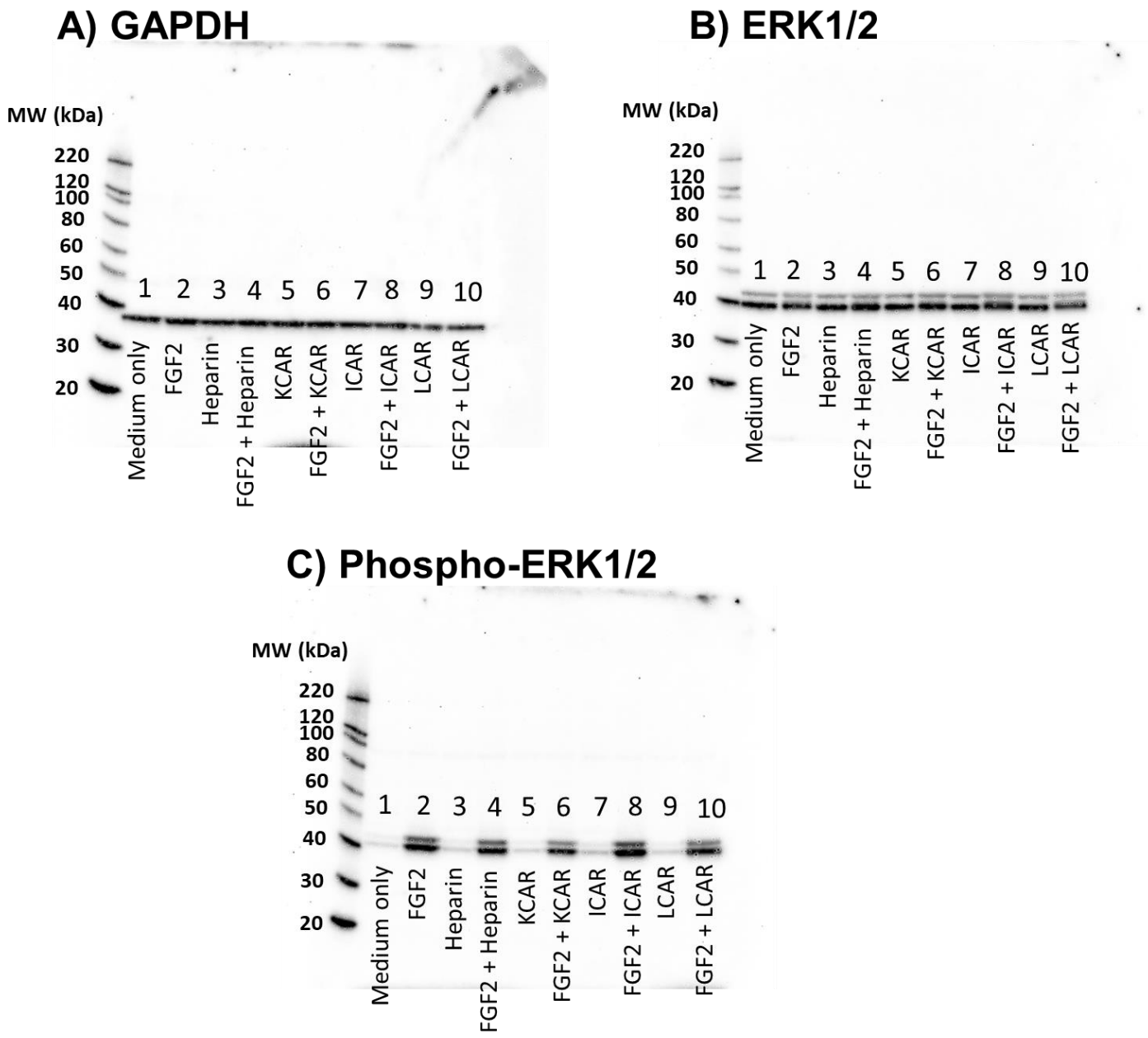


Figure 4.6. Western blot membranes showing the presence of A) GAPDH, B) ERK1/2 and C) phospho-ERK1/2 in fibroblasts incubated in various conditions. The proteins were detected using anti-GAPDH, anti-ERK1/2, and anti-phospho-ERK1/2 antibodies, respectively. In each membrane, the test conditions consist of 1: Medium only; 2: FGF2 (2 ng/mL); 3: Heparin (0.1 µg/mL); 4: FGF2 (2 ng/mL) and heparin (0.1 µg/mL); 5: kappa-carrageenan (KCAR, 0.1 µg/mL); 6: FGF2 (2 ng/mL) and kappa-carrageenan (0.1 µg/mL); 7: iota-carrageenan (ICAR, 0.1 µg/mL); 8: FGF2 (2 ng/mL) and iota-carrageenan (0.1 µg/mL); 9: lambda-carrageenan (LCAR, 0.1 µg/mL); 10: FGF2 (2 ng/mL) and lambda-carrageenan (0.1 µg/mL).

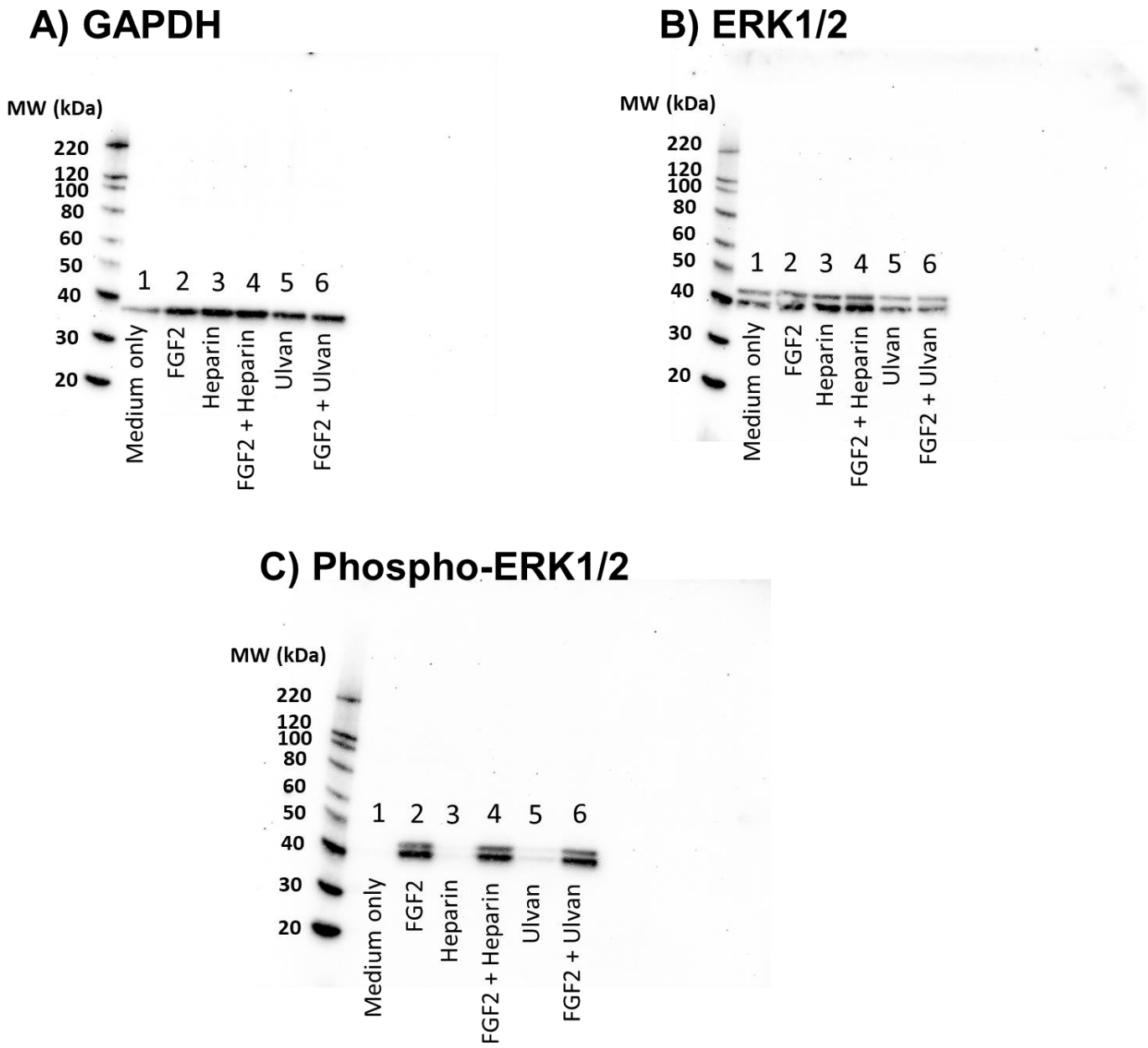


Figure 4.7. Western blot membranes showing the presence of A) GAPDH, B) ERK1/2 and C) phospho-ERK1/2 in fibroblasts incubated in various conditions. The proteins were detected using anti-GAPDH, anti-ERK1/2, and anti-phospho-ERK1/2 antibodies, respectively. In each membrane, the test conditions consist of 1: Medium only; 2: FGF2 (2 ng/mL); 3: Heparin (0.1 µg/mL); 4: FGF2 (2 ng/mL) and heparin (0.1 µg/mL); 5: ulvan (0.1 µg/mL); 6: FGF2 (2 ng/mL) and ulvan (0.1 µg/mL).

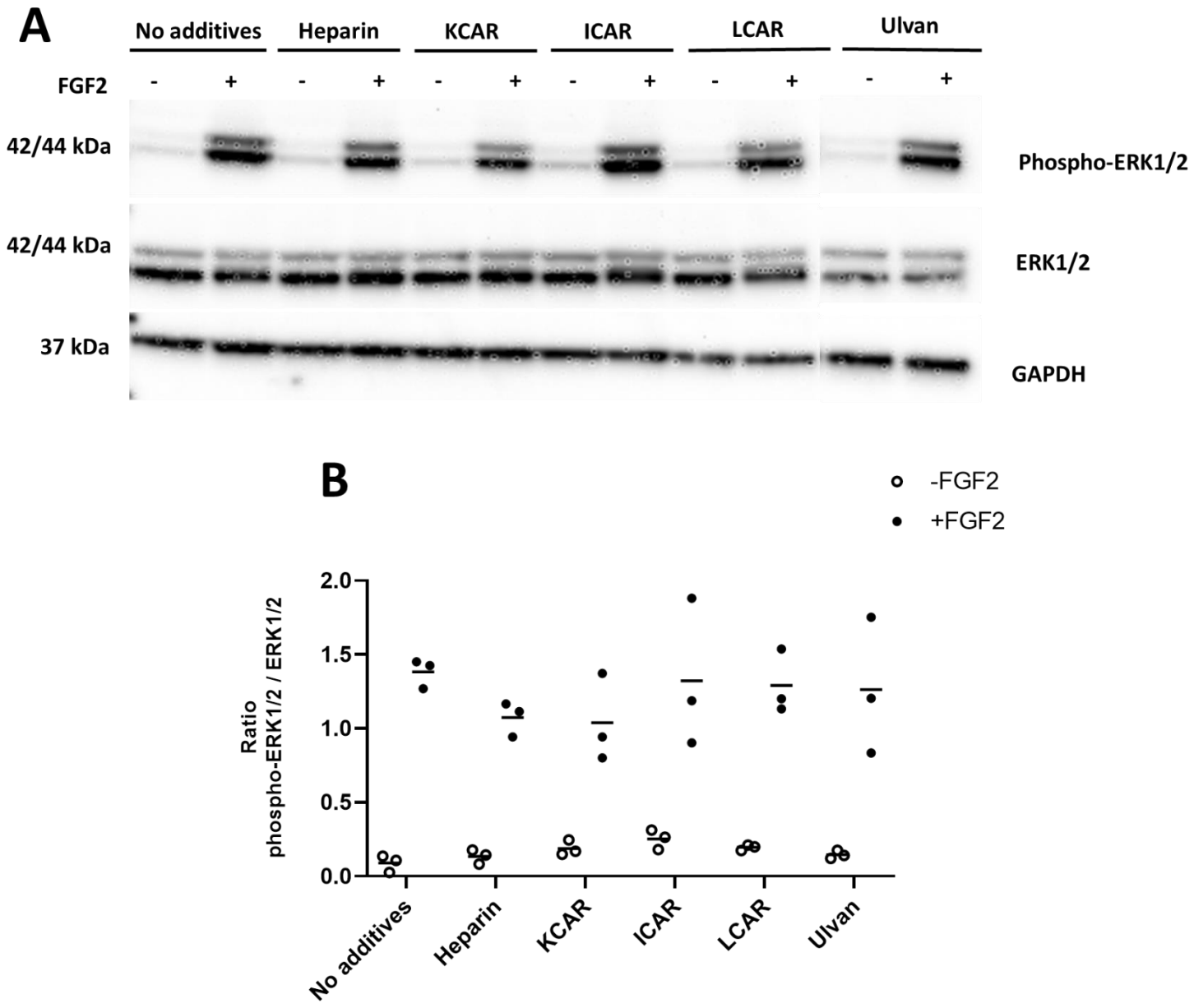


Figure 4.8. FGF2 signalling of various SPs in fibroblasts determined by western blotting. A) Western blot bands showing the presence of GAPDH, ERK1/2 and phospho-ERK1/2 in fibroblasts incubated in various SPs with or without FGF2 using anti-GAPDH, anti-ERK1/2, and anti-phospho-ERK1/2 antibodies, respectively. B) Analysis of intensity of western blot bands in figure A showing the ratio of phospho-ERK1/2 and ERK1/2. All data points are normalised to GAPDH band intensity.

4.3.3. Cell migration assay

To understand the ability of SPs to induce cellular responses, migration activity of endothelial cells and fibroblasts was investigated as migration is an important behavioural characteristic of cells during the wound healing process. Migrated cells were stained with phalloidin and DAPI to visualise actin filaments and nuclei, respectively. Figure 4.9 and Figure 4.10 illustrate the merged images and grey scale images of phalloidin channel of migrated endothelial cells after 8-hour incubation in the presence of various SPs with or without FGF2. Regarding cell morphology, cells were elongated when incubated in the presence of FGF2 which are indicated by yellow arrows, while cells that were not exposed to FGF2 appeared to be in round shape which are indicated by white arrows. Interestingly, cells incubated with either ICAR or ulvan even in the presence of FGF2 did not have an elongated morphology but maintained in round shape. Moreover, several cells exposed to LCAR with FGF2 showed the tendency to aggregate together while being elongated. These results demonstrate that different types of SPs exhibited varying effects on the morphological aspect of migrated cells.

The number of migrated endothelial cells was quantified by counting the number of cell nuclei in four areas of each Transwell insert. Figure 4.11 shows the number of migrated endothelial cells following exposure to the SPs with or without FGF2. Cells incubated with FGF2 only show a significantly higher ($p<0.001$) number of migrated cells than cells incubated with FGF2 in the presence of either heparin, KCAR, ICAR, or ulvan. Moreover, cells exposed to FGF2 and LCAR migrated to a significantly higher ($p<0.01$) extent than cells exposed to FGF2 in the presence of either heparin, ICAR, or ulvan. In the cell receptor activation results, LCAR had significantly lower ability to activate FGFR1c in endothelial cells compared to heparin, while in the migration assay

it had higher migratory effect than heparin. This indicates besides SPs other biological molecules that were present in the culture medium could have been involved in modulating the migration of endothelial cells.

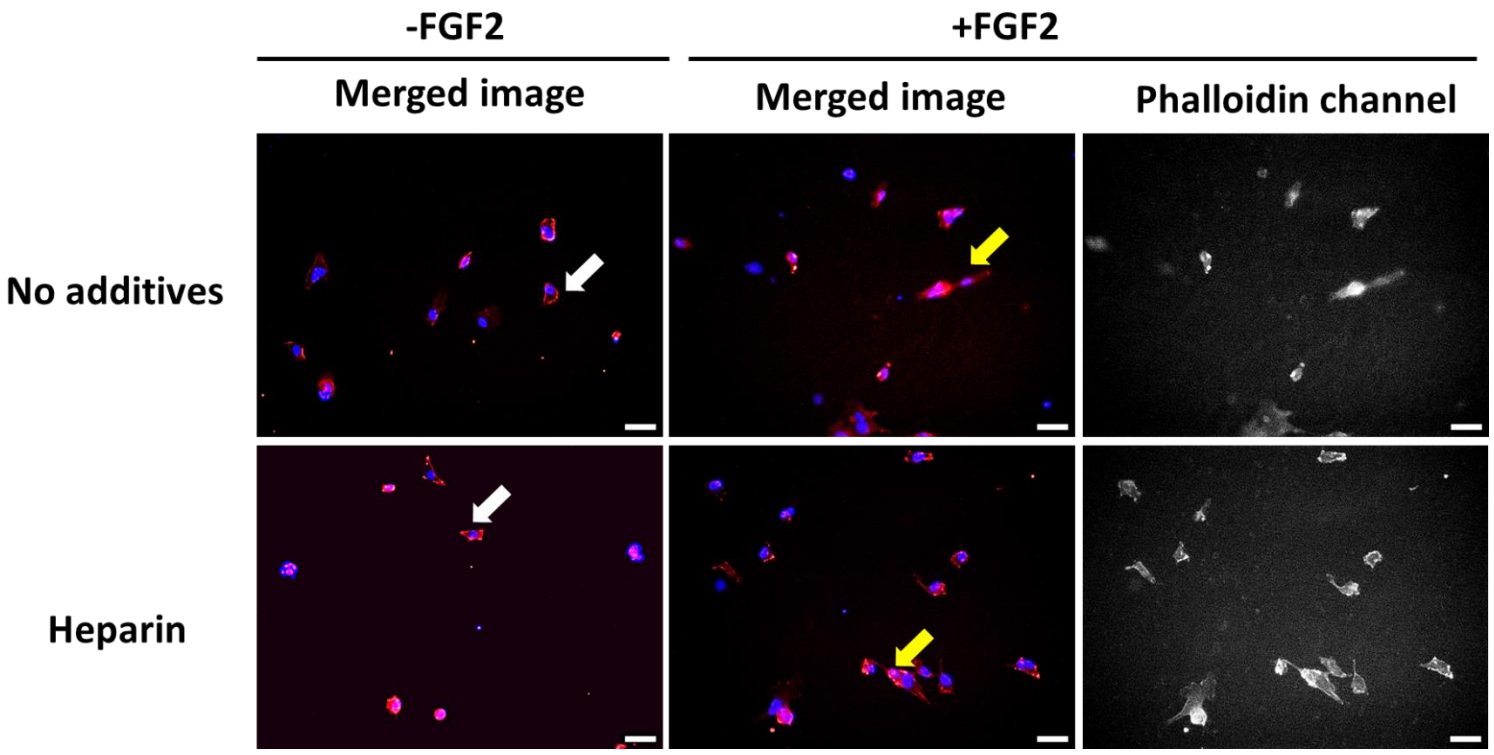


Figure 4.9. Merged images and grey scale images of phalloidin channel of migrated endothelial cells in Transwell migration assay. Cells were incubated in EBM supplemented with 2 % FBS or heparin in the presence or absence of FGF2 for 24 hours and stained with Phalloidin (red) and DAPI (blue) for morphology observation. White arrows indicate rounded cells, while yellow arrows indicate elongated cells. Scale bars = 50 µm.

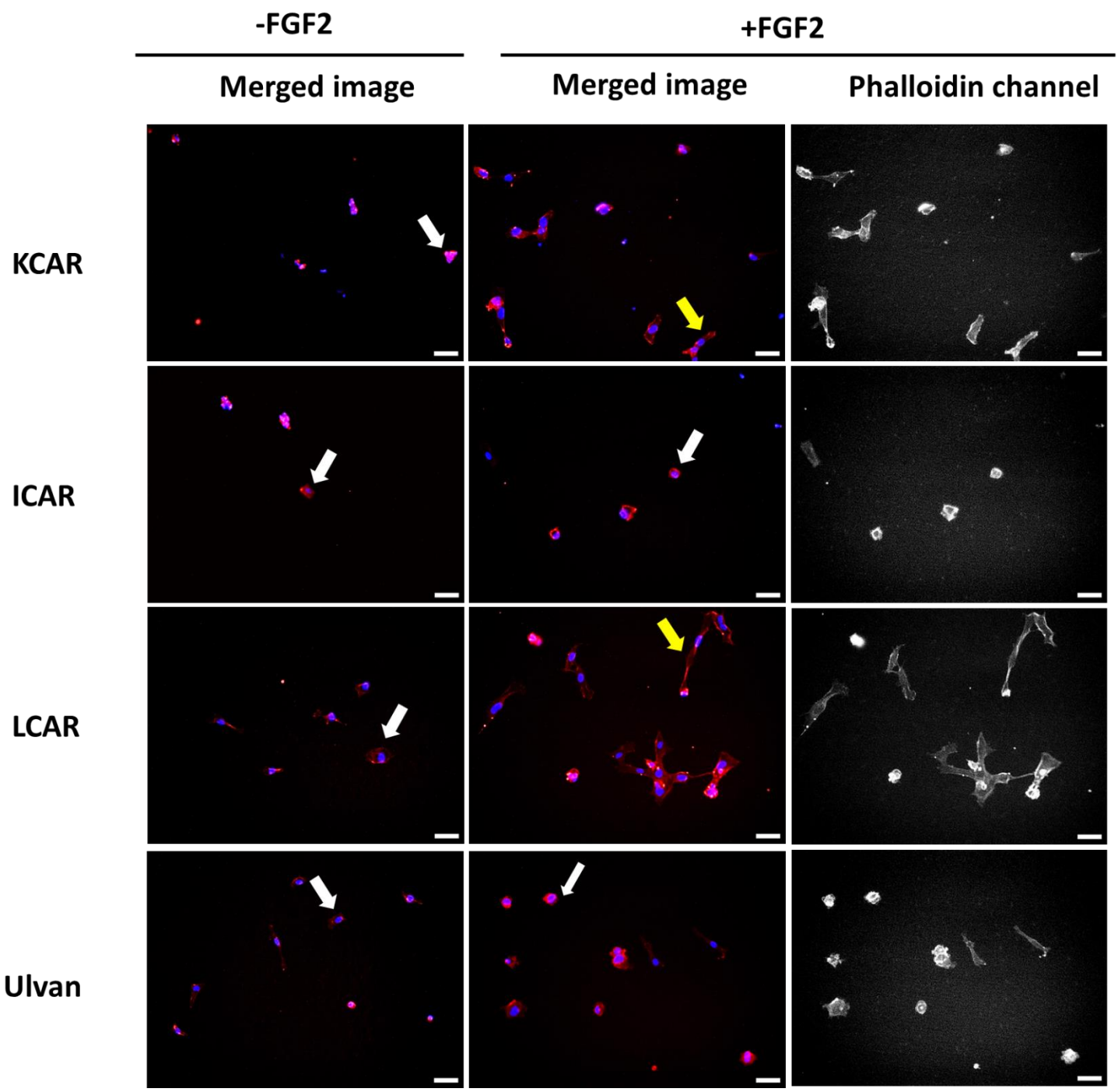


Figure 4.10. Merged images and grey scale images of phalloidin channel of migrated endothelial cells in Transwell migration assay. Cells were incubated in various SPs in the presence or absence of FGF2 for 24 hours and stained with Phalloidin (red) and DAPI (blue) for morphology observation. White arrows indicate rounded cells, while yellow arrows indicate elongated cells. Scale bars = 50 µm.

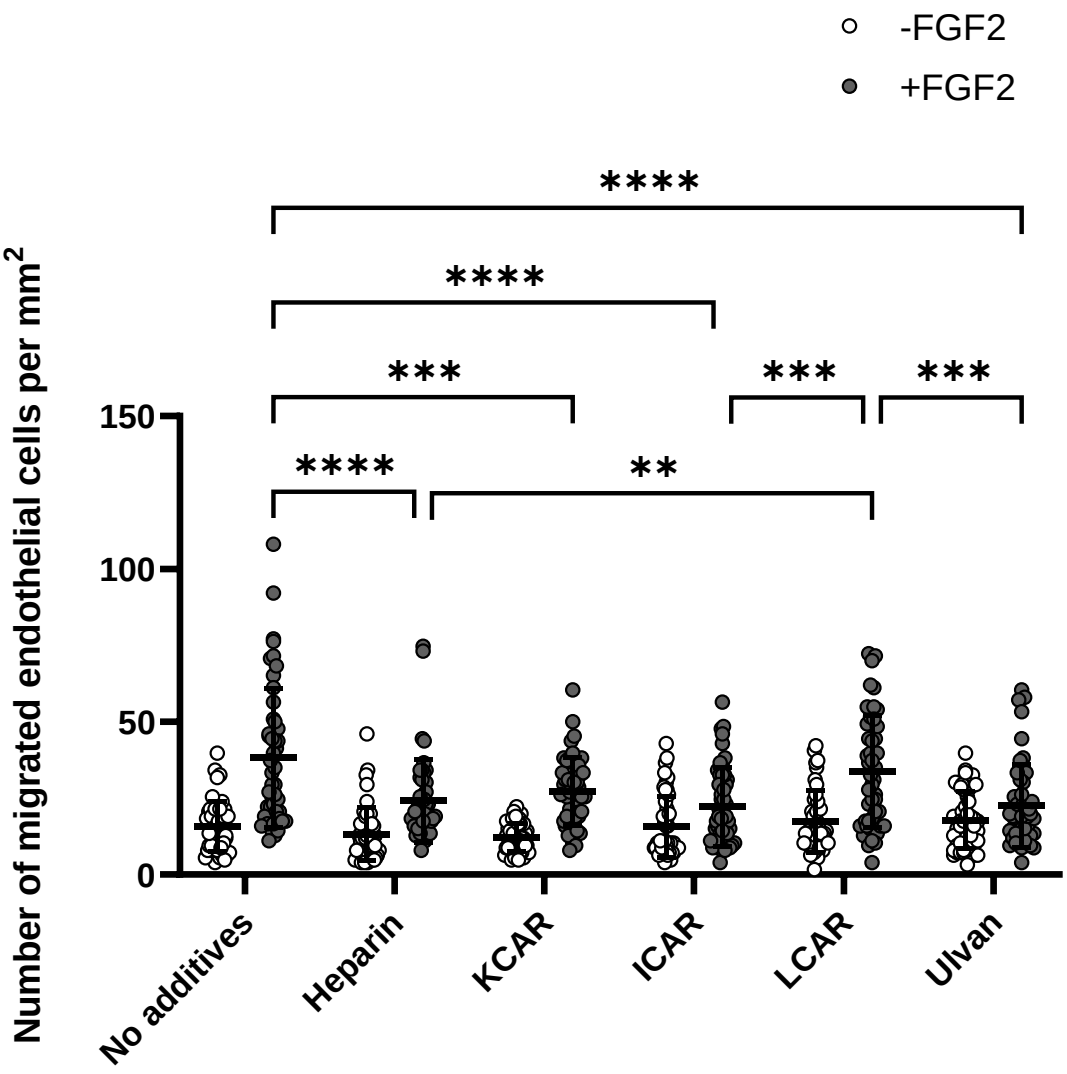


Figure 4.11. Quantification of number of migrated endothelial cells in Transwell migration assay. In each Transwell insert, DAPI-stained cell nuclei were counted in four areas (top, bottom, left, right). Three inserts were used for each test condition and the experiment was repeated four times. * represents statistically significant difference (**p<0.01, ****p<0.0001).

In addition to endothelial cells, the effect of SPs on the migration of fibroblasts was also evaluated with morphology of migrated cells showed in Figure 4.12 and Figure 4.13. A majority of the cells incubated with FGF2 only or FGF2 and heparin were rounded (white arrows) with a few being elongated (yellow arrows). Meanwhile, almost all cells were elongated when exposed to FGF2 and the SPs as indicated by the yellow arrows. A few cells incubated with FGF2 and KCAR or FGF2 and LCAR tended to aggregate when migrating, whereas cells cultured with FGF2 and ICAR or FGF2 and ulvan were distributed across the Transwell insert membrane.

The number of migrated fibroblasts was quantified and shown in Figure 4.14. In the presence of FGF2, all SPs except KCAR demonstrated a significantly lower ($p<0.05$) ability in stimulating the migration of fibroblasts as compared to heparin. Even though the growth factor signalling activity of ICAR, LCAR, and ulvan in fibroblasts did not show a significant difference to heparin, they had lower migratory effect on those cells than heparin. In comparison to endothelial cells, more migrated fibroblasts were seen given the two cell types were incubated in similar concentration of SPs and FGF2. This indicates the variation in the ability of SPs to modulate the migration activity of different cell types.

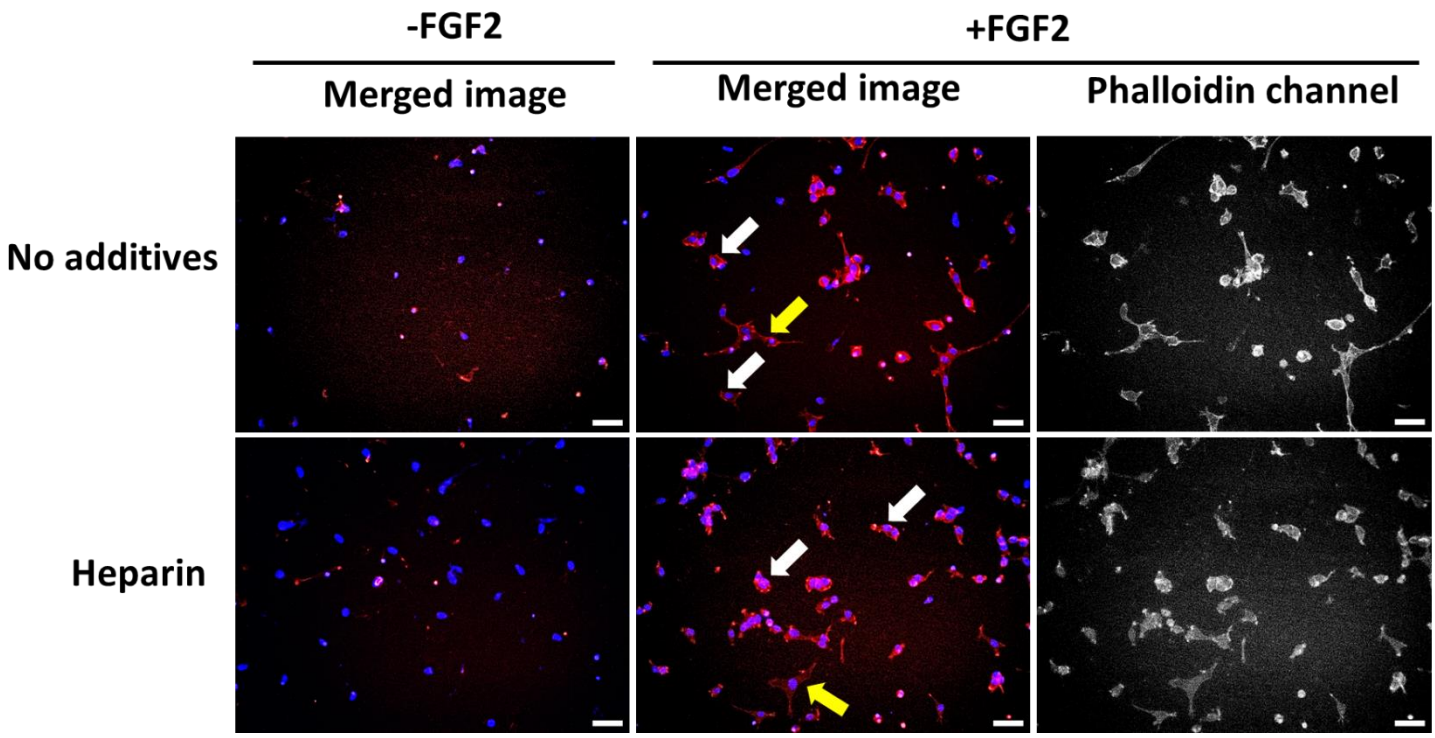


Figure 4.12. Merged images and grey scale images of phalloidin channel of migrated fibroblasts in Transwell migration assay. Cells were incubated in DMEM or heparin in the presence or absence of FGF2 for 8 hours and stained with Phalloidin (red) and DAPI (blue) for morphology observation. White arrows indicate rounded cells, while yellow arrows indicate elongated cells. Scale bars = 50 µm.

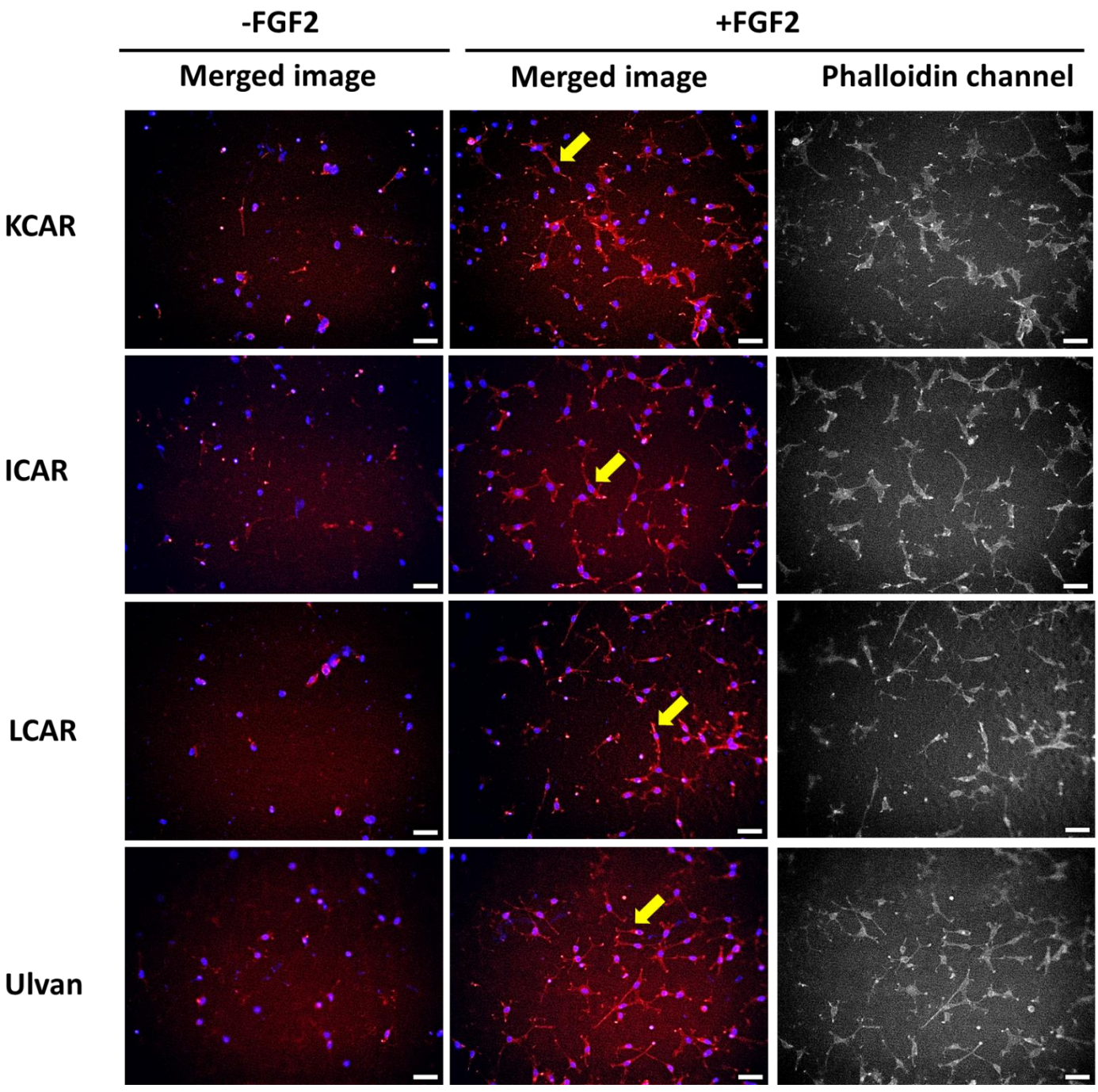


Figure 4.13. Merged images and grey scale images of phalloidin channel of migrated fibroblasts in Transwell migration assay. Cells were incubated in various SPs in the presence or absence of FGF2 for 8 hours and stained with Phalloidin (red) and DAPI (blue) for morphology observation. White arrows indicate rounded cells, while yellow arrows indicate elongated cells. Scale bars = 50 µm.

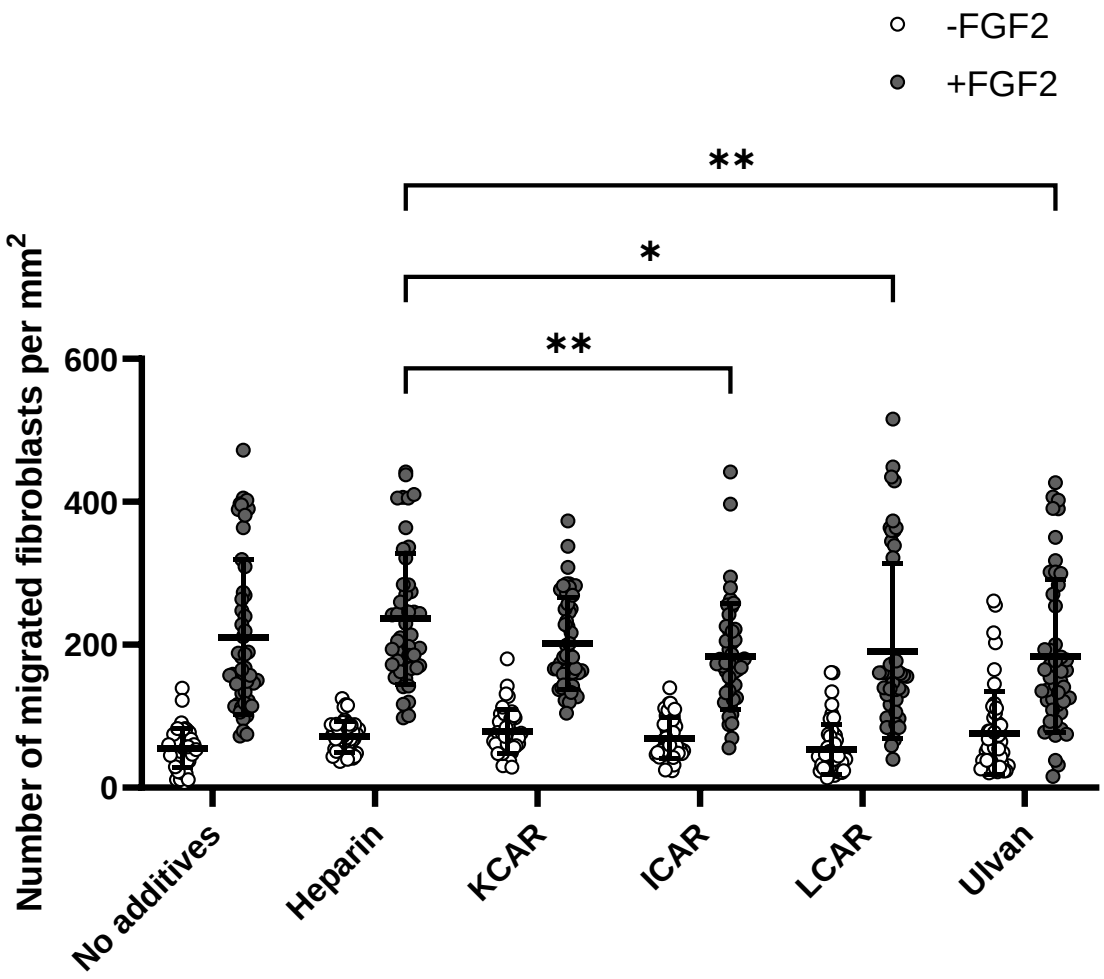


Figure 4.14. Quantification of number of migrated fibroblasts in Transwell migration assay. In each Transwell insert, DAPI-stained cell nuclei were counted in four areas (top, bottom, left, right). Three inserts were used for each test condition and the experiment was repeated four times. * represents statistically significant difference (*p<0.05, **p<0.01).

4.3.4. Cell spheroid sprouting assay

To understand the potential angiogenetic effect of SPs, endothelial cells were used to form spheroids which were encapsulated in fibrin gels and incubated with SPs for sprout formation. The number of sprouts, and cumulative sprout length of each spheroid was used as a measure of sprout formation and was quantified using ImageJ software as illustrated in Figure 4.15. Each sprout of a spheroid was marked with a point for counting (Figure 4.15A), while a line was manually drawn along the sprout and the length of sprout was determined by measuring the length of the line (Figure 4.15B). Figure 4.16 and Figure 4.17 show the sprouting activity of endothelial cell spheroids after 24-hour incubation. More sprouts were observed in spheroids incubated in the presence of FGF2 compared to samples without FGF2. Spheroids incubated in FGF2 and heparin formed thicker sprouts compared to sprouts formed in FGF2 only or FGF2 and the SPs. Most of the sprouts in all incubation conditions were relatively straight and remained separate with no sign of branching sprouts being observed.

In order to compare between samples, the number of sprouts per spheroid and cumulative sprout length of each spheroid were quantitatively analysed. Figure 4.18A shows the number of sprouts per spheroid. Spheroids incubated with FGF2 only or FGF2 with heparin formed a significantly greater ($p<0.05$) number of sprouts than spheroids incubated with FGF2 and the SPs. This trend correlated with the growth factor signalling activity of SPs in endothelial cells where SPs induced a lower level of receptor activation in the cells compared to heparin. No significant differences in the number of sprouts per spheroid were observed between the SPs investigated. Figure 4.18B demonstrates the measured cumulative sprout length of each spheroid. Spheroids incubated with FGF2 and heparin produced sprouts with significantly longer ($p<0.0001$)

cumulative length than spheroids incubated in FGF2 only or FGF2 and the SPs. In addition, the cumulative sprout length of spheroids exposed to FGF2 and either KCAR or ulvan was significantly shorter than that of spheroids cultured in FGF2 only. Together these results indicate that in the presence of FGF2 SPs reduced the sprouting activity of endothelial spheroids as compared to heparin, which was illustrated by lower number of sprouts and shorter sprout length.

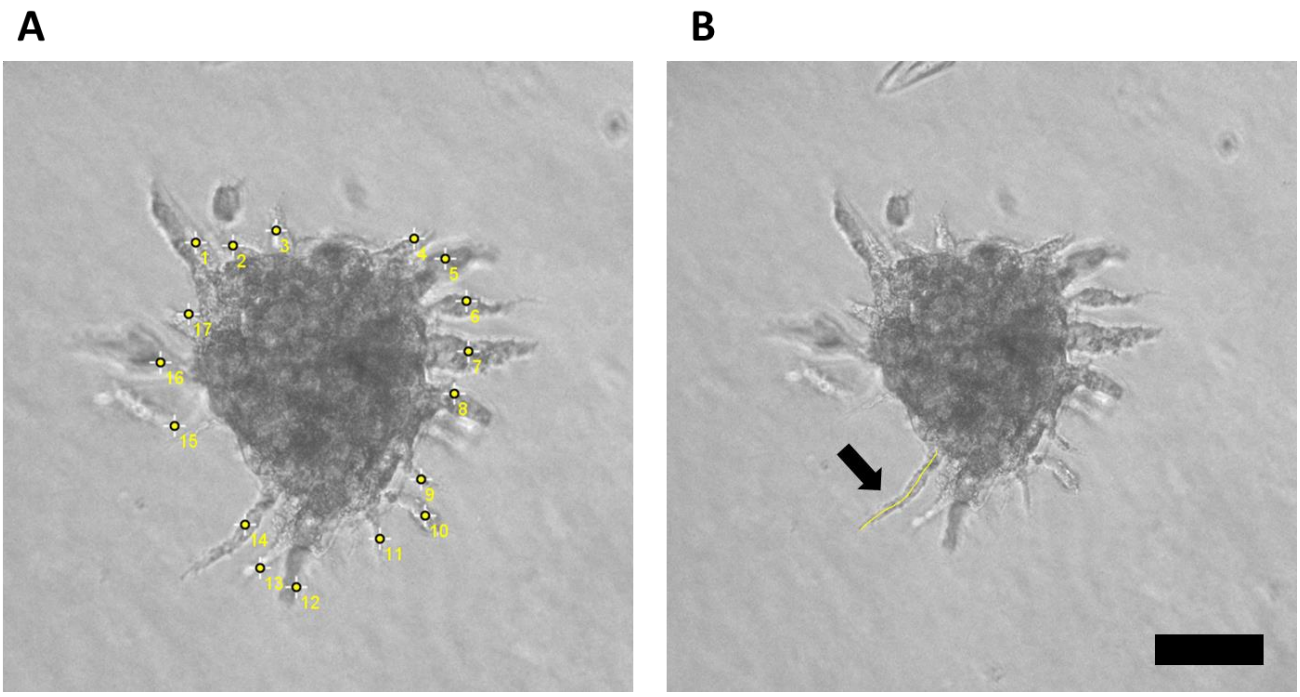


Figure 4.15. Sprouting analysis of a spheroid. A) Each sprout on the spheroid was marked with a point for counting. B) Black arrow showing a line drawn along the sprout for measuring sprout length. Scale bar = 50 µm.

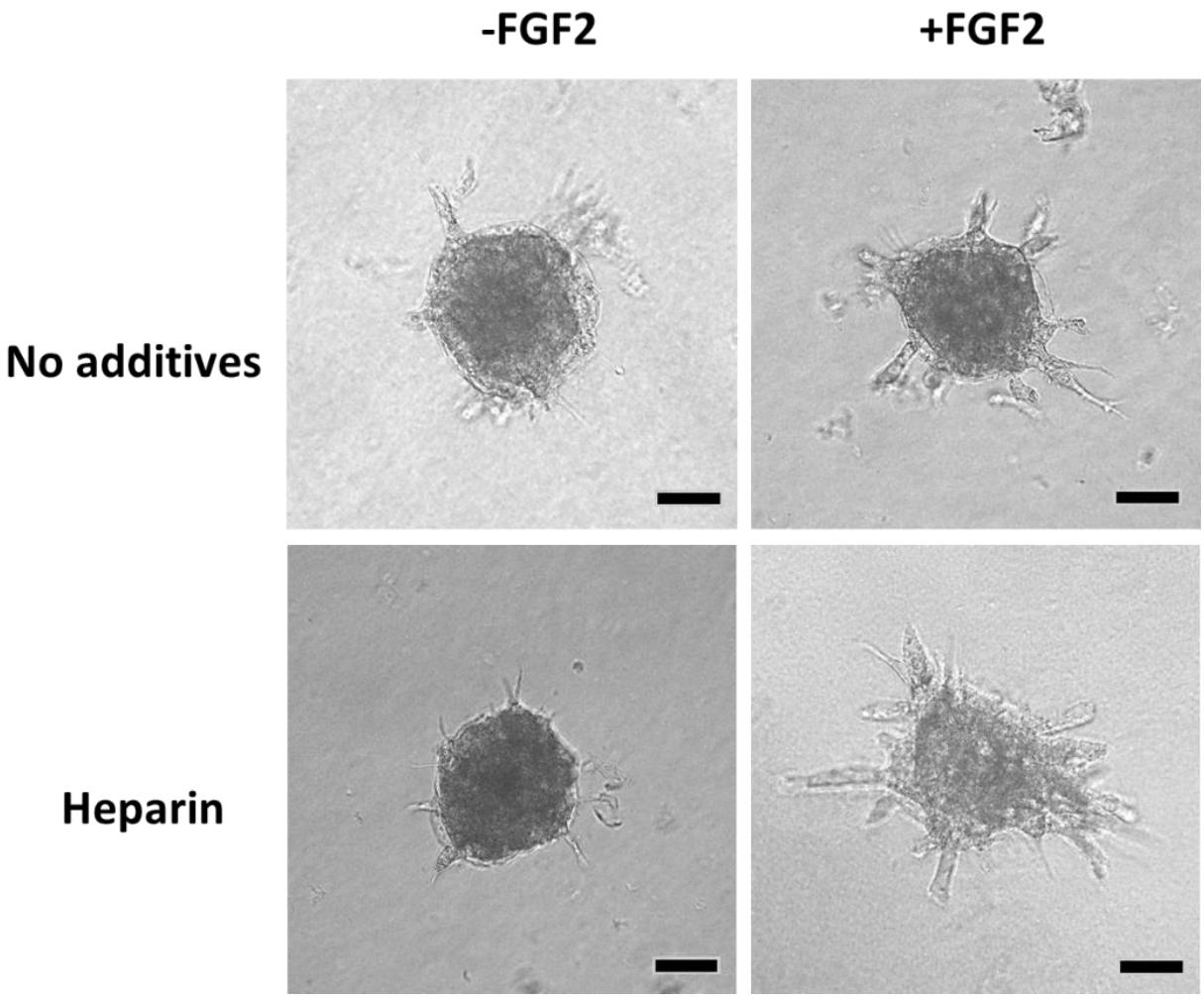


Figure 4.16. Sprouting of endothelial cell spheroids embedded in fibrin gels. The spheroids were incubated in EBM supplemented with 2 % FBS or heparin in the presence or absence of FGF2 for 24 hours for sprout formation. Scale bars = 50 µm.

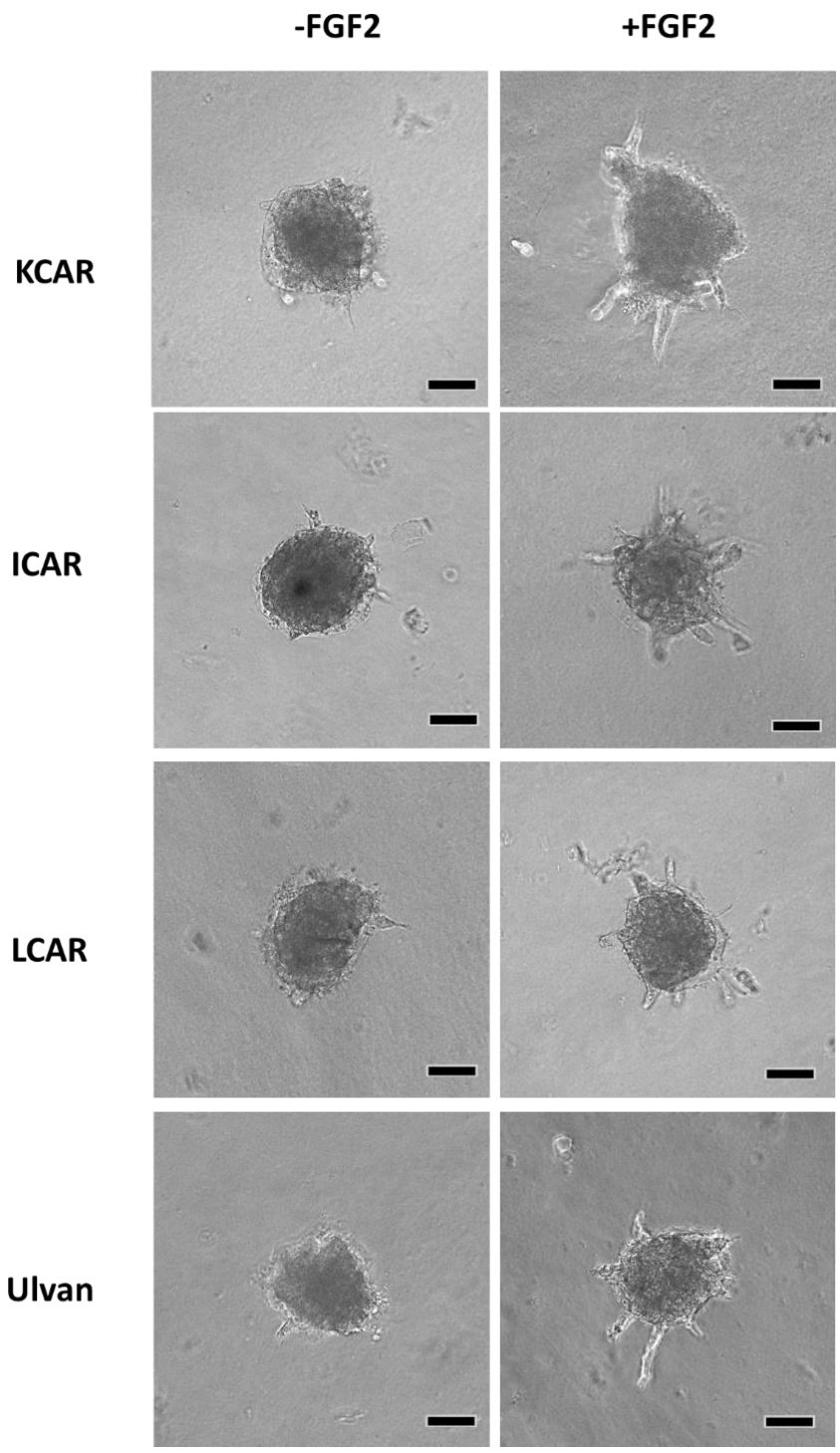


Figure 4.17. Sprouting of endothelial cell spheroids embedded in fibrin gels. The spheroids were incubated in various SPs in the presence or absence of FGF2 for 24 hours for sprout formation. Scale bars = 50 µm.

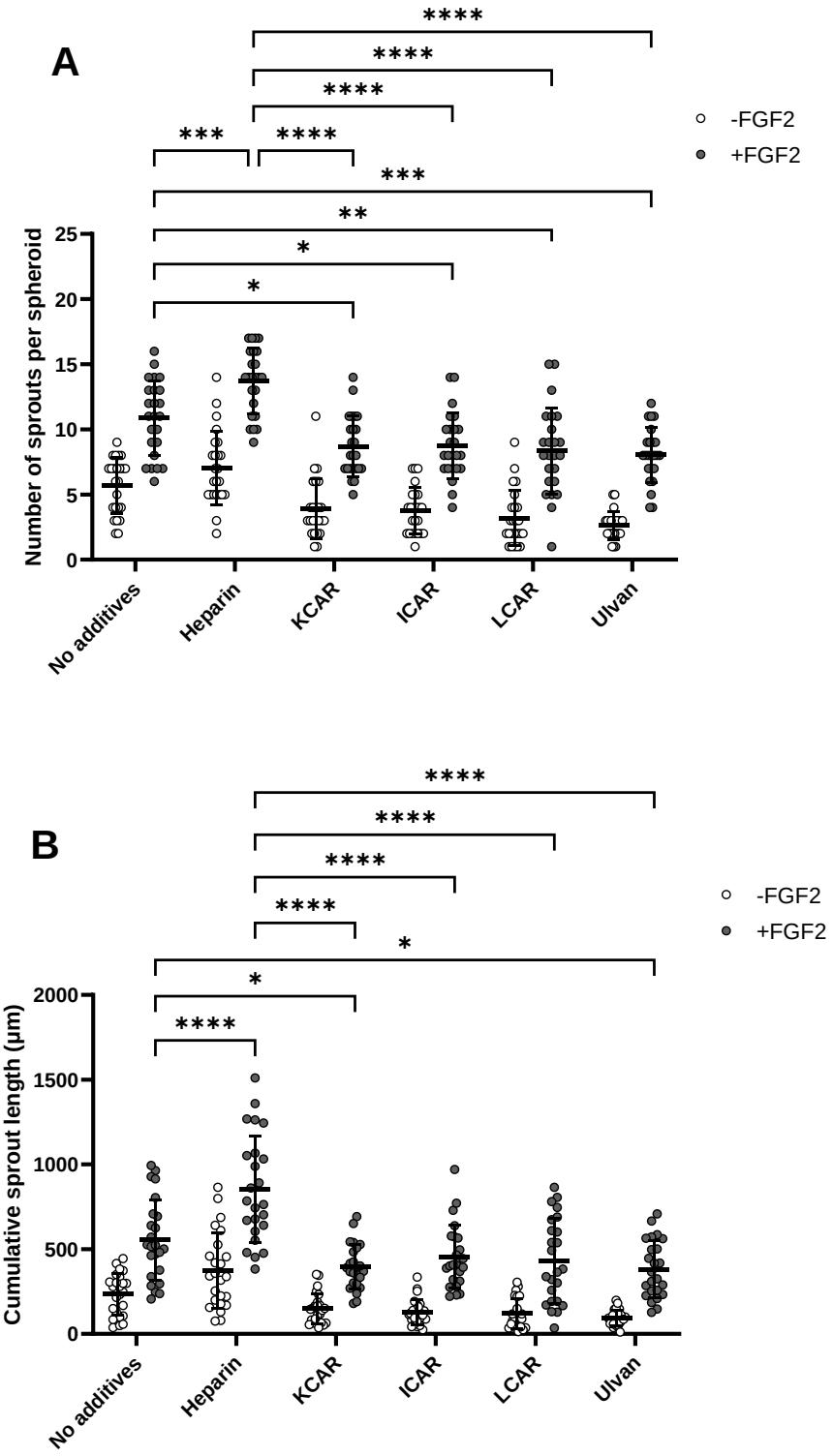


Figure 4.18. Measurement of A) sprout number and B) sprout length of endothelial cell spheroids. For each test condition, six spheroids were used for quantifying sprout number and sprout length. Cumulative sprout length indicates the total length of all counted sprouts in each spheroid. The whole experiment was repeated four times. * represents statistically significant difference (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001).

4.4. Discussion

The aim of this chapter was to investigate the ability of seaweed derived SPs to signal FGF2 and modulate FGF2-induced cellular responses including cell migration and cell sprouting. It was hypothesised that different sulphation characteristics of SPs could alter FGF2 signalling and downstream cell migration and cell sprouting.

In Chapter 3, it was shown that degree of sulphation and position of sulphate groups of SPs had an impact on the level of FGF2 binding, and LCAR was reported to have significantly higher amount of bound FGF2 than other SPs. Different levels of FGF2 binding could affect the ability of SPs to signal the growth factor as they may facilitate the binding of FGF2 to its receptor by acting as a co-factor of FGF2 or inhibit FGF2 from binding to its receptor due to the strong binding between the growth factor and SPs. In this chapter, growth factor signalling was evaluated using the BaF32 assay and cell receptor activation assay, and results show that different SPs had distinct signalling activity. Regarding the BaF32 assay, heparin was shown to induce the proliferation of the BaF32 cells, indicating the ability of heparin to form a ternary complex with FGF2 and FGFR1c. This phenomenon was also observed in LCAR but not KCAR, ICAR, and ulvan. From the elemental analysis results, LCAR and ICAR had higher degree of sulphation than KCAR and ulvan but only LCAR had the ability to form a ternary complex with FGF2 and FGFR1c, indicating not only degree of sulphation but position of sulphate groups also plays an essential role in growth factor signalling. It has been reported that the interaction between heparan sulphate (HS) and FGF2 requires the presence of 2-O-sulphate groups on the iduronic acid units and N-sulphate groups on the glucosamine units, while the interaction with FGFR1c requires additional 6-O-sulphate

groups on the glucosamine units of HS [222-224]. Compared with HS, the repeating disaccharide unit of LCAR has a 2-O-sulphate group on one of the galactose units and 2,6-O-sulphate groups on the other galactose unit, while ICAR had a 4-O-sulphate group on the galactose unit and 2-O-sulphate group on the anhydrogalactose unit. It was likely that the similarity in the position of sulphate groups between LCAR and HS enabled LCAR to signal FGF2. BaF32 assay results revealed that LCAR signalled FGF2 at a lower level than heparin. It has been reported that HS oligosaccharides prepared from porcine mucosal HS exhibited lower levels of FGF2 signalling as compared to heparin [222]. This further indicates that LCAR potentially mimicked HS but not heparin regarding the capacity to signal FGF2. The effect of different sulphation characteristics of HS mimetics on growth factor signalling has also been reported in the literature. Lord et al. [225] synthesised sulphated chitosan-arginine as HS mimetics where N-sulphate groups were attached to the C2 position of the glucosamine units, and the compound was shown to signal FGF2 as evaluated in the BaF32 assay. The group of Leali et al. [226] created sulphated derivatives of the *Escherichia coli* K5 polysaccharides with varying degree of sulphation and position of sulphate groups. The highly sulphated derivative with 2,3-O-sulphate groups on glucuronic acid units and N,6-O-sulphate groups on glucosamine units were shown to interact with FGF2, though these materials inhibited the binding of growth factor to native HS and its associated receptor in endothelial cells. Similar trends were also observed in a study by Wu et al. [227] in which propylene glycol alginate sodium sulphate oligosaccharides with high degree of sulphation had lower level of FGF2 signalling as compared to the ones with lower degree of sulphation, though the effect of position of sulphate groups was not mentioned. These reported findings indicated the ability of HS mimetics to signal FGF2 can be controlled by the degree of sulphation and

position of sulphate groups, and the present study showed that LCAR could mimic the signalling activity of HS possibly due to similarities in the position of sulphate groups. It should be noted that results from BaF32 assay are usually compared within each material but not between different material types to determine whether each material could form a ternary complex and signal the growth factor.

The growth factor signalling activity of SPs was further evaluated in endothelial cells and fibroblasts by the phosphorylation of ERK1/2. Comparing the effect of SPs on FGF2 signalling between these two cell types would provide further insights into the underlying mechanism of cellular behaviours in different types of cells. Results in this study show that there were differences in the ratio of phospho-ERK1/2 and ERK1/2 detected in endothelial cells and fibroblasts when they were incubated with FGF2 only. It has been demonstrated that HS from endothelial cells exhibited a lower binding affinity for FGF2 as compared to HS from fibroblasts [228], which leads to the lower level of FGF receptor activation. This is in line with the findings in the present study where endothelial cells incubated with exogenous FGF2 resulted in lower level of phospho-ERK1/2 than that detected in fibroblasts. Furthermore, the level of expression of HS proteoglycans has been shown to vary in different cell and tissue types, which could contribute to the differences in the level of FGF2 signalling between two cell types evaluated [229, 230]. When the seaweed-based SPs were added to both cell types in the presence of FGF2, the level of phospho-ERK1/2 was not significantly altered compared to cells cultured in FGF2 only, demonstrating that they did not interfere with the FGF2 signalling occurring within these cells.

In the growth factor receptor activation assay, cells were exposed to FGF2 and SPs for ten minutes prior to the detection of phospho-ERK1/2. One study by Zhu et al. [231] reported that

the phosphorylation of ERK1/2 in fibroblasts reached the maximum level after 10-minute incubation with 0.01 or 0.03 ng/mL of FGF2. Another study by Ferreras et al. [232] also showed that a maximum level of phospho-ERK1/2 was detected in endothelial cells after 10-minute exposure to 20 ng/mL of FGF2 in the presence of HS. Results reported here demonstrated the SPs could activate the FGF receptor after ten-minute incubation in the presence of 2 ng/mL of FGF2, and it is essential to know whether the SPs could induce the phosphorylation of ERK1/2 in a time-dependent manner. Therefore, the growth factor receptor activation activity of SPs should be tested at different time points in future studies to fully understand this biological function. In addition to incubation time, the concentration of SPs is also a contributing factor to the activation of growth factor receptor. Cole et al. [66] created synthetic heparan sulphate oligosaccharides containing 8 to 12 saccharide residues with 2-O-sulphate groups on iduronic acids and N-sulphate groups on glucosamine units, and tested their effect on FGF2-induced receptor activation in endothelial cells using various concentrations of the oligosaccharides. The results showed that the HS materials could induce the phosphorylation of ERK1/2 at the concentration of 0.1 μg/mL and the level of phospho-ERK1/2 decreased with the increase of HS concentration. This confirms the concentration of SPs used in our study was appropriate for investigating how different SPs impacted growth factor receptor activation. Future studies can focus on varying the concentration of SPs to understand the influence of various concentrations on the activation of growth factor receptor.

After the initiation of growth factor signalling, cellular activities were induced as a result of signal transduction occurring inside the cells. Cell migration and cell sprouting were investigated in this study using fibroblasts and endothelial cells which are two types that contribute to events that

occur during angiogenesis. Migration assay results show that in the same incubation conditions, the number of migrated fibroblasts was higher than endothelial cells. Detailed examination of cell migration velocity by Tefft et al. [233] showed that fibroblasts migrated at a significantly higher velocity than endothelial cells, supporting results presented that higher number of migrated fibroblasts was observed as compared to endothelial cells. Results presented in the current study showed that SPs did not interfere with the migration of fibroblasts in the presence of FGF2 as compared to cells incubated in FGF2 only. In addition to FGF2, it has been shown that other growth factors such as platelet derived growth factor (PDGF) and epidermal growth factor (EGF) were also involved in regulating the migration of fibroblasts [234-237]. Particularly, PDGF was shown to be produced by fibroblasts [238] and therefore was likely to be potentiated by the presence of SPs. The migration activity of endothelial cells was modulated by heparin and SPs in a different manner as compared to fibroblasts. The number of migrated endothelial cells incubated in heparin and FGF2 was shown to be significantly lower than cells incubated in FGF2 alone. This could be because heparin acted as a growth factor sink due to high degree of sulphation which bound FGF2 with high affinity and prevented it from signalling FGF2 receptor to induce cell migration. Furthermore, results presented showed that the SPs investigated had no impact on the FGF2-induced phosphorylation of ERK1/2; however, all SPs except LCAR significantly decreased the number of migrated endothelial cells in the presence of FGF2, as compared to FGF2 only. The difference in the structure of SPs may have influenced their ability to modulate endothelial cell migration. ATR-FTIR results in Chapter 3 showed that LCAR contained 6-O-sulphate groups in its chemical backbone. Previous studies have demonstrated that the presence of 6-O-sulphate groups in heparin-like oligosaccharides or HS played an

essential role in promoting the migration of endothelial cells [239, 240], which supports the findings in the present study that LCAR did not cause a decrease in the number of migrated endothelial cells in the presence of FGF2. The differences in the migration between endothelial cells and fibroblasts could also be attributed to the differences in experimental setup. While fibroblasts were incubated in medium without foetal bovine serum (FBS), endothelial cells were treated in medium containing 2 % FBS. The addition of FBS at a low concentration was to maintain the viability of endothelial cells, however, FBS could also be a source of other biological molecules such as vascular endothelial growth factors which could have been modulated by SPs and altered the migration of endothelial cells [241].

In addition to Transwell migration assay, scratch wound assay can also be used to monitor the migration of cells. In scratch wound assay, cells are usually wounded by creating a cell-free zone in a confluent monolayer of cells using a pipette tip, and the migration of cells from the edges into the centre of the zone is recorded [242]. Meanwhile, in Transwell migration assay, cells are allowed to migrate following a chemotactic gradient [243]. The extracellular matrix environment of wounded cells may be different from intact cells in Transwell migration assay due to proteolytic degradation of proteins and GAGs as a response to injury [244]. Together, these factors may contribute to differences in cell migration evaluated by the two assays mentioned.

Sprouting is a crucial activity of endothelial cells during angiogenesis. The effect of SPs on sprouting activity of endothelial cell spheroids was investigated. The sprouting assay results indicate that the seaweed-based SPs exhibited inhibitory effect on the angiogenic activity of FGF2 as evidenced by the significant reduction in the number of sprouts and cumulative sprout length of spheroids incubated with SPs in the presence of FGF2. It has been shown that FGF2-induced

angiogenesis in endothelial cells were regulated by ERK1/2, p38, and c-Jun N-terminal kinase (JNK) signalling pathways [245, 246]. Results in the present study showed that the seaweed derived SPs did not significantly decrease the level of FGF2-induced phosphorylation of ERK1/2 in endothelial cells, implying that they may have modulated the p38 and JNK signalling pathways which resulted in the reduced level of sprouting in endothelial spheroids. In future studies, the impact of SPs on p38 and JNK signalling pathways can be investigated to understand the molecular mechanisms of inhibitory effects of SPs on FGF2-induced angiogenesis. Regarding the position of sulphate groups of SPs, it was shown in Chapter 3 that KCAR, ICAR and ulvan did not contain 6-O-sulphate groups, whereas LCAR had a small amount of 6-O-sulphate groups which was demonstrated by the small peak associated with the sulphate groups at C6 position of galactose residues of LCAR in its ATR-FTIR spectra. Previous studies have reported that reducing or desulphating 6-O-sulphation in HS resulted in the inhibition of FGF2-induced angiogenesis [212, 247], indicating that the absence or low level of 6-O-sulphate groups in the SPs may contribute to the reduced level of endothelial sprouting. This characteristic of SPs could be used in developing materials for anti-angiogenesis applications. Specifically, the SPs can be utilised for the development of anti-angiogenic drugs that prevent the growth of blood vessels of tumours in cancer treatment. However, the inhibitory effect of SPs on cancer cells should be investigated in future studies prior to clinical use of these materials.

4.5. Conclusion

This study demonstrated that the marine-derived SPs did not influence FGF2-induced growth factor receptor activation in endothelial cells and fibroblasts; however, trends in how the SPs modulated this activity were different between the two cell types. Specifically, among the four types of SPs, LCAR was shown to stimulate FGF2 signalling in the BaF32 proliferation assay and promote the migration of endothelial cells, but it did not have the same effect on fibroblasts. All types of SPs tested exhibited the inhibitory effect on the sprouting activity of endothelial cells. These findings indicate that different sulphation characteristics of SPs impacted the ability of SPs to signal growth factor and induce cellular responses in different cell types. Further studies can be conducted to understand the mechanism underlying the variability in the biological functions of these materials.

Chapter 5

Effect of varying sulphur level within each type of SPs on growth factor signalling

5.1. Introduction

Carrageenan and ulvan are two types of sulphated polysaccharides (SPs) derived from seaweed that possess heterogenous and complex chemical structures. These properties vary not only due to difference in seaweed species and environmental factors, but also extraction and purification methods used to obtain the SPs [248]. The structural differences of carrageenan and ulvan are identified in terms of molecular weight, degree of sulphation, and position of sulphate groups along the polysaccharide backbone, and the degree of complexity and heterogeneity that is present, based on factors stated above. This heterogeneity has posed a challenge in determining the relationship between structure and function since the biological activity of SPs is highly dependent on their structure [249-252]. As shown in results in Chapter 3 and 4, variation in the chemical structure of carrageenan and ulvan resulted in high variability in the level of growth factor binding, intracellular signalling, and cellular responses.

The polysaccharide chains of carrageenan and ulvan are naturally decorated with varying amount of sulphate groups at different positions [253]. To study the structure-function relationship of these SPs, other researchers have performed chemical modification of native SPs to alter their sulphate content or position of sulphate groups, and changes in the biological activities of the modified SPs were reported. Opoku et al. [254] synthesised oversulphated kappa-carrageenan (KCAR) to obtain the sulphate percent of 28.5 %, and this KCAR derivative was shown to have higher anticoagulant activity than the native KCAR whose sulphate percent was 16 %. The research group of Araujo and others [255] conducted selective sulphation of carrageenan and demonstrated that sulphation at C2 of 3,6-anhydrogalactose and C6 of-galactose unit of carrageenan enhanced the anticoagulant activity. Regarding ulvan, it was reported that

sulphated ulvan with higher sulphate content (32.8 %) increased *in vivo* antioxidant activity as compared to native ulvan with lower sulphate content (19.5 %) [256]. Although chemical modification of native SPs could help to elucidate the relationship between structure and function of SPs, this method required complex chemical reactions and the natural polymer chains of SPs could also be affected.

In Chapter 3 and 4, carrageenan and ulvan were shown to bind and signal fibroblast growth factor-2 (FGF2) and induce cellular responses with a noticeable level of variability. It is hypothesised that these results arise from structural variability within each of the SPs evaluated which resulted in varying biological responses to the materials. One of the methods to evaluate the polyanionic nature of SPs is anion exchange chromatography (AEC). This method has been utilised to separate low molecular weight fucoidan, which is a type of sulphated polysaccharides extracted from brown seaweed, into different fractions based on the level of negative charge, and these fractions had distinct capabilities to signal different FGFs in cells expressing FGF receptor 1c [138]. In comparison to fucoidan, carrageenan and ulvan are extracted from red and green seaweed, respectively, thus expected to exhibit different biological functions. However, the effect of fractionated carrageenan and ulvan on the ability to activate the signalling of growth factors remains to be investigated.

In this regard, it was hypothesised that each type of carrageenan or ulvan may contain fractions with distinct sulphur percent and these fractions exhibit different ability to signal growth factor. By understanding the variation in the level of sulphur within each SP and how it can affect the growth factor signalling activity, we can gain more insights into the relationship between structure and function of SPs as well as the underlying reason for the variability in the biological

functions of the SPs. Hence, the focus of this chapter was on the separation and characterisation of carrageenan and ulvan to obtain information regarding the sulphur content of different SP fractions, and the evaluation of how they modulate FGF2 signalling. Anion exchange chromatography was used for the separation of the SPs into different fractions based on the level of negative charge. The obtained fractions were characterised using X-ray photoelectron spectroscopy to determine the sulphur percent and relative sulphate amount. BaF32 assay was employed to examine the ability of fractionated SPs to signal FGF2.

5.2. Materials and methods

Heparin sodium salt from porcine intestinal mucosa (H3393), kappa-carrageenan (22048), iota-carrageenan from red seaweed *Eucheuma spinosum* (C1138), and lambda-carrageenan (22049) were purchased from Sigma-Aldrich (Castle Hill, Australia). Ulvan from green seaweed *Ulva armoricana* (YU11689) was obtained from Biosynth Ltd (United Kingdom). Human fibroblast growth factor-2 (FGF2, PHG0024) was supplied by Thermo Fisher in lyophilised form. The growth factor was reconstituted with MilliQ water at 0.1 mg/mL prior to being used in experiments. All other chemicals were purchased from Sigma-Aldrich (Castle Hill, Australia) unless stated otherwise.

5.2.1. Anion exchange chromatography

To evaluate the variation in sulphur content within heparin, kappa-carrageenan, iota-carrageenan, lambda-carrageenan, or ulvan, anion exchange chromatography was first used to separate each type of SPs into different fractions based on the levels of negative charge. A chromatography column attached to a BioLogic LP system (Biorad) was used for the separation process. The column was packed with an ion-exchange chromatography resin containing positively charged diethylaminoethyl functional groups (DEAE-Sepharose Fast Flow). To begin with, each SP was dissolved in running buffer (20 mM Tris base, 10 mM EDTA, pH 7.5) at the concentration of 1 mg/mL. Next, the DEAE column was equilibrated with running buffer for three column volumes at the flow rate of 2 mL/min before 50 mL of each SP solution was injected into the column. Any unbound material was removed using the running buffer until obtaining a stable baseline absorbance. Each SP was then eluted from the column at the flow rate of 2 mL/min using

running buffer containing various concentrations of NaCl (0.25, 0.5, or 0.75 M) to obtain separate fractions based on negative charge level. Following sample collection, the column was regenerated with running buffer that contained 2 M NaCl and stored in ethanol 20 % at 4°C. The collected SP fractions were dialysed (2 kDa cut-off) against MilliQ water for three days and lyophilised to obtain the dry products. All fractions were labelled as described in Table 5.1.

Table 5.1. Description of names of fractions obtained from anion exchange chromatography.

Fraction name	Description
HEP0.25	Heparin fraction eluted by running buffer containing 0.25 M NaCl
HEP0.5	Heparin fraction eluted by running buffer containing 0.5 M NaCl
HEP0.75	Heparin fraction eluted by running buffer containing 0.75 M NaCl
KCAR0.25	Kappa-carrageenan fraction eluted by running buffer containing 0.25 M NaCl
KCAR0.5	Kappa-carrageenan fraction eluted by running buffer containing 0.5 M NaCl
ICAR0.25	Iota-carrageenan fraction eluted by running buffer containing 0.25 M NaCl
ICAR0.5	Iota-carrageenan fraction eluted by running buffer containing 0.5 M NaCl
LCAR0.25	Lambda-carrageenan fraction eluted by running buffer containing 0.25 M NaCl
LCAR0.5	Lambda-carrageenan fraction eluted by running buffer containing 0.5 M NaCl
Ulvan0.25	Ulvan fraction eluted by running buffer containing 0.25 M NaCl
Ulvan0.5	Ulvan fraction eluted by running buffer containing 0.5 M NaCl
Ulvan0.75	Ulvan fraction eluted by running buffer containing 0.75 M NaCl

5.2.2. X-ray photoelectron spectroscopy (XPS)

To determine the sulphur content of SP fractions, an X-ray Photoelectron Spectrometer (Kratos, Manchester, UK) was employed to obtain the XPS spectra of the fractionated SPs. This instrument was equipped with a monochromated Al Kα source (source energy 1486.69 eV) that was operated at a power of 150 W with a charger neutraliser. A piece of double-sided tape, approximately 8 mm x 8 mm, was placed on the XPS platen before the samples were tipped on the tape. A spatula was used to spread the samples out to achieve a single layer of adhered particles. Any excess particles not adhered to the tape were removed from the platen by a brush. A pass energy of 160 eV and a 1 eV step size was used to obtain the survey spectra of SP fractions for detecting and quantifying the atomic percent of sulphur in the fractions. High-resolution C1s spectra of were also acquired at a pass energy of 20 eV and a step size of 0.1 eV. Each sample was measured in triplicate and repeated three times, and obtained data was transformed into VAMAS format and subsequently analysed using CasaXPS software, version 2.3.16. A Shirley background and Gaussian-broadened Lorentzian functions (30 GL) were utilised to curve fitted the high-resolution C1s spectra. The constraint placed on the full-width-half-maximum of all components was based on that of the C–C, C–H peak. Charge correction was conducted for all spectra according to the relative position of the C–C, C–H peak to 285 eV.

5.2.3. Growth factor signalling activity of SP fractions.

The ability of SP fractions to signal FGF2 was evaluated using the BaF32 cell assay. Information regarding BaF32 cell line and how the assay was performed is detailed as follows.

5.2.3.1. Culture and maintenance of murine interleukin-3 dependent pro-B cell line (BaF32 cells)

BaF32 cells are a myeloid B cell line that is dependent on Interleukin-3 (IL-3)-dependent and deficient in heparan sulphate proteoglycan. These cells have been transfected with fibroblast growth factor receptor (FGFR) 1c. In the presence of FGF and heparin/heparan sulphate (HS), FGF signalling is induced due to the formation of ternary complexes between heparin/HS, FGF, and FGFR on the cell surface which leads to the proliferation of BaF32 cells. Meanwhile, in the absence of the ternary complex being formed, the BaF32 cells do not proliferate. Therefore, a proliferation assay using BaF32 cells was utilised to evaluate the FGF2 signalling activity of SP fractions.

BaF32 cells transfected with FGFR1c were kindly provided by Professor David Ornitz (Washington University). The cells were cultured in complete RPMI1640 medium (Media Preparation Unit, University of Melbourne) that contained 10 % (v/v) Foetal Bovine Serum (FBS), 1 % (v/v) Penicillin-Streptomycin (PS) and 10 % (v/v) WEHI-3BD conditioned medium containing IL-3 at 37 °C with 5 % CO_2. BaF32 cells were passaged at 90% confluency and transferred into a new flask at a concentration of 10^6 cells per mL in 10 mL. To eliminate untransfected cells prior to the assay, 200 µg/mL G418 antibiotic solution was added into the complete medium during the culture of the BaF32 cells for two weeks. Medium was changed every four days, and the viability of the cells was checked using Trypan blue.

WEHI-3BD cells were kindly provided by the Ludwig Institute (Melbourne, Australia) and were cultured in RPMI1640 that contained 10 % (v/v) FBS and 1 % (v/v) PS at 37 °C with 5 % CO_2. The

collection of the conditioned medium of WEHI-3BD cells was performed twice per week and the collect medium was preserved at -20 °C. The medium was defrosted and filtered prior to adding to BaF32 cell culture medium.

5.2.3.2. BaF32 proliferation assay

BaF32 cells expressing the FGF receptor FGFR1c were cultured in RPMI 1640 medium that contained 10 % (v/v) FBS, 10 % (v/v) WEHI-3BD conditioned medium and 1 % (v/v) PS. Prior to the assay, cells were starved in RPMI1640 medium without WEHI-3BD conditioned medium for 24 hours before being seeded into a 96-well plate at a concentration of 2 x 10^4 cells per well. Cells were incubated with phenol red-free RPMI medium (Thermo Fisher) only, and test samples including 30 nM of each SP fraction either in the presence or absence of 0.3 nM FGF2. Cells treated with heparin fractions and FGF2 were utilised as a positive control while cells exposed to heparin fractions, FGF2, or medium only were regarded as negative controls. Test substances in medium without cells were also used for background absorbance readings. Cells were incubated for 72 hours at 37 °C and 5 % CO_2. Thereafter, the Cell Titer 96 Aqueous One Solution Reagent (MTS reagent, Promega) was added to each well and incubated for six hours at 37 °C, followed by the measurement of absorbance at 490 nm to assess BaF32 cell proliferation as a measure of growth factor signalling activity of SP fractions.

5.2.4. Statistical analysis

Significant differences were determined by two-way analysis of variance (ANOVA) followed by Bonferroni's multiple comparison test using GraphPad Prism 9.5.1 (GraphPad Software, San

Diego, CA, USA). Differences between groups were considered significant at $p < 0.05$ and indicated in the figures as *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, and ****$p < 0.0001$.

5.3. Results

5.3.1. Anion exchange chromatography of sulphated polysaccharides

Anion exchange chromatography (AEC) was performed to separate each type of sulphated polysaccharides (SPs) into different fractions based on the level of negative charge. The chromatography resin used in this technique contained positively charged groups which bound to negatively charged groups present in the SPs such as carboxyl (-COOH) or sulphate (OSO_3^-) groups via ionic bonding. A series of running buffers with increasing concentration of NaCl was used for stepwise elution of SPs, in which bound material with a net low negative charge level was eluted at low NaCl concentration (0.25 M) due to low ionic bonding strength between the SPs and diethylaminoethyl (DEAE) resin, while fractions with a medium or high overall negative charge level were eluted at medium (0.5 M) or high (0.75 M) NaCl concentration, respectively, as a result of higher ionic strength between SPs and DEAE resin. Figure 5.1 to Figure 5.5 demonstrate the AEC profiles of heparin and different types of seaweed derived SPs. In each chromatogram, absorbance of the eluted sample and conductivity of the buffer was recorded. Increasing the concentration of NaCl in the buffers resulted in the increase of conductivity. When a buffer containing higher NaCl concentration was introduced, the increase of absorbance value indicated samples eluting off the column which were collected as one fraction. The fraction was collected until absorbance reading of the eluting solution returned to baseline value (~0 AU).

Three AEC experiments were conducted for each material to compare the reproducibility of the AEC separation process. Figure 5.1A, 5.1B, and 5.1C show the AEC profiles of heparin which correspond to three AEC repeats. Three factions were collected using buffer that contained 0.25, 0.5, and 0.75 M NaCl, respectively, indicating heparin contained fractions with three distinct levels of negative charge. The peak corresponding to the fraction collected at 0.5 M NaCl buffer was smaller than the other two peaks at 0.25 and 0.75 M NaCl, implying that in the bulk heparin sample there were more polysaccharide chains with low or high level of negative charge than polysaccharide chains with medium level of negative charge. The area under each peak detected in each AEC run was quantified using OriginPro 2019b Software (OriginLab Corporation) as a measure of relative amount of each SP fraction collected. The area of each peak was presented as the percent of total peak area in each AEC repeat. Figure 5.1D shows the quantitative analysis of peak area in the AEC run of heparin. The level of HEP0.25 fraction appeared to be similar in three repeats and accounted for approximately 41 % of the total peak area in each AEC run. The level of HEP0.5 fraction in the first run was higher than that in the second and third run, while the level of HEP0.75 fraction obtained from the first run was lower than the other two runs.

Regarding ulvan, there were three peaks appearing in each of the repeating AEC profiles shown in Figure 5.2A, 5.2B, and 5.2C, which corresponded to three fractions collected in each run. The peak at 0.25 M NaCl was broader compared to the peaks at 0.5 and 0.75 M NaCl. Furthermore, the peak at 0.5 M NaCl had the highest absorbance while the peak at 0.75 M NaCl had the lowest absorbance among the three peaks obtained. The peak area percent of ulvan fractions is shown in Figure 5.2D. The amount of Ulvan0.25 was the highest among three fractions obtained, which constituted from 65 to 75 % of the total peak area in three AEC runs. The amount of Ulvan0.5

and Ulvan0.75 fractions collected in the first run was less than that obtained in the second and third run. AEC profiles of KCAR (Figure 5.3A, 5.3B, and 5.3C), ICAR (Figure 5.4A, 5.4B, and 5.4C), and LCAR (Figure 5.5A, 5.5B, and 5.5C) show that two peaks were detected at 0.25 and 0.5 M NaCl, indicating two fractions were collected for each type of carrageenan. The peaks at 0.25 M NaCl appeared to be larger than the peaks at 0.5 M NaCl, which suggested the majority of molecular chains of carrageenan contained low level of negative charge. The obtained amount of KCAR and LCAR fractions appeared to be relatively consistent throughout three runs as shown in Figure 5.3D and Figure 5.5D, respectively. Meanwhile, the AEC separation of ICAR revealed that the amount of ICAR0.25 fraction in the third run was lower than in the first two runs and thus higher amount of ICAR0.5 fraction was achieved as shown in Figure 5.4D. Together these findings show that heparin and each type of seaweed derived SPs were composed of fractions with varying levels of negative charge demonstrating the heterogeneity of these materials. Results also reveal slight variability in the amount of fractionated SPs collected in each AEC run.

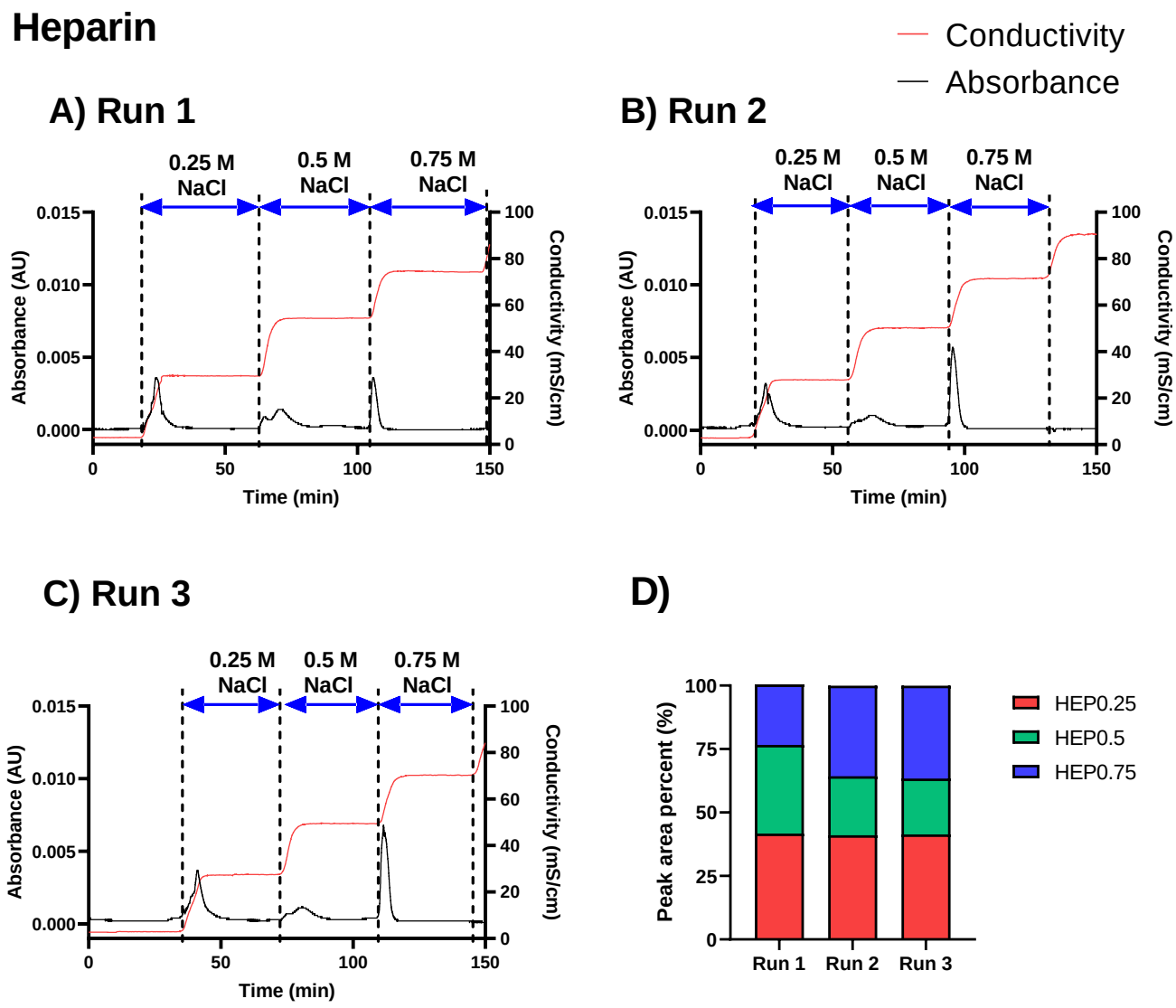


Figure 5.1. Anion exchange chromatography (AEC) profiles of heparin. Three repeats were conducted to test the reproducibility of the separation process, and three resulting AEC profiles were shown in A) Run 1, B) Run 2, and C) Run 3. D) The area under each peak detected in each AEC run was quantified and presented as the percent of total peak area in each run. DEAE column was equilibrated with running buffer (20 mM Tris base, 10 mM EDTA, pH 7.5). 50 mL of heparin (1 mg/mL) was loaded into the column at the flow rate of 2 mL/min. Unbound materials were washed off with running buffer. HEP0.25, HEP0.5, and HEP0.75 fractions were eluted by running buffer containing 0.25 M, 0.5 M, and 0.75 M NaCl, respectively.

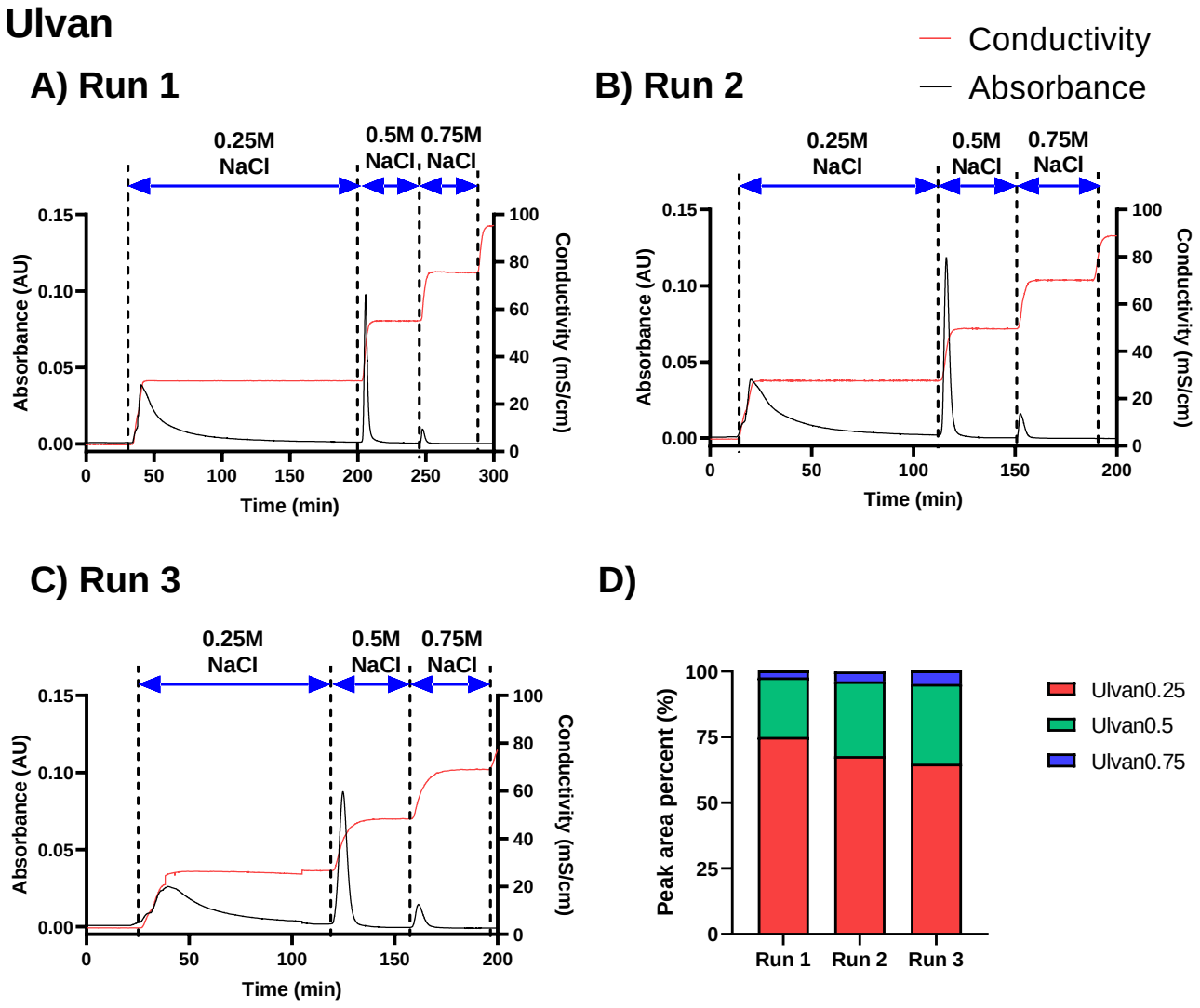


Figure 5.2. Anion exchange chromatography (AEC) profiles of ulvan. Three repeats were conducted to test the reproducibility of the separation process, and three resulting AEC profiles were shown in A) Run 1, B) Run 2, and C) Run 3. D) The area under each peak detected in each AEC run was quantified and presented as the percent of total peak area in each run. DEAE column was equilibrated with running buffer (20 mM Tris base, 10 mM EDTA, pH 7.5). 50 mL of ulvan (1 mg/mL) was loaded into the column at the flow rate of 2 mL/min. Unbound materials were washed off with running buffer. Ulvan0.25, Ulvan0.5, and Ulvan0.75 fractions were eluted by running buffer containing 0.25 M, 0.5 M, and 0.75 M NaCl, respectively.

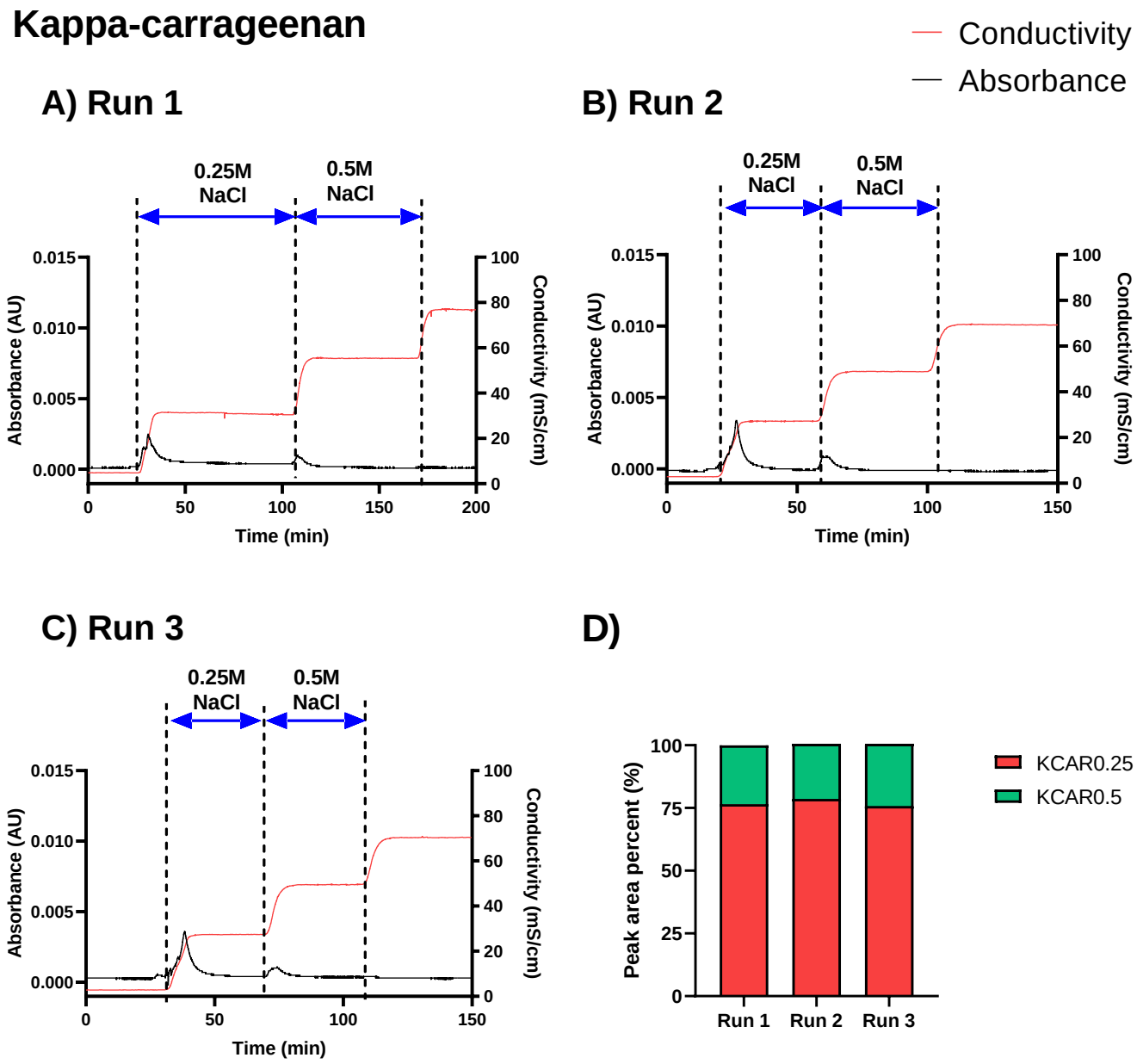


Figure 5.3. Anion exchange chromatography (AEC) profiles of kappa-carrageenan (KCAR). Three repeats were conducted to test the reproducibility of the separation process, and three resulting AEC profiles were shown in A) Run 1, B) Run 2, and C) Run 3. D) The area under each peak detected in each AEC run was quantified and presented as the percent of total peak area in each run. DEAE column was equilibrated with running buffer (20 mM Tris base, 10 mM EDTA, pH 7.5). 50 mL of KCAR (1 mg/mL) was loaded into the column at the flow rate of 2 mL/min. Unbound materials were washed off with running buffer. KCAR0.25 and KCAR0.5 fractions were eluted by running buffer containing 0.25 M and 0.5 M NaCl, respectively.

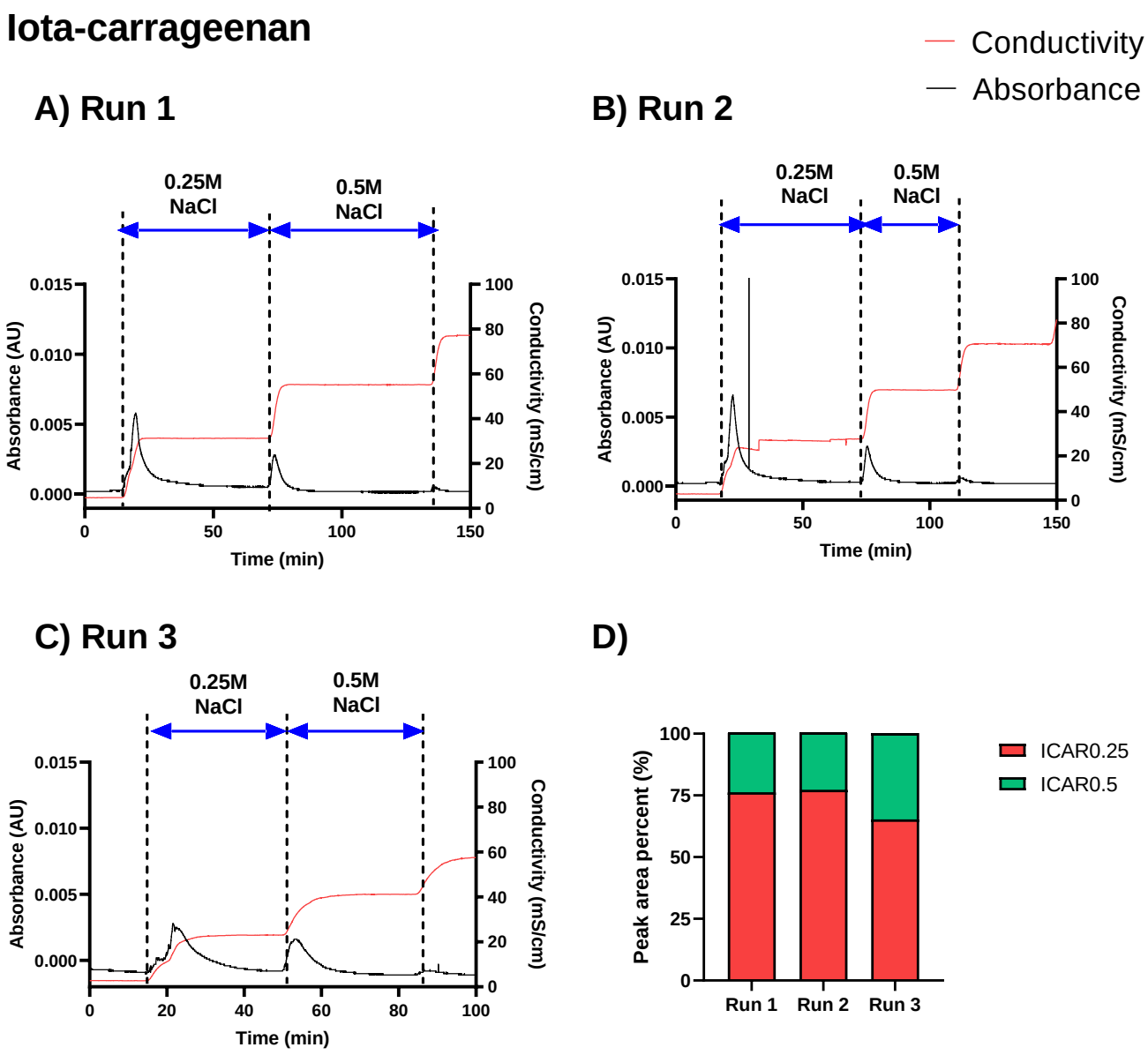


Figure 5.4. Anion exchange chromatography (AEC) profiles of iota-carrageenan (ICAR). Three repeats were conducted to test the reproducibility of the separation process, and three resulting AEC profiles were shown in A) Run 1, B) Run 2, and C) Run 3. D) The area under each peak detected in each AEC run was quantified and presented as the percent of total peak area in each run. DEAE column was equilibrated with running buffer (20 mM Tris base, 10 mM EDTA, pH 7.5). 50 mL of ICAR (1 mg/mL) was loaded into the column at the flow rate of 2 mL/min. Unbound materials were washed off with running buffer. ICAR0.25 and ICAR0.5 fractions were eluted by running buffer containing 0.25 M and 0.5 M NaCl, respectively.

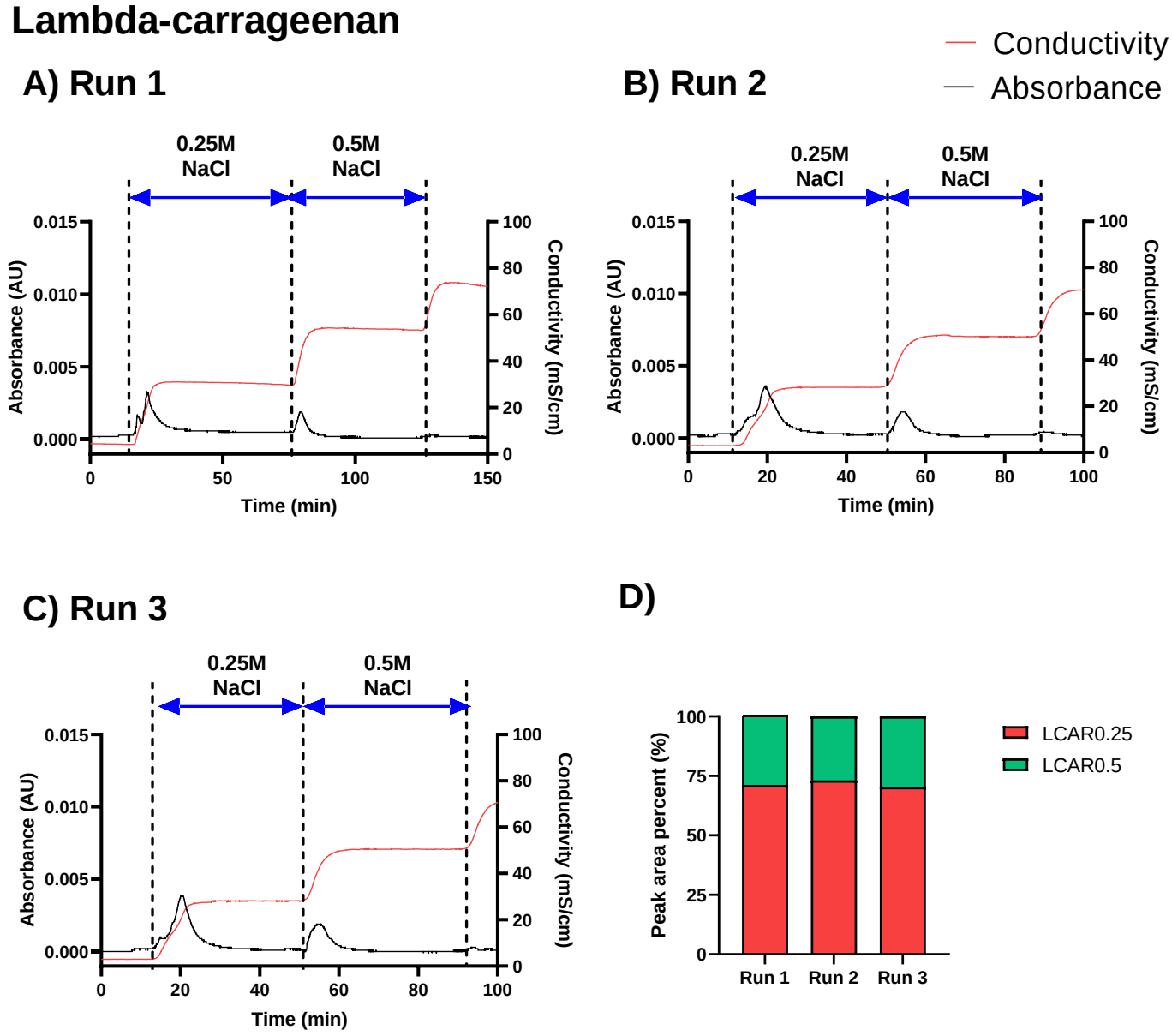


Figure 5.5. Anion exchange chromatography (AEC) profiles of lambda-carrageenan (LCAR). Three repeats were conducted to test the reproducibility of the separation process, and three resulting AEC profiles were shown in A) Run 1, B) Run 2, and C) Run 3. D) The area under each peak detected in each AEC run was quantified and presented as the percent of total peak area in each run. DEAE column was equilibrated with running buffer (20 mM Tris base, 10 mM EDTA, pH 7.5). 50 mL of LCAR (1 mg/mL) was loaded into the column at the flow rate of 2 mL/min. Unbound materials were washed off with running buffer. LCAR0.25 and LCAR0.5 fractions were eluted by running buffer containing 0.25 M and 0.5 M NaCl, respectively.

5.3.2. Sulphur content and growth factor signalling of fractionated SPs

X-ray photoelectron spectroscopy (XPS) was utilised to determine the sulphur content of the fractionated SPs. The effect of different sulphur content of fractionated SPs on FGF2 signalling was evaluated using the BaF32 assay.

5.3.2.1. Sulphur content and growth factor signalling of fractionated heparin

The atomic percent of sulphur of the heparin fractions was shown in Figure 5.6A which increased following the increase in the level of negative charge of these fractions. Specifically, HEP0.75 contained the highest sulphur percent which was 3.8 ± 0.2 %, followed by HEP0.5 with 1.6 ± 0.3 %. The sulphur percent of HEP0.25 was below the limit of detection. The ratio between sulphur and carbon (S/C ratio) was calculated to understand the relative amount of sulphate of the heparin fractions, as shown in Figure 5.6B. The S/C ratio of HEP0.75 was 0.066 ± 0.005, which was 2.5 times higher than that of HEP0.5 (0.026 ± 0.006), confirming that HEP0.75 contained more sulphate groups than HEP0.5. In addition to quantifying sulphur percent, high resolution spectra of carbon (C1s spectra) were obtained to understand the chemical bonding of fractionated heparin. Figure 5.6C to 5.6E show the C1s spectra of HEP0.25, HEP0.5, and HEP0.75, respectively. In all spectra, three peaks were fitted at different binding energies, which corresponded to three main types of carbon linkages: C-C/C-H at ~285 eV, C-O/C-O-C at ~286.6 eV, and C=O at ~288.5 eV. As sulphate groups are linked to the carbon atoms demonstrated in the chemical structure of the repeating disaccharide unit of heparin in Figure 5.6F, varying area of such peaks between heparin fractions may be indicative of variation in the degree of sulphation of fractionated heparin. Table 5.2 shows the percent of peak area for each heparin fraction. In terms of C-C/C-H and C-O/C-O-C peaks, there were no significant differences between three heparin fractions.

Regarding the O-C=O peak, HEP0.75 had a significantly lower peak area than HEP0.5 ($p<0.05$) and HEP0.25 ($p<0.01$), indicating HEP0.75 had less O-C=O linkages than the other two fractions.

The ratio of the sum of C-O/C-O-C and O-C=O peak area to C-C/C-H peak area (ratio of peak area) was calculated to determine the ratio of oxygen-containing linkages to carbon-carbon/carbon-hydrogen linkages which contain no oxygen. The O/C ratio of HEP0.25, HEP0.5, and HEP0.75 was also calculated to understand the ratio of oxygen to carbon atoms and the correlation with the ratio of peak area. Regarding the ratio of peak area, no significant differences were observed between HEP0.25, HEP0.5, and HEP0.75. Interestingly, these fractions were shown to have the same O/C ratio which was around 0.5. This indicates a similar trend between the O/C ratio and the ratio of peak area of heparin fractions.

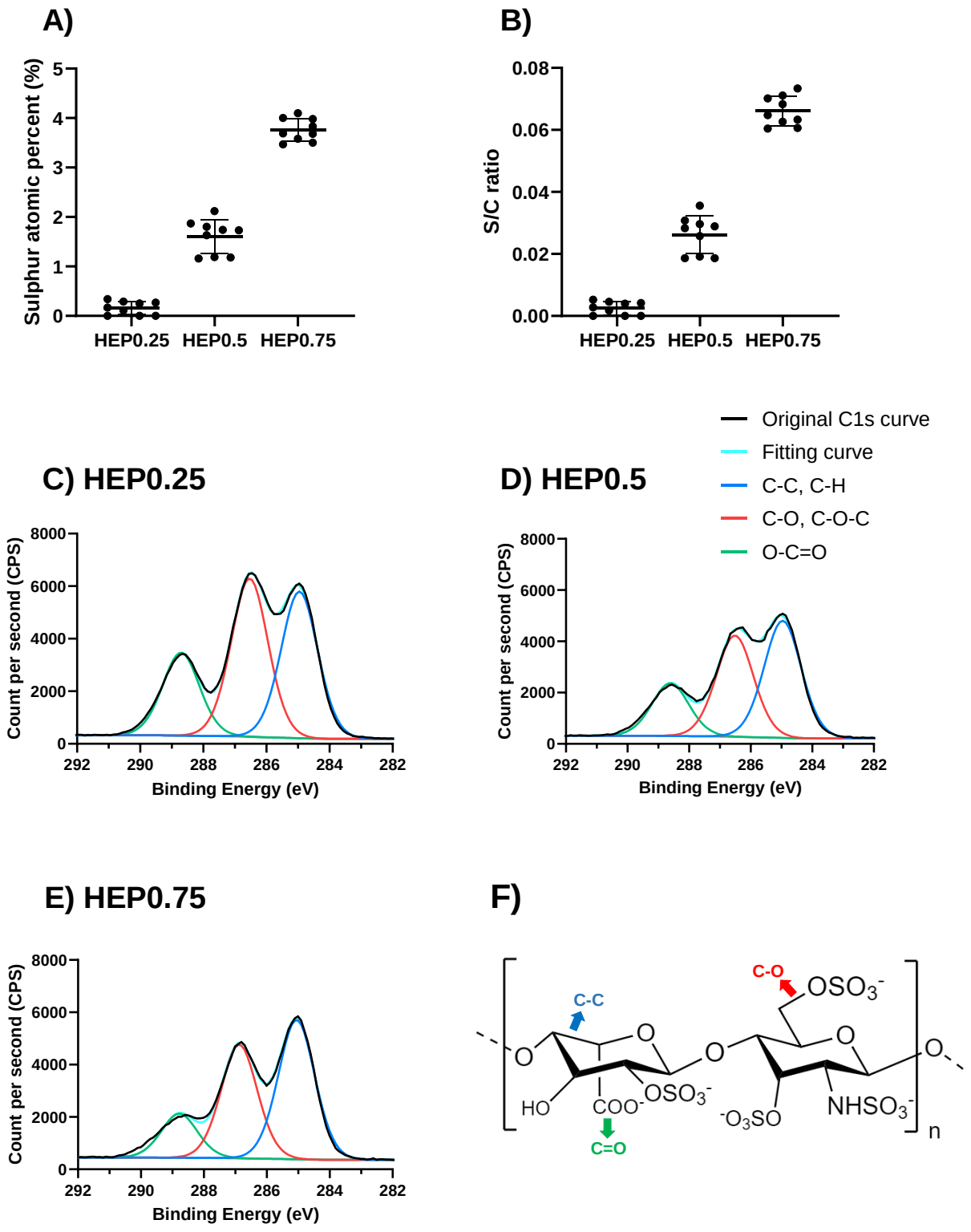


Figure 5.6. Characterisation of sulphur content of fractionated heparin (HEP). A) Sulphur atomic percent and B) Ratio of sulphur and carbon percent (S/C ratio) of three heparin fractions obtained from anion exchange chromatography. High resolution C1s spectra of C) HEP0.25, D) HEP0.5, and E) HEP0.75. F) Structure of a repeating disaccharide unit of heparin with typical chemical bonds being shown to illustrate the corresponding carbon linkages detected in the C1s spectra.

Table 5.2. Percent (%) of peak area in C1s spectra and O/C ratio of HEP0.25, HEP0.5, and HEP0.75 fractions. Each sample was measured three times. The O/C ratio was calculated from the atomic percent of oxygen and carbon of each sample.

Sample	Repeat	% of peak area in C1s spectra			[(2) + (3)]/(1) ratio	O/C ratio
		C-C, C-H (1)	C-O, C-O-C (2)	O-C=O (3)		
HEP0.25	1	30.43	45.2	24.37	2.3	0.5
	2	42.58	37.62	19.81	1.3	0.4
	3	38.91	40.09	21	1.6	0.4
HEP0.5	1	42.23	38.1	19.67	1.4	0.5
	2	40.67	39.48	19.85	1.5	0.5
	3	47.02	34.29	18.69	1.1	0.4
HEP0.75	1	44.57	39.96	15.47	1.2	0.5
	2	50.09	36.53	13.38	1.0	0.5
	3	45.6	38.5	15.9	1.2	0.5

The ability of fractionated heparin to signal growth factor was evaluated using the BaF32 assay. The proliferation of FGFR1c-transfected BaF32 cells when exposed to FGF2 and heparin fractions was utilised as a measure of growth factor signalling activity. Figure 5.7 shows the proliferation of BaF32 cells via absorbance measurement at 490 nm. Cells cultured in FGF2 only or any of the heparin fractions in absence of FGF2 did not proliferate compared to cells cultured in medium only. Meanwhile, cells exposed to HEP0.5 or HEP0.75 in the presence of FGF2 showed a significant increase ($p<0.01$ and 0.0001, respectively) in absorbance compared to cells incubated with HEP0.5 or HEP0.75 only, indicating the formation of ternary complexes between each of these fractions with FGF2 and FGFR1c which led to the proliferation of BaF32 cells. Cells exposed to HEP0.25 and FGF2 did not proliferate significantly compared to cells treated with HEP0.25 only, implying that this fraction did not promote the formation of ternary complex with FGF2 and

FGFR1c. Together these findings suggest that heparin contained fractions with varying sulphur content, and this variation had an impact on the growth factor signalling activity of such fractions.

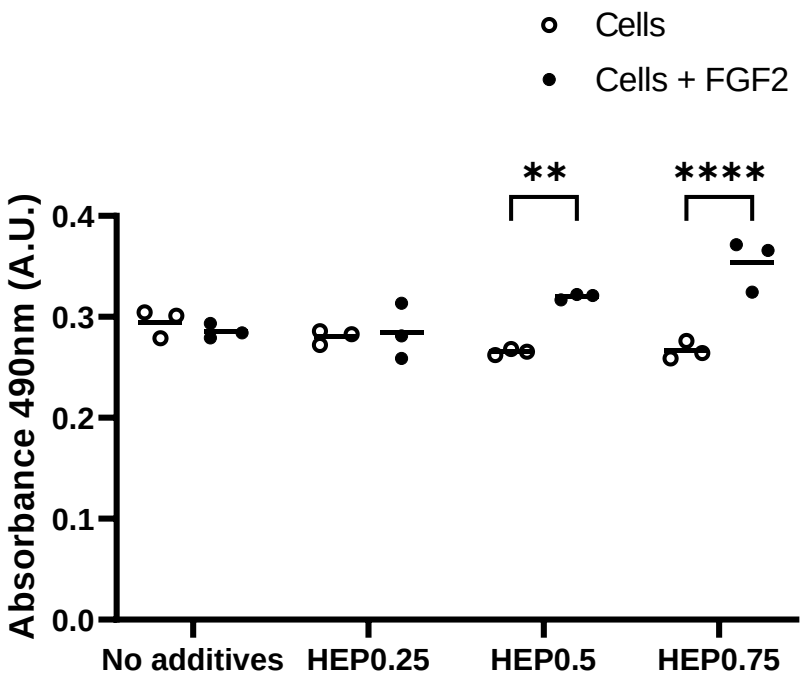


Figure 5.7. FGF2 signalling activity of fractionated heparin determined via the proliferation of BaF32 cells transfected with FGFR1c in the presence of fractionated heparin and FGF2. * represents statistically significant difference (**p<0.01, ****p<0.0001).

5.3.2.2. Sulphur content and growth factor signalling of fractionated kappa-carrageenan

X-ray photoelectron spectroscopy analysis was performed to understand the variation in the sulphur content of fractionated kappa-carrageenan (KCAR). Figure 5.8A shows the sulphur atomic percent of KCAR0.25 and KCAR0.5. The sulphur percent of KCAR0.25 was found to be 0.4 ± 0.2 % while that of KCAR0.5 was 0.2 ± 0.1 %. High variability of sulphur percent was observed in both fractions showing the structural heterogeneity within each of them. The ratio between sulphur and carbon (S/C ratio) of these fractions shown in Figure 5.8B indicates that KCAR0.25 and KCAR0.5 had different amount of sulphate groups in their chemical backbones. High resolution C1s spectra of fractionated KCAR was also acquired and shown in Figure 5.8C and Figure 5.8D. The chemical structure of the repeating disaccharide unit of KCAR was shown in Figure 5.8E to illustrate the chemical bonds analysed in the C1s spectra. There was a difference in the height of C-C/C-H peak between KCAR0.25 and KCAR0.5, which may indicate a variation in the amount of C-C/C-H linkages between the two fractions. Table 5.3 shows the percent of peak area for each KCAR fraction. Statistically, there were no significant differences in the percent of peak area of C-C/C-H, C-O/C-O-C, and O-C=O peaks between two KCAR fractions. Similar trends were observed in the ratio of peak area and the O/C ratio. Regarding KCAR0.5, it was observed that higher ratio of peak area correlated with higher O/C ratio.

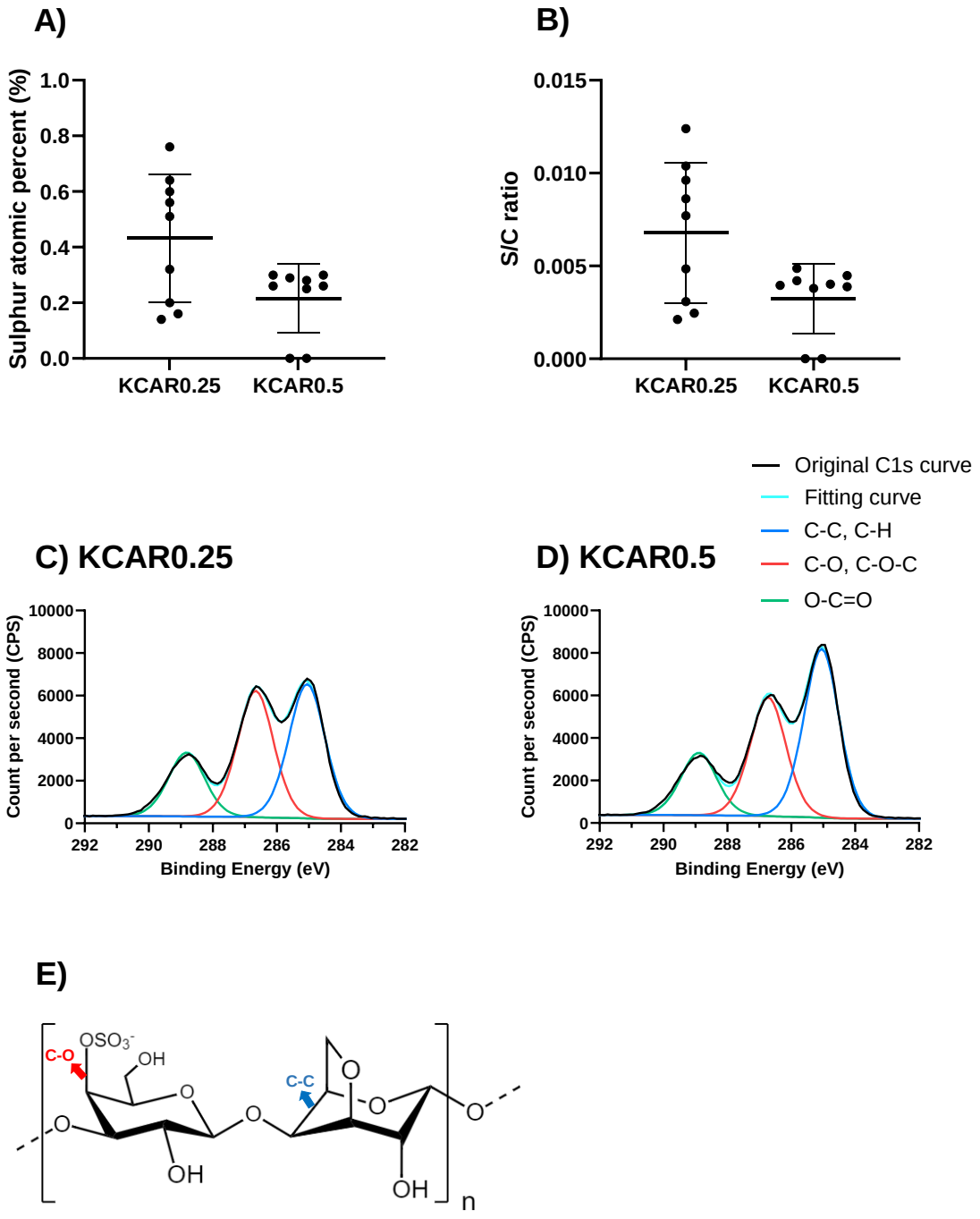


Figure 5.8. Characterisation of sulphur content of fractionated kappa-carrageenan (KCAR). A) Sulphur atomic percent and B) Ratio of sulphur and carbon percent (S/C ratio) of two KCAR fractions obtained from anion exchange chromatography. High resolution C1s spectra of C) KCAR0.25 and D) KCAR0.5. E) Structure of a repeating disaccharide unit of KCAR with typical chemical bonds being shown to illustrate the corresponding carbon linkages detected in the C1s spectra.

Table 5.3. Percent (%) of peak area in C1s spectra and O/C ratio of KCAR0.25 and KCAR0.5 fractions. Each sample was measured three times. The O/C ratio was calculated from the atomic percent of oxygen and carbon of each sample.

Sample	Repeat	% of peak area in C1s spectra			[(2) + (3)]/(1) ratio	O/C ratio
		C-C, C-H (1)	C-O, C-O-C (2)	O-C=O (3)		
KCAR0.25	1	46.37	35.75	17.88	1.2	0.4
	2	43.19	38.34	18.47	1.3	0.4
	3	35.41	42.29	22.3	1.8	0.4
KCAR0.5	1	59.56	26.94	13.5	0.7	0.3
	2	32.42	44.05	23.53	2.1	0.5
	3	42.15	37.58	20.27	1.4	0.4

The ability of fractionated KCAR to signal growth factor was evaluated using the BaF32 assay as demonstrated in Figure 5.9. Cells incubated in KCAR0.25 showed a significant proliferation compared to cells incubated in medium, HEP0.25, and HEP0.5. Cells cultured in KCAR0.5 also had a significant proliferation as compared to cells cultured in two heparin fractions. Cells incubated with either KCAR0.25 or KCAR0.5 in the presence of FGF2 did not show a significant difference in proliferation compared to cells incubated with FGF2 only. Moreover, cell proliferation did not increase in the presence of KCAR0.25 and FGF2 compared to KCAR0.25 only. It was also interesting to see that the number of cells incubated with KCAR0.5 in the presence of FGF2 was significantly lower ($p<0.01$) as compared to cells cultured in KCAR0.5 only. Together these results suggest that KCAR was composed of fractions with different sulphur content and distinct trends in the interaction with FGF2 that impacted the formation of ternary complex with FGFR1c.

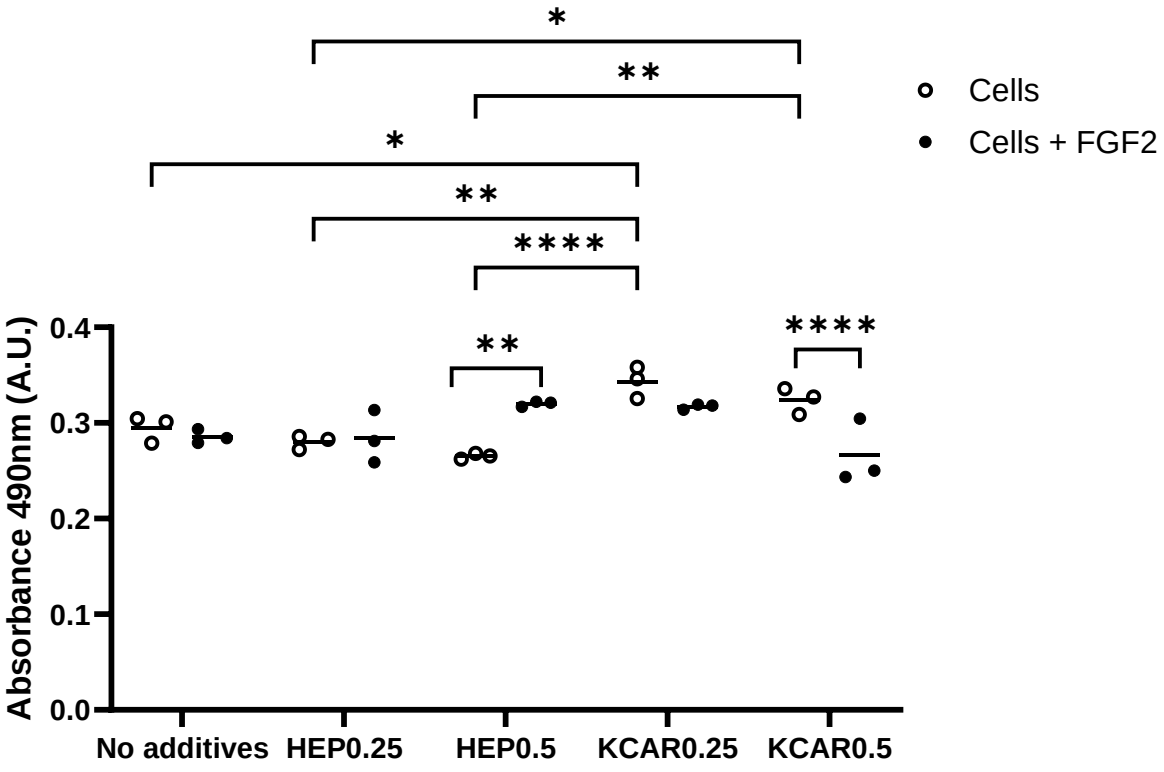


Figure 5.9. FGF2 signalling activity of fractionated KCAR determined via the proliferation of BaF32 cells transfected with FGFR1c in the presence of fractionated KCAR and FGF2. * represents statistically significant difference (*p<0.05, **p<0.01, ****p<0.0001).

5.3.2.3. Sulphur content and growth factor signalling of fractionated iota-carrageenan

Figure 5.10A and Figure 5.10B shows the sulphur atomic percent and the ratio between sulphur and carbon (S/C ratio) of fractionated iota-carrageenan (ICAR) determined by X-ray photoelectron spectroscopy. There was a high variation in the sulphur percent of ICAR0.25 in which sulphur could not be detected in some measurements. The sulphur percent of ICAR0.5 was determined to be 0.3 ± 0.1 % while the S/C ratio was 0.005 ± 0.001. High resolution C1s spectra of ICAR0.25 and ICAR0.5 were illustrated in Figure 5.10C and Figure 5.10D, respectively. The chemical structure of the repeating disaccharide unit of ICAR was shown in Figure 5.10E to illustrate the chemical bonds analysed in the C1s spectra. C-O/C-O-C peak at 286.6 eV appeared to be different between the two fractions, indicating a variation in the level of C-O/C-O-C linkages between the two fractions. Table 5.4 shows the percent of peak area for each ICAR fraction. There were no statistically significant differences in the percent of peak area of C-C/C-H, C-O/C-O-C, and O-C=O peaks between two ICAR fractions. Similar trends were observed in the ratio of peak area and the O/C ratio. Both ICAR0.25 and ICAR0.5 showed that an increase in the ratio of peak area correlated with an increase in the O/C ratio.

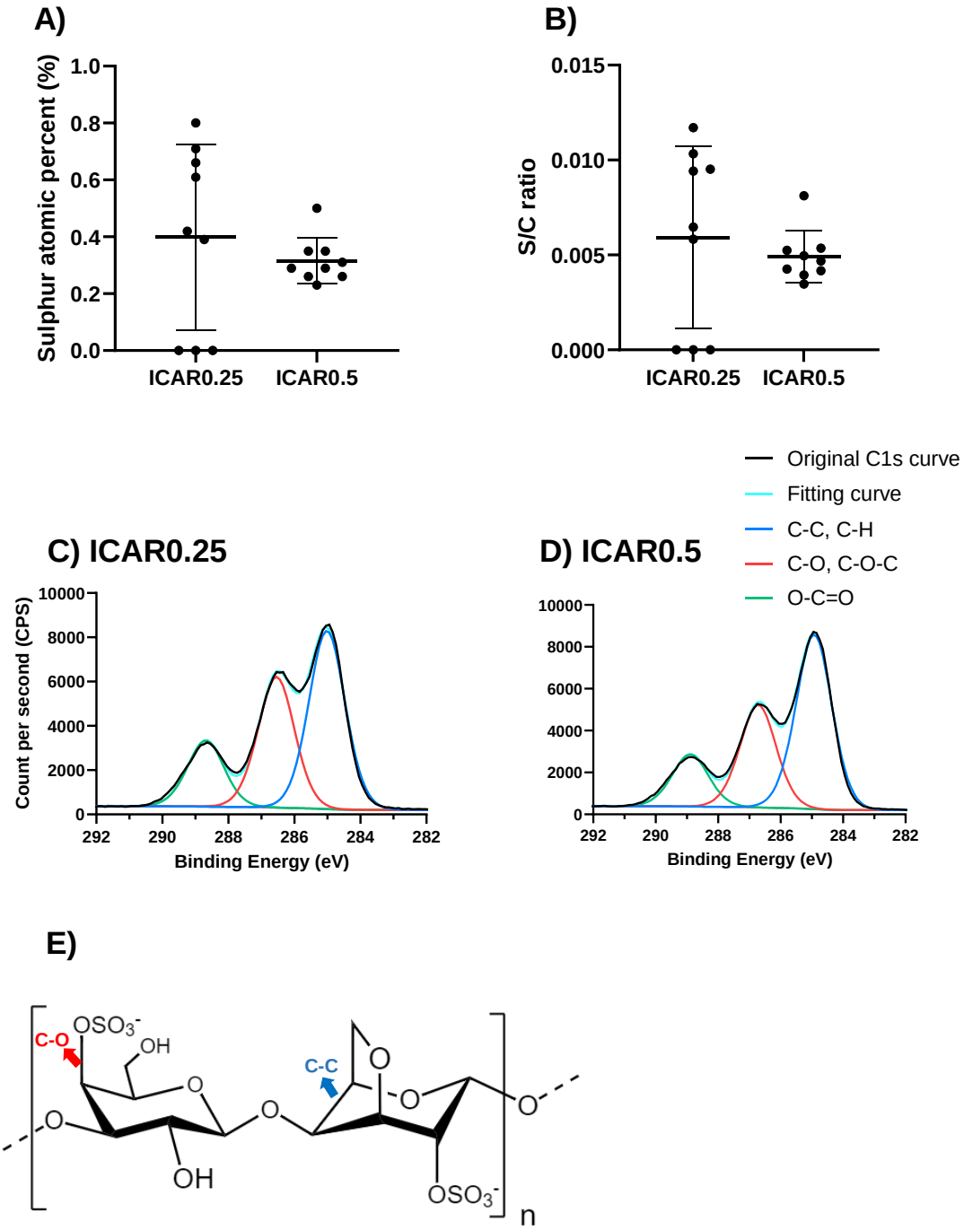


Figure 5.10. Characterisation of sulphur content of fractionated iota-carrageenan (ICAR). A) Sulphur atomic percent and B) Ratio of sulphur and carbon percent (S/C ratio) of two ICAR fractions obtained from anion exchange chromatography. High resolution C1s spectra of C) ICAR0.25 and D) ICAR0.5. E) Structure of a repeating disaccharide unit of ICAR with typical chemical bonds being shown to illustrate the corresponding carbon linkages detected in the C1s spectra.

Table 5.4. Percent (%) of peak area in C1s spectra and O/C ratio of ICAR0.25 and ICAR0.5 fractions. Each sample was measured three times. The O/C ratio was calculated from the atomic percent of oxygen and carbon of each sample.

Sample	Repeat	% of peak area in C1s spectra			[(2) + (3)]/(1) ratio	O/C ratio
		C-C, C-H (1)	C-O, C-O-C (2)	O-C=O (3)		
ICAR0.25	1	43.31	37.81	18.78	1.3	0.4
	2	55.6	31.72	12.68	0.8	0.3
	3	43.58	35.92	20.49	1.3	0.4
ICAR0.5	1	55.95	29.47	14.58	0.8	0.3
	2	35.99	41.34	22.67	1.8	0.4
	3	60.43	26.48	13.09	0.7	0.3

The ability of fractionated ICAR to signal growth factor was evaluated using BaF32 proliferation assay. Figure 5.11 shows that an increase in the proliferation of BaF32 cells when cultured with fractionated ICAR and FGF2 was not observed as compared to fractionated ICAR only, implying that these fractions could not form the ternary complex with FGF2 and FGFR1c. Together these results demonstrate a variation in the sulphur content of ICAR fractions and it did not have an impact on the ability of these fractions to signal FGF2.

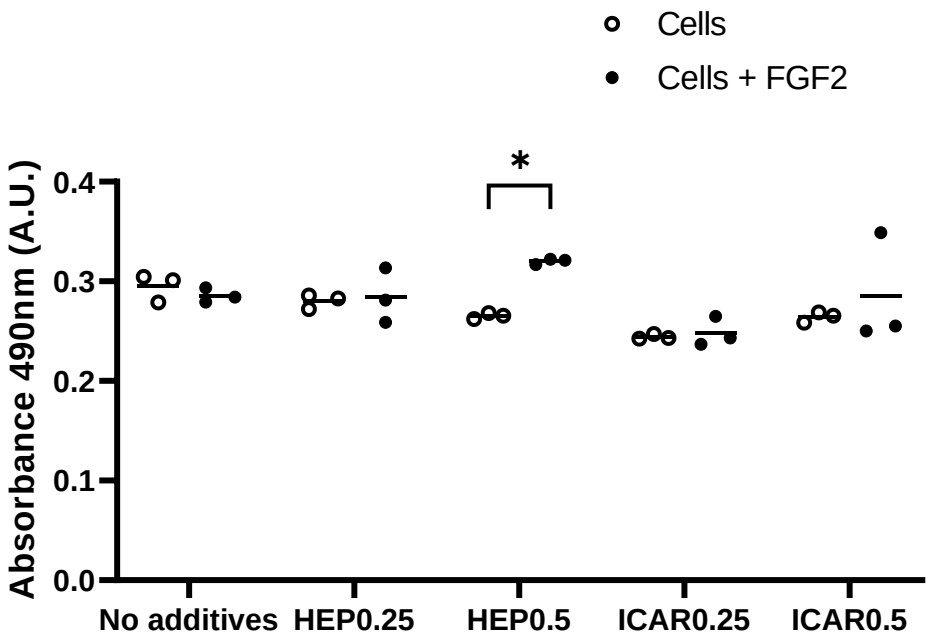


Figure 5.11. FGF2 signalling activity of fractionated ICAR determined via the proliferation of BaF32 cells transfected with FGFR1c in the presence of fractionated ICAR and FGF2. * represents statistically significant difference (*p<0.05).

5.3.2.4. Sulphur content and growth factor signalling of fractionated lambda-carrageenan

The sulphur content and ratio between sulphur and carbon (S/C) ratio of lambda-carrageenan (LCAR) was analysed using X-ray photoelectron spectroscopy (XPS). Figure 5.12A illustrates the sulphur atomic percent of LCAR0.25 and LCAR0.5. Regarding LCAR0.25, there was a high variability in the sulphur percent with several measurements showing no sulphur detected. LCAR0.5 had the sulphur percent of 0.5 ± 0.1 % and the S/C ratio of 0.008 ± 0.002. High resolution C1s spectra of both fractions were shown in Figure 5.12C and Figure 5.12D. The chemical structure of the repeating disaccharide unit of LCAR was shown in Figure 5.12E to illustrate the chemical bonds analysed in the C1s spectra. It was noticed that the height of C-C/C-H and C-O/C-O-C peaks of LCAR0.5 was different from that of LCAR0.25, indicating the difference in the amount of these linkages in the two fractions. Table 5.5 shows the percent of peak area for each LCAR fraction. There were no statistically significant differences in the percent of peak area of C-C/C-H, C-O/C-O-C, and O-C=O peaks between two LCAR fractions. Similar trends were observed in the ratio of peak area and the O/C ratio. An increase in the ratio of peak area of LCAR0.25 and LCAR0.5 was associated with an increase in the O/C ratio.

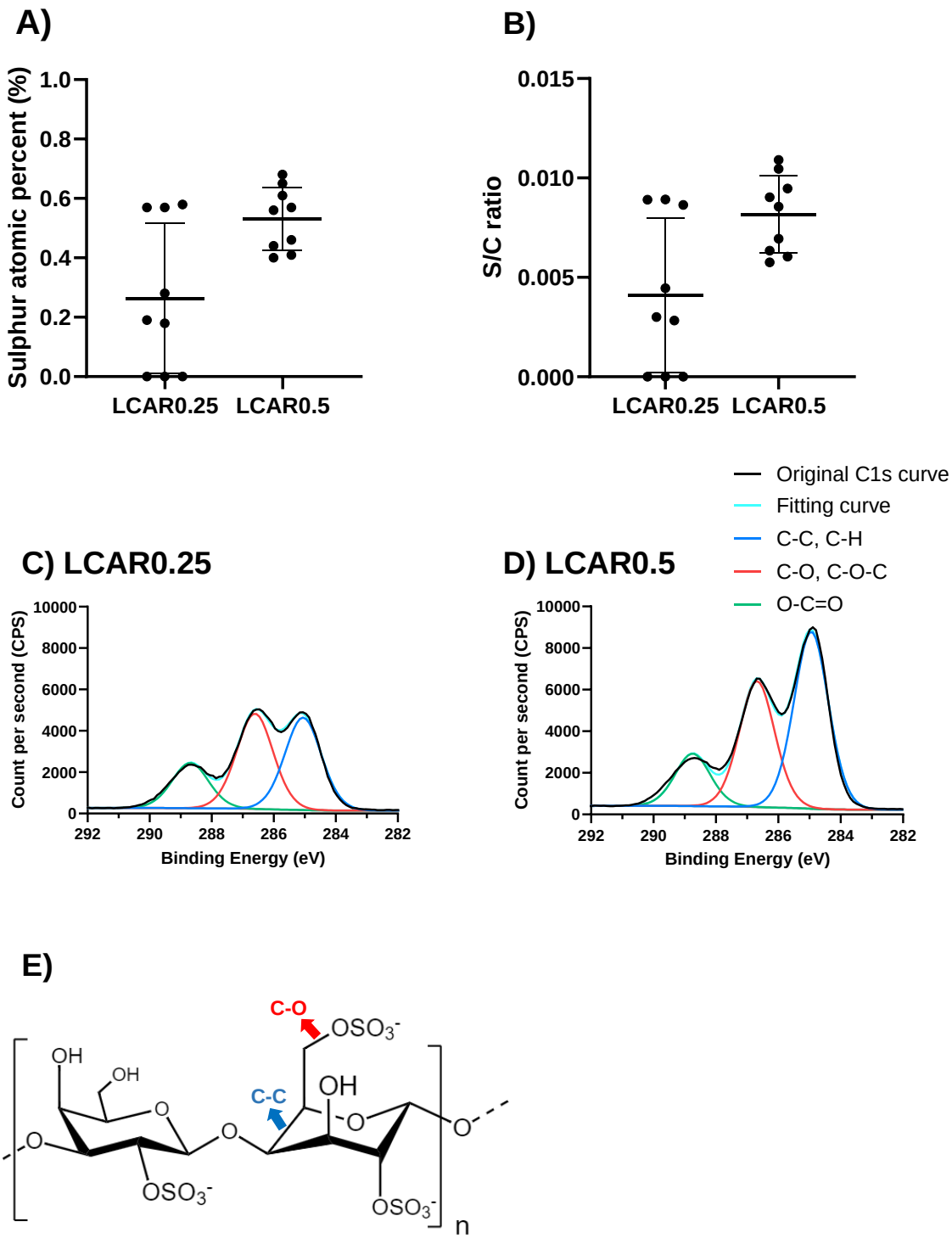


Figure 5.12. Characterisation of sulphur content of fractionated lambda-carrageenan (LCAR). A) Sulphur atomic percent and B) Ratio of sulphur and carbon percent (S/C ratio) of two LCAR fractions obtained from anion exchange chromatography. High resolution C1s spectra of C) LCAR0.25 and D) LCAR0.5. E) Structure of a repeating disaccharide unit of LCAR with typical chemical bonds being shown to illustrate the corresponding carbon linkages detected in the C1s spectra.

Table 5.5. Percent (%) of peak area in C1s spectra and O/C ratio of LCAR0.25 and LCAR0.5 fractions. Each sample was measured three times. The O/C ratio was calculated from the atomic percent of oxygen and carbon of each sample.

Sample	Repeat	% of peak area in C1s spectra			[(2) + (3)]/(1) ratio	O/C ratio
		C-C, C-H (1)	C-O, C-O-C (2)	O-C=O (3)		
LCAR0.25	1	34.97	42.25	22.79	1.9	0.4
	2	32.71	46.78	20.51	2.1	0.4
	3	53.79	31.38	14.82	0.9	0.3
LCAR0.5	1	52.46	33.64	13.9	0.9	0.3
	2	50.03	35.95	14.02	1.0	0.4
	3	40.47	39.62	19.91	1.5	0.4

Growth factor signalling ability of fractionated LCAR was investigated using BaF32 assay and results were presented in Figure 5.13. Cells incubated in either LCAR0.25 or LCAR0.5 showed a significant proliferation as compared to cells incubated in medium, HEP0.25, or HEP0.5. Cells exposed to LCAR0.25 in the presence of FGF2 did not show any significant difference compared to cells incubated in LCAR0.25 only. Interestingly, a significant increase in the proliferation of BaF32 cells when treated with LCAR0.25 and FGF2 was observed as compared to FGF2 only. LCAR0.5 in the presence of FGF2 significantly ($p<0.0001$) reduced the number of cells compared to cells cultured in LCAR0.5 only, indicating LCAR0.5 interacting with FGF2 that inhibited the interaction with FGFR1c. In Chapter 4, it was shown that unfractionated LCAR could signal FGF2. Results in the current study demonstrate that LCAR consisted of fractions with varying level of sulphur that induced different trends in FGF2 signalling.

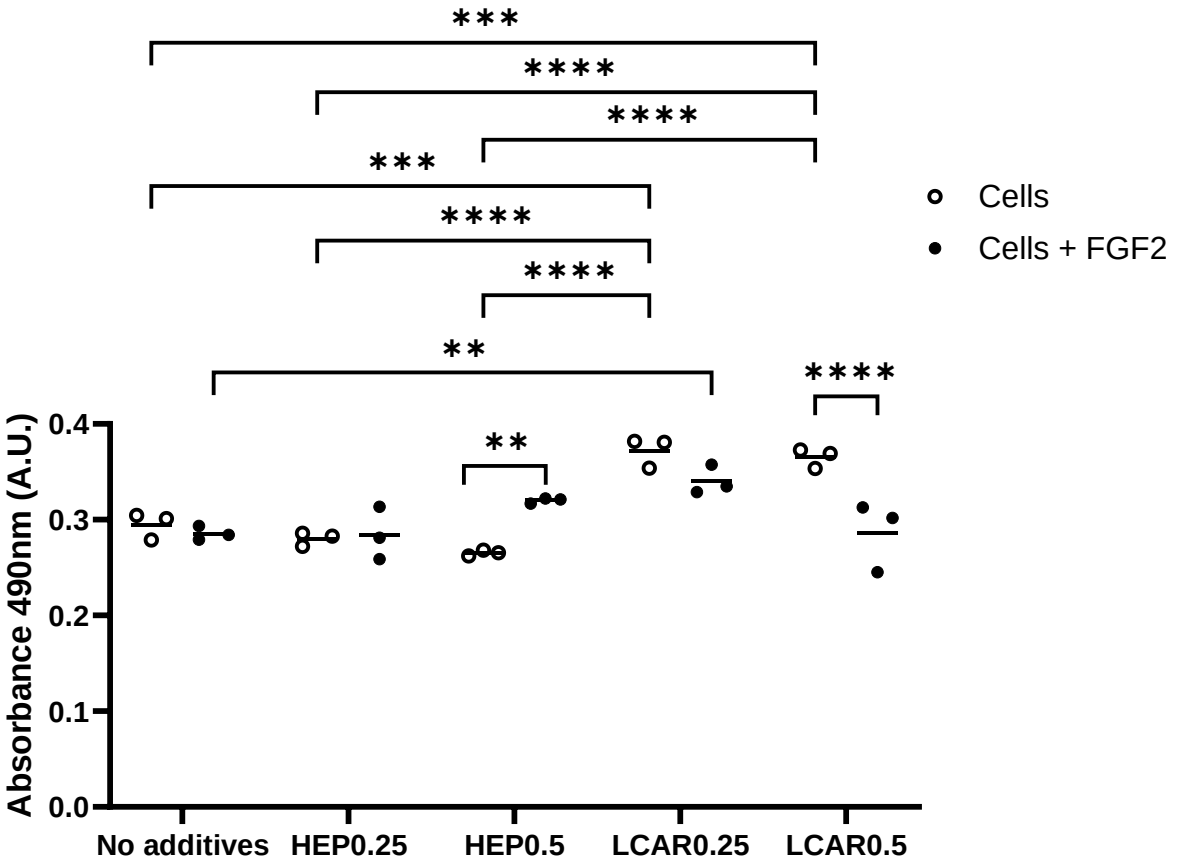


Figure 5.13. FGF2 signalling activity of fractionated LCAR determined via the proliferation of BaF32 cells transfected with FGFR1c in the presence of fractionated LCAR and FGF2. * represents statistically significant difference (**p<0.01,***p<0.001, ****p<0.0001).

5.3.2.5. Sulphur content and growth factor signalling of fractionated ulvan

X-ray photoelectron spectroscopy was employed to investigate the variation in sulphur content of ulvan fractions. Figure 5.14A shows three ulvan fractions had different atomic percent of sulphur. Ulvan0.25 was found to contain no sulphur in its structure, while the sulphur percent of Ulvan0.5 was 0.8 ± 0.2 % which was lower than Ulvan0.75 (1.9 ± 0.6 %). Notably, high variability was observed in the sulphur percent of Ulvan0.75. The ratio between sulphur and carbon (S/C ratio) was calculated and presented in Figure 5.14B. Ulvan0.75 possessed an S/C ratio of 0.030 ± 0.010 which was higher than that of Ulvan0.5 (0.011 ± 0.003), indicating that Ulvan0.75 had more sulphate groups in its chemical structure than Ulvan0.5. Figure 5.14C to Figure 5.14E show the high resolution C1s spectra of three ulvan fractions. The chemical structure of one type of repeating disaccharide unit of ulvan was shown in Figure 5.14F to illustrate the chemical bonds examined in the C1s spectra. It is noticeable that the intensity of C-C/C-H peak of Ulvan0.5 was higher than Ulvan0.25 and Ulvan0.75, which may indicate that Ulvan0.5 contained more C-C/C-H linkages than the other two fractions. Table 5.6 shows the percent of peak area for each ulvan fraction. Regarding the C-C/C-H peak, Ulvan0.25 had a significantly lower peak area than Ulvan0.5 and Ulvan0.75. Meanwhile, it had a significantly higher area of O-C=O peak than the other two fractions. In terms of the C-O/C-O-C peak, Ulvan0.25 had a significantly higher peak area than Ulvan0.5 but no significant difference to Ulvan0.75. The ratio of peak area of Ulvan0.25 was shown to be significantly higher than Ulvan0.5 and Ulvan0.75, whereas no significant differences in the O/C ratio were observed between the three fractions.

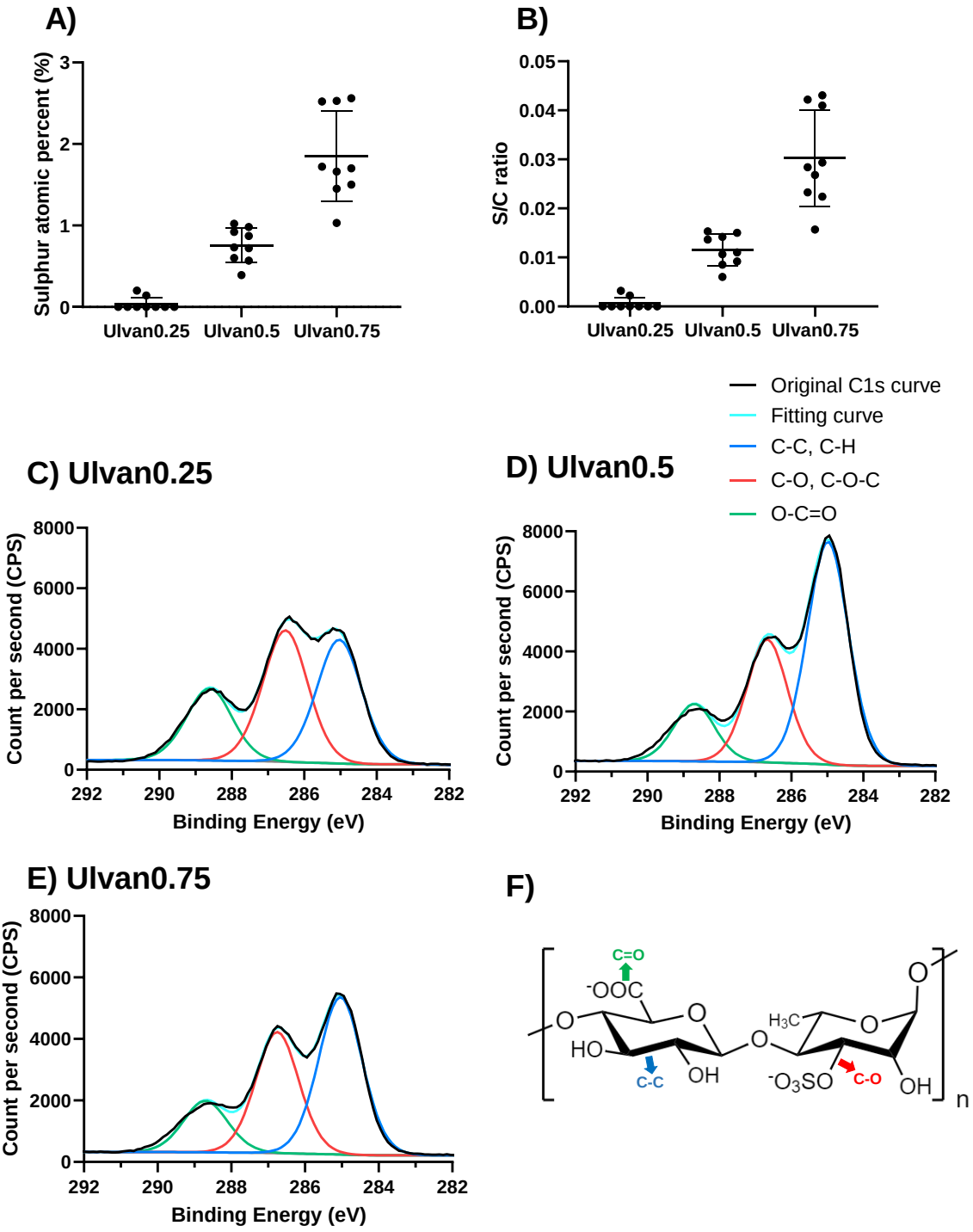


Figure 5.14. Characterisation of sulphur content of fractionated ulvan. A) Sulphur atomic percent and B) Ratio of sulphur and carbon percent (S/C ratio) of three ulvan fractions obtained from anion exchange chromatography. High resolution C1s spectra of C) Ulvan0.25, D) Ulvan0.5, and E) Ulvan0.75. F) Structure of a type of repeating disaccharide unit of ulvan with typical chemical bonds being shown to illustrate the corresponding carbon linkages detected in the C1s spectra.

Table 5.6. Percent (%) of peak area in C1s spectra and O/C ratio of Ulvan0.25, Ulvan0.5, and Ulvan0.75 fractions. Each sample was measured three times. The O/C ratio was calculated from the atomic percent of oxygen and carbon of each sample.

Sample	Repeat	% of peak area in C1s spectra			[(2) + (3)]/(1) ratio	O/C ratio
		C-C, C-H (1)	C-O, C-O-C (2)	O-C=O (3)		
Ulvan0.25	1	35.83	42.31	21.87	1.8	0.4
	2	38.55	38.14	23.31	1.6	0.4
	3	39.34	38.9	21.76	1.5	0.4
Ulvan0.5	1	56.65	30.72	12.63	0.8	0.3
	2	58.77	26.5	14.73	0.7	0.3
	3	51.37	33.25	15.38	0.9	0.4
Ulvan0.75	1	52.11	32.98	14.9	0.9	0.4
	2	48.62	36.66	14.72	1.1	0.4
	3	44.2	38.49	17.31	1.3	0.5

Figure 5.15 illustrates the BaF32 assay results which demonstrate the growth factor signalling activity of fractionated ulvan. Cells exposed to Ulvan0.5 showed a significant reduction in proliferation as compared to cells exposed to medium, HEP0.25, and HEP0.75. Cells cultured in Ulvan0.75 also had a significant decrease in proliferation compared to cells cultured in medium, HEP0.25, HEP0.5, and HEP0.75. Ulvan0.25 in the presence of FGF2 did not show any effect on the proliferation of BaF32 cells compared to Ulvan0.25 only, whereas Ulvan0.5 and Ulvan0.75 in the presence of FGF2 both significantly increased the number of BaF32 cells compared to Ulvan0.5 and Ulvan0.75 only. Moreover, the number of cells cultured in Ulvan0.5 and Ulvan0.75 in the presence of FGF2 was significantly higher than cells cultured in FGF2 only. In contrast, cells incubated with Ulvan0.25 and FGF2 showed a significant decrease in proliferation as compared to FGF2 only. This implies the ability of Ulvan0.5 and Ulvan0.75 to form ternary complex with

FGF2 and FGFR1c that induced the proliferation of BaF32 cells. Together these results show that ulvan had different fractions with varying sulphur content, and this variation affected the ability of fractionated ulvan to signal FGF2.

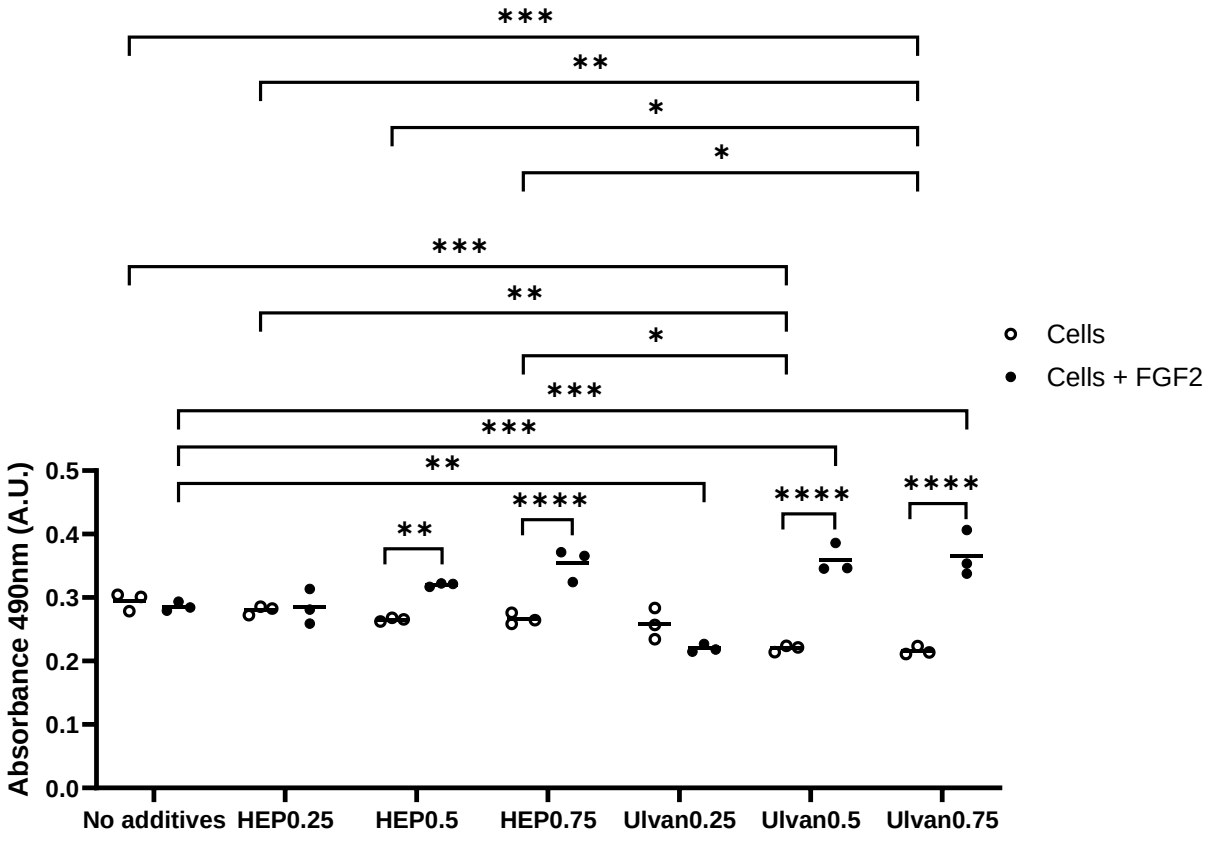


Figure 5.15. FGF2 signalling activity of fractionated ulvan determined via the proliferation of BaF32 cells transfected with FGFR1c in the presence of fractionated ulvan and FGF2. * represents statistically significant difference (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001).

5.4. Discussion

This chapter aimed to separate carrageenan and ulvan into different fractions based on the level of negative charge, followed by characterising the sulphur content of the SP fractions and evaluating how the different fractions modulated FGF2 signalling. It was hypothesised that the SPs contained different fractions with varying amount of sulphur, and this variation could affect the ability of the fractionated SPs to signal FGF2.

Chapter 3 demonstrated the degree of sulphation and position of sulphate groups that decorated the backbone of carrageenan and ulvan were heterogeneous, making it difficult to accurately determine which structural characteristic contributes to the downstream biological functions. Chapter 4 showed a noticeable variation in the biological functions for all SPs in terms of FGF2 signalling, FGFR activation, cell migration, and cell sprouting. It was hypothesised that structural variability within each SP led to varying biological responses to the materials. Based on this, heparin, carrageenan, and ulvan were separated into different fractions based on the overall level of negative charge using anion exchange chromatography (AEC). Results showed that each type of carrageenan yielded two fractions when eluted at 0.25 and 0.5 M NaCl, while heparin and ulvan resulted in three fractions eluted at 0.25, 0.5, and 0.75 M NaCl. In carrageenan, sulphate groups (OSO_3^-) are the main source of negative charge [257], while the source of negative charge of heparin and ulvan comes from not only sulphate groups but also the presence of carboxyl (COO^-) groups [168, 258]. Elemental analysis results from chapter 3 showed that heparin had higher degree of sulphation than carrageenan and ulvan. High resolution C1s spectra analysis via XPS also revealed that ulvan had higher number of O-C=O linkages, a component of carboxyl groups, than carrageenan. These findings demonstrate that heparin and ulvan had higher amount

of negatively charged groups than carrageenan, which will lead to higher level of negative charge and increase the binding strength between the chromatography column and the SP. This could explain why the separation of heparin and ulvan required a high concentration of NaCl (0.75 M) to elute fractions containing high level of negative charge.

The sulphur percent of SP fractions was quantified using XPS analysis. Regarding heparin and ulvan, the sulphur percent of the fractions increased with the increase of salt concentration at which they were eluted. This trend was reported in a previous study [259] where SPs from green algae were fractionated using AEC and the fraction eluted at 1 M NaCl had higher level of sulphate (17.7 %) than the fraction eluted at 0.5 M NaCl (10.5 %). Azm et al. [260] also utilised AEC to fractionate SPs extracted from green seaweed and reported that the increase in sulphate level of the fractions was proportional to the increase in the concentration of salt from 0.2 M to 0.8 M that was used in eluting buffers. However, this trend was not observed in the separation of carrageenan. Fractions eluted at low salt concentration (0.25 M NaCl) showed variability in the detected sulphur percent while the sulphur percent of fractions eluted at medium salt concentration (0.5 M NaCl) was more repeatable. Some carrageenan fractions eluted at 0.25 M NaCl buffer had similar sulphur percent to fractions eluted at 0.5 M NaCl buffer. This could be due to the variation in the level of negative charge within the 0.25 M fractions with some sub-fractions having higher level of negative charge being eluted when the buffer was just switched to 0.5 M NaCl. In future studies, buffers containing an increased number of NaCl concentrations can be utilised to separate the SPs into fractions with more definite level of negative charge. It was shown that the sulphur percent of KCAR0.5, ICAR0.5, and LCAR0.5 was increased following the increase in theoretical number of sulphate groups in these types of carrageenan [261],

indicating that the crude form of carrageenan contained a heterogeneous population of fractions with various sulphur content, and fractions with certain amount of sulphur could be obtained by fractionating the crude carrageenan using a salt gradient as demonstrated in the AEC experiment.

Growth factor signalling activity of fractionated SPs was evaluated in this chapter. It was shown that the variation in the sulphur content of SP fractions influenced their ability to signal FGF2. KCAR and LCAR fractions eluted at 0.5 M NaCl were shown to reduce the proliferation of BaF32 cells in the presence of FGF2 compared to cells treated with the fractions only, while KCAR and LCAR fractions eluted at 0.25 M NaCl had no impact on FGF2 signalling. It was hypothesised that KCAR0.5 and LCAR0.5 may disrupt the formation of ternary complexes with FGF2 and FGFR1c which in turn affected FGF2 signalling. Results from a previous study showed that SP fractions eluted at higher salt concentrations (1.2 M and 2 M NaCl) had lower ability to induce the proliferation of BaF32 cells as compared to fractions eluted at lower salt concentration (0.5 M NaCl); however, the reason for this phenomenon was not mentioned [262]. In the present study, no fractions were obtained at salt concentrations higher than 0.75 M NaCl as demonstrated in the full AEC profiles of all samples evaluated, which can be found in the Appendix. The difference in the salt concentration used to elute the SP fractions in the two studies could be the reason for the different trend in how SP fractions modulated FGF signalling.

The sulphur percent of KCAR0.5 and LCAR0.5 was lower than that of HEP0.5 despite being eluted at the same salt concentration (0.5 M NaCl). HEP0.5 was shown to enhance the proliferation of BaF32 cells in the presence of FGF2 while KCAR0.5 and LCAR0.5 did not. The difference in the sulphur content between SP fractions and the heparin fraction may lead to the difference in the ability of these materials to signal FGF2. It was reported that a low sulphated fraction of fucoidan

induced a lower level of FGF/FGFR1c signalling in BaF32 cells as compared to the high sulphated fraction [263, 264], which supports the current results that KCAR0.5 and LCAR0.5 exhibited the inhibitory effect on the FGF2 signalling in BaF32 cells. Regarding ulvan, the fractions eluted at 0.5 M and 0.75 M NaCl demonstrated the capacity to signal FGF2 which was reflected by the significant increase in the number of BaF32 cells. This may be explained that the level of sulphate groups in the Ulvan0.5 and Ulvan0.75 was appropriate to facilitate the formation of ternary complexes with FGF2 and FGFR1c. Furthermore, this also shows that only a small amount of sulphur in the fractionated ulvan (0.8 % in Ulvan0.5 and 1.9 % in Ulvan0.75), as compared to the sulphur percent of HEP0.5 (1.6 %) and HEP0.75 (3.8 %), could also modulate the signalling of FGF2. This may indicate other factors, such as the position of sulphate groups, could also be involved in the growth factor signalling activity of the ulvan fractions.

Differences in BaF32 assay results of unfractionated SPs in Chapter 4 and fractionated SPs in Chapter 5 were observed. It was shown that unfractionated LCAR could promote FGF2 signalling while other SPs could not. Meanwhile, this trend changed when the SPs were separate into different fractions with varying level of sulphur. Specifically, fractionated KCAR, LCAR, and ulvan had the ability to modulate FGF2 signalling while fractionated ICAR was unable to induce this biological activity. By separating the SPs into different fractions, it was shown that each SP contained populations of structure with varying amount of sulphur, and this structural heterogeneity was a contributing factor that regulated the growth factor signalling activity of the SPs that could not be observed when these SPs were used in unfractionated forms. Understanding the structural heterogeneity of SPs is the first step towards the clinical application of these materials. Characterising the structure of SPs at fraction level will provide more insights

into the structural mechanism underlying the biological functions of SPs, which helps to overcome the challenge of heterogeneous structure of recombinant HS materials that may cause undesirable effects. This also facilitates preclinical trials of SPs including *in vitro* and *in vivo* examinations for a wide range of biomedical applications such as wound healing and cancer treatment.

5.5. Conclusion

This study reveals that varying sulphur content within each type of seaweed derived SPs had an impact on the capability of growth factor signalling. KCAR and LCAR fractions with higher sulphur content demonstrated inhibitory effect on FGF2 signalling compared to fractions with lower sulphur content, whereas ICAR fractions did not show any signalling ability. Ulvan fractions were shown to mimic the growth factor signalling activity of heparin fractions that were eluted at similar salt concentrations. Overall, the findings in this chapter provide an approach to modulate growth factor signalling by using SP fractions containing various level of sulphur. This creates opportunities to take advantage of the natural variation of the seaweed derived SPs as an alternative strategy to chemical modifications of polysaccharides for achieving particular sulphation characteristics.

Chapter 6

Thesis summary and future directions

6.1. Summary of main findings, significance, and contributions to research field

Heparan sulphate (HS) is a biological molecule that binds and signals growth factors and plays essential roles in wound healing and tissue regeneration [265]. The development of HS biomimetic materials will help to surmount the constraints of existing HS mimetics and have potential to provide a rich source of materials for future exploration in therapeutic applications. Moreover, the results presented will enhance our knowledge of the relationship between material structure and biological function. This thesis utilised sulphated polysaccharides (SPs) that were extracted from different types of seaweeds and investigated their ability to mimic the biological functions of HS. Specifically, this study examined the effect of structural changes of SPs on growth factor binding and downstream cellular activities.

The seaweed derived SPs evaluated in this thesis possess similar structural characteristics to HS in terms of the presence of sulphate groups along their polysaccharide backbone. The chemical structure of SPs was heterogeneous and varies between different seaweed species. Given the biological function of SPs is dependent on their structure, it is crucial to understand the variation in the structure of SPs. Chapter 3 demonstrated that the degree of sulphation and position of sulphate groups varied between different types of SPs. Particularly, this variation was observed both in SPs extracted from the same type of seaweed (three types of carrageenan from red seaweed) and SPs extracted from different types of seaweed (carrageenan and ulvan from red and green seaweed, respectively). Data presented emphasised the heterogeneity of the chemical structure of SPs. In addition, results also revealed the effect of varying sulphation of SPs on the

interaction with fibroblast growth factor-2 (FGF2) in which a higher degree of sulphation resulted in a higher amount of FGF2 bound. Lambda-carrageenan (LCAR) was shown to have significantly higher amount of FGF2 bound as compared to kappa-carrageenan (KCAR), iota-carrageenan (ICAR), and ulvan. Interestingly, the difference in the amount of FGF2 bound between ICAR and LCAR demonstrated that having comparable degree of sulphation was not equivalent to having similar amount of growth factor bound. The position of sulphate groups could also determine the quantity of growth factor bound to SPs. Together this shows the ability to modulate growth factor binding by varying both degree of sulphation and position of sulphate groups.

The ability of SPs to bind FGF2 is a prerequisite for them to modulate the downstream signalling activity of the growth factor which is crucial for cells to induce biological responses [42]. In Chapter 4, the ability of SPs to regulate FGF2 signalling was evaluated using different methods. One method was using a cell line (BaF32 cells) that is deficient in HS proteoglycan and has been transfected with FGF receptor (FGFR) 1c to investigate the formation of SPs-FGF2-FGR1c ternary complex as a measure of growth factor signalling. Lambda-carrageenan (LCAR), which had similar position of sulphate groups to heparin, was shown to be capable of forming the ternary complex with FGF2 and FGFR1c, resulting in the proliferation of BaF32 cells which was indicative of successful signalling of FGF2. Meanwhile, KCAR, ICAR, and ulvan did not have the ability to signal FGF2 as demonstrated in the BaF32 assay. The result once again indicated the significance of appropriate position of sulphate groups of SPs for growth factor signalling. Another method was detecting the presence of phospho-ERK1/2 in endothelial cells and fibroblasts when exposed to SPs and FGF2 as an indirect measure of the activation of FGFR. SPs were shown to insignificantly modify the FGF2-induced phosphorylation of ERK1/2; however, the level of FGFR activation was

different between two cell types. Moreover, SPs also showed various modulatory effects on cell migration, in which they decreased the FGF2-induced migration of endothelial cells while had no impact on the FGF2-induced migration of fibroblasts. It was also demonstrated that kappa-, iota-, lambda-carrageenan, and ulvan exhibited inhibitory effect on the sprouting behaviour of endothelial cells. This gave new insights into the use of these SPs as therapeutic materials for anti-angiogenesis applications. This chapter also showed that SPs exhibited their modulating effect on growth factor signalling and cellular responses in a highly variable manner. Hence, it is essential to look into the structural variation within each type of SPs to further understand how this may cause the variability in the biological function of each SP.

This study hypothesised that the structural variation within each type of SPs contributed to the variability in their biological functions regarding modulating the binding and signalling of growth factor and regulating cellular activities. Therefore, in Chapter 5, each type of SPs was separated into different fractions based on the overall level of negative charge, followed by the characterisation of sulphur percent of the fractionated SPs. It was shown that each type of SPs contained different fractions with varying sulphur content. This property was similar to the structural characteristic of HS in which the chemical backbone of HS consists of unsulphated and sulphated domains. This provides further insights into the chemical structure of the SPs evaluated to determine if they can replicate specific sulphation characteristics of HS. Data presented also revealed that varying sulphur content of fractionated SPs affected their ability to signal FGF2 as demonstrated by BaF32 cell assay. Interestingly, some fractionated SPs such as KCAR0.5 and LCAR0.5 exhibited inhibitory effect on growth factor signalling while this phenomenon was not observed in the unfractionated SPs. This suggests that we need to evaluate the variation in

sulphation of these SPs at a fraction level to better understand the relationship between structure and function.

This thesis investigated the chemical structure of carrageenan and ulvan and compared their modulating functions in growth factor binding and signalling, and cellular activities. The findings in this study not only help to advance knowledge in the structure-function relationship of different types of seaweed derived SPs, but also demonstrate a new approach in the development of HS biomimetic materials by using seaweed derived SPs with appropriate sulphation properties for targeting specific applications. While the current synthetic HS mimetics require complex synthesis steps to produce HS containing sulphate groups at a specific position and the yield is relatively low, the seaweed derived SPs are available in abundant quantity and possess the same characteristics without requiring any chemical synthesis or modification, not to mention their ability to bind, signal growth factor and modulate cellular responses. Taking into account these factors, the SPs have potential for a sustainable and scalable production of HS mimetic materials for biomedical applications.

6.2. Limitations and future directions

While this thesis has provided new insights into the relationship between structure and function of the seaweed derived SPs, there are a few limitations regarding the design of experiments that can be enhanced in future studies to achieve a more comprehensive research outcome.

The interaction between SPs and growth factor was investigated using surface plasmon resonance (SPR) which monitored the interaction between the two biomolecules on a surface in a flow condition. In this experiment, the chemical backbone of SPs was modified with thiol groups

(-SH) at the reducing end to allow immobilisation on the gold surface of the SPR sensor chip, which resulted in SPs being tethered to a surface rather than being free in solution. While this type of immobilisation replicates the natural presentation of HS on the cell surface or in the extracellular matrix where HS chains are covalently attached to the core protein of HS proteoglycans [266], it may also affect the conformational flexibility of the SPs which in turn influences the binding kinetics of SPs. To address this matter, isothermal titration calorimetry (ITC) could be used as a complementary method to measure the binding of SPs and growth factor. This technique monitors the interaction between the two molecules in a static solution environment which could facilitate the interaction between the SPs and FGF2 in their native state [267, 268]. Utilising two methods would help us understand how SPs modulate growth factor binding in different testing environments and potentially different conformation states. Furthermore, information regarding the binding affinity and stoichiometry could be obtained to gain more insight into the mechanism of binding. Although SPR results from this thesis show that LCAR had higher level of FGF2 binding compared to KCAR, ICAR, and ulvan, all SPs should be investigated using ITC to obtain a comprehensive understanding of growth factor interaction of different SPs.

Regarding the study of growth factor receptor activation, the concentration of SPs and growth factor, and the incubation time remained constant throughout the experiments presented in this thesis. It has been reported that varying the concentration of HS and FGF2 as well as their incubation time resulted in altered level of FGF2 signalling [269, 270]. Therefore, different concentrations of SPs could be used to investigate whether SPs exhibit their growth factor signalling activity in a concentration-dependent manner. Varying incubation time could also be

implemented to evaluate if the signalling activity of SPs could be sustained over time. This is important as to enhance the short lifetime issue of growth factors when they are used in the body's physiological environment [271]. Throughout this thesis, the modulating effect of SPs was evaluated for FGF2 only. There are other growth factors that are vital in wound and tissue repair, such as vascular endothelial growth factor (VEGF) and heparin-binding epidermal growth factor–like growth factor (HB-EGF), both of which are known to interact with HS [272, 273]. The ability of SPs to bind and signal these growth factors should be examined in future studies to explore the modulating effect of the SPs on a wide range of growth factors and give further insight into the structure-function relationship between SPs and biomolecules.

This thesis has demonstrated the relationship between sulphation of SPs and their biological functions, though it is known that SP sulphation is not the only structural component that modulates their interaction with biomolecules [274]. In future studies, the SPs can be separated into different fractions based on molecular weight using size exclusion chromatography [275]. It has been shown that different molecular weights of heparin/HS could alter the binding and signalling of VEGF and FGF [222, 276]. Understanding the effect of varying molecular weight of the SPs on growth factor binding and signalling would provide a comprehensive tool to control the function of these materials through altering their structural characteristics. From the application perspective, gaining insights into the structure-function relationship of SPs will form the basis of the development of SPs-based biomaterials that can be customised for specific applications such as wound repair and tissue regeneration. In terms of clinical translation, the use of fractionated SPs can be considered an appropriate option since it can reduce the variability

in the biological functions of unfractionated SPs and provide results with high level of reproducibility in preclinical trials.

The SPs used in this thesis were extracted from macroalgae including red and green seaweed. In future research, other marine organisms can be explored to produce SPs. It has been reported that microalgae are capable of producing bioactive compounds such as polysaccharides, lipids, and proteins [277]. Moreover, they are easy to cultivate, and the harvesting process is not dependent on season or climate [277, 278], making them a sustainable and scalable source of materials for the production of SPs. A recent study found that SPs extracted from microalgae with molecular weight of 190.94 kDa and sulphate content of 16.83 % had the ability in modulating the p38 mitogen-activated protein kinase (MAPK) and c-Jun N-terminal kinase (JNK) signalling pathways that were involved in the apoptosis of a cancer cell line [279], demonstrating the potential to mimic HS regarding the regulation of cellular activities in cancer [280]. Therefore, utilising microalgae to produce SPs can be a promising approach for identifying more structures of SPs with distinct biological functions.

This thesis investigated the effect of SPs on the migration activity of endothelial cells and fibroblasts and the sprouting activity of endothelial cells. Other cell types such as keratinocytes and mesenchymal stem cells can also be used to understand how SPs may impact the re-epithelialisation and differentiation of these cells, respectively. This is important as these cellular activities are critical events in the wound healing process that are modulated by HS [281-284], and determining the role of SPs in regulating various cellular activities would advance our understanding of the ability of SPs to mimic various biological functions of HS. In addition, future studies can focus on investigating the effect of SP structures on different signalling pathways that

underlie cellular responses, including phosphoinositide 3-kinase-Akt (PI3K/AKT) and phospholipase Cγ1 (PLCγ1) [285, 286]. Gaining insights into how SPs control intracellular signalling would be beneficial to achieving desirable cellular behaviours for the treatment of diseases and body injuries.

Data presented showed that SPs could bind FGF2 and facilitate the growth factor signalling. In future work, the SPs can be utilised in combination with other natural polymers (chitosan, gelatin) or synthetic polymers (polyvinyl alcohol, polycaprolactone) for the fabrication of biomaterials that aid the delivery of growth factors in different therapeutic applications. The effect of sulphation of SPs on their function was unveiled in this thesis, and it is interesting to see how this new knowledge can elevate the functionality of SPs when they are being employed in a biomaterial system. Moreover, *in vivo* studies should be undertaken to confirm the results observed *in vitro*. This can involve the use of animal models to better understand the impact of structural variation of SPs in a physiological environment. In addition, the function of endogenous HS in *in vitro* and *in vivo* models can be inhibited by using heparanase or anti-HSPG blocking antibodies in order to investigate the intrinsic mechanism of SPs. The concentration of HS inhibitors and the selection of appropriate experimental controls should be optimised to obtain the most accurate results.

6.3. Conclusion

Designing materials that mimic the function of biological molecules in the human body requires a fundamental understanding of their structure-function relationship. This thesis explored the effect of different sulphation characteristics of marine derived SPs on growth factor binding and cellular activities. The findings showed that SPs modulated growth factor binding, intracellular signalling, and downstream cellular response in a structure-dependent manner. Moreover, each type of SPs was shown to contain fractions with varying sulphur content, which may contribute to the variability of the biological functions of SPs. Together this work has contributed new knowledge to the field of HS biomimetic materials by presenting a thorough study of chemical structure and biological function of SPs. The next step in the successful development of new functional materials for clinical applications would be characterising the position of sulphate groups and the molecular weight of fractionated SPs and investigating their ability to signal different growth factors and induce cell and tissue responses through *in vitro* and *in vivo* studies.

References

1. Malavaki, C.J., et al., *Heparan sulfate: biological significance, tools for biochemical analysis and structural characterization.* Biomed Chromatogr, 2011. **25**(1-2): p. 11-20.
2. Shriver, Z., et al., *Heparin and Heparan Sulfate: Analyzing Structure and Microheterogeneity*. 2012, Springer Berlin Heidelberg. p. 159-176.
3. Sarrazin, S., W.C. Lamanna, and J.D. Esko, *Heparan sulfate proteoglycans.* Cold Spring Harbor Perspectives in Biology, 2011. **3**(7): p. 1-33.
4. Li, J.P. and M. Kusche-Gullberg, *Heparan Sulfate: Biosynthesis, Structure, and Function.* Int Rev Cell Mol Biol, 2016. **325**: p. 215-73.
5. Pongener, I., et al., *Developments in the Chemical Synthesis of Heparin and Heparan Sulfate.* Chem Rec, 2021. **21**(11): p. 3238-3255.
6. Li, J., et al., *Synthesis of 3-O-Sulfated Heparan Sulfate Oligosaccharides Using 3-O-Sulfotransferase Isoform 4.* ACS Chem Biol, 2021. **16**(10): p. 2026-2035.
7. Thacker, B.E. and S.T. Sharfstein, *Metabolic engineering of mammalian cells to produce heparan sulfates.* Emerg Top Life Sci, 2018. **2**(3): p. 443-452.
8. Caputo, H.E., J.E. Straub, and M.W. Grinstaff, *Design, synthesis, and biomedical applications of synthetic sulphated polysaccharides.* Chem Soc Rev, 2019. **48**(8): p. 2338-2365.
9. Campo, V.L., et al., *Carrageenans: Biological properties, chemical modifications and structural analysis - A review*, in *Carbohydrate Polymers*. 2009. p. 167-180.
10. Ale, M.T. and A.S. Meyer, *Fucoidans from brown seaweeds: an update on structures, extraction techniques and use of enzymes as tools for structural elucidation.* RSC advances, 2013. **3**(22 p.8131-8141): p. 8141-8131.
11. Marc, L. and R. Audrey, *Structure and Functional Properties of Ulvan, a Polysaccharide from Green Seaweeds.* Biomacromolecules, 2007. **8**(6): p. 1765-1774.
12. Kang, J., et al., *Insights into the structure-bioactivity relationships of marine sulfated polysaccharides: A review.* Food Hydrocolloids, 2021.

13. Patel, S., *Therapeutic importance of sulfated polysaccharides from seaweeds: updating the recent findings.* 3 Biotech, 2012. **2**(3): p. 171-185.
14. Werner, S. and R. Grose, *Regulation of wound healing by growth factors and cytokines.* Physiological reviews, 2003.
15. Broughton, G., 2nd, J.E. Janis, and C.E. Attinger, *The basic science of wound healing.* Plast Reconstr Surg, 2006. **117**(7 Suppl): p. 12S-34S.
16. Ellis, S., E.J. Lin, and D. Tartar, *Immunology of Wound Healing.* Curr Dermatol Rep, 2018. **7**(4): p. 350-358.
17. Grazul-Bilska, A.T., et al., *Wound healing: the role of growth factors.* Drugs Today (Barc), 2003. **39**(10): p. 787-800.
18. Velnar, T., T. Bailey, and V. Smrkoli, *The Wound Healing Process: an Overview of the Cellular and Molecular Mechanisms.* JOURNAL OF INTERNATIONAL MEDICAL RESEARCH, 2009. **37**(5): p. 1528-1542.
19. Häkkinen, L., H. Larjava, and L. Koivisto, *Granulation tissue formation and remodeling.* Endodontic Topics, 2012. **24**(1): p. 94-129.
20. Greaves, N.S., et al., *Current understanding of molecular and cellular mechanisms in fibroplasia and angiogenesis during acute wound healing.* Journal of Dermatological Science, 2013. **72**(3): p. 206-217.
21. Velnar, T. and L. Gradisnik, *Tissue Augmentation in Wound Healing: the Role of Endothelial and Epithelial Cells*, in *Medical archives (Sarajevo, Bosnia and Herzegovina)*. 2018, Academy of Medical Sciences of Bosnia and Herzegovina: Bosnia and Herzegovina. p. 444-448.
22. Tonnesen, M.G., X. Feng, and R.A. Clark, *Angiogenesis in wound healing.* J Investig Dermatol Symp Proc, 2000. **5**(1): p. 40-6.
23. Johnson, K.E. and T.A. Wilgus, *Vascular Endothelial Growth Factor and Angiogenesis in the Regulation of Cutaneous Wound Repair*, in *Advances in wound care*. 2014, Mary Ann Liebert, Inc: United States. p. 647-661.
24. Yifan, T., et al., *Myofibroblasts: Function, Formation, and Scope of Molecular Therapies for Skin Fibrosis*, in *Biomolecules*. 2021, MDPI AG. p. 1095-1095.

25. Lee, K., E.A. Silva, and D.J. Mooney, *Growth factor delivery-based tissue engineering: general approaches and a review of recent developments.* JOURNAL OF THE ROYAL SOCIETY INTERFACE, 2011. **8**(55): p. 153-170.
26. Babensee, J.E., L.V. McIntire, and A.G. Mikos, *Growth Factor Delivery for Tissue Engineering.* Pharmaceutical Research: An Official Journal of the American Association of Pharmaceutical Scientists, 2000. **17**(5): p. 497-504.
27. Barrientos, S., et al., *Growth factors and cytokines in wound healing.* Wound Repair & Regeneration, 2008. **16**(5): p. 585-601.
28. Zarei, F. and M. Soleimaninejad, *Role of growth factors and biomaterials in wound healing.* Artificial Cells, Nanomedicine & Biotechnology, 2018. **46**: p. 906-911.
29. Xie, Y., et al., *FGF/FGFR signaling in health and disease.* Signal Transduct Target Ther, 2020. **5**(1): p. 181.
30. Ornitz, D.M. and N. Itoh, *The fibroblast growth factor signaling pathway.* Wiley Interdisciplinary Reviews: Developmental Biology, 2015. **4**(3): p. 215-266.
31. Xie, J., et al., *Effects of basic fibroblast growth factor on the expression of extracellular matrix and matrix metalloproteinase-1 in wound healing*, in *Clinical and experimental dermatology*. 2008, Blackwell Scientific Publications: England. p. 176-182.
32. Ye-Rang, Y., et al., *Fibroblast Growth Factors: Biology, Function, and Application for Tissue Regeneration*, in *Journal of Tissue Engineering*. 2010, SAGE Publishing.
33. Beenken, A. and M. Mohammadi, *The FGF family: biology, pathophysiology and therapy.* Nat Rev Drug Discov, 2009. **8**(3): p. 235-53.
34. Goetz, R. and M. Mohammadi, *Exploring mechanisms of FGF signalling through the lens of structural biology.* Nat Rev Mol Cell Biol, 2013. **14**(3): p. 166-80.
35. Kim, S.H., J. Turnbull, and S. Guimond, *Extracellular matrix and cell signalling: the dynamic cooperation of integrin, proteoglycan and growth factor receptor.* J Endocrinol, 2011. **209**(2): p. 139-51.
36. Kowitsch, A., G. Zhou, and T. Groth, *Medical application of glycosaminoglycans: a review.* J Tissue Eng Regen Med, 2018. **12**(1): p. e23-e41.

37. Schlessinger, J., et al., *Crystal Structure of a Ternary FGF-FGFR-Heparin Complex Reveals a Dual Role for Heparin in FGFR Binding and Dimerization.* Molecular Cell, 2000. **6**(3): p. 743-750.

38. Teven, C.M., et al., *Fibroblast growth factor (FGF) signaling in development and skeletal diseases.* Genes & Diseases, 2014. **1**(2): p. 199-213.

39. Shapiro, P., *Ras-MAP Kinase Signaling Pathways and Control of Cell Proliferation: Relevance to Cancer Therapy.* Critical Reviews in Clinical Laboratory Sciences, 2002. **39**(4/5): p. 285.

40. Johnson, G.L. and R. Lapadat, *Mitogen-Activated Protein Kinase Pathways Mediated by ERK, JNK, and p38 Protein Kinases*, in *Science*. 2002. p. 1911-1912.

41. Lax, I., et al., *The Docking Protein FRS2α Controls a MAP Kinase-Mediated Negative Feedback Mechanism for Signaling by FGF Receptors.* Molecular Cell, 2002. **10**(4): p. 709-719.

42. Turner, N. and R. Grose, *Fibroblast growth factor signalling: from development to cancer*, in *Nature reviews. Cancer*. 2010, Nature Pub. Group: England. p. 116-129.

43. Tomlinson, D.C., et al., *FGFR1-induced Epithelial to mesenchymal transition through MAPK/PLCγ/COX-2-mediated mechanisms.* PLoS ONE, 2012. **7**(6).

44. Bix, G. and R.V. Iozzo, *Novel interactions of perlecan: Unraveling perlecan's role in angiogenesis.* MICROSCOPY RESEARCH AND TECHNIQUE, 2008. **71**(5): p. 339-348.

45. Schaefer, L. and R.M. Schaefer, *Proteoglycans: from structural compounds to signaling molecules.* Cell Tissue Res, 2010. **339**(1): p. 237-46.

46. Perrimon, N. and M. Bernfield, *Cellular functions of proteoglycans--an overview.* Semin Cell Dev Biol, 2001. **12**(2): p. 65-7.

47. Billings, P.C. and M. Pacifici, *Interactions of signaling proteins, growth factors and other proteins with heparan sulfate: mechanisms and mysteries.* Connective Tissue Research, 2015. **56**(4): p. 272-280.

48. Esko, J.D. and U. Lindahl, *Molecular diversity of heparan sulfate.* Journal of Clinical Investigation, 2001. **108**(2): p. 169-173.

49. Poulain, F.E. and H.J. Yost, *Heparan sulfate proteoglycans: a sugar code for vertebrate development?* Development (09501991), 2015. **142**(20): p. 3456-3467.

50. Lane, D.A. and U. Lindahl, *Heparin : chemical and biological properties, clinical applications*. 1989: Edward Arnold.

51. Carlsson, P., et al., *Heparin/Heparan Sulfate Biosynthesis: PROCESSIVE FORMATION OF N-SULFATED DOMAINS*.* Journal of Biological Chemistry, 2008. **283**(29): p. 20008-20014.

52. Xu, D. and J.D. Esko, *Demystifying heparan sulfate-protein interactions.* Annu Rev Biochem, 2014. **83**: p. 129-57.

53. Sun, L., P. Chopra, and G.-J. Boons, *Chemoenzymatic Synthesis of Heparan Sulfate Oligosaccharides having a Domain Structure.* Angewandte Chemie, 2022. **134**(47).

54. Oduah, E.I., R.J. Linhardt, and S.T. Sharfstein, *Heparin: Past, Present, and Future.* Pharmaceuticals (Basel), 2016. **9**(3).

55. Casu, B., et al., *The structure of heparin oligosaccharide fragments with high anti-(factor Xa) activity containing the minimal antithrombin III-binding sequence Chemical and13C nuclear-magnetic-resonance studies.* Biochemical Journal, 1981. **197**(3): p. 599-609.

56. Miller, T., et al., *Molecular engineering of glycosaminoglycan chemistry for biomolecule delivery.* Acta Biomater, 2014. **10**(4): p. 1705-19.

57. Jin, W., et al., *Interactions of fibroblast growth factors with sulfated galactofucan from Saccharina japonica.* Int J Biol Macromol, 2020. **160**: p. 26-34.

58. Schultz, V., et al., *Heparan Sulfate Domains Required for Fibroblast Growth Factor 1 and 2 Signaling through Fibroblast Growth Factor Receptor 1c.* The Journal of biological chemistry, 2017. **292**(6): p. 2495-2509.

59. Robinson, C.J., et al., *VEGF165-binding sites within heparan sulfate encompass two highly sulfated domains and can be liberated by K5 lyase.* Journal of Biological Chemistry, 2006. **281**(3): p. 1731-1740.

60. Stringer, S., *The role of heparan sulphate proteoglycans in angiogenesis*. 2006, Portland Press Ltd.

61. Robinson, C.J. and S.E. Stringer, *The splice variants of vascular endothelial growth factor (VEGF) and their receptors.* Journal of cell science, 2001. **114**(5): p. 853-865.

62. Thacker, B.E., et al., *Heparan sulfate 3-O-sulfation: A rare modification in search of a function.* Matrix Biology, 2014. **35**: p. 60-72.

63. Warda, M., et al., *Isolation and characterization of heparan sulfate from various murine tissues.* Glycoconjugate Journal: Official Journal of the International Glycoconjugate Organization, 2006. **23**(7-8): p. 555-563.

64. Brooke, L.F., et al., *Can We Produce Heparin/Heparan Sulfate Biomimetics Using 'Mother-Nature' as the Gold Standard?*, in *Molecules*. 2015, MDPI AG. p. 4254-4276.

65. Mende, M., et al., *Chemical Synthesis of Glycosaminoglycans.* Chem Rev, 2016. **116**(14): p. 8193-255.

66. Cole, C.L., et al., *Synthetic Heparan Sulfate Oligosaccharides Inhibit Endothelial Cell Functions Essential for Angiogenesis.* PLoS ONE, 2010. **5**(7): p. 1-15.

67. Farrugia, B.L., et al., *Chitosan-Based Heparan Sulfate Mimetics Promote Epidermal Formation in a Human Organotypic Skin Model.* Advanced Functional Materials, 2018. **28**(36).

68. Peschel, D., et al., *Modulation of osteogenic activity of BMP-2 by cellulose and chitosan derivatives.* Acta Biomaterialia, 2012. **8**(1): p. 183-193.

69. Cao, L., et al., *Vascularization and bone regeneration in a critical sized defect using 2- N ,6-O- sulfated chitosan nanoparticles incorporating BMP-2.* Biomaterials, 2014. **35**(2): p. 684-698.

70. Purcell, B.P., et al., *Incorporation of sulfated hyaluronic acid macromers into degradable hydrogel scaffolds for sustained molecule delivery.* Biomaterials science, 2014. **2**(5 p.693-702): p. 702-693.

71. Mhanna, R., et al., *Sulfated Alginate as a Mimic of Sulfated Glycosaminoglycans: Binding of Growth Factors and Effect on Stem Cell Behavior.* Advanced Biosystems, 2017. **1**(7).

72. Shen, T., et al., *Sulfated chitosan rescues dysfunctional macrophages and accelerates wound healing in diabetic mice.* Acta Biomaterialia, 2020. **117**: p. 192-203.

73. Sahraneshin-Samani, F., et al., *Regioselective sulfated chitosan produces a biocompatible and antibacterial wound dressing with low inflammatory response.* Biomaterials Advances, 2022. **139**.

74. Tong, M., et al., *Diabetes-impaired wound healing is improved by matrix therapy with heparan sulfate glycosaminoglycan mimetic OTR4120 in rats*. 2012, American Diabetes Association. p. 2633.

75. Barritault, D., et al., *RGTA® or ReGeneraTing Agents mimic heparan sulfate in regenerative medicine: from concept to curing patients.* Glycoconjugate Journal, 2017. **34**(3): p. 325-338.

76. Liu, J. and R.J. Linhardt, *Chemoenzymatic synthesis of heparan sulfate and heparin.* Natural product reports, 2014. **31**(12): p. 1676-1685.

77. Fu, L., M. Suflita, and R.J. Linhardt, *Bioengineered heparins and heparan sulfates.* Advanced drug delivery reviews, 2016. **97**: p. 237-249.

78. Li, J., et al., *Chemoenzymatic Synthesis of Heparan Sulfate Mimetic Glycopolymers and Their Interactions with the Receptor for Advanced Glycation End-Product.* ACS Macro Letters, 2019. **8**(12): p. 1570-1574.

79. Lord, M.S. and J.M. Whitelock, *Recombinant production of proteoglycans and their bioactive domains.* The FEBS Journal, 2013. **280**(10): p. 2490-2510.

80. Rnjak-Kovacina, J., et al., *Recombinant Domain V of Human Perlecan Is a Bioactive Vascular Proteoglycan.* Biotechnology Journal, 2017. **12**(12).

81. Viklund, L., et al., *Expression and Characterization of Minican, a Recombinant Syndecan-1 with Extensively Truncated Core Protein.* Biochemical and Biophysical Research Communications, 2002. **290**(1): p. 146-152.

82. Lord, M.S., et al., *Bioengineered human heparin with anticoagulant activity.* Metab Eng, 2016. **38**: p. 105-114.

83. Mano, J.F., *Biomimetic approaches for biomaterials development*. 2013: John Wiley & Sons.

84. Fratzl, P., *Biomimetic materials research: what can we really learn from nature's structural materials?* Journal of the Royal Society Interface, 2007. **4**(15): p. 637-642.

85. Ahmed, S. and S. Ikram, *Chitosan based scaffolds and their applications in wound healing.* Achievements in the life sciences, 2016. **10**(1): p. 27-37.

86. Farokhi, M., et al., *Alginate based scaffolds for cartilage tissue engineering: a review.* International Journal of Polymeric Materials and Polymeric Biomaterials, 2020. **69**(4): p. 230-247.

87. Ngo, D.-H. and S.-K. Kim, *Sulfated polysaccharides as bioactive agents from marine algae.* International Journal of Biological Macromolecules, 2013. **62**: p. 70-75.

88. Silva, T.H., et al., *Materials of marine origin: a review on polymers and ceramics of biomedical interest.* International materials reviews, 2012. **57**(5): p. 276-306.

89. Maria Filomena de Jesus, R., M. Alcina Maria Bernardo de, and M. Rui Manuel Santos Costa de, *Marine Polysaccharides from Algae with Potential Biomedical Applications*, in *Marine Drugs*. 2015, MDPI AG. p. 2967-3028.

90. Falshaw, R., H.J. Bixler, and K. Johndro, *Structure and performance of commercial kappa-2 carrageenan extracts: I. Structure analysis.* Food hydrocolloids, 2001. **15**(4): p. 441-452.

91. Guo, Z., et al., *Carrageenan oligosaccharides: A comprehensive review of preparation, isolation, purification, structure, biological activities and applications.* Algal Research, 2022. **61**.

92. Zhou, G., et al., *In vivo antitumor and immunomodulation activities of different molecular weight lambda-carrageenans from Chondrus ocellatus.* Pharmacological research, 2004. **50**(1): p. 47-53.

93. Barabanova, A.O., et al., *Structure and properties of carrageenan-like polysaccharide from the red alga Tichocarpus crinitus (Gmel.) Rupr. (Rhodophyta, Tichocarpaceae).* Journal of applied phycology, 2008. **20**(6): p. 1013-1020.

94. Vera, J., et al., *Seaweed polysaccharides and derived oligosaccharides stimulate defense responses and protection against pathogens in plants.* Marine Drugs, 2011. **9**(12): p. 2514-2525.

95. Yegappan, R., et al., *Injectable angiogenic and osteogenic carrageenan nanocomposite hydrogel for bone tissue engineering.* International journal of biological macromolecules, 2019. **122**: p. 320-328.

96. Popa, E.G., et al., *Chondrogenic potential of injectable κ-carrageenan hydrogel with encapsulated adipose stem cells for cartilage tissue-engineering applications*, in *Journal*

of tissue engineering and regenerative medicine. 2015, John Wiley & Sons: England. p. 550-563.

97. Rocha, P.M., et al., *Encapsulation of adipose-derived stem cells and transforming growth factor-β1 in carrageenan-based hydrogels for cartilage tissue engineering.* Journal of Bioactive and Compatible Polymers, 2011. **26**(5): p. 493-507.

98. Dev, A., et al., *κ-carrageenan-C-phycocyanin based smart injectable hydrogels for accelerated wound recovery and real-time monitoring.* Acta Biomaterialia, 2020. **109**: p. 121-131.

99. Jaiswal, L., S. Shankar, and J.-W. Rhim, *Carrageenan-based functional hydrogel film reinforced with sulfur nanoparticles and grapefruit seed extract for wound healing application.* Carbohydrate Polymers, 2019. **224**.

100. Cunha, L. and A. Grenha, *Sulfated Seaweed Polysaccharides as Multifunctional Materials in Drug Delivery Applications.* Mar Drugs, 2016. **14**(3).

101. Jiao, G., et al., *Chemical structures and bioactivities of sulfated polysaccharides from marine algae.* Marine Drugs, 2011. **9**(2): p. 196-223.

102. Kang, J., et al., *Insights into the structure-bioactivity relationships of marine sulfated polysaccharides: A review.* Food Hydrocolloids, 2022. **123**.

103. Fitton, J.H., D.N. Stringer, and S.S. Karpiniec, *Therapies from fucoidan: An update.* Marine drugs, 2015. **13**(9): p. 5920-5946.

104. Ruixiang, Z., et al., *Fucoidan: Structure and Bioactivity*, in *Molecules*. 2008, MDPI AG. p. 1671-1695.

105. Jin, Z., Y.-X. Han, and X.-R. Han, *Degraded Iota-Carrageenan Can Induce Apoptosis in Human Osteosarcoma Cells Via the Wnt/β-Catenin Signaling Pathway.* Nutrition and cancer, 2013. **65**(1): p. 126-131.

106. Ustyuzhanina, N.E., et al., *Fucoidans: pro- or antiangiogenic agents?* Glycobiology (Oxford), 2014. **24**(12): p. 1265-1274.

107. Hayashi, K., et al., *Defensive effects of a fucoidan from brown alga Undaria pinnatifida against herpes simplex virus infection.* International immunopharmacology, 2008. **8**(1): p. 109-116.

108. Sezer, A.D., et al., *Preparation of Fucoidan-Chitosan Hydrogel and Its Application as Burn Healing Accelerator on Rabbits.* Biological and Pharmaceutical Bulletin, 2008. **31**(12): p. 2326-2333.

109. Jin, G. and G.H. Kim, *Rapid-prototyped PCL/fucoidan composite scaffolds for bone tissue regeneration: design, fabrication, and physical/biological properties.* Journal of materials chemistry, 2011. **21**(44): p. 1771-17718.

110. Huang, Y.-C. and T.-J. Liu, *Mobilization of mesenchymal stem cells by stromal cell-derived factor-1 released from chitosan/tripolyphosphate/fucoidan nanoparticles.* Acta biomaterialia, 2012. **8**(3): p. 1048-1056.

111. Kidgell, J.T., et al., *Ulvan: A systematic review of extraction, composition and function.* Algal research (Amsterdam), 2019. **39**: p. 101422.

112. Ivanova, V., et al., *Isolation of a polysaccharide with antiviral effect from Ulva lactuca.* Preparative biochemistry, 1994. **24**(2 p.83-97): p. 97-83.

113. Mao, W., et al., *Sulfated polysaccharides from marine green algae Ulva conglobata and their anticoagulant activity.* Journal of applied phycology, 2006. **18**(1): p. 9-14.

114. Qi, H., et al., *Antihyperlipidemic activity of high sulfate content derivative of polysaccharide extracted from Ulva pertusa (Chlorophyta).* Carbohydrate polymers, 2012. **87**(2): p. 1637-1640.

115. Alves, A., et al., *Processing ulvan into 2D structures: Cross-linked ulvan membranes as new biomaterials for drug delivery applications.* International journal of pharmaceutics, 2012. **426**(1-2): p. 76-81.

116. Mariia, K., et al., *Novel chitosan-ulvan hydrogel reinforcement by cellulose nanocrystals with epidermal growth factor for enhanced wound healing: In vitro and in vivo analysis.* International Journal of Biological Macromolecules, 2021. **183**: p. 435-446.

117. Ren, Y., et al., *hUC-MSCs lyophilized powder loaded polysaccharide ulvan driven functional hydrogel for chronic diabetic wound healing.* Carbohydrate Polymers, 2022. **288**.

118. Silva, T.H., et al., *Marine algae sulfated polysaccharides for tissue engineering and drug delivery approaches.* Biomatter (Austin, TX), 2012. **2**(4): p. 278-289.

119. Colliec-Jouault, S. and A. Zykwinska, *Marine Glycosaminoglycans (GAGs) and GAG-Mimetics: Applications in Medicine and Tissue Engineering*, in *Extracellular Sugar-Based Biopolymers Matrices*. 2019, Springer. p. 625-648.

120. Kolsi, R.B.A., et al., *Structural characterization and functional properties of antihypertensive Cymodocea nodosa sulfated polysaccharide.* Carbohydrate polymers, 2016. **151**: p. 511-522.

121. Otero, P., et al., *Seaweed polysaccharides: emerging extraction technologies, chemical modifications and bioactive properties*, in *Critical Reviews in Food Science & Nutrition*. 2023. p. 1901-1929.

122. Bi, D., et al., *Immune Activation of RAW264.7 Macrophages by Low Molecular Weight Fucoidan Extracted from New Zealand Undaria pinnatifida.* J Agric Food Chem, 2018. **66**(41): p. 10721-10728.

123. Campo, V.L., et al., *Carrageenans: Biological properties, chemical modifications and structural analysis – A review.* Carbohydrate Polymers, 2009. **77**(2): p. 167-180.

124. Liu, X., et al., *Anticoagulant Properties of a Green Algal Rhamnan-type Sulfated Polysaccharide and Its Low-molecular-weight Fragments Prepared by Mild Acid Degradation.* Marine drugs, 2018. **16**(11): p. 445.

125. Koyanagi, S., et al., *Oversulfation of fucoidan enhances its anti-angiogenic and antitumor activities.* Biochemical pharmacology, 2003. **65**(2): p. 173-179.

126. Jin, W., et al., *Interactions of fibroblast growth factors with sulfated galactofucan from Saccharina japonica.* International Journal of Biological Macromolecules, 2020.

127. Niu, T.-T., et al., *Modulation of the binding of basic fibroblast growth factor and heparanase activity by purified λ-carrageenan oligosaccharides.* Carbohydrate polymers, 2015. **125**: p. 76-84.

128. Jin, W., et al., *Structural characterization and anti-lung cancer activity of a sulfated glucurono-xylo-rhamnan from Enteromorpha prolifera.* Carbohydrate Polymers, 2020: p. 116143.

129. Soeda, S., et al., *Oversulfated fucoidan inhibits the basic fibroblast growth factor-induced tube formation by human umbilical vein endothelial cells: its possible mechanism of action.* Biochimica et Biophysica Acta (BBA)-Molecular Cell Research, 2000. **1497**(1): p. 127-134.

130. Soeda, S., Y. Shibata, and H. Shimeno, *Inhibitory effect of oversulfated fucoidan on tube formation by human vascular endothelial cells.* Biological and Pharmaceutical Bulletin, 1997. **20**(11): p. 1131-1135.

131. Liu, F., et al., *Fucoidan extract derived from Undaria pinnatifida inhibits angiogenesis by human umbilical vein endothelial cells.* Phytomedicine, 2012. **19**(8-9): p. 797-803.

132. Cong, Q., et al., *Structural characterization and effect on anti-angiogenic activity of a fucoidan from Sargassum fusiforme.* Carbohydrate polymers, 2016. **136**: p. 899-907.

133. Matsubara, K., et al., *Effects of middle molecular weight fucoidans on in vitro and ex vivo angiogenesis of endothelial cells.* International journal of molecular medicine, 2005. **15**(4): p. 695-699.

134. Matou, S., et al., *Effect of fucoidan on fibroblast growth factor-2-induced angiogenesis in vitro.* Thrombosis research, 2002. **106**(4-5): p. 213-221.

135. Lake, A.C., et al., *Low molecular weight fucoidan increases VEGF165-induced endothelial cell migration by enhancing VEGF165 binding to VEGFR-2 and NRP1.* Journal of Biological Chemistry, 2006. **281**(49): p. 37844-37852.

136. Zemani, F., et al., *Low-molecular-weight fucoidan enhances the proangiogenic phenotype of endothelial progenitor cells.* Biochemical Pharmacology, 2005. **70**(8): p. 1167-1175.

137. Kim, B.S., et al., *Fucoidan/FGF-2 induces angiogenesis through JNK-and p38-mediated activation of AKT/MMP-2 signalling.* Biochemical and Biophysical Research Communications, 2014. **450**(4): p. 1333-1338.

138. Geng, L., et al., *Comparative study of the effect of different fucoidans from Sargassum maclurei and Saccharina japonica on FGFs/FGFR signaling activation in BaF3 cells.* International Journal of Biological Macromolecules, 2018. **107**: p. 2429-2435.

139. Chen, A., et al., *The structure property and endothelial protective activity of fucoidan from Laminaria japonica.* International journal of biological macromolecules, 2017. **105**: p. 1421-1429.

140. Sun, C., et al., *Sulfated polysaccharides interact with fibroblast growth factors and protect from denaturation.* FEBS open bio, 2019. **9**(8): p. 1477-1487.

141. Kidgell, J.T., et al., *The molecular weight of ulvan affects the in vitro inflammatory response of a murine macrophage.* International Journal of Biological Macromolecules, 2020. **150**: p. 839-848.

142. Zeng, H.-Y. and Y.-C. Huang, *Basic fibroblast growth factor released from fucoidan-modified chitosan/alginate scaffolds for promoting fibroblasts migration.* Journal of Polymer Research, 2018. **25**(3): p. 83.

143. Ren, X., et al., *Growth Factor Engineering Strategies for Regenerative Medicine Applications.* Front Bioeng Biotechnol, 2019. **7**: p. 469.

144. Vasita, R. and D.S. Katti, *Growth factor-delivery systems for tissue engineering: a materials perspective.* Expert Rev Med Devices, 2006. **3**(1): p. 29-47.

145. Subbiah, R. and R.E. Guldberg, *Materials Science and Design Principles of Growth Factor Delivery Systems in Tissue Engineering and Regenerative Medicine.* Adv Healthc Mater, 2019. **8**(1): p. e1801000.

146. Santo, V.t.E.r., et al., *Carrageenan-Based Hydrogels for the Controlled Delivery of PDGF-BB in Bone Tissue Engineering Applications.* Biomacromolecules, 2009. **10**(6): p. 1392-1401.

147. Purnama, A., et al., *Fucoidan in a 3D scaffold interacts with vascular endothelial growth factor and promotes neovascularization in mice.* Drug Deliv Transl Res, 2015. **5**(2): p. 187-97.

148. Huang, Y.C. and Y.T. Yang, *Effect of basic fibroblast growth factor released from chitosan-fucoidan nanoparticles on neurite extension.* J Tissue Eng Regen Med, 2016. **10**(5): p. 418-27.

149. Mariia, K., et al., *Novel chitosan-ulvan hydrogel reinforcement by cellulose nanocrystals with epidermal growth factor for enhanced wound healing: In vitro and in vivo analysis.* Int J Biol Macromol, 2021. **183**: p. 435-446.

150. Qiao, M., et al., *Recent advances in biotechnology for heparin and heparan sulfate analysis.* Talanta, 2020. **219**: p. 121270.

151. Rabenstein, D.L., *Heparin and heparan sulfate: structure and function.* Nat Prod Rep, 2002. **19**(3): p. 312-31.

152. Shriver, Z., et al., *Heparin and heparan sulfate: analyzing structure and microheterogeneity.* Handb Exp Pharmacol, 2012(207): p. 159-76.

153. Ledin, J., et al., *Heparan sulfate structure in mice with genetically modified heparan sulfate production.* J Biol Chem, 2004. **279**(41): p. 42732-41.

154. van den Born, J., et al., *Novel heparan sulfate structures revealed by monoclonal antibodies.* J Biol Chem, 2005. **280**(21): p. 20516-23.

155. Taylor, S.L., et al., *By-Products of Heparin Production Provide a Diverse Source of Heparin-like and Heparan Sulfate Glycosaminoglycans.* Scientific Reports, 2019. **9**(1).

156. Warda, M., et al., *Isolation and characterization of heparan sulfate from various murine tissues.* Glycoconj J, 2006. **23**(7-8): p. 555-63.

157. Mhanna, R., et al., *Sulfated Alginate as a Mimic of Sulfated Glycosaminoglycans: Binding of Growth Factors and Effect on Stem Cell Behavior.* Adv Biosyst, 2017. **1**(7): p. e1700043.

158. Doncel-Perez, E., et al., *Synthesis, physicochemical characterization and biological evaluation of chitosan sulfate as heparan sulfate mimics.* Carbohydr Polym, 2018. **191**: p. 225-233.

159. Purcell, B.P., et al., *Incorporation of Sulfated Hyaluronic Acid Macromers into Degradable Hydrogel Scaffolds for Sustained Molecule Delivery.* Biomater Sci, 2014. **2**: p. 693-702.

160. Zhou, Z., et al., *Chemical synthesis and pharmacological properties of heparin pentasaccharide analogues.* Eur J Med Chem, 2022. **234**: p. 114256.

161. Zhang, X., et al., *Chemoenzymatic synthesis of heparan sulfate and heparin oligosaccharides and NMR analysis: paving the way to a diverse library for glycobiologists.* Chem Sci, 2017. **8**(12): p. 7932-7940.

162. Wander, R., et al., *Structural and substrate specificity analysis of 3-O-sulfotransferase isoform 5 to synthesize heparan sulfate.* ACS Catal, 2021. **11**(24): p. 14956-14966.

163. Janani, M., C. Ramalingam, and S. Sivaramakrishnan, *Sulfated polysaccharides and its commercial applications in food industries-a review*, in *Journal of Food Science & Technology*. 2021. p. 2453-2466.

164. Priyan Shanura Fernando, I., et al., *Algal polysaccharides: potential bioactive substances for cosmeceutical applications.* Critical Reviews in Biotechnology, 2019. **39**(1): p. 99-113.

165. Isuru, W., P. Ratih, and K. Se-Kwon, *Biological activities and potential health benefits of sulfated polysaccharides derived from marine algae*, in *Carbohydrate Polymers*. 2011. p. 14-21.

166. Cunha, L. and A. Grenha, *Sulfated Seaweed Polysaccharides as Multifunctional Materials in Drug Delivery Applications.* MARINE DRUGS, 2016. **14**(3).

167. Li, L., et al., *Carrageenan and its applications in drug delivery.* Carbohydr Polym, 2014. **103**: p. 1-11.

168. Kidgell, J.T., et al., *Ulvan: A systematic review of extraction, composition and function.* Algal Research, 2019. **39**.

169. Farooq, M., et al., *The Role of Fibroblast Growth Factor (FGF) Signaling in Tissue Repair and Regeneration.* Cells, 2021. **10**(11).

170. Coombe, D.R. and W.C. Kett, *Heparan sulfate-protein interactions: therapeutic potential through structure-function insights.* Cellular and Molecular Life Sciences CMLS, 2005. **62**(4): p. 410-424.

171. Seo, J.H., et al., *Facile and rapid direct gold surface immobilization with controlled orientation for carbohydrates.* Bioconjug Chem, 2007. **18**(6): p. 2197-201.

172. Wahlström, N., et al., *Composition and structure of cell wall ulvans recovered from Ulva spp. along the Swedish west coast.* Produktionssystem för alger med högvärdiga tillämpningar (Sweaweed) Carbohydrate Polymers, 2020. **233**.

173. Rabenstein, D.L., *Heparin and heparan sulfate: structure and function.* NATURAL PRODUCT REPORTS, 2002. **19**(3): p. 312-331.

174. Sushko, N.I., et al., *Vibrational spectra of heparins.* Journal of Applied Spectroscopy, 1994. **61**(5-6): p. 704-707.

175. Grant, D., et al., *Infrared spectroscopy of heparins suggests that the region 750-950 cm-1 is sensitive to changes in iduronate residue ring conformation.* Biochemical Journal, 1991. **275**(1): p. 193-197.

176. Thanh, T.T., et al., *Structure and cytotoxic activity of ulvan extracted from green seaweed Ulva lactuca.* Int J Biol Macromol, 2016. **93**(Pt A): p. 695-702.

177. Trivedi, N., et al., *An integrated process for the extraction of fuel and chemicals from marine macroalgal biomass.* Sci Rep, 2016. **6**: p. 30728.

178. Gómez-Ordóñez, E. and P. Rupérez, *FTIR-ATR spectroscopy as a tool for polysaccharide identification in edible brown and red seaweeds.* Food Hydrocolloids, 2011. **25**(6): p. 1514-1520.

179. Korva, H., et al., *Spectroscopic study of natural and synthetic polysaccharide sulfate structures.* Starch - Stärke, 2016. **68**(9-10): p. 854-863.

180. Pereira, L., et al., *Identification of selected seaweed polysaccharides (phycocolloids) by vibrational spectroscopy (FTIR-ATR and FT-Raman).* Food Hydrocolloids, 2009. **23**(7): p. 1903-1909.

181. Lefebvre, J., et al., *Experimental methods in chemical engineering: X-ray photoelectron spectroscopy-XPS.* CANADIAN JOURNAL OF CHEMICAL ENGINEERING, 2019.

182. Carneiro de Oliveira, J., et al., *Grafting Diels-Alder moieties on cellulose nanocrystals through carbamation.* Carbohydrate Polymers, 2020. **250**.

183. Rabee, A.I.M., et al., *Acidity-reactivity relationships in catalytic esterification over ammonium sulfate-derived sulfated zirconia.* Catalysts, 2017. **7**(7).

184. Rodiahwati, W., T.C. Brown, and B.W. Greatrex, *Formation of levoglucosenone and furfural from three different carrageenans via acid-catalyzed pyrolysis in polyethylene glycol.* Bioresource Technology Reports, 2023. **23**.

185. Rochas, C., M. Lahaye, and W. Yaphe, *Sulfate Content of Carrageenan and Agar Determined by Infrared Spectroscopy.* Botanica Marina, 1986. **29**(4): p. 335-340.

186. Yan, Y., et al., *Structure comparison and anti-coagulant effects of the glycosaminoglycans from porcine duodenum and jejunum.* Food Chemistry, 2022. **373**.

187. Keire, D.A., et al., *Assay of possible economically motivated additives or native impurities levels in heparin by 1H NMR, SAX-HPLC, and anticoagulation time approaches*, in *Journal of Pharmaceutical and Biomedical Analysis*. 2010. p. 656-664.

188. Kuwabara, K., et al., *Cellular interaction and cytotoxicity of the Iowa mutation of apolipoprotein A-I (ApoA-IIowa) amyloid mediated by sulfate moieties of heparan sulfate.* Journal of Biological Chemistry, 2015. **290**(40): p. 24210-24221.

189. Wang, J., et al., *Regioselective sulfation of Artemisia sphaerocephala polysaccharide: Characterization of chemical structure.* Carbohydrate polymers, 2015. **133**: p. 327-320.

190. Rodríguez Sánchez, R.A., M.C. Matulewicz, and M. Ciancia, *NMR spectroscopy for structural elucidation of sulfated polysaccharides from red seaweeds.* International Journal of Biological Macromolecules, 2022. **199**: p. 386-400.

191. Chi, Y., et al., *Structural characterization of ulvan extracted from Ulva clathrata assisted by an ulvan lyase.* Carbohydrate Polymers, 2020. **229**.

192. Yang, C., et al., *Heparan sulfate glycomimetics via iterative assembly of "clickable" disaccharides.* Chemical Science, 2023. **14**(13): p. 3514-3522.

193. Mhanna, R., et al., *Sulfated Alginate as a Mimic of Sulfated Glycosaminoglycans: Binding of Growth Factors and Effect on Stem Cell Behavior*, in *Advanced biosystems*. 2017, Wiley-VCH Verlag GmbH & Co: Germany. p. e1700043.

194. Malaeb, W., et al., *The sulfation of biomimetic glycosaminoglycan substrates controls binding of growth factors and subsequent neural and glial cell growth.* Biomaterials Science (2047-4830), 2019. **7**(10): p. 4283-4298.

195. Miura, T., et al., *Highly sulfated hyaluronic acid maintains human induced pluripotent stem cells under feeder-free and bFGF-free conditions.* Biochemical & Biophysical Research Communications, 2019. **518**(3): p. 506-512.

196. Zhang, K., et al., *Synthesis and Bioactivity of Cellulose Derivatives.* Macromolecular symposia, 2009. **280**(1): p. 28-35.

197. Doncel-Pérez, E., et al., *Synthesis, physicochemical characterization and biological evaluation of chitosan sulfate as heparan sulfate mimics.* Carbohydrate Polymers, 2018. **191**: p. 225-233.

198. Sargison, L., et al., *Variability in the composition of porcine mucosal heparan sulfates.* Carbohydrate Polymers, 2022. **282**.

199. Zhang, F., et al., *Comparison of the interactions of different growth factors and glycosaminoglycans.* Molecules, 2019. **24**(18).

200. Guo, S. and L.A. Dipietro, *Factors affecting wound healing*, in *Journal of dental research*. 2010, Sage: United States. p. 219-229.

201. Ronca, R., et al., *The potential of fibroblast growth factor/fibroblast growth factor receptor signaling as a therapeutic target in tumor angiogenesis.* EXPERT OPINION ON THERAPEUTIC TARGETS, 2015. **19**(10): p. 1361-1377.

202. Tonnesen, M.G., X. Feng, and R.A.F. Clark, *Angiogenesis in Wound Healing*, in *The Journal of Investigative Dermatology Symposium Proceedings*. 2000. p. 40-46.

203. Veith, A.P., et al., *Therapeutic strategies for enhancing angiogenesis in wound healing.* Advanced Drug Delivery Reviews, 2019. **146**: p. 97-125.

204. Francesca, C., R. Chiara, and M. Monica, *Role of fibroblasts in wound healing and tissue remodeling on Earth and in space*, in *Frontiers in Bioengineering and Biotechnology*. 2022, Frontiers Media S.A.

205. Bainbridge, P., *Wound healing and the role of fibroblasts.* Journal of Wound Care, 2013. **22**(8): p. 407-411.

206. Yong Huan, S., et al., *Distribution of fibroblast growth factors and their roles in skin fibroblast cell migration.* Molecular Medicine Reports, 2016. **14**(4): p. 3336-3342.

207. Matsuo, I. and C. Kimura-Yoshida, *Extracellular modulation of Fibroblast Growth Factor signaling through heparan sulfate proteoglycans in mammalian development.* Current Opinion in Genetics & Development, 2013. **23**(4): p. 399-407.

208. Farooq, M., et al., *The Role of Fibroblast Growth Factor (FGF) Signaling in Tissue Repair and Regeneration.* CELLS, 2021. **10**(11): p. 3242.

209. Yuan, X., et al., *Cinnamaldehyde accelerates wound healing by promoting angiogenesis via up-regulation of PI3K and MAPK signaling pathways.* Laboratory Investigation, 2018. **98**(6): p. 783-798.

210. Geng, K., et al., *Electrical stimulation facilitates the angiogenesis of human umbilical vein endothelial cells through MAPK/ERK signaling pathway by stimulating FGF2 secretion.* American Journal of Physiology - Cell Physiology, 2019. **317**(2): p. C277-C286.

211. Roskoski, J.R., *ERK1/2 MAP kinases: Structure, function, and regulation.* Pharmacological Research, 2012. **66**(2): p. 105-143.

212. Ferreras, C., et al., *Endothelial Heparan Sulfate 6-O-Sulfation Levels Regulate Angiogenic Responses of Endothelial Cells to Fibroblast Growth Factor 2 and Vascular Endothelial Growth Factor*.* Journal of Biological Chemistry, 2012. **287**(43): p. 36132-36146.

213. Ganesan, P., et al., *Marine algal carotenoids inhibit angiogenesis by down-regulating FGF-2-mediated intracellular signals in vascular endothelial cells.* Molecular and Cellular Biochemistry: An International Journal for Chemical Biology in Health and Disease, 2013. **380**(1-2): p. 1-9.

214. Faulkner, J., X. Song, and L. Wang, *Application of a Mutant Cell Library to Determine the Structure-Function Relationship of Heparan Sulfate in Facilitating FGF2-FGFR1 Signaling*, in *Methods in molecular biology (Clifton, N.J.)*. 2022, Humana Press: United States. p. 637-644.

215. Song, Y.-Y., et al., *The role of the ERK signaling pathway in promoting angiogenesis for treating ischemic diseases.* FRONTIERS IN CELL AND DEVELOPMENTAL BIOLOGY, 2023. **11**: p. 1164166.

216. Forsten-Williams, K., C.C. Chua, and M.A. Nugent, *The kinetics of FGF-2 binding to heparan sulfate proteoglycans and MAP kinase signaling.* Journal of Theoretical Biology, 2005. **233**(4): p. 483-499.

217. Miller, G.J., et al., *Efficient chemical synthesis of heparin-like octa-, deca- and dodecasaccharides and inhibition of FGF2- and VEGF165-mediated endothelial cell functions.* Chemical Science, 2013. **4**(8): p. 3218-3222.

218. Han, G., et al., *Different influence of sulfated chitosan with different sulfonic acid group sites on HUVECs behaviors.* Journal of Biomaterials Science -- Polymer Edition, 2020. **31**(10): p. 1237-1253.

219. Matou, S., et al., *Effect of an oversulfated exopolysaccharide on angiogenesis induced by fibroblast growth factor-2 or vascular endothelial growth factor in vitro.* Biochemical Pharmacology, 2005. **69**(5): p. 751-759.

220. Andrew, C.A., et al., *Fundamentals of FGF19 & FGF21 action in vitro and in vivo*, in *PLoS ONE*. 2012, Public Library of Science (PLoS). p. e38438-e38438.

221. Presta, M., et al., *Fibroblast growth factor/fibroblast growth factor receptor system in angiogenesis.* Cytokine and Growth Factor Reviews, 2005. **16**(2): p. 159-178.

222. Guimond, S.E. and J.E. Turnbull, *Fibroblast growth factor receptor signalling is dictated by specific heparan sulphate saccharides.* Current Biology, 1999. **9**(22): p. 1343.

223. Turnbull, J.E., et al., *Identification of the basic fibroblast growth factor binding sequence in fibroblast heparan sulfate.* Journal of Biological Chemistry, 1992. **267**(15): p. 10337-10341.

224. Schultz, V., et al., *Heparan Sulfate Domains Required for Fibroblast Growth Factor 1 and 2 Signaling through Fibroblast Growth Factor Receptor 1c.* Journal of Biological Chemistry, 2017. **292**(6): p. 2495-2509.

225. Lord, M.S., et al., *Synthesis and characterization of water soluble biomimetic chitosans for bone and cartilage tissue regeneration.* J Mater Chem B, 2014. **2**(38): p. 6517-6526.

226. Leali, D., et al., *Fibroblast Growth Factor-2 Antagonist Activity and Angiostatic Capacity of Sulfated Escherichia coli K5 Polysaccharide Derivatives.* Journal of Biological Chemistry, 2001. **276**(41): p. 37900-37908.

227. Wu, J., et al., *Anticoagulant and FGF/FGFR signal activating activities of the heparinoid propylene glycol alginate sodium sulfate and its oligosaccharides.* Carbohydrate polymers, 2016. **136**: p. 648-641.

228. Pye, D.A. and S. Kumar, *Endothelial and fibroblast cell-derived heparan sulphate bind with differing affinity to basic fibroblast growth factor.* Biochemical and Biophysical Research Communications, 1998. **248**(3): p. 889-895.

229. Kim, C.W., et al., *Members of the syndecan family of heparan sulfate proteoglycans are expressed in distinct cell-, tissue-, and development-specific patterns*, in *Molecular biology of the cell*. 1994, American Society for Cell Biology: United States. p. 797-805.

230. Lories, V., et al., *Differential expression of cell surface heparan sulfate proteoglycans in human mammary epithelial cells and lung fibroblasts.* Journal of Biological Chemistry, 1992. **267**(2): p. 1116-1122.

231. Hongyan, Z., et al., *The heparan sulfate co-receptor and the concentration of fibroblast growth factor-2 independently elicit different signaling patterns from the fibroblast growth factor receptor.* Cell Communication & Signaling, 2010. **8**: p. 14-23.

232. Ferreras, C., et al., *Endothelial heparan sulfate 6-O-sulfation levels regulate angiogenic responses of endothelial cells to fibroblast growth factor 2 and vascular endothelial growth factor.* J Biol Chem, 2012. **287**(43): p. 36132-46.

233. Tefft, J.B., C.S. Chen, and J. Eyckmans, *Reconstituting the dynamics of endothelial cells and fibroblasts in wound closure.* APL Bioengineering, 2021. **5**(1).

234. Maheshwari, G., et al., *Biophysical integration of effects of epidermal growth factor and fibronectin on fibroblast migration*, in *Biophysical journal*. 1999, Cell Press: United States. p. 2814-2823.

235. Seppä, H., et al., *Platelet-Derived Growth Factor Is Chemotactic for Fibroblasts.* The Journal of Cell Biology, 1982. **92**(2): p. 584-588.

236. Pierce, G.F., et al., *Role of platelet-derived growth factor in wound healing.* Journal of Cellular Biochemistry, 1991. **45**(4): p. 319-326.

237. Kanazawa, S., et al., *bFGF Regulates PI3-Kinase-Rac1-JNK Pathway and Promotes Fibroblast Migration in Wound Healing.* PLoS ONE, 2010. **5**(8): p. 1-12.

238. Heldin, C.-H. and B. Westermark, *Mechanism of action and in vivo role of platelet-derived growth factor.* Physiological Reviews, 1999. **79**(4): p. 1283.

239. Miller, G.J., et al., *Efficient chemical synthesis of heparin-like octa-, deca- and dodecasaccharides and inhibition of FGF2- and VEGF165-mediated endothelial cell functions.* Chemical Science, 2013. **4**(8): p. 3218-3222.

240. Williamson, K.A., et al., *Age-related impairment of endothelial progenitor cell migration correlates with structural alterations of heparan sulfate proteoglycans.* Aging Cell, 2013. **12**(1): p. 139-147.

241. Soga, N., et al., *Rac regulates vascular endothelial growth factor stimulated motility*, in *Cell communication & adhesion*. 2001, Taylor & Francis: England. p. 1-13.

242. Keren, I.H. and L.H. Renee, *Cell Migration and Invasion Assays as Tools for Drug Discovery*, in *Pharmaceutics*. 2011, MDPI AG. p. 107-124.

243. Monsuur, H.N., et al., *Methods to study differences in cell mobility during skin wound healing in vitro.* Journal of Biomechanics, 2016. **49**(8): p. 1381-1387.

244. Johan van, N., et al., *Heparan Sulfate Proteoglycan Mimetics Promote Tissue Regeneration: An Overview*, in *Tissue Regeneration*, D. Jamie, Editor. 2012, IntechOpen: Rijeka. p. Ch. 4.

245. Tanaka, K., M. Abe, and Y. Sato, *Roles of extracellular signal-regulated kinase 1/2 and p38 mitogen-activated protein kinase in the signal transduction of basic fibroblast growth factor in endothelial cells during angiogenesis.* Japanese Journal of Cancer Research, 1999. **90**(6): p. 647-654.

246. Kaikai, S., et al., *Critical role of c-Jun N-terminal kinase in regulating bFGF-induced angiogenesis in vitro*, in *The Journal of Biochemistry*. 2011. p. 189-197.

247. Lundin, L., et al., *Selectively desulfated heparin inhibits fibroblast growth factor-induced mitogenicity and angiogenesis.* Journal of Biological Chemistry, 2000. **275**(32): p. 24653-24660.

248. Kang, J., et al., *Insights into the structure-bioactivity relationships of marine sulfated polysaccharides: A review.* J Food Hydrocolloids, 2022. **123**: p. 107049.

249. Wang, W., et al., *Preparation and anti-influenza A virus activity of κ-carrageenan oligosaccharide and its sulphated derivatives.* Food Chemistry, 2012. **133**(3): p. 880-888.

250. Li, H., et al., *Structural characterization of an anticoagulant-active sulfated polysaccharide isolated from green alga Monostroma latissimum.* Carbohydrate Polymers, 2011. **85**(2): p. 394-400.

251. Ponce, N.M.A., et al., *Fucoidans from the phaeophyta Scytosiphon lomentaria: Chemical analysis and antiviral activity of the galactofucan component.* Carbohydrate Research, 2019. **478**: p. 18-24.

252. Jiao, L., et al., *Antitumor and Immunomodulating Activity of Polysaccharides from Enteromorpha intestinalis*, in *Biotechnology and bioprocess engineering*. 2010, Korean Society for Biotechnology and Bioengineering. p. 421-428.

253. Otero, P., et al., *Seaweed polysaccharides: Emerging extraction technologies, chemical modifications and bioactive properties.* Crit Rev Food Sci Nutr, 2023. **63**(13): p. 1901-1929.

254. Opoku, G., X. Qiu, and V. Doctor, *Effect of oversulfation on the chemical and biological properties of kappa carrageenan.* Carbohydrate Polymers, 2006. **65**(2): p. 134-138.

255. de Araújo, C.A., et al., *Selective sulfation of carrageenans and the influence of sulfate regiochemistry on anticoagulant properties.* Carbohydrate Polymers, 2013. **91**(2): p. 483-491.

256. Qi, H. and Y. Sun, *Antioxidant activity of high sulfate content derivative of ulvan in hyperlipidemic rats.* International journal of biological macromolecules, 2015. **76**: p. 329-326.

257. Li, L., et al., *Carrageenan and its applications in drug delivery.* Carbohydrate Polymers, 2014. **103**: p. 1-11.

258. Capila, I. and R.J. Linhardt, *Heparin-protein interactions*, in *Angewandte Chemie (International ed. in English)*. 2002, Wiley-VCH: Germany. p. 391-412.

259. Karnjanapratum, S. and S. You, *Molecular characteristics of sulfated polysaccharides from Monostroma nitidum and their in vitro anticancer and immunomodulatory activities.* International Journal of Biological Macromolecules, 2011. **48**(2): p. 311-318.

260. El Azm, N.A., et al., *Production of bioactive compounds from the sulfated polysaccharides extracts of ulva lactuca: post-extraction enzymatic hydrolysis followed by ion-exchange chromatographic fractionation.* Molecules, 2019. **24**(11).

261. Jiao, G., et al., *Chemical structures and bioactivities of sulfated polysaccharides from marine algae.* Mar Drugs, 2011. **9**(2): p. 196-223.

262. Hou, N., et al., *Polysaccharides and their depolymerized fragments from Costaria costata: Molecular weight and sulfation-dependent anticoagulant and FGF/FGFR signal activating activities.* Int J Biol Macromol, 2017. **105**(Pt 2): p. 1511-1518.

263. Chen, A., et al., *The structure property and endothelial protective activity of fucoidan from Laminaria japonica.* Int J Biol Macromol, 2017. **105**(Pt 2): p. 1421-1429.

264. Geng, L., et al., *Comparative study of the effect of different fucoidans from Sargassum maclurei and Saccharina japonica on FGFs/FGFR signaling activation in BaF3 cells*, in *International journal of biological macromolecules*. 2018, Elsevier: Netherlands. p. 2429-2435.

265. Olczyk, P., Ł. Mencner, and K. Komosinska-Vassev, *Diverse Roles of Heparan Sulfate and Heparin in Wound Repair*, in *BioMed research international*. 2015, Hindawi Pub. Co: United States. p. 549417.

266. Sarrazin, S., W.C. Lamanna, and J.D. Esko, *Heparan sulfate proteoglycans*, in *Cold Spring Harbor perspectives in biology*. 2011, Cold Spring Harbor Laboratory Press: United States.

267. Powell, A.K., et al., *Interactions of heparin/heparan sulfate with proteins: Appraisal of structural factors and experimental approaches*, in *Glycobiology*. 2004. p. 17R-30R.

268. Saxena, K., et al., *Influence of Heparin Mimetics on Assembly of the FGF·FGFR4 Signaling Complex.* Journal of Biological Chemistry, 2010. **285**(34): p. 26628-26640.

269. Hongyan, Z., et al., *The heparan sulfate co-receptor and the concentration of fibroblast growth factor-2 independently elicit different signaling patterns from the fibroblast growth factor receptor.* Cell Communication & Signaling, 2010. **8**: p. 14-23.

270. Ashikari-Hada, S., et al., *Specific inhibition of FGF-2 signaling with 2-O-sulfated octasaccharides of heparan sulfate.* Glycobiology, 2009. **19**(6): p. 644-654.

271. Lee, K., E.A. Silva, and D.J. Mooney, *Growth factor delivery-based tissue engineering: general approaches and a review of recent developments*, in *Journal of the Royal Society, Interface*. 2011, Royal Society: England. p. 153-170.

272. Bao, P., et al., *The Role of Vascular Endothelial Growth Factor in Wound Healing.* Journal of Surgical Research, 2009. **153**(2): p. 347-358.

273. Shirakata, Y., et al., *Heparin-binding EGF-like growth factor accelerates keratinocyte migration and skin wound healing.* Journal of Cell Science, 2005. **118**(11): p. 2363-2370.

274. Fu, Y., et al., *Structure-activity relationships of bioactive polysaccharides extracted from macroalgae towards biomedical application: A review.* Carbohydrate Polymers, 2023.

275. Wang, Z., et al., *Molecular Mass Characterization of Glycosaminoglycans with Different Degrees of Sulfation in Bioengineered Heparin Process by Size Exclusion Chromatography*, in *Current analytical chemistry*. 2012, Bentham Science Publishers: United Arab Emirates. p. 506-511.

276. Wijelath, E., et al., *Multiple mechanisms for exogenous heparin modulation of vascular endothelial growth factor activity.* Journal of Cellular Biochemistry, 2010. **111**(2): p. 461-468.

277. Rui Manuel Santos Costa de, M., R. Maria Filomena de Jesus, and M. Alcina Maria Miranda Bernardo de, *Bioactivity and Applications of Sulphated Polysaccharides from Marine Microalgae*, in *Marine Drugs*. 2013, MDPI AG. p. 233-252.

278. Jia Sen, T., et al., *A review on microalgae cultivation and harvesting, and their biomass extraction processing using ionic liquids*, in *Bioengineered*. 2020, Taylor & Francis Group. p. 116-129.

279. Srimongkol, P., et al., *Sulfated polysaccharides derived from marine microalgae, Synechococcus sp. VDW, inhibit the human colon cancer cell line Caco-2 by promoting cell apoptosis via the JNK and p38 MAPK signaling pathway.* Algal Research, 2023. **69**.

280. Sasisekharan, R., et al., *Roles of heparan-sulphate glycosaminoglycans in cancer.* Nature Reviews Cancer, 2002. **2**(7): p. 521.

281. Pastar, I., et al., *Epithelialization in Wound Healing: A Comprehensive Review*, in *Advances in wound care*. 2014, Mary Ann Liebert, Inc: United States. p. 445-464.

282. Maxson, S., et al., *Concise review: role of mesenchymal stem cells in wound repair*, in *Stem cells translational medicine*. 2012, Oxford University Press: England. p. 142-149.

283. Ojeh, N., et al., *Ectopic Expression of Syndecan-1 in Basal Epidermis Affects Keratinocyte Proliferation and Wound Re-Epithelialization.* Journal of Investigative Dermatology, 2008. **128**(1): p. 26-34.

284. Christian, D., et al., *Heparan Sulfate Mediates the Proliferation and Differentiation of Rat Mesenchymal Stem Cells.* Stem Cells & Development, 2009. **18**(4): p. 661-670.

285. Maffucci, T., et al., *A Phosphoinositide 3-Kinase/Phospholipase Cgamma1 Pathway Regulates Fibroblast Growth Factor-Induced Capillary Tube Formation.* PLoS ONE, 2009. **4**(12): p. 1-10.

286. Chen, J., et al., *The Key Regulatory Roles of the PI3K/Akt Signaling Pathway in the Functionalities of Mesenchymal Stem Cells and Applications in Tissue Regeneration.* Tissue Engineering Part B: Reviews, 2013. **19**(6): p. 516-528.

Appendix

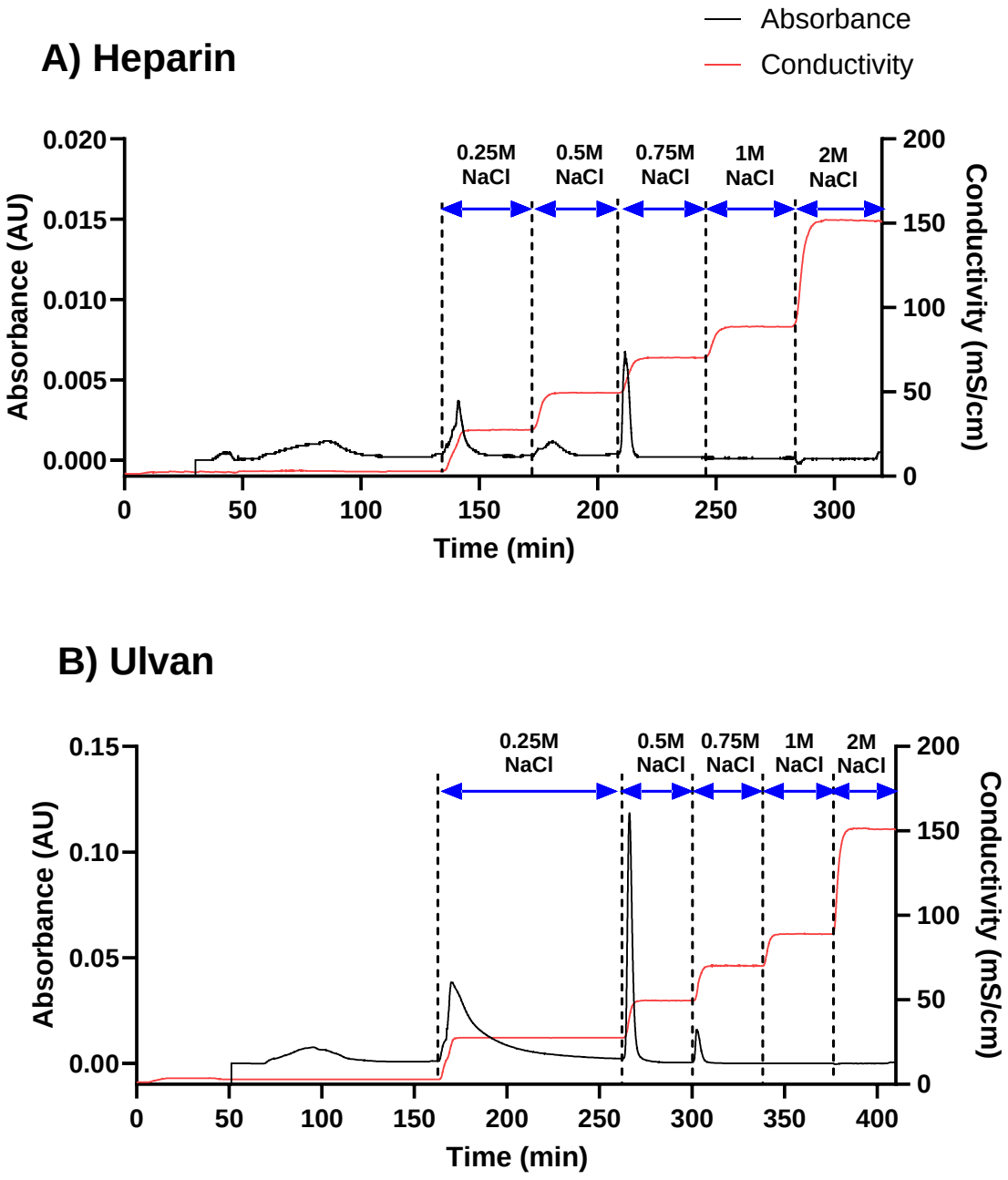


Figure A.1. Full AEC profiles of A) heparin and B) ulvan showing the fractions eluted at salt concentrations of 0.25 M, 0.5 M, 0.75 M, 1 M, and 2 M NaCl.

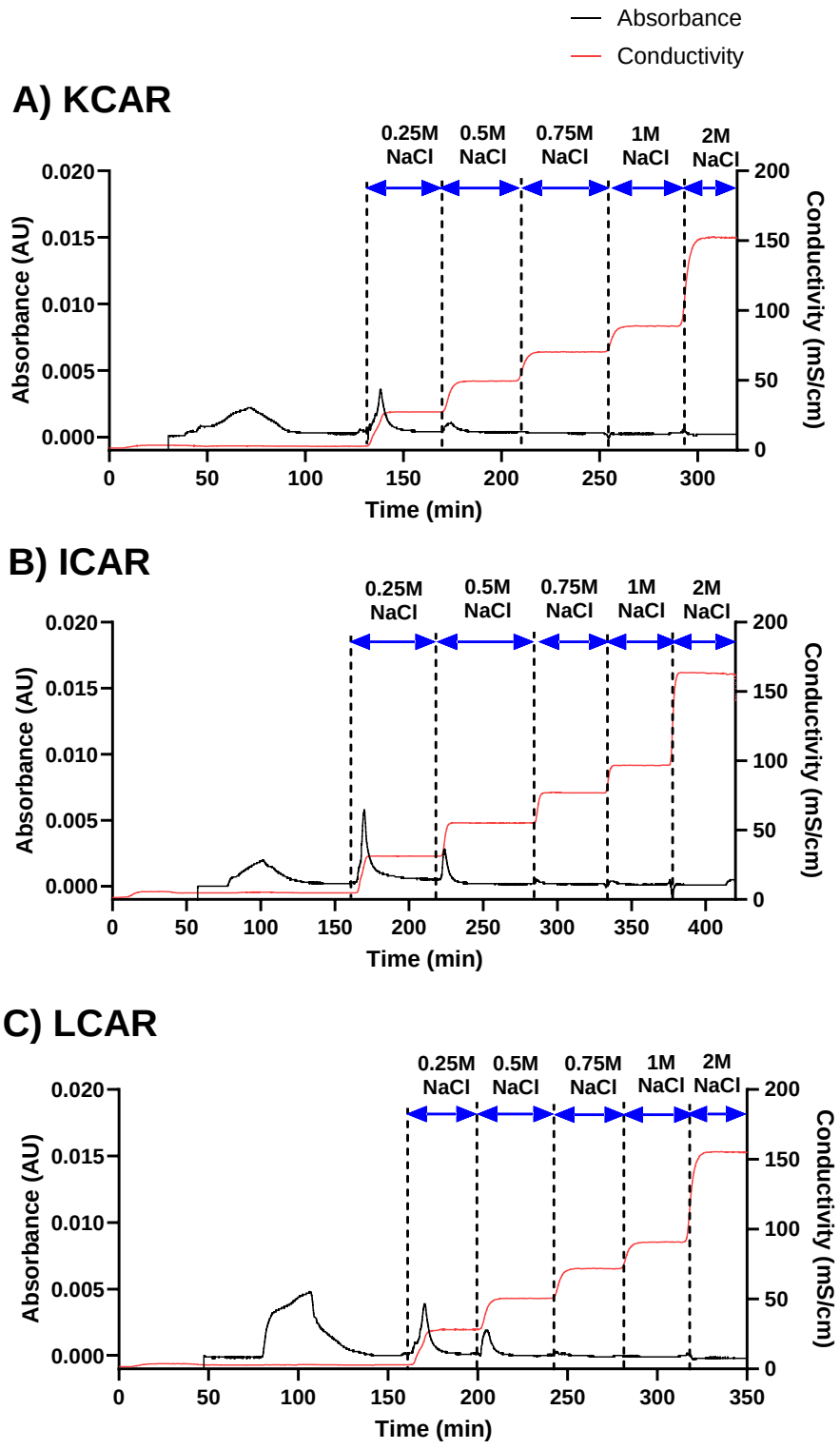


Figure A.2. Full AEC profiles of A) KCAR, B) ICAR, and C) LCAR showing the fractions eluted at salt concentrations of 0.25 M, 0.5 M, 0.75 M, 1 M, and 2 M NaCl.